Cache-Only Temporal Refinement of S2M2-S on SCARED

Current Findings, Negative Results, and Next Research Direction

ARGOS Internal Research Note

August 17, 2026

Abstract

This document summarizes the current experimental findings on temporal refinement for surgical stereo depth/disparity estimation within the ARGOS project. The current benchmark evaluates cached S2M2-S predictions on an audited SCARED temporal-GT sequence, comparing raw predictions, fixed causal exponential moving average (EMA), no-flow adaptive EMA variants, RAFT-based motion-compensated EMA, and StereoAnyVideo as an external video-model comparison. The main finding is that a very simple fixed EMA with $\alpha = 0.35$ is an unexpectedly strong deployment baseline, substantially improving temporal stability and depth error over raw S2M2-S at negligible cost. No-RAFT adaptive heuristics based on raw-vs-memory disagreement and disparity gradients did not outperform the fixed EMA baseline. RAFT-warped EMA remains the best quality method, but its online cost is significantly higher. These results suggest that simple adaptive rules without explicit motion compensation are insufficient, and that the next meaningful research direction should focus either on lightweight flow-based motion compensation, such as RAFT-Small, or on distilling a RAFT-based temporal teacher into a lightweight flow-free recurrent student.

1 Context and Research Question

The broader goal is to build a practical temporal refinement pipeline for surgical stereo reconstruction. The target setting is not generic video depth estimation, but lightweight and causal temporal refinement for surgical scenes, where deployment constraints matter.

The current research question is:

> *Can we improve the temporal consistency and geometric accuracy of S2M2-S disparity predictions using lightweight causal temporal fusion, without running a heavy video model or optical flow estimator online?*

This question emerged after observing that a full RAFT-warped EMA method improves quality, but is too expensive for a lightweight deployment scenario. The key tension is therefore:

motion-aware temporal fusion improves quality vs. **online optical flow is expensive**.

2 Dataset and Cached Inputs

The current benchmark is cache-only: no model inference is rerun, and no training is performed.

2.1 Audited SCARED Temporal-GT Sequence

The audited sequence is:

```
dataset/SCARED/curated/temporal_gt/test_dataset_9_keyframe_3
```

The audit found:

- 130 left/right RGB frames.
- 130 ground-truth disparity maps.
- 130 ground-truth depth maps.
- 130 valid masks.
- 130 calibration files.
- 129 adjacent frame pairs.
- 103 valid frames retained after validity filtering.
- 102 valid temporal pairs used for temporal metrics.

2.2 Prediction Caches

The benchmark uses cached prediction folders:

```
results/03_temporal_refinement/evaluation/gt_temporal_test_dataset_9_keyframe_3/
    predictions/S2M2-S_512
```

```
results/03_temporal_refinement/evaluation/gt_temporal_test_dataset_9_keyframe_3/
    predictions/StereoAnyVideo_384x640
```

The RAFT flow cache is used only for metric computation and for evaluating already-defined RAFT-based baselines:

```
results/04_dataset_derivatives/SCARED/temporal_gt_flow_cache/test_dataset_9_keyframe_3/
    raft
```

For no-RAFT adaptive methods, the flow cache is explicitly **not** used in the prediction formula.

3 Benchmark Design

3.1 Cache-Only Evaluation

The benchmark is deliberately cache-only. It does not run:

- S2M2-S inference.
- StereoAnyVideo inference.
- RAFT inference.
- Training.
- Dataset modification.
- Prediction-cache modification.

This design isolates the effect of the temporal fusion/refinement rule.

3.2 Main Evaluated Methods

The evaluated families are:

1. **S2M2-S raw**: cached single-frame S2M2-S predictions.
2. **Fixed EMA**: causal temporal smoothing with fixed global α.
3. **RAFT-warped EMA**: motion-compensated EMA using cached RAFT flow.
4. **Confidence/occlusion reset**: RAFT-based warped EMA with reset logic.
5. **Conservative adaptive EMA with RAFT**: RAFT-based adaptive variant.
6. **StereoAnyVideo**: cached video-model comparison, not ground truth.
7. **No-RAFT adaptive EMA**: new lightweight adaptive heuristics without flow.

4 Temporal Fusion Methods

4.1 Fixed Causal EMA

The fixed EMA baseline is:

$$\hat{d}_t = \alpha d_t + (1-\alpha)\hat{d}_{t-1}, \tag{1}$$

where d_t is the raw S2M2-S disparity prediction at time t, and $\hat{d}_t$ is the refined disparity.
The best fixed EMA found so far is:

$$\alpha = 0.35. \tag{2}$$

This means the method strongly reuses temporal memory while still incorporating the current prediction.

4.2 RAFT-Warped EMA

The RAFT-warped EMA uses optical flow to warp the previous refined disparity into the current frame before fusion:

$$\hat{d}_t = \alpha d_t + (1-\alpha)W(\hat{d}_{t-1}, F_{t-1\to t}), \tag{3}$$

where $W(\cdot)$ denotes warping and $F_{t-1\to t}$ is the forward optical flow.
This method is useful as a motion-compensated reference, but it requires online flow if deployed directly.

4.3 No-RAFT Adaptive EMA: Raw-vs-Memory Disagreement

The first no-flow adaptive variant uses pixel-wise disagreement between the current raw prediction and previous temporal memory:

$$r_t = \text{clip}\left(\frac{|d_t - \hat{d}_{t-1}|}{s_{\text{diff}}}, 0, 1\right), \tag{4}$$

$$\alpha_t = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min})r_t, \tag{5}$$

$$\hat{d}_t = \alpha_t d_t + (1-\alpha_t)\hat{d}_{t-1}. \tag{6}$$

The intended interpretation is:

- If raw and memory agree, trust memory more.
- If raw and memory disagree, trust raw more.

The tested sweep was:

- $\alpha_{\min} \in \{0.25, 0.30, 0.35, 0.40\}$.
- $\alpha_{\max} \in \{0.70, 0.80, 0.90\}$.
- $s_{\text{diff}} \in \{2.0, 3.0, 5.0, 8.0\}$.

4.4 No-RAFT Adaptive EMA: Disagreement + Disparity Gradient

The second no-flow adaptive variant adds disparity-gradient protection:

$$r_{\text{diff}} = \text{clip}\left(\frac{|d_t - \hat{d}_{t-1}|}{s_{\text{diff}}}, 0, 1\right), \tag{7}$$

$$r_{\text{grad}} = \text{clip}\left(\frac{\|\nabla d_t\|}{s_{\text{grad}}}, 0, 1\right), \tag{8}$$

$$r_t = \frac{w_{\text{diff}} r_{\text{diff}} + w_{\text{grad}} r_{\text{grad}}}{w_{\text{diff}} + w_{\text{grad}}}, \tag{9}$$

$$\alpha_t = \alpha_{\min} + (\alpha_{\max} - \alpha_{\min}) r_t, \tag{10}$$

$$\hat{d}_t = \alpha_t d_t + (1 - \alpha_t)\hat{d}_{t-1}. \tag{11}$$

The intended interpretation is:

- Smooth more on flat/stable tissue regions.
- Trust raw predictions more near disparity edges and discontinuities.

The tested sweep was:

- $\alpha_{\min} \in \{0.25, 0.30, 0.35\}$.
- $\alpha_{\max} \in \{0.75, 0.85, 0.90\}$.
- $s_{\text{diff}} \in \{3.0, 5.0\}$.
- $s_{\text{grad}} \in \{3.0, 5.0, 8.0\}$.
- $w_{\text{grad}} \in \{0.25, 0.50, 1.00\}$.

5 Metrics

The benchmark reports both geometric and temporal metrics.

5.1 Geometric Metrics

The main geometric metrics are:

- Depth MAE in millimeters.
- Disparity MAE in pixels.
- Disparity RMSE in pixels.

- Bad-1px, Bad-2px, Bad-3px.
- Bad-1mm, Bad-2mm, Bad-5mm.
- Valid frame count.
- Valid pixel coverage.

5.2 Temporal Metrics

The temporal diagnostics are:

- Raw temporal disparity difference:
$$|\hat{d}_t - \hat{d}_{t-1}|.$$
- Motion-compensated temporal MAE:
$$|\hat{d}_t - W(\hat{d}_{t-1}, F_{t-1\to t})|.$$

The second metric uses RAFT flow as a **metric-only** tool for no-RAFT methods. It does not imply that no-RAFT methods use flow in their prediction formula.

5.3 Runtime Metrics

Runtime is reported fairly according to the expected deployment cost:

- No-flow methods:
$$T_{\text{online}} = T_{\text{S2M2}} + T_{\text{postprocess}}.$$
- Forward-flow methods:
$$T_{\text{online}} = T_{\text{S2M2}} + T_{\text{flow,fwd}} + T_{\text{postprocess}}.$$
- Forward-backward-flow methods:
$$T_{\text{online}} = T_{\text{S2M2}} + T_{\text{flow,fwd}} + T_{\text{flow,bwd}} + T_{\text{postprocess}}.$$

The current measured averages are:

- S2M2-S inherited runtime: 60.97 ms.
- RAFT forward runtime: 71.63 ms.
- RAFT backward runtime: 68.84 ms.
- S2M2-S peak VRAM: 371.33 MB.
- RAFT peak VRAM: 3429.09 MB.
- StereoAnyVideo cached runtime: 146.30 ms.
- StereoAnyVideo peak VRAM: 10 132.18 MB.

6 Main Results

6.1 Key Method Comparison

Table 1 summarizes the most important rows from the benchmark.

Table 1: Main comparison of representative methods on the audited SCARED temporal-GT sequence. Lower is better for all error and temporal metrics.

Method	Depth MAE mm	Disp MAE px	MC Temp MAE px	Runtime ms	Peal
S2M2-S raw	2.5840	8.3835	1.7244	60.97	
Fixed EMA $\alpha = 0.35$	**2.4516**	**8.2498**	**1.4950**	**63.16**	
Fixed EMA $\alpha = 0.50$	2.4902	8.2511	1.5754	63.11	
RAFT-warped EMA $\alpha = 0.50$	2.4377	8.2217	1.4826	157.03	
Conf./occ. reset $\alpha = 0.50$	2.5160	8.2827	1.6218	212.78	
Conservative adaptive EMA with RAFT	2.4612	8.2077	1.5246	214.40	
StereoAnyVideo	2.5874	8.2502	1.6618	146.30	
Best no-RAFT adaptive diff+grad	2.4955	8.2813	1.5575	70.87	

6.2 Best No-RAFT Adaptive Variant

The best no-RAFT adaptive variant observed in the sweep is approximately:

- Method: adaptive no-RAFT diff+grad.
- $\alpha_{\min} = 0.25$.
- $\alpha_{\max} = 0.75$.
- $s_{\text{diff}} = 3.0$.
- $s_{\text{grad}} = 8.0$.
- $w_{\text{grad}} = 1.0$.

Its metrics are:

- Depth MAE: 2.4955 mm.
- Disparity MAE: 8.2813 px.
- Motion-compensated temporal MAE: 1.5575 px.
- Estimated online runtime: 70.87 ms.
- Peak VRAM: 371 MB.

6.3 Fixed EMA vs Best No-RAFT Adaptive

Table 2 shows the decisive comparison.

Table 2: Fixed EMA remains better than the best no-RAFT adaptive variant.

Method	Depth MAE mm	Disp MAE px	MC Temp MAE px	Runtime ms
Fixed EMA $\alpha = 0.35$	**2.4516**	**8.2498**	**1.4950**	**63.16**
Best no-RAFT adaptive diff+grad	2.4955	8.2813	1.5575	70.87

The conclusion is direct:

No-RAFT adaptive EMA did not outperform fixed EMA.

The adaptive method is worse in depth accuracy, disparity accuracy, temporal consistency, and runtime.

7 Interpretation

7.1 Fixed EMA Is a Strong Baseline, Not a Novel Method

The fixed EMA result is important but not itself a strong methodological novelty. It shows that S2M2-S predictions contain temporally coherent signal that can be improved with simple causal smoothing.

Compared with S2M2-S raw, fixed EMA $\alpha = 0.35$ improves:

- Depth MAE from 2.5840 mm to 2.4516 mm.
- Disparity MAE from 8.3835 px to 8.2498 px.
- Motion-compensated temporal MAE from 1.7244 px to 1.4950 px.

This is a very strong deployment baseline because it adds only about 2.2 ms of post-processing and does not require flow or a video model.

However, fixed EMA is too simple to be claimed as the main methodological contribution.

7.2 Why No-RAFT Adaptive Failed

The no-RAFT adaptive rules attempted to use local risk estimates to decide how much memory to keep. The underlying assumption was:

> If raw and memory disagree, trust raw more; if they agree, trust memory more.

This assumption is too weak in surgical temporal stereo because disagreement can mean many different things:

- True scene motion.
- Camera motion.
- Tissue deformation.
- S2M2-S flicker.
- Specular instability.
- Disparity-edge motion.
- Occlusion/disocclusion.
- Noise or invalid predictions.

Without explicit motion compensation or learned temporal context, the adaptive rule cannot reliably distinguish these cases. As a result, it often uses too much raw prediction or too much memory in the wrong regions. The fixed EMA, although simpler, is more robust because it does not overreact to noisy risk maps.

7.3 RAFT-Warped EMA Is the Quality Reference

RAFT-warped EMA remains the strongest quality reference:

- Depth MAE: 2.4377 mm.
- Disparity MAE: 8.2217 px.
- Motion-compensated temporal MAE: 1.4826 px.

Compared with fixed EMA, the gain is real but small:

- Depth MAE improves by about 0.014 mm.
- MC temporal MAE improves by about 0.012 px.

The cost difference is much larger:

- Fixed EMA runtime: about 63 ms.
- RAFT-warped EMA runtime: about 157 ms in the latest fair accounting.
- Fixed EMA peak VRAM: about 371 MB.
- RAFT-warped EMA peak VRAM: about 3429 MB.

This means RAFT motion compensation is useful, but full RAFT is not ideal for lightweight deployment.

7.4 StereoAnyVideo Is Not Competitive Here

StereoAnyVideo is useful as a cached video-model comparison, but on this audited sequence it is not superior to fixed EMA:

- Depth MAE is worse than fixed EMA.
- MC temporal MAE is worse than fixed EMA.
- Runtime is higher.
- VRAM is much higher.

This is an important empirical observation: a general video stereo/depth model is not automatically better than a strong single-frame stereo model plus simple temporal smoothing in this surgical setting.

8 Current Scientific Status

The current findings can be summarized as:

1. S2M2-S is a strong single-frame baseline.
2. Fixed EMA is a surprisingly strong temporal baseline.
3. No-RAFT adaptive heuristics do not beat fixed EMA.
4. RAFT-warped EMA is the best quality reference but too expensive online.
5. StereoAnyVideo is not superior on this audited sequence.

Therefore:

> **The current work has strong empirical insights, but the final methodological novelty is not yet fixed.**

9 Implications for Novelty

9.1 What Is Not Enough

The following are not sufficient as the main novelty:

- Fixed EMA alone.
- Raw-vs-memory adaptive EMA alone.
- Disparity-gradient-protected adaptive EMA alone.

These are valuable baselines and diagnostics, but not enough for a strong method paper.

9.2 Potential Novel Direction 1: Fast Flow-Based Temporal Fusion

The first next direction is to replace full RAFT with a faster optical-flow backend and evaluate whether motion compensation can become practical.

A minimal next experiment is:

- RAFT-Small with 6 iterations.
- RAFT-Small with 12 iterations.
- Warped EMA with $\alpha = 0.50$.

This would answer:

> *Is full RAFT necessary, or can RAFT-Small recover most of the temporal benefit at much lower cost?*

The comparison table should include:

- S2M2-S raw.
- Fixed EMA $\alpha = 0.35$.
- Best no-RAFT adaptive.
- RAFT-Small 6-iters warped EMA.
- RAFT-Small 12-iters warped EMA.
- Full RAFT warped EMA.
- StereoAnyVideo.

This is a clean engineering/scientific baseline, but it is probably not the strongest novelty because it still uses optical flow online.

9.3 Potential Novel Direction 2: RAFT Teacher to Lightweight Student

The more novel direction is to use the expensive RAFT-based method as an offline teacher and train a lightweight recurrent student for online inference.

The core idea is:

offline RAFT-based temporal teacher $\rightarrow$ online flow-free lightweight student.

At training time:

- Use raw S2M2-S predictions.
- Use RAFT-warped EMA outputs as temporal teacher targets or regularizers.
- Use ground-truth disparity where available.

At inference time:

- Use S2M2-S raw prediction.
- Use a small ConvGRU or alpha/reset fusion module.
- Do not run RAFT.

A safe student design would predict pixel-wise fusion weights rather than an unconstrained residual:

$$\alpha_t = f_\theta(d_t, \hat{d}_{t-1}, I_t, h_{t-1}), \tag{12}$$

$$\hat{d}_t = \alpha_t d_t + (1 - \alpha_t)\hat{d}_{t-1}. \tag{13}$$

The student could be supervised with a mixed loss:

$$\mathcal{L} = \lambda_{\text{gt}}\mathcal{L}_{\text{gt}} + \lambda_{\text{teacher}}\mathcal{L}_{\text{teacher}} + \lambda_{\text{raw}}\mathcal{L}_{\text{raw guard}} + \lambda_{\text{tv}}\mathcal{L}_{\text{alpha smooth}}. \tag{14}$$

A reasonable initial weighting would be:

- $\lambda_{\text{gt}} = 1.0$.
- $\lambda_{\text{teacher}} = 0.2$.
- $\lambda_{\text{raw}} = 0.05$.
- $\lambda_{\text{tv}} = 0.01$.

The key claim would be:

> *A lightweight recurrent student can approximate motion-compensated temporal behavior learned from an expensive RAFT teacher without requiring optical flow at inference.*

This is stronger as a methodological novelty, but it requires more data and a proper train/validation protocol.

10 Recommended Next Steps

10.1 Immediate Decision

The no-RAFT adaptive branch should be closed as a negative result:

> *Simple no-flow adaptive fusion based on raw-memory disagreement and disparity gradients does not improve over fixed EMA.*

The fixed EMA $\alpha = 0.35$ should remain the deployment baseline.

10.2 Next Controlled Experiment

The next low-effort controlled experiment should be:

> **Evaluate RAFT-Small 6/12 iterations as a lightweight motion-compensation backend for warped EMA.**

This is the cleanest way to determine whether the current cost barrier comes from full RAFT specifically, or from online optical flow in general.

10.3 Next Novel Method Direction

If RAFT-Small still does not provide a satisfactory quality/runtime trade-off, the most meaningful method direction is:

> **RAFT-warped EMA teacher → lightweight causal recurrent student.**

This would turn the current empirical observations into a coherent method paper.

11 Current Takeaway

The current takeaway is:

> *Fixed causal EMA is a very strong deployment baseline for temporal refinement of S2M2-S on the audited SCARED sequence. Simple no-flow adaptive heuristics do not outperform it. RAFT-based motion compensation provides the best quality, but at significantly higher runtime and memory cost. The next meaningful step is therefore either to test lightweight flow backends such as RAFT-Small, or to distill the RAFT-based temporal behavior into a lightweight flow-free recurrent student.*

A Relevant Commands

A.1 Inspect Output Folder

```
ls -lh results/03_temporal_refinement/scared_s2m2_temporal_baselines_v2_no_raft_adaptive
```

A.2 Inspect Summary CSV

```
cat results/03_temporal_refinement/scared_s2m2_temporal_baselines_v2_no_raft_adaptive/
    summary.csv
```

A.3 Inspect Adaptive Sweep

```
cat results/03_temporal_refinement/scared_s2m2_temporal_baselines_v2_no_raft_adaptive/
    adaptive_sweep_summary.csv
```

A.4 Inspect Videos

```
ls -lh results/03_temporal_refinement/scared_s2m2_temporal_baselines_v2_no_raft_adaptive/
    videos/
```

Table 3: Comparison between fixed EMA, no-RAFT adaptive EMA, RAFT-Small, full RAFT, and StereoAnyVideo. Lower is better for all error and temporal metrics.

Method	**Depth MAE** mm	**Disp MAE** px	**MC Temp MAE** px	**Runtime** ms	**Peak VRAM** MB
S2M2-S raw	2.5840	8.3835	1.7244	60.97	371
Fixed EMA $\alpha = 0.35$	2.4516	8.2498	1.4950	63.13	371
Best no-RAFT adaptive	2.4955	8.2813	1.5575	74.25	371
RAFT-Small 6-iters warped EMA	**2.4360**	**8.2189**	**1.4819**	117.72	3393
RAFT-Small 12-iters warped EMA	2.4371	8.2206	1.4822	110.42	3394
Full RAFT warped EMA	2.4377	8.2217	1.4826	139.28	3429
StereoAnyVideo	2.5874	8.2502	1.6618	146.30	10132

B Canonical Benchmark Folder

```
results/03_temporal_refinement/scared_s2m2_temporal_baselines_v2_no_raft_adaptive
```

C Files Involved

```
scripts/temporal_refinement/lib/temporal_baselines.py
scripts/temporal_refinement/eval_scripts/benchmark_scared_s2m2_temporal_baselines.py
scripts/temporal_refinement/lib/video_qualitative.py
```

C.1 RAFT-Small Lightweight Flow Results

After the failure of the no-RAFT adaptive EMA variants to outperform the fixed EMA baseline, we evaluated RAFT-Small as a lightweight optical-flow backend for motion-compensated temporal fusion. The goal was to determine whether the quality improvement of full RAFT-warped EMA depends on the full RAFT model, or whether a smaller and faster RAFT variant is sufficient.

The evaluated RAFT-Small variants were:

- RAFT-Small with 6 update iterations.
- RAFT-Small with 12 update iterations.

Both were used inside the same warped EMA formulation:

$$\hat{dt} = \alpha d_t + (1 - \alpha) W(\hat{dt} - 1, F_{t-1 \to t}), \tag{15}$$

with $\alpha = 0.50$, where $F_{t-1\to t}$ is estimated by RAFT-Small rather than full RAFT.

The key result is that RAFT-Small essentially matches full RAFT quality while substantially reducing optical-flow runtime. In fact, the 6-iteration RAFT-Small variant slightly outperformed full RAFT in the current benchmark, although the difference is small enough to be treated as practically equivalent.

The most important observation is that full RAFT is not necessary to obtain the motion-compensated temporal gain. RAFT-Small with 6 iterations achieved:

- Depth MAE: 2.4360 mm.
- Disparity MAE: 8.2189 px.
- Motion-compensated temporal MAE: 1.4819 px.
- Estimated online runtime: 117.72 ms.
- Peak VRAM: approximately 3393 MB.

Compared with full RAFT-warped EMA, RAFT-Small 6-iters is practically equivalent in quality and faster in runtime:

- Full RAFT forward-flow runtime: 71.63 ms.
- RAFT-Small 6-iters forward-flow runtime: 33.09 ms.
- RAFT-Small 12-iters forward-flow runtime: 41.70 ms.

However, the memory reduction is limited:

- Full RAFT peak VRAM: approximately 3429 MB.
- RAFT-Small peak VRAM: approximately 3393 MB.

Therefore, the main advantage of RAFT-Small is reduced runtime, not reduced VRAM.

Interpretation. RAFT-Small 6-iters is currently the best quality method in the benchmark. It slightly improves over fixed EMA and full RAFT, while being faster than full RAFT. However, it is still much more expensive than fixed EMA in both runtime and memory. Compared with fixed EMA $\alpha = 0.35$, RAFT-Small 6-iters improves Depth MAE by approximately 0.0156 mm and motion-compensated temporal MAE by approximately 0.0131 px, but increases runtime from approximately 63.13 ms to 117.72 ms and peak VRAM from approximately 371 MB to approximately 3393 MB.

This creates a clear quality–deployment trade-off:

> *Fixed EMA remains the best cheap deployment baseline, while RAFT-Small 6-iters is the best motion-compensated quality baseline.*

The remaining research opportunity is to close the gap between these two extremes: obtain quality close to RAFT-Small while keeping runtime and memory closer to fixed EMA.

D Experiment Paths and Artifacts

This section records the canonical paths of the current ARGOS temporal-refinement experiments.

D.1 Audited Input Sequence

```
dataset/SCARED/curated/temporal_gt/test_dataset_9_keyframe_3
```

This folder contains the audited SCARED temporal-GT sequence used for all current cache-only temporal-refinement experiments.

D.2 Cached S2M2-S Predictions

```
results/03_temporal_refinement/evaluation/gt_temporal_test_dataset_9_keyframe_3/
    predictions/S2M2-S_512
```

This folder contains the cached S2M2-S predictions used as the raw disparity input for the temporal-refinement benchmark.

D.3 Cached StereoAnyVideo Predictions

```
results/03_temporal_refinement/evaluation/gt_temporal_test_dataset_9_keyframe_3/
    predictions/StereoAnyVideo_384x640
```

This folder contains the cached StereoAnyVideo predictions used as a video-model comparison. StereoAnyVideo is treated as a comparison/teacher candidate, not as ground truth.

D.4 Full RAFT Flow Cache

```
results/04_dataset_derivatives/SCARED/temporal_gt_flow_cache/test_dataset_9_keyframe_3/
    raft
```

This folder contains the full RAFT optical-flow cache. For no-RAFT adaptive methods, this cache is used only for motion-compensated metric computation, not for prediction.

D.5 No-RAFT Adaptive Benchmark Output

```
results/03_temporal_refinement/scared_s2m2_temporal_baselines_v2_no_raft_adaptive
```

This folder contains the no-RAFT adaptive EMA benchmark. It includes the fixed EMA baselines, no-flow adaptive sweeps, summary CSVs, and qualitative videos.

Important files:

```
summary.csv
adaptive_sweep_summary.csv
per_frame_metrics.csv
per_pair_temporal_metrics.csv
method_config.json
cache_audit.json
README.md
run.log
videos/
```

The main conclusion from this experiment is that no-RAFT adaptive EMA variants did not outperform fixed EMA $\alpha = 0.35$.

D.6 RAFT-Small Benchmark Output

```
results/03_temporal_refinement/scared_s2m2_temporal_baselines_v3_raft_small
```

This folder contains the RAFT-Small benchmark comparing RAFT-Small 6-iters, RAFT-Small 12-iters, full RAFT, fixed EMA, best no-RAFT adaptive, S2M2-S raw, and StereoAnyVideo.

Important files:

```
summary.csv
per_frame_metrics.csv
per_pair_temporal_metrics.csv
flow_cache_audit.json
method_config.json
README.md
run.log
videos/
```

The main conclusion from this experiment is that RAFT-Small 6-iters matches or slightly improves over full RAFT-warped EMA while reducing forward-flow runtime substantially. However, its VRAM cost remains close to full RAFT.

D.7 Code Files

The current benchmark code is organized around the following files:

```
scripts/temporal_refinement/lib/temporal_baselines.py
scripts/temporal_refinement/eval_scripts/benchmark_scared_s2m2_temporal_baselines.py
scripts/temporal_refinement/lib/video_qualitative.py
scripts/temporal_refinement/lib/flow_cache.py
scripts/temporal_refinement/eval_scripts/build_scared_temporal_gt_flow_cache.py
```

The temporal baselines are implemented in:

```
scripts/temporal_refinement/lib/temporal_baselines.py
```

The main cache-only benchmark entry point is:

```
scripts/temporal_refinement/eval_scripts/benchmark_scared_s2m2_temporal_baselines.py
```

The flow-cache builder is:

```
scripts/temporal_refinement/eval_scripts/build_scared_temporal_gt_flow_cache.py
```

D.8 Artifact-Aware Evaluation of RAFT-Small

The RAFT-Small experiment initially appeared highly favorable under standard global metrics. RAFT-Small 6-iters warped EMA achieved the best numerical performance among the evaluated methods:

- Depth MAE: 2.4360 mm.
- Disparity MAE: 8.2189 px.
- Motion-compensated temporal MAE: 1.4819 px.

However, qualitative video inspection revealed strong ghosting, blur, and loss of sharpness. This motivated a new artifact-aware evaluation pass. The goal was to test whether the numerically best method was also visually reliable enough for deployment.

The artifact-aware benchmark added the following metrics:

- **Edge sharpness ratio**, measuring whether disparity edges are preserved or blurred.
- **Boundary disparity MAE**, measuring error specifically near strong GT disparity boundaries.
- **Motion-region ghosting score**, measuring disagreement with current-frame evidence in dynamic regions.
- **GT-based ghosting error**, measuring whether ghosting regions are also wrong relative to GT.
- **Occlusion/disocclusion MAE**, measuring error in regions where temporal warping is unreliable.
- **Temporal lag rate**, measuring whether the prediction resembles the previous GT frame more than the current GT frame.
- **RGB–disparity edge alignment**, measuring whether disparity edges remain aligned with image edges.

Table 4 summarizes the main artifact-aware findings.

The results show that RAFT-Small remains the best method under global numerical metrics, but it is not artifact-free. In particular, RAFT-Small 6 produces higher GT-based ghosting error than fixed EMA in motion regions:

- Fixed EMA ghosting GT error at $\tau = 2$: 10.6972 px.
- RAFT-Small 6 ghosting GT error at $\tau = 2$: 11.2446 px.

Table 4: Artifact-aware comparison of temporal-refinement methods. Lower is better for error, ghosting, lag, and runtime. Edge sharpness values closer to 1 indicate stronger preservation of raw disparity edges.

Method	Depth MAE mm	Disp MAE px	MC Temp MAE px	EdgeSharp RawEdges	Boundary MAE px	GhostGT $\tau = 2$ px	Lag Rate	Runtime ms
S2M2 raw	2.5840	8.3835	1.7244	1.0000	13.4354	13.6517	0.5196	60.97
Fixed EMA	2.4516	8.2498	1.4950	0.6523	13.2133	**10.6972**	0.6667	62.59
Best adaptive	2.4955	8.2813	1.5575	0.8589	13.3289	11.0833	0.6373	73.19
RAFT-Small 6	**2.4360**	**8.2189**	**1.4819**	0.7022	13.2474	11.2446	0.6471	120.60
RAFT-Small 12	2.4371	8.2206	1.4822	0.7025	13.2494	11.2468	0.6471	110.24
Full RAFT	2.4377	8.2217	1.4826	0.7024	13.2501	11.2488	0.6471	140.00
SAV	2.5874	8.2502	1.6618	0.9020	**12.7958**	12.6591	0.5098	146.30

This confirms the qualitative observation: motion-compensated smoothing can reduce global error while still producing stale or ghosted geometry in dynamic regions.

The edge-sharpness results also show that all temporal smoothing methods reduce high-frequency disparity structure. Raw S2M2-S has an edge-sharpness ratio of 1.0 by definition. Fixed EMA drops to 0.6523 on strong raw edges, while RAFT-Small variants remain around 0.702. This suggests that RAFT-based warping preserves more edge structure than fixed EMA, but still introduces substantial smoothing compared with the raw prediction.

StereoAnyVideo has the best boundary MAE and relatively high edge preservation, but its global depth and temporal metrics are worse. This indicates that it may preserve some boundary structure, but does not provide the best metric geometry on this sequence.

Overall, the artifact-aware benchmark changes the interpretation of RAFT-Small:

> *RAFT-Small is the best numerical motion-compensated baseline, but should be treated as an artifact-prone teacher or upper bound rather than a direct deployment method.*

This finding motivates selective distillation rather than naive teacher copying.

D.9 From Artifact-Aware Benchmarking to Selective Distillation

The artifact-aware results suggest that naive distillation from RAFT-Small would be risky. Although RAFT-Small improves global Depth MAE and motion-compensated temporal MAE, it can introduce ghosting and blur in visually critical regions. Therefore, the future student should not copy RAFT-Small everywhere.

The next step is to build an oracle teacher-selection benchmark. The oracle will choose, for each pixel or region, the best source among:

- raw S2M2-S,
- fixed EMA,
- best no-RAFT adaptive EMA,
- RAFT-Small warped EMA,
- full RAFT warped EMA,
- StereoAnyVideo.

This oracle is not a deployable method because it uses ground truth. Its purpose is to estimate the headroom available for a learned student. If the oracle significantly improves over RAFT-Small while avoiding artifact-prone regions, this would justify training an artifact-aware student.

The intended student direction is:

Use RAFT-Small as an offline motion-compensated teacher only where it is artifact-safe, while preserving raw or fixed-EMA behavior near boundaries, occlusions, and dynamic regions.

This leads to the following research hypothesis:

A lightweight causal student can approach the numerical performance of RAFT-Small while avoiding the ghosting, blur, and lag revealed by artifact-aware diagnostics.

The resulting method would not simply distill RAFT. It would perform artifact-aware selective temporal distillation.

D.10 Oracle Teacher Selection: Meaning and Interpretation

After the artifact-aware evaluation, we introduced an oracle teacher-selection benchmark. The goal of this experiment is not to define a deployable method, but to estimate the theoretical headroom available if a future model could learn when to trust each prediction source.

An oracle is an upper-bound baseline that uses privileged information, in this case the ground-truth disparity, to select the best prediction source. Therefore, it cannot be used at inference time. Its purpose is diagnostic: it answers the question of whether different methods contain complementary local information.

Given a set of candidate predictions

$$\mathcal{M} = \{d^{raw}, d^{fixed}, d^{adaptive}, d^{raft\text{-}small}, d^{raft}, d^{sav}\},$$

the pixel-wise oracle selects, for each valid pixel p at time t, the method with the lowest absolute disparity error with respect to ground truth:

$$m^*(t,p) = \arg\min_{m\in\mathcal{M}} |d_m(t,p) - d_{gt}(t,p)| \,.$$

The oracle prediction is then constructed as

$$d_{oracle}(t,p) = d_{m^*(t,p)}(t,p).$$

This means that the oracle may choose S2M2-S raw in one pixel, fixed EMA in another pixel, RAFT-Small in another region, and StereoAnyVideo elsewhere. Such a method is not deployable because it uses the ground truth to make the selection. However, it is extremely useful because it tells us whether a learned selector or student model has meaningful potential.

Key finding. The best single temporal baseline before oracle selection was RAFT-Small 6-iters warped EMA:

$$\text{Depth MAE}_{raft\text{-}small} = 2.4360\,\text{mm}.$$

The all-source pixel-wise oracle achieved:

$$\text{Depth MAE}_{oracle} = 2.1395\,\text{mm}.$$

This corresponds to a headroom of:

$$\Delta_{\text{oracle}} = 2.4360 - 2.1395 = 0.2965\,\text{mm}.$$

This is a substantial gap. It shows that no single prediction source is locally optimal across the whole surgical scene. Even though RAFT-Small is the best single method globally, other sources are better in many local regions.

Table 5: Oracle teacher-selection results. Oracle methods use ground truth for source selection and are therefore upper bounds, not deployable methods. Lower is better for all error, ghosting, and lag metrics.

Method	Depth MAE mm	Disp MAE px	MC Temp px	Boundary MAE px	GhostGT $\tau = 2$ px	Lag Rate
RAFT-Small 6 warped EMA	2.4360	8.2189	1.4819	13.2474	11.2446	0.6471
Oracle raw/fixed/RAFT-Small	2.2079	7.3824	**1.3547**	11.8713	10.6597	0.1176
Oracle all sources	**2.1395**	**7.0901**	1.4585	**11.0762**	**10.2597**	**0.0000**
Region edge-aware oracle	2.5046	8.2610	2.0156	13.4354	10.8990	0.5490
Artifact-safe RAFT selector	2.2365	7.4697	1.3887	12.0842	10.7794	0.1275

Oracle selection distribution. The all-source oracle selected the candidate methods with the following average proportions:

$$\begin{aligned} \text{StereoAnyVideo} &: 29.59\%, \\ \text{Fixed EMA} &: 22.44\%, \\ \text{Raw S2M2-S} &: 17.63\%, \\ \text{RAFT-Small} &: 9.00\%, \\ \text{Full RAFT} &: 7.99\%. \end{aligned}$$

This result is important because RAFT-Small, despite being the best global baseline, is selected in only a small fraction of pixels. Conversely, StereoAnyVideo is selected very frequently despite having weak global depth performance. This suggests that StereoAnyVideo may preserve useful local structure or boundary information, while fixed EMA and raw S2M2-S retain useful information in other regions.

Therefore, the problem should not be framed as choosing a single best teacher. Instead, the results motivate a multi-teacher selection strategy.

> *The best temporal refinement strategy is likely not a single global method, but a learned selective fusion mechanism that decides when to trust raw current-frame evidence, temporal smoothing, motion-compensated refinement, or video-model priors.*

Interpretation. The oracle results reveal substantial headroom above the best single temporal baseline. The all-source oracle improves Depth MAE from 2.4360 mm to 2.1395 mm, and Disparity MAE from 8.2189 px to 7.0901 px. This confirms that the evaluated prediction sources are complementary.

However, the pixel-wise oracle should not be interpreted as a deployable visual reconstruction method. Since it switches prediction source independently at each pixel, it can introduce artificial spatial discontinuities. This is reflected by the very high edge-sharpness ratios observed for the oracle outputs. These high values do not necessarily indicate better visual quality; instead, they may indicate source-switching artifacts.

Therefore, the oracle should be interpreted only as a theoretical upper bound. Its role is to quantify whether a learned, regularized, artifact-aware selector could improve over any individual teacher.

Artifact-safe selector. The artifact-safe RAFT selector provides a more constrained upper bound. It uses RAFT-Small only where it is more accurate than raw and fixed EMA, and where artifact diagnostics indicate that RAFT-Small is safe. This selector achieves:

$$\text{Depth MAE} = 2.2365\,\text{mm},$$

while selecting RAFT-Small in only:

$$15.88\%$$

of pixels.

This is highly informative. It suggests that motion-compensated RAFT-Small behavior is useful, but only in selected regions. Copying RAFT-Small everywhere would be suboptimal and may reproduce ghosting or blur artifacts.

The correct direction is therefore:

> *artifact-aware multi-teacher selective distillation, not naive RAFT-Small distillation.*

D.11 Current Evidence and Dataset Limitations

The current results should be interpreted as a controlled pilot benchmark, not as a final general claim. All temporal-refinement, artifact-aware, and oracle-selection results reported so far are based on a single audited SCARED temporal-GT sequence:

```
dataset/SCARED/curated/temporal_gt/test_dataset_9_keyframe_3
```

After validity filtering, the benchmark contains:

$$103$$

valid frames and:

$$102$$

valid temporal pairs.

This is sufficient to reveal important failure modes and to motivate the next methodological direction, but it is not sufficient to claim generality across surgical scenes, anatomies, lighting conditions, camera motions, datasets, or sensing setups.

The correct interpretation is therefore:

> *On this audited sequence, global metrics favor RAFT-Small, artifact-aware metrics reveal ghosting and blur, and oracle teacher selection shows substantial headroom for selective multi-teacher fusion.*

The current results do not yet prove that the same oracle selection distribution, artifact behavior, or teacher complementarity will hold across all surgical datasets. In particular, the high selection rate of StereoAnyVideo, the artifact-prone behavior of RAFT-Small, and the magnitude of the oracle headroom must be validated on additional sequences.

A stronger benchmark protocol should include:

- multiple SCARED temporal-GT sequences;
- sequence-level splits;
- evaluation across different motion, occlusion, and boundary regimes;
- additional surgical datasets where possible;
- global geometric metrics;
- artifact-aware diagnostics;
- runtime and memory evaluation under a fair deployment model.

Thus, the current benchmark is best viewed as a diagnostic proof of concept. It identifies a clear research direction, but multi-sequence validation is required before making broad claims.

Next methodological implication. The oracle benchmark motivates the construction of a lightweight student model trained with selective multi-teacher supervision. The future student should learn to approximate the oracle selection behavior without access to ground truth at inference time. It should learn when to preserve raw current-frame evidence, when to use fixed temporal smoothing, and when to imitate motion-compensated teacher behavior.

This leads to the following hypothesis:

> *A lightweight causal student can approach the headroom exposed by oracle teacher selection while avoiding the ghosting, blur, and discontinuities of direct temporal smoothing or naive teacher copying.*

D.12 Rectified Temporal-GT Conversion from Raw SCARED Keyframes

To extend the temporal evaluation beyond the initial audited sequence, we investigated whether additional raw SCARED keyframes could be converted into temporally consistent stereo sequences with dense geometric supervision. The raw SCARED keyframes contain a vertically stacked stereo video, per-frame calibration metadata, and per-frame 3D scene point maps.

A first conversion extracted the top and bottom halves of the stereo video as left and right views, and derived depth supervision directly from the third channel of the scene point maps. For each valid point, disparity was initially computed as

$$d = \frac{f_x B}{Z}, \tag{16}$$

where f_x is the focal length, B is the stereo baseline, and Z is the depth value from the scene point map.

This initial conversion produced visually plausible depth and validity overlays. For `dataset_1_keyframe_1`, the converted sequence contained 197 frames with complete left/right images, depth maps, disparity maps, validity masks, and calibration files. The median depth was approximately 70.3 mm, the median disparity was approximately 61.3 px, and the mean valid-pixel ratio was 46.3%. However, when S2M2-S at 512 px input width was evaluated on this unrectified conversion, the results were catastrophic:

$$\text{Disp. MAE} = 58.74\,\text{px}, \qquad \text{Bad-1px} \approx 99.99\%, \qquad \text{Depth MAE} = 2798.65\,\text{mm}. \tag{17}$$

Since the image shapes, prediction counts, masks, and calibration files were all valid, this failure was not caused by an I/O mismatch.

We traced the issue to a stereo geometry mismatch. The previous working temporal-GT sequence, `test_dataset_9_keyframe_3`, had been generated in rectified stereo coordinates and stored the rectified calibration matrices $\mathbf{P}_1$, $\mathbf{P}_2$, $\mathbf{R}_1$, $\mathbf{R}_2$, and $\mathbf{Q}$. In contrast, the newly extracted `dataset_1_keyframe_1` sequence was still in the raw top/bottom camera geometry. This is incompatible with standard stereo networks such as S2M2, RAFT-Stereo, and CREStereo, which assume rectified stereo pairs where disparity corresponds to horizontal displacement.

We therefore implemented a rectified temporal-GT converter. The converter first splits the vertically stacked raw stereo video into left and right views, then uses the per-frame calibration parameters $\mathbf{K}_L$, $\mathbf{K}_R$, $\mathbf{D}_L$, $\mathbf{D}_R$, $\mathbf{R}$, and $\mathbf{T}$ to compute stereo rectification using OpenCV. Rectification maps are produced with `cv2.stereoRectify` and `cv2.initUndistortRectifyMap`. The left and right images are then remapped into a common rectified coordinate system. The 3D scene points are projected into the rectified left camera, using a minimum-depth scatter operation to resolve many-to-one projections. The final rectified depth map, validity mask, and disparity map are saved in the same coordinate system as the rectified left image. Rectified disparity is then computed as

$$d_{\text{rect}} = \frac{f_{x,\text{rect}} B_{\text{rect}}}{Z_{\text{rect}}}. \tag{18}$$

The resulting rectified sequence is saved with the same layout convention as the previously validated temporal-GT sequence:

Table 6: Effect of stereo rectification on S2M2-S@512 evaluation for `dataset_1_keyframe_1`. The unrectified conversion is geometrically incompatible with standard disparity-based stereo models, while the rectified conversion produces meaningful errors.

Conversion	Disp. MAE ↓	Disp. RMSE ↓	Bad-1px ↓	Bad-3px ↓	Depth MAE ↓
Unrectified raw split	58.74 px	59.70 px	99.99%	99.99%	2798.65 mm
Rectified + GT reprojection	0.891 px	1.687 px	33.55%	1.27%	0.942 mm

```
gt/DepthL_float32
gt/Disparity_float32
gt/ValidMask
```

Each frame also stores a rectified calibration file containing $\mathbf{P}_1$, $\mathbf{P}_2$, $\mathbf{R}_1$, $\mathbf{R}_2$, $\mathbf{Q}$, f_x, and the stereo baseline, together with the original calibration parameters. The unrectified conversion was kept as an audit artifact and was not overwritten.

To validate the rectified conversion, we reran S2M2-S at 512 px input width on the rectified version of `dataset_1_keyframe_1`. The same model, evaluated on the rectified sequence, improved dramatically:

$$\text{Disp. MAE} = 0.891\,\text{px}, \qquad \text{Disp. RMSE} = 1.687\,\text{px}, \tag{19}$$

$$\text{Bad-1px} = 33.55\%, \qquad \text{Bad-2px} = 6.56\%, \qquad \text{Bad-3px} = 1.27\%, \tag{20}$$

$$\text{Depth MAE} = 0.942\,\text{mm}. \tag{21}$$

The prediction count was 197/197, the prediction, ground-truth disparity, and valid-mask shapes all matched at 1024×1280, no resizing was required for metric computation, and no warnings were produced. This confirmed that the previous catastrophic error was caused by missing rectification rather than by invalid raw data.

Finally, we implemented a batch wrapper for rectified temporal-GT conversion. The wrapper scans raw SCARED folders using the strict pattern

```
dataset/SCARED/raw/extracted/<dataset>/<dataset>/keyframe_*
```

selects keyframes containing `rgb.mp4`, `frame_data.tar.gz`, `scene_points.tar.gz`, and stereo reference images, and calls the validated rectified converter for each candidate. Test datasets are skipped by default because the available raw test folders do not contain the required temporal scene-point archives. The wrapper produces per-sequence summaries, a conversion CSV, a skipped-sequence CSV, a batch summary JSON, and a README. This keeps dataset preparation, model inference, and temporal benchmarking as separate stages of the pipeline.

This validation establishes the correct preprocessing path for multi-sequence SCARED temporal evaluation:

$$\begin{aligned}&\text{raw vertical stereo video} \rightarrow \text{split left/right} \rightarrow \text{stereo rectification}\\ &\qquad \rightarrow \text{GT reprojection} \rightarrow \text{rectified depth/disparity/mask} \rightarrow \text{stereo model evaluation.}\end{aligned} \tag{22}$$

D.13 Towards Teacher-Oriented Lightweight Temporal Stereo Refinement

The rectified multi-sequence evaluation shows that a strong single-frame stereo foundation model, such as S2M2, can produce geometrically plausible disparities on individual frames, but still suffers from temporal instability and artifact-sensitive failure modes in surgical video. In particular, the model may preserve local stereo geometry while exhibiting frame-to-frame flickering, boundary degradation, and inconsistent behavior around tissue edges, specularities, occlusion-like regions, and rapidly changing surgical structures.

This motivates moving beyond a purely frame-wise stereo benchmark. Instead of asking whether a single model is globally better, we consider the problem of *local teacher selection*: different temporal or refinement strategies may be preferable in different spatial and temporal regimes. A raw S2M2 prediction may preserve sharp boundaries, an exponential moving average may reduce flicker in stable regions, a motion-aware refinement may better handle temporal consistency, and a video-based teacher may provide smoother behavior across difficult clips. However, no single candidate is expected to dominate everywhere.

Oracle teacher selection as an upper bound. To quantify the potential benefit of local teacher selection, we use an oracle analysis. Given a set of candidate disparity predictions

$$\mathcal{D}_t = \{D_t^{(1)}, D_t^{(2)}, \ldots, D_t^{(K)}\},$$

and a ground-truth disparity map D_t^{gt}, the pixel-wise oracle selects, for each pixel p, the candidate with the lowest absolute disparity error:

$$k^*(p,t) = \arg \min_{k \in \{1,\ldots,K\}} \left| D_t^{(k)}(p) - D_t^{gt}(p) \right|.$$

The oracle disparity is therefore

$$D_t^{oracle}(p) = D_t^{(k^*(p,t))}(p).$$

This oracle is not deployable, since it requires ground truth at inference time. Its purpose is diagnostic: it measures the maximum achievable gain if a perfect selector could choose the best local teacher. In previous small-scale oracle experiments, the oracle improved over the raw S2M2 baseline, showing that there is local headroom. Importantly, the oracle analysis also showed that the best candidate depends on the region type: raw predictions may be preferable near edges, smoothing may help in stable tissue regions, and motion-aware or video teachers may help in temporally unstable regions. This indicates that the problem is not simply solved by selecting one globally superior model.

Artifact-aware teacher analysis. To understand where each teacher is useful, we evaluate not only standard geometric metrics, but also temporal and artifact-aware metrics. These include standard disparity and depth errors:

$$\text{MAE}_{disp} = \frac{1}{|\Omega|} \sum_{p \in \Omega} \left| D_t(p) - D_t^{gt}(p) \right|,$$

as well as bad-pixel rates, depth MAE, and RMSE. In addition, we consider artifact-sensitive diagnostics such as boundary disparity error, edge sharpness ratio, temporal disparity variation, motion-region error proxies, lag proxies, RGB-disparity edge correlation, and ghosting/occlusion indicators when the required motion information is available.

For the new streaming rectified benchmark, the evaluation is deliberately performed without dense prediction caches. Predictions are computed on the fly and immediately consumed for metric computation. Temporal metrics are computed using only the previous valid frame in memory. Flow-dependent metrics, such as motion-compensated temporal error, are not reported unless optical flow is explicitly computed; otherwise, same-pixel no-flow proxies are used and clearly reported as such.

From oracle to lightweight distillation. The oracle itself is not intended as the final method. Instead, it provides supervision for a deployable lightweight refinement module. The central question becomes:

Can a small online module recover part of the oracle/teacher headroom without running heavy teachers at infere

Heavy teachers such as RAFT-based motion compensation or video stereo models can be useful offline, but they are too expensive for a real-time surgical pipeline. Therefore, we use them only during training or analysis to generate targets. At inference time, the goal is to keep the system close to:

$$\text{S2M2 raw prediction} \quad + \quad \text{lightweight temporal refinement head.}$$

A practical formulation is to train a small refinement head that predicts either a residual correction, a confidence/artifact mask, or blending weights among cheap online candidates. For residual distillation, the module predicts

$$\Delta D_t = D_t^{target} - D_t^{S2M2},$$

where D_t^{target} may be an oracle prediction, a selected teacher, or a teacher-refined disparity. The final refined disparity is then

$$D_t^{refined} = D_t^{S2M2} + \Delta \hat{D}_t.$$

Alternatively, the module may predict a confidence map C_t and apply the correction only where the raw stereo prediction is unreliable:

$$D_t^{refined} = D_t^{S2M2} + C_t \cdot \Delta \hat{D}_t.$$

Another lightweight option is to predict blending weights over cheap online candidates:

$$D_t^{refined} = w_{raw} D_t^{raw} + w_{ema} D_t^{ema} + w_{adaptive} D_t^{adaptive},$$

where

$$w_{raw} + w_{ema} + w_{adaptive} = 1.$$

In this setting, expensive teachers such as RAFT or video stereo are used only offline to supervise the desired behavior, while the online system uses only fast S2M2-derived candidates.

DINOv3 as a tissue- and artifact-aware prior. A promising extension is to condition the lightweight refinement module on dense foundation features extracted from a vision backbone such as DINOv3. The goal is not to replace stereo geometry with monocular depth. Instead, DINOv3 features can provide semantic and structural priors that are difficult to obtain from disparity alone. In surgical scenes, these features may encode tissue appearance, instrument/tissue separation, specular regions, boundaries, texture changes, and other cues correlated with stereo failure modes.

The proposed architecture is therefore:

$$\begin{aligned} (I_t^L, I_t^R) &\rightarrow \text{S2M2} \rightarrow D_t^{S2M2}, \\ I_t^L &\rightarrow \text{DINOv3 encoder} \rightarrow F_t^{DINO}, \\ (D_t^{S2M2}, D_{t-1}^{S2M2}, F_t^{DINO}, \phi_t) &\rightarrow \text{lightweight refinement head} \rightarrow (D_t^{refined}, C_t), \end{aligned}$$

where ϕ_t denotes additional cheap features such as disparity gradients, RGB gradients, temporal disparity differences, edge maps, and artifact proxies.

This design separates geometry from semantic/artifact reasoning. S2M2 remains responsible for stereo matching, while DINOv3 provides dense tissue-aware features that help the refinement head decide where to preserve sharp stereo boundaries, where to smooth flicker, and where to reduce confidence. The DINOv3 encoder can be frozen and evaluated at reduced resolution, while only a small decoder or refinement head is trained. This keeps the online cost significantly lower than running heavy temporal teachers.

Minimal experimental plan. Before training a full refinement model, we first test whether DINOv3 features are predictive of S2M2 failure modes. A small diagnostic head can be trained to predict high-error or high-flicker regions from DINO features and simple stereo features:

$$(F_t^{DINO}, D_t^{S2M2}, \nabla D_t^{S2M2}, |D_t^{S2M2} - D_{t-1}^{S2M2}|) \rightarrow \hat{E}_t,$$

where $\hat{E}_t$ is a predicted error, artifact, or low-confidence map. If these features correlate with disparity error, boundary error, or temporal instability, then DINO-guided refinement is justified.

The final method can then be trained using oracle or teacher-derived targets:

$$\mathcal{L} = \lambda_{disp}\mathcal{L}_{disp} + \lambda_{temp}\mathcal{L}_{temp} + \lambda_{edge}\mathcal{L}_{edge} + \lambda_{conf}\mathcal{L}_{conf} + \lambda_{distill}\mathcal{L}_{distill}.$$

Here, $\mathcal{L}_{distill}$ encourages the lightweight model to recover part of the oracle or teacher improvement, while $\mathcal{L}_{edge}$ discourages oversmoothing across tissue boundaries and $\mathcal{L}_{temp}$ penalizes unwanted flickering.

Scientific interpretation. This reframes the contribution from simply benchmarking stereo models to understanding and exploiting local teacher complementarity. The key hypothesis is that surgical stereo errors are structured: they depend on tissue appearance, boundaries, motion, specularities, occlusions, and temporal instability. Heavy teachers and oracle selection reveal where improvements are possible, while a DINO-guided lightweight module aims to distill this behavior into a deployable real-time refinement stage.

In summary, the proposed direction is:

$$\text{fast stereo foundation model} + \text{offline teacher/oracle analysis} + \text{DINO-guided lightweight distillation} \rightarrow \text{temporally}$$

D.14 Rectified Multi-Sequence Evaluation and Teacher-Oriented Distillation

After validating the rectified SCARED temporal-GT conversion on a single clean sequence, we extended the evaluation to the full rectified multi-sequence subset. The final rectified temporal-GT dataset contains 27 complete sequences and 22,950 frames. An integrity audit identified 20,621 frames suitable for evaluation after filtering low-valid or suspicious frames, while 2,329 frames were skipped. All evaluated sequences share a consistent rectified stereo representation with calibrated projection matrices and rectified ground-truth disparity/depth maps.

Streaming S2M2-S evaluation. We evaluated S2M2-S in a streaming, no-cache setting. Predictions were computed on the fly and immediately consumed for metric computation, avoiding dense prediction storage. The evaluation was performed over all 27 sequences and 20,621 valid frames, without saving per-frame prediction arrays. This avoided an estimated 100.69 GiB of prediction cache storage.

The full S2M2-S artifact-temporal evaluation produced the following aggregate results:

$$\begin{aligned}\mathrm{MAE}_{disp} &= 6.829 \text{ px},\\ \text{Bad-3px} &= 42.72\%,\\ \mathrm{MAE}_{depth} &= 3.544 \text{ mm}.\end{aligned}$$

The runtime was approximately 6,521.3 seconds for the full evaluation, with a median runtime of 62.45 ms per frame and a peak GPU memory usage of 395 MB.

Compared with the initial clean sequence, where S2M2-S achieved a disparity MAE of 0.891 px and a Bad-3px rate of 1.27%, the full multi-sequence benchmark reveals substantially broader degradation. This indicates that the original single-sequence result was not representative of the full rectified SCARED temporal setting.

Artifact-aware and temporal diagnostics. In addition to standard geometry metrics, we computed temporal and artifact-aware diagnostics. Since the full streaming run avoids optical flow, all temporal metrics are reported as same-pixel, no-flow proxies. Motion-compensated metrics, flow-based ghosting, and flow-based occlusion metrics are intentionally not computed in this pass.

The main temporal and artifact-aware aggregate results were:

$$\begin{aligned}
\text{TemporalPairs} &= 20{,}594, \\
\text{TemporalCoverage} &= 1.0, \\
\Delta D_{temporal}^{samepixel} &= 1.648 \text{ px}, \\
\text{MotionMismatch}^{samepixel} &= 2.026 \text{ px}, \\
\text{BoundaryMAE}_{disp} &= 8.982 \text{ px}, \\
\text{LagRate} &= 0.656.
\end{aligned}$$

These diagnostics show that S2M2-S is not only affected by frame-wise geometric errors, but also by temporal instability, boundary-sensitive errors, and lag/flicker-like behavior. In particular, the boundary disparity MAE is higher than the global disparity MAE, suggesting that naive temporal smoothing may reduce flicker but risks degrading sharp tissue boundaries.

Failure taxonomy and clip selection. Using the full S2M2-S artifact-temporal run, we generated a distillation planning set. Sequences were grouped using quantile-based thresholds into:

{clean core, high boundary error, high temporal flicker, high motion mismatch, low-valid stress, catastrophic g

From this taxonomy, six compact contiguous clips were selected for future teacher/oracle distillation. The selected clips cover clean and stress cases and contain 502 frames in total. This subset is designed to be small enough for offline teacher generation while still representing the dominant failure modes observed in the full benchmark.

Teacher/oracle target generation. To measure the local headroom available from temporal and video teachers, we generated selected-clip teacher/oracle distillation targets. Unlike the full streaming evaluation, this stage is allowed to use heavier teachers offline, but only on the selected clips. The online method is not expected to run these teachers at inference time.

For each selected frame, the following candidate predictions were generated:

$$\mathcal{D}_t = \{D_t^{raw}, D_t^{ema}, D_t^{adaptive}, D_t^{raft}, D_t^{sav}\},$$

where D_t^{raw} is the raw S2M2-S prediction, D_t^{ema} is a fixed EMA candidate with $\alpha = 0.35$, $D_t^{adaptive}$ is the best previous no-RAFT adaptive candidate, D_t^{raft} is a RAFT-Small warped EMA candidate, and D_t^{sav} is a StereoAnyVideo prediction.

StereoAnyVideo was processed in temporal chunks of 32 frames. The last short chunk was padded by repeating the last valid frame, and padded outputs were discarded after inference. This allowed SAV to be used consistently on clips longer than its internal temporal window.

Given the set of candidates, the pixel-wise oracle selects the candidate with minimum ground-truth disparity error:

$$k^*(p,t) = \arg\min_k \left| D_t^{(k)}(p) - D_t^{gt}(p) \right|,$$

and the oracle disparity is:

$$D_t^{oracle}(p) = D_t^{(k^*(p,t))}(p).$$

We compute three oracle variants:

$$\begin{aligned}
D_t^{oracle,no\text{-}flow} &= \text{Oracle}\left(D_t^{raw}, D_t^{ema}, D_t^{adaptive}\right), \\
D_t^{oracle,raft} &= \text{Oracle}\left(D_t^{raw}, D_t^{ema}, D_t^{raft}\right), \\
D_t^{oracle,all} &= \text{Oracle}\left(D_t^{raw}, D_t^{ema}, D_t^{adaptive}, D_t^{raft}, D_t^{sav}\right).
\end{aligned}$$

The final selected-clip teacher/oracle run processed all 6 clips and 502 frames. RAFT-Small and SAV were available for all clips. The target outputs were saved as compressed downsampled arrays at scale 0.25, using float16 for dense maps and uint8 for masks and labels. No full-resolution teacher cache was created.

The selected-clip aggregate results were:

$$\begin{aligned}\mathrm{MAE}_{raw} &= 13.2051 \text{ px},\\ \mathrm{MAE}_{oracle,no\text{-}flow} &= 11.9220 \text{ px},\\ \mathrm{MAE}_{oracle,all} &= 8.9466 \text{ px}.\end{aligned}$$

Therefore, the no-flow oracle improves over raw S2M2-S by:

$$13.2051 - 11.9220 = 1.2831 \text{ px},$$

corresponding to approximately:

$$\frac{1.2831}{13.2051} \times 100 \approx 9.7\%.$$

The all-source oracle improves over raw S2M2-S by:

$$13.2051 - 8.9466 = 4.2585 \text{ px},$$

corresponding to approximately:

$$\frac{4.2585}{13.2051} \times 100 \approx 32.2\%.$$

This result confirms that local teacher selection exposes substantial headroom. Cheap no-flow candidates provide a moderate improvement, while the inclusion of RAFT-Small and StereoAnyVideo as offline teachers provides a much larger upper bound.

Motivation for lightweight distillation. The oracle itself is not deployable, since it uses ground truth to select the best teacher. Similarly, RAFT-Small and StereoAnyVideo are too expensive to run as part of the intended online surgical stereo pipeline. Their role is therefore limited to offline supervision.

The selected-clip teacher/oracle experiment motivates a lightweight distillation strategy. The goal is to train a small online refinement module that recovers part of the oracle improvement without running heavy teachers at inference time. A natural formulation is residual distillation:

$$\Delta D_t^{target} = D_t^{oracle,all} - D_t^{raw},$$

where the model predicts:

$$\Delta \hat{D}_t = f_\theta\left(I_t^L, D_t^{raw}, D_{t-1}^{raw}, \phi_t\right),$$

with ϕ_t denoting cheap temporal and artifact features such as disparity gradients, RGB gradients, temporal disparity differences, valid masks, and boundary indicators.

The refined disparity can then be written as:

$$D_t^{refined} = D_t^{raw} + C_t \cdot \Delta \hat{D}_t,$$

where C_t is a predicted confidence or correction mask. This formulation allows the network to correct unstable or artifact-prone regions while preserving raw S2M2 predictions where they are already reliable.

A second formulation is blending-weight distillation:

$$D_t^{refined} = w_{raw} D_t^{raw} + w_{ema} D_t^{ema} + w_{adaptive} D_t^{adaptive},$$

with

$$w_{raw} + w_{ema} + w_{adaptive} = 1.$$

In this case, the heavy teachers are used only to define the desired behavior during training, while the online model blends only cheap S2M2-derived candidates.

Compressed temporal context. To improve temporal consistency without online optical flow or video teachers, we also consider a compressed causal temporal context window. Instead of providing only the current disparity and the immediately previous frame, the lightweight refiner may receive a small history:

$$\mathcal{C}_t = \left[D_t^{raw}, D_{t-1}^{raw}, D_{t-2}^{raw}, D_{t-3}^{raw}\right],$$

stored or reconstructed at reduced resolution. Additional temporal statistics can be computed from this window:

$$\begin{aligned}
\mu_t &= \mathrm{mean}(\mathcal{C}_t), \\
m_t &= \mathrm{median}(\mathcal{C}_t), \\
\sigma_t^2 &= \mathrm{var}(\mathcal{C}_t), \\
\Delta_t^{prev} &= |D_t^{raw} - D_{t-1}^{raw}|, \\
\Delta_t^{median} &= |D_t^{raw} - m_t|.
\end{aligned}$$

These compressed temporal features can help the refiner distinguish true geometric change from flicker. The design remains causal and lightweight, making it more suitable for surgical robotics than online RAFT or full video stereo.

DINO-guided feature probing. In parallel, we identified a second direction based on dense self-supervised visual features. Rather than using DINO as a monocular depth model, we treat DINO features as tissue-, boundary-, and artifact-aware priors. The hypothesis is that frozen DINO features extracted from the left image encode appearance cues correlated with S2M2 failure modes, such as tissue boundaries, specularities, instrument regions, and texture changes.

The proposed diagnostic probe is:

$$\left(F_t^{DINO}, D_t^{raw}, \nabla D_t^{raw}, |D_t^{raw} - D_{t-1}^{raw}|\right) \rightarrow \hat{E}_t,$$

where F_t^{DINO} are frozen DINO features and $\hat{E}_t$ is a predicted error, artifact, or low-confidence map. The probe should first be evaluated on a small balanced subset of frames selected from the full S2M2 artifact-temporal run. The goal is to test whether DINO features can predict high-error, high-boundary-error, high-flicker, or high-motion-mismatch regions.

If the probe is successful, DINO features can be integrated into the lightweight refinement architecture:

$$\begin{aligned}
(I_t^L, I_t^R) &\rightarrow \mathrm{S2M2} \rightarrow D_t^{raw}, \\
I_t^L &\rightarrow \mathrm{DINO} \rightarrow F_t^{DINO}, \\
(D_t^{raw}, \mathcal{C}_t, F_t^{DINO}, \phi_t) &\rightarrow \mathrm{TinyRefiner} \rightarrow (D_t^{refined}, C_t).
\end{aligned}$$

This keeps S2M2 responsible for stereo geometry, while DINO provides structural and tissue-aware context for confidence prediction and residual correction.

Summary. The current results support the following research direction:

$$\text{fast stereo foundation model} + \text{offline heavy teachers} + \text{oracle target generation} + \text{lightweight distillation} \rightarrow \text{tempor}$$

The full S2M2-S evaluation shows that a fast stereo foundation model alone is not sufficient on multi-sequence surgical video. The selected-clip oracle analysis shows that RAFT-Small and StereoAnyVideo provide substantial offline teacher headroom, reducing disparity MAE from 13.2051 px to 8.9466 px on the selected distillation clips. The next step is therefore not to deploy heavy teachers, but to distill their local behavior into a lightweight online temporal refinement module, optionally guided by compressed temporal context and DINO-based tissue/artifact features.

D.15 From Selected-Clip Oracle Distillation to Full-Scale Refiner Training

The selected-clip teacher/oracle experiment demonstrated that heavy video teachers, especially StereoAnyVideo, expose substantial local headroom over raw S2M2 predictions. However, training a lightweight refiner only on the selected oracle clips is not sufficient for generalization. The selected distillation set contains only 6 clips and 502 frames, and early tiny-refiner experiments showed that a model can learn the oracle residual on validation-like data but fails to generalize reliably to held-out failure modes.

For this reason, we separate the refinement problem into two stages. The first stage uses the full rectified temporal-GT dataset to learn a general geometric correction of S2M2. The second stage uses the selected teacher/oracle clips to inject failure-mode-specific and temporally informed behavior from heavy offline teachers such as StereoAnyVideo.

Full-scale S2M2–GT target generation. We generated a large-scale supervised training target dataset using all valid frames from the rectified temporal-GT SCARED subset. Unlike the selected-clip teacher/oracle generation, this stage does not use RAFT, StereoAnyVideo, DINO, or any other heavy teacher. Instead, it uses only S2M2-S predictions and the available ground-truth disparity.

For each valid stereo frame pair, we run S2M2-S once on the rectified left/right images:

$$(I_t^L, I_t^R) \xrightarrow{\text{S2M2}} D_t^{raw}.$$

The raw S2M2 disparity, ground-truth disparity, and valid mask are then downsampled to the target training resolution using the same valid-mask-aware downsampling procedure introduced for the corrected oracle targets. The training target is the residual between the low-resolution ground truth and the low-resolution raw S2M2 prediction:

$$\Delta D_t^{GT} = D_t^{GT} - D_t^{raw}.$$

Thus, the large-scale training set contains compact low-resolution tensors:

$$\left\{D_t^{raw},\ D_t^{GT},\ M_t,\ \Delta D_t^{GT}\right\},$$

where M_t is the downsampled valid mask. Full-resolution S2M2 predictions are not cached.

The generated full-scale target dataset contains:

$$\begin{aligned} N_{\text{seq}} &= 27, \\ N_{\text{frames}} &= 20{,}621, \\ N_{\text{shards}} &= 27, \\ \text{storage} &= 2.9\text{ GB}. \end{aligned}$$

The raw low-resolution S2M2 metrics over this dataset are:

$$\begin{aligned} \text{MAE}_{raw} &= 7.025\text{ px}, \\ \text{Bad-1px}_{raw} &= 73.57\%, \\ \text{Bad-3px}_{raw} &= 44.15\%, \\ \overline{r}_{valid} &= 45.53\%. \end{aligned}$$

The target generation took approximately 3650.5 seconds, corresponding to about 60.8 minutes, with an average runtime of 58.18 ms per frame. This is comparable to the previous full S2M2 streaming evaluation, but instead of discarding the prediction after metric computation, the pipeline stores only compact low-resolution supervision targets.

Geometry-first lightweight refinement. The first full-scale refiner is formulated as a geometric residual correction module. Given the raw S2M2 prediction, the model predicts a residual correction:

$$\widehat{\Delta D}_t = f_\theta(D_t^{raw}, M_t),$$

and the refined disparity is:

$$D_t^{refined} = D_t^{raw} + \widehat{\Delta D}_t.$$

The model is supervised directly with the ground-truth residual:

$$\mathcal{L}_{GT} = \frac{1}{|M_t|} \sum_{p \in M_t} \rho \left(\widehat{\Delta D}_t(p) - \Delta D_t^{GT}(p) \right),$$

where $\rho(\cdot)$ is a robust regression loss such as masked L_1 or Charbonnier loss.

This stage is intentionally teacher-free. Its goal is not yet to imitate StereoAnyVideo or RAFT, but to answer a simpler and more fundamental question: can a small residual model systematically improve S2M2 on held-out surgical sequences when trained on the full 20k-frame temporal-GT target set?

The expected evaluation compares raw and refined predictions against the saved low-resolution ground truth:

$$\begin{aligned} \mathrm{MAE}_{raw} &= \left| D_t^{raw} - D_t^{GT} \right|, \\ \mathrm{MAE}_{refined} &= \left| D_t^{refined} - D_t^{GT} \right|. \end{aligned}$$

A successful first-stage refiner should satisfy:

$$\mathrm{MAE}_{refined} < \mathrm{MAE}_{raw},$$

and should also reduce Bad-1px and Bad-3px rates on held-out sequences.

Why this stage does not yet solve temporal consistency. The full-GT residual refiner is primarily a frame-wise geometric correction module. If the input is only the current raw disparity D_t^{raw}, the model can correct systematic spatial errors, but it has limited ability to reason about temporal instability, flicker, lag, or frame-to-frame inconsistency. Therefore, this first stage should be interpreted as a geometry refinement baseline rather than a complete temporal refinement method.

The temporal problem requires the model to observe a causal history of S2M2 predictions:

$$\mathcal{C}_t = \left[D_{t-3}^{raw}, D_{t-2}^{raw}, D_{t-1}^{raw}, D_t^{raw} \right].$$

From this context, additional temporal features can be derived:

$$\begin{aligned} \mu_t &= \mathrm{mean}(\mathcal{C}_t), \\ m_t &= \mathrm{median}(\mathcal{C}_t), \\ \sigma_t^2 &= \mathrm{var}(\mathcal{C}_t), \\ \Delta_t^{prev} &= |D_t^{raw} - D_{t-1}^{raw}|, \\ \Delta_t^{median} &= |D_t^{raw} - m_t|. \end{aligned}$$

A temporal refiner can then be written as:

$$\widehat{\Delta D}_t = f_\theta \left(D_t^{raw}, D_{t-1}^{raw}, D_{t-2}^{raw}, D_{t-3}^{raw}, \phi_t \right),$$

where ϕ_t may include temporal statistics, valid masks, disparity gradients, RGB gradients, or artifact indicators.

This formulation allows the model to distinguish true geometric change from unstable prediction flicker. In deployment, the system remains causal:

$$(I_t^L, I_t^R) \to \mathrm{S2M2} \to D_t^{raw} \to \mathrm{TemporalRefiner} \to D_t^{refined}.$$

Role of StereoAnyVideo and oracle distillation. StereoAnyVideo is not intended to be run online in the final lightweight pipeline. Instead, it is used as an offline teacher. The selected-clip oracle analysis showed that StereoAnyVideo is selected in a large fraction of pixels and provides most of the additional all-source oracle headroom over raw S2M2, especially in difficult failure modes such as catastrophic geometry, motion mismatch, and low-valid stress cases.

However, the teacher/oracle clips are too few to train a general model from scratch. Therefore, the proposed strategy is:

$$\text{pretrain on full GT targets} \quad \rightarrow \quad \text{fine-tune or distill on selected oracle/SAV targets.}$$

The full-GT stage provides broad generalization, while the selected teacher/oracle stage provides failure-mode-specific supervision.

A hybrid objective can be formulated as:

$$\mathcal{L} = \mathcal{L}_{GT}^{full} + \lambda_{teacher}\mathcal{L}_{oracle}^{selected} + \lambda_{temp}\mathcal{L}_{temporal},$$

where:

$$\mathcal{L}_{GT}^{full}$$

is computed over the full 20k-frame S2M2–GT target dataset,

$$\mathcal{L}_{oracle}^{selected}$$

is computed over the selected clips using the all-available oracle residual, and

$$\mathcal{L}_{temporal}$$

encourages frame-to-frame stability.

The selected oracle residual is:

$$\Delta D_t^{oracle} = D_t^{oracle,all} - D_t^{raw},$$

where $D_t^{oracle,all}$ is computed from the local best candidate among raw S2M2, fixed EMA, adaptive no-RAFT, RAFT-Small, and StereoAnyVideo:

$$D_t^{oracle,all}(p) = D_t^{(k^*)}(p), \quad k^* = \arg\min_k \left| D_t^{(k)}(p) - D_t^{GT}(p) \right|.$$

This allows the lightweight model to learn when to behave like raw S2M2, when to apply a small geometric correction, and when to imitate the behavior of a stronger temporal/video teacher.

Experimental roadmap. The refinement experiments are organized as a sequence of controlled ablations:

- **v1:** $D_t^{raw} \rightarrow \Delta D_t^{GT}$ frame-wise geometry refiner,
- **v2:** $D_{t-k:t}^{raw} \rightarrow \Delta D_t^{GT}$ causal temporal context refiner,
- **v3:** $\text{v2} + \mathcal{L}_{oracle}^{selected}$ SAV/oracle teacher distillation,
- **v4:** $\text{v3} + F_t^{DINO}$ semantic/artifact-aware refinement if useful.

The first stage establishes whether a lightweight residual model can improve S2M2 on held-out surgical sequences. The temporal stage then evaluates whether compressed causal context reduces flicker and motion mismatch. Finally, the SAV/oracle distillation stage tests whether heavy offline video teachers can improve difficult failure modes without being deployed at inference time.

D.16 Intermediate Findings and Motivation for Temporal Refinement

The full-scale S2M2–GT target dataset enabled the first generalizable training experiments for a lightweight residual refiner. This dataset contains 27 sequences and 20,621 valid frames, stored as compact low-resolution sequence shards. Each target contains the raw S2M2 disparity, the downsampled ground-truth disparity, the valid mask, and the residual target

$$\Delta D_t^{GT} = D_t^{GT} - D_t^{raw}.$$

This setup allows the refiner to be trained without rerunning S2M2, RAFT, StereoAnyVideo, or DINO during training.

The first full-GT refiner was an unconstrained residual model:

$$D_t^{refined} = D_t^{raw} + f_\theta(D_t^{raw}, M_t).$$

Although this model reduced the training error, it did not generalize reliably. On the held-out test split, the improvement was marginal:

$$4.335 \text{ px} \rightarrow 4.228 \text{ px},$$

while on the validation split the model degraded the raw prediction:

$$4.677 \text{ px} \rightarrow 5.280 \text{ px}.$$

The Bad-3px rate also worsened substantially on validation:

$$44.42\% \rightarrow 56.16\%.$$

This indicates that an unconstrained residual model tends to overcorrect regions where S2M2 is already accurate.

To address this, we evaluated a conservative gated formulation:

$$D_t^{refined} = D_t^{raw} + G_t \cdot R_t,$$

where $G_t \in [0, 1]$ is a learned correction gate and R_t is a bounded residual proposal. The goal was to preserve the identity mapping where raw S2M2 is reliable and only apply corrections in high-error regions.

A first gated model initialized the gate close to zero, but during training the gate progressively opened on both good and bad pixels. This caused validation degradation, suggesting that gate sparsity alone is insufficient. A second safe-gated variant introduced explicit gate supervision using the raw S2M2 error:

$$G_t^{target}(p) = \begin{cases} 0, & |D_t^{raw}(p) - D_t^{GT}(p)| < 1 \text{ px}, \\ 1, & |D_t^{raw}(p) - D_t^{GT}(p)| \geq 3 \text{ px}. \end{cases}$$

This made the model safer: at the first epoch it remained almost identical to raw S2M2,

$$4.700959 \text{ px} \rightarrow 4.701217 \text{ px},$$

but subsequent epochs again opened the gate and degraded validation performance. The gate classifier achieved only weak discrimination between good and bad regions, with AUC values around 0.54–0.59.

These experiments suggest that frame-wise raw disparity alone is not sufficient to robustly predict where corrections should be applied. In other words, the model cannot reliably infer from D_t^{raw} alone whether a local structure is a true geometric surface, a temporal flicker artifact, a boundary error, or a failure mode caused by low valid-mask support. This motivates moving from frame-wise refinement to temporal-context refinement.

Conclusion from v1 experiments. The v1 experiments establish three important findings:

(1) A lightweight refiner can technically learn residual corrections,
(2) unconstrained correction risks damaging already-correct pixels,
(3) single-frame disparity is insufficient for reliable correction gating.

Therefore, the next refinement stage must provide the model with temporal context.

D.17 Temporal-Context Refiner

The next model uses a causal window of recent S2M2 predictions:

$$\mathcal{C}_t = \left[D_{t-3}^{raw}, D_{t-2}^{raw}, D_{t-1}^{raw}, D_t^{raw}\right].$$

From this window, the model can derive temporal consistency cues:

$$\begin{aligned}
\mu_t &= \mathrm{mean}(\mathcal{C}_t),\\
m_t &= \mathrm{median}(\mathcal{C}_t),\\
\sigma_t^2 &= \mathrm{var}(\mathcal{C}_t),\\
\Delta_t^{prev} &= |D_t^{raw} - D_{t-1}^{raw}|,\\
\Delta_t^{median} &= |D_t^{raw} - m_t|.
\end{aligned}$$

The temporal refiner is then formulated as:

$$(\widehat{\Delta D}_t, G_t) = f_\theta\left(D_t^{raw}, D_{t-1}^{raw}, D_{t-2}^{raw}, D_{t-3}^{raw}, \mu_t, m_t, \sigma_t^2, \Delta_t^{prev}, \Delta_t^{median}, M_t\right),$$

with final prediction:

$$D_t^{refined} = D_t^{raw} + G_t \cdot \widehat{\Delta D}_t.$$

This formulation gives the model access to the information that was missing from the frame-wise variants. If a region is stable across recent frames, the gate can remain closed and preserve the raw S2M2 prediction. If a region flickers, collapses, or deviates strongly from the recent temporal context, the gate can open and apply a correction.

This stage directly addresses the temporal nature of the surgical stereo problem while remaining lightweight and causal. Heavy video teachers such as StereoAnyVideo are still not used online. Instead, they remain offline teachers for a later distillation stage after the temporal refiner has established a generalizable baseline.

D.18 Safe Refinement via Abstention and Hard-Pixel Crop Training

The previous frame-wise and temporal-context residual refiners showed that dense correction is unsafe. Even when a learned gate is introduced, the model tends to open the gate on both good and bad pixels, degrading regions where raw S2M2 is already accurate. A failure analysis confirmed that the main limitation is not model capacity alone, but unsafe correction. In particular, pixels with raw error above 20 px represent only 8.9% of valid pixels but contribute 51.4% of the total MAE. As a result, full-frame residual training is dominated by outliers and encourages corrections that are too aggressive for easier validation sequences.

To address this, we introduce a conservative abstention-based refiner. Instead of always modifying the raw S2M2 prediction, the model predicts both a bad-pixel probability and a residual proposal:

$$(P_t^{bad}, \widehat{\Delta D}_t) = f_\theta\left(D_{t-3:t}^{raw}, M_t, \phi_t\right),$$

where $D_{t-3:t}^{raw}$ is a causal temporal context window of raw S2M2 predictions, M_t is the valid mask, and ϕ_t contains additional low-cost features such as temporal variance, temporal difference, and spatial disparity gradients.

The model output is interpreted as a correction policy rather than a dense refinement field. The default action is to preserve the raw S2M2 prediction:

$$D_t^{refined}(p) = D_t^{raw}(p).$$

A residual correction is applied only when the predicted probability of raw S2M2 failure exceeds a threshold:

$$D_t^{refined}(p) = \begin{cases} D_t^{raw}(p) + \widehat{\Delta D}_t(p), & P_t^{bad}(p) \geq \tau, \\ D_t^{raw}(p), & P_t^{bad}(p) < \tau. \end{cases}$$

A soft variant is also available during training:

$$D_t^{soft}(p) = D_t^{raw}(p) + P_t^{bad}(p)\,\widehat{\Delta D}_t(p).$$

The residual proposal is bounded to avoid large destructive corrections:

$$\widehat{\Delta D}_t = s \cdot \tanh(R_t),$$

where s is a fixed residual scale. This prevents the model from producing arbitrarily large disparity shifts.

Balanced hard-pixel crop training. Rather than training only on full frames, which are dominated by a small number of large-error outliers, the abstention model is trained on balanced crops. Crops are sampled from multiple categories:

hard/outlier crops,
good/stable crops,
boundary/high-gradient crops,
temporal-unstable crops,
random crops.

This ensures that the model sees difficult regions often enough to learn a correction policy, while also seeing enough good regions to learn when to abstain.

The bad-pixel target is derived directly from the raw S2M2 error:

$$Y_t^{bad}(p) = \mathbb{1}\left[|D_t^{raw}(p) - D_t^{GT}(p)| \geq 3 \text{ px}\right].$$

Good pixels are defined as:

$$Y_t^{good}(p) = \mathbb{1}\left[|D_t^{raw}(p) - D_t^{GT}(p)| < 1 \text{ px}\right].$$

The training objective combines bad-pixel detection, residual correction, and preservation of already-correct regions:

$$\mathcal{L} = \lambda_{det}\mathcal{L}_{det} + \lambda_{res}\mathcal{L}_{res} + \lambda_{ref}\mathcal{L}_{ref} + \lambda_{pres}\mathcal{L}_{pres} + \lambda_{sparse}\mathcal{L}_{sparse}.$$

Here, $\mathcal{L}_{det}$ is a BCE or focal loss for bad-pixel detection, $\mathcal{L}_{res}$ supervises the residual proposal on hard pixels, $\mathcal{L}_{ref}$ compares the soft refined prediction to ground truth, $\mathcal{L}_{pres}$ penalizes changes on raw-good pixels, and $\mathcal{L}_{sparse}$ discourages modifying too many pixels.

Preliminary result. The abstention model uses a 4-frame causal context and 16 input channels. The current implementation contains 194,818 parameters and is trained on balanced 96×96 crops. Unlike previous residual refiners, this model can improve validation MAE and Bad-3px while modifying only a small fraction of pixels.

At epoch 5, the validation result was:

$$\begin{aligned} \mathrm{MAE}_{raw} &= 5.1996 \text{ px}, \\ \mathrm{MAE}_{refined} &= 5.1538 \text{ px}, \\ \text{Bad-3px}_{raw} &= 38.878\%, \\ \text{Bad-3px}_{refined} &= 38.449\%. \end{aligned}$$

Only 3.73% of pixels were modified at the selected threshold. This is the first refiner variant that improves both MAE and Bad-3px without broadly damaging already-correct S2M2 regions.

This result supports the hypothesis that surgical stereo refinement should be formulated as a conservative correction policy rather than a dense residual prediction problem. The refiner should abstain by default and intervene only in regions that are likely to contain S2M2 failure.

D.19 Staged Abstention as Safe Refinement Policy

The previous refinement experiments showed that improving S2M2 predictions is not simply a matter of increasing model capacity. Dense residual refinement and jointly trained gated refinement both suffered from unsafe correction behavior: the model learned to reduce large training errors, but also modified regions where raw S2M2 was already accurate. This resulted in overcorrection and introduced new Bad-3px errors on reliable pixels.

A failure analysis confirmed this behavior. The full target dataset is strongly affected by high-error outliers: pixels with raw S2M2 error above 20 px account for only 8.9% of valid pixels, but contribute 51.4% of the total MAE. As a result, standard full-frame residual training is dominated by a small fraction of extreme-error pixels. The model is encouraged to apply broad corrections, even though most pixels should remain close to the raw S2M2 prediction.

This motivated a change in formulation. Instead of treating refinement as dense residual prediction, we formulate it as a staged detection-and-correction policy. The model should first identify where S2M2 is likely to fail, and only then apply a bounded correction. The default behavior is abstention, i.e. preserving the raw S2M2 prediction.

The final policy is:

$$D_t^{refined}(p) = \begin{cases} D_t^{raw}(p) + \widehat{\Delta D}_t(p), & P_t^{bad}(p) \geq \tau, \\ D_t^{raw}(p), & P_t^{bad}(p) < \tau, \end{cases}$$

where $P_t^{bad}(p)$ is the predicted probability that raw S2M2 is erroneous at pixel p, $\widehat{\Delta D}_t(p)$ is the predicted residual correction, and τ is a validation-calibrated threshold. The residual is bounded:

$$\widehat{\Delta D}_t = s \cdot \tanh(R_t),$$

where s is a fixed residual scale. This prevents the refiner from applying arbitrarily large disparity shifts.

Staged training. Unlike the previous jointly trained gated models, the abstention refiner is trained in two stages.

First, we train a bad-pixel detector:

$$P_t^{bad} = f_\theta^{det}\left(D_{t-3:t}^{raw}, M_t, \phi_t\right),$$

where $D_{t-3:t}^{raw}$ is a causal temporal context window, M_t is the valid mask, and ϕ_t contains low-cost temporal and spatial features such as temporal variance, temporal difference, and disparity gradients. The detector target is defined directly from the raw S2M2 error:

$$Y_t^{bad}(p) = \mathbb{1}\left[|D_t^{raw}(p) - D_t^{GT}(p)| \geq 3 \text{ px}\right].$$

The detector is optimized independently using a balanced crop sampling strategy and a BCE/focal detection loss.

Second, we train the residual correction head:

$$\widehat{\Delta D}_t = f_\theta^{res}\left(D_{t-3:t}^{raw}, M_t, \phi_t\right),$$

using the ground-truth residual:

$$\Delta D_t^{GT} = D_t^{GT} - D_t^{raw}.$$

During this stage, the detector is initially frozen, so that residual learning does not destabilize the correction policy. The residual loss is concentrated on hard pixels, while a strong preservation loss discourages modifications on already-correct pixels:

$$\mathcal{L}_{pres} = \sum_{p:\ |D_t^{raw}(p) - D_t^{GT}(p)| < 1} \left| D_t^{refined}(p) - D_t^{raw}(p) \right|.$$

This explicitly encourages identity behavior where S2M2 is reliable.

Balanced crop training. The staged abstention model is trained on balanced crops rather than only full frames. Crops are sampled from hard-error regions, good/stable regions, high-gradient boundary regions, temporally unstable regions, and random regions. This prevents a small number of large-error pixels from dominating the objective, while still exposing the model to the failure cases that require correction.

Result. The staged abstention model is the first refiner variant that improves both validation and held-out test performance while preserving reliable S2M2 regions. The model uses a 4-frame causal temporal context, 16 input channels, and 194,818 parameters. The best residual epoch was epoch 7, with a calibrated correction threshold of $\tau = 0.5$.

On the validation split:

$$\begin{aligned}
\text{MAE}_{raw} &= 5.1996 \text{ px},\\
\text{MAE}_{refined} &= 5.1167 \text{ px},\\
\text{Bad-3px}_{raw} &= 38.878\%,\\
\text{Bad-3px}_{refined} &= 38.479\%.
\end{aligned}$$

On the held-out test split:

$$\begin{aligned}
\text{MAE}_{raw} &= 4.6690 \text{ px},\\
\text{MAE}_{refined} &= 4.5637 \text{ px},\\
\text{Bad-3px}_{raw} &= 33.536\%,\\
\text{Bad-3px}_{refined} &= 33.230\%.
\end{aligned}$$

The test MAE improvement is:

$$4.6690 - 4.5637 = 0.1053 \text{ px},$$

corresponding to a relative improvement of:

$$\frac{0.1053}{4.6690} \times 100 \approx 2.25\%.$$

The Bad-3px rate improves by:

$$33.536 - 33.230 = 0.306$$

percentage points.

Although the absolute improvement is modest, the safety behavior is substantially better than previous refiners. On the test split, the model modifies 21.45% of pixels and introduces new Bad-3px errors on only 0.151% of raw-good pixels. The bad-pixel detector also generalizes better than the jointly trained gates, achieving:

$$\text{AUC}_{test} = 0.768, \qquad \text{AP}_{test} = 0.693.$$

Interpretation. This result supports the central hypothesis that surgical stereo refinement should not be formulated as unconstrained dense residual prediction. Since raw S2M2 is already reliable in many regions, the refiner must be conservative. The successful formulation is therefore:

$$\text{detect failure} \quad \rightarrow \quad \text{correct only when necessary} \quad \rightarrow \quad \text{otherwise preserve S2M2.}$$

Compared with dense residual and jointly trained gated variants, staged abstention provides a safer refinement mechanism. It reduces MAE and Bad-3px on held-out sequences while avoiding widespread degradation of good raw predictions. This makes it a stronger foundation for subsequent extensions, including larger abstention backbones, teacher/oracle fine-tuning with StereoAnyVideo, and semantic or artifact-aware features.

Final staged-abstention analysis. We further analyzed the best staged-abstention refiner on validation and held-out test sequences. The final model contains 194,818 parameters and uses a 4-frame causal temporal context with 16 input channels. The refiner is extremely light: a refiner-only benchmark reports approximately 1.08 ms per frame on CUDA with batch size 64.

On the held-out test split, the staged abstention policy improves both MAE and Bad-3px:

$$\begin{aligned} \text{MAE}_{raw} &= 4.6690 \text{ px}, \\ \text{MAE}_{refined} &= 4.5637 \text{ px}, \\ \text{Bad-3px}_{raw} &= 33.536\%, \\ \text{Bad-3px}_{refined} &= 33.230\%. \end{aligned}$$

The refiner modifies 21.46% of valid pixels. Importantly, it modifies 40.33% of raw bad pixels but only 8.74% of raw good pixels, indicating that the learned policy is selective rather than densely corrective. The fraction of raw-good pixels converted into new Bad-3px errors is only 0.152%, showing that the model largely preserves reliable S2M2 regions.

The detector also generalizes to the held-out test split, achieving:

$$\text{AUC}_{test} = 0.768, \qquad \text{AP}_{test} = 0.693.$$

Across validation and test sequences, 4 sequences improve, 3 show mixed behavior, and only 1 degrades. This confirms that the staged abstention strategy is more robust than previous dense residual or jointly gated refiners.

These results support the hypothesis that surgical stereo refinement should be formulated as a conservative detection-and-correction policy. Since raw S2M2 is already accurate in many regions, the safest online strategy is not to predict a dense correction everywhere, but to detect likely failure regions and apply a bounded residual only there:

$$D_t^{refined}(p) = \begin{cases} D_t^{raw}(p) + \widehat{\Delta D}_t(p), & P_t^{bad}(p) \geq \tau, \\ D_t^{raw}(p), & P_t^{bad}(p) < \tau. \end{cases}$$

This preserves the real-time character of the original S2M2 pipeline while adding a lightweight, temporally informed correction module.

D.20 Hybrid Oracle Fine-Tuning

The staged-abstention refiner provided the first safe improvement over raw S2M2. However, its evaluation on the selected teacher/oracle clips showed that most of the available oracle headroom remained unused. On the selected 502-frame subset, raw S2M2 obtained an MAE of 11.3231 px, while the v3.1 staged-abstention refiner reduced it to 11.0421 px. The all-available oracle reached 6.8062 px, meaning that v3.1 recovered only 6.22% of the available oracle gap.

At the same time, the detector remained strong on these difficult clips, with AUC/AP of 0.842/0.845. This suggests that the model can identify many unreliable S2M2 regions, but that its residual head does not yet exploit the teacher/oracle signal sufficiently. We therefore introduced a hybrid oracle fine-tuning stage.

The goal of this stage is not to imitate StereoAnyVideo globally. Instead, we use the all-available oracle as a local teacher only where it is clearly better than raw S2M2. The oracle selects among the available candidates, including raw S2M2, simple temporal baselines, RAFT-Small, and StereoAnyVideo. This prevents the model from treating any single teacher as universally correct.

Let D^{raw} be the raw S2M2 disparity, D^{GT} the ground-truth disparity, and D^{oracle} the all-available oracle disparity. Oracle supervision is activated only on valid hard pixels where the oracle improves over raw S2M2:

$$\mathcal{M}^{oracle}(p) = \mathbb{1}\left[|D^{raw}(p) - D^{GT}(p)| - |D^{oracle}(p) - D^{GT}(p)| > \epsilon\right] \cdot \mathbb{1}\left[|D^{raw}(p) - D^{GT}(p)| \geq 3\right].$$

The oracle residual target is:

$$\Delta D^{oracle}(p) = D^{oracle}(p) - D^{raw}(p).$$

During hybrid fine-tuning, training batches are mixed between the full-GT target set and the selected oracle clips:

$$\mathcal{B} = 0.75\,\mathcal{B}_{GT} + 0.25\,\mathcal{B}_{oracle}.$$

The full-GT batches preserve the general safety behavior of the v3.1 model, while the oracle batches expose the residual head to difficult failure modes with large teacher headroom.

The hybrid objective combines GT supervision, oracle residual supervision, good-pixel preservation, sparsity, and a new-Bad3 safety term:

$$\mathcal{L} = \lambda_{GT}\mathcal{L}_{GT} + \lambda_{oracle}\mathcal{L}_{oracle} + \lambda_{pres}\mathcal{L}_{pres} + \lambda_{sparse}\mathcal{L}_{sparse} + \lambda_{newbad}\mathcal{L}_{newbad}.$$

The preservation term discourages changes on pixels where raw S2M2 is already accurate:

$$\mathcal{L}_{pres} = \sum_{p:\ |D^{raw}(p) - D^{GT}(p)| < 1} \left| D^{refined}(p) - D^{raw}(p) \right|.$$

The new-Bad3 penalty discourages the model from converting raw-good pixels into new Bad-3px errors:

$$\mathcal{L}_{newbad} = \sum_{p:\ |D^{raw}(p) - D^{GT}(p)| < 3} \max\left(0, |D^{refined}(p) - D^{GT}(p)| - 3\right).$$

Unfrozen hybrid fine-tuning. The first hybrid variant, v3.2, fine-tuned both the detector and the residual head. This made the model substantially more conservative, but it also produced an identity-collapse failure mode. After the first epochs, the preservation, sparsity, and new-Bad3 losses pushed the detector probabilities downward. The validation threshold then moved to a very conservative value, causing almost no pixels to be modified. As a result, v3.2 was safer than v3.1 but recovered less oracle headroom.

On the selected clips, v3.2 reduced the frame-mean MAE to 11.0780 px, recovering 5.43% of the oracle gap. This is below the v3.1 gap recovery of 6.22%. However, v3.2 reduced the new-Bad3 rate from raw-good pixels from 4.79% to 2.33%, and reduced the modified-pixel fraction from 52.27% to 14.27%. This confirmed that the hybrid objective improved safety, but was too conservative when the detector was allowed to adapt.

Frozen-detector hybrid fine-tuning. To prevent detector collapse, we froze the v3.1 detector and fine-tuned only the residual correction pathway. This produces the v3.2b/v3.2c frozen-detector hybrid variants. The key idea is to preserve the already well-calibrated detection policy and adapt only the correction magnitude:

$$\text{keep where-to-correct fixed} \quad \rightarrow \quad \text{improve how-to-correct.}$$

Table 7: Hybrid oracle fine-tuning on selected teacher/oracle clips. All selected metrics are reported as frame means.

Method	MAE ↓	Oracle gap ↑	Bad-3 ↓	New Bad-3 ↓	Modified ↓
v3.1 staged abstention	11.0421	6.22%	51.280	4.79%	52.27%
v3.2 hybrid oracle	11.0780	5.43%	51.032	2.33%	14.27%
v3.2b frozen-detector hybrid	11.0075	6.99%	50.71	5.04%	18.43%
v3.2c frozen-detector hybrid, long	**11.0054**	**7.03%**	**50.698**	5.36%	18.43%

In v3.2b, freezing the detector prevented the identity collapse observed in v3.2. The correction threshold remained stable at $\tau = 0.7$, and the modified-pixel fraction remained approximately constant throughout training. This yielded a better trade-off than both v3.1 and v3.2: v3.2b improved the selected-frame MAE to 11.0075 px and recovered 6.99% of the oracle gap, while modifying only 18.43% of pixels.

We then extended the same recipe with a longer cosine-decay schedule, producing v3.2c. The model was trained for 32 epochs with the detector frozen, residual learning rate 3×10^{-4}, sparsity weight 0.02, and the same 75/25 full-GT/oracle crop mixture. The best checkpoint was selected at epoch 29. The improvement over v3.2b was small but consistent, indicating that this recipe had reached a plateau.

The final selected-frame metrics are:

$$
\begin{aligned}
\mathrm{MAE}_{raw} &= 11.3231 \text{ px},\\
\mathrm{MAE}_{v3.1} &= 11.0421 \text{ px},\\
\mathrm{MAE}_{v3.2} &= 11.0780 \text{ px},\\
\mathrm{MAE}_{v3.2b} &= 11.0075 \text{ px},\\
\mathrm{MAE}_{v3.2c} &= 11.0054 \text{ px}.
\end{aligned}
$$

The corresponding oracle-gap recovery is:

$$
\begin{aligned}
\mathrm{Gap}_{v3.1} &= 6.22\%,\\
\mathrm{Gap}_{v3.2} &= 5.43\%,\\
\mathrm{Gap}_{v3.2b} &= 6.99\%,\\
\mathrm{Gap}_{v3.2c} &= 7.03\%.
\end{aligned}
$$

On the full-GT held-out test split, v3.2c remains better than raw S2M2:

$$\mathrm{MAE}_{raw} = 4.6690 \text{ px}, \qquad \mathrm{MAE}_{v3.2c} = 4.6145 \text{ px}.$$

The full-GT Bad-3px rate also improves slightly:

$$\text{Bad-3}_{raw} = 33.536\%, \qquad \text{Bad-3}_{v3.2c} = 33.441\%.$$

Comparison. Table 7 summarizes the selected-clip comparison. All selected-clip values are reported as frame means to allow fair comparison between v3.1, v3.2, v3.2b, and v3.2c.

Interpretation. The hybrid oracle experiments show that the detector is no longer the main bottleneck. When the detector is allowed to adapt during hybrid fine-tuning, the model becomes overly conservative and collapses toward the identity solution. Freezing the detector preserves the useful staged-abstention policy and allows the residual head to absorb a small amount of oracle information.

The final v3.2c model is the best practical hybrid variant. It recovers more oracle headroom than v3.1 while modifying substantially fewer pixels. However, the gain over v3.2b is marginal, and the total recovered oracle gap remains only 7.03%. This indicates that the current residual head and loss formulation still cannot exploit most of the available oracle/SAV headroom. Further progress will likely require a different residual-learning strategy, stronger failure-mode conditioning, or targeted handling of the pathological clips that dominate the frame-mean new-Bad3 metric.

D.21 Final Hybrid-Oracle Refiner

The final hybrid-oracle variant, v3.2c, is the strongest practical model in the selected failure-clip setting. It builds on the staged-abstention v3.1 baseline and fine-tunes only the residual correction pathway while keeping the v3.1 detector frozen. This prevents the identity-collapse failure mode observed in v3.2, where preservation and sparsity losses pushed the detector probabilities down and caused the model to stop correcting.

On the selected teacher/oracle clips, v3.2c improves over both raw S2M2 and the v3.1 staged-abstention baseline. Raw S2M2 obtains a frame-mean MAE of 11.3231 px, v3.1 reduces it to 11.0421 px, and v3.2c further reduces it to 11.0054 px. The all-available oracle achieves 6.8062 px, so v3.2c recovers 7.03% of the available oracle gap, compared with 6.22% for v3.1:

$$\begin{aligned}\mathrm{MAE}_{raw} &= 11.3231 \text{ px},\\ \mathrm{MAE}_{v3.1} &= 11.0421 \text{ px},\\ \mathrm{MAE}_{v3.2c} &= 11.0054 \text{ px},\\ \mathrm{MAE}_{oracle} &= 6.8062 \text{ px}.\end{aligned}$$

At the same time, v3.2c is substantially more selective than v3.1. The modified pixel fraction decreases from 52.27% in v3.1 to 18.43% in v3.2c. This shows that the frozen-detector hybrid strategy recovers more oracle headroom while applying corrections to a much smaller subset of pixels.

On the full-GT held-out test split, v3.2c remains better than raw S2M2:

$$\mathrm{MAE}_{raw} = 4.6690 \text{ px}, \qquad \mathrm{MAE}_{v3.2c} = 4.6145 \text{ px}.$$

The Bad-3px rate also improves slightly:

$$\text{Bad-3}_{raw} = 33.536\%, \qquad \text{Bad-3}_{v3.2c} = 33.441\%.$$

However, v3.2c is slightly weaker than the v3.1 staged-abstention baseline on the full-GT test split, where v3.1 reaches 4.5637 px MAE. This indicates a trade-off: hybrid oracle fine-tuning improves selected failure clips and selectivity, but slightly sacrifices general full-GT refinement performance.

Failure-mode limitation. A pathological-clip analysis shows that the remaining safety issue is not a simple denominator artifact. The correlation between raw-good pixel count and new-Bad3 percentage is weak (-0.321), whereas the correlation between modified-pixel percentage and new-Bad3 percentage is strong (0.784). This indicates that elevated frame-mean new-Bad3 is caused by genuine over-correction on specific content, rather than by noisy ratios on frames with few raw-good pixels.

The issue is concentrated in two selected clips: one dominated by high temporal flicker and one dominated by high boundary error. On these two clips, the abstention gate opens on 39.6% of pixels on average, compared with 9.3% on the other four clips. The corresponding new-Bad3 rate is 15.7% on the two pathological clips and only 0.89% on the remaining clips. Among the 50 worst frames by new-Bad3, 39 still show a net MAE improvement, while 11 are net regressions. Thus, the model is not globally unsafe, but it is too aggressive on specific failure modes.

Interpretation. v3.2c should therefore be considered the current best hybrid-oracle refiner. It improves selected failure clips more than v3.1, recovers more oracle headroom, and modifies far fewer pixels. Its limitation is mode-dependent safety: high temporal flicker and high boundary error can cause the abstention gate to open too broadly. The detector is no longer the main bottleneck; the remaining bottleneck is failure-mode-conditioned residual correction under abstention constraints.

Modern damping refiner: v4-tiny. We first evaluated whether a moderately larger modern encoder–decoder could improve over the tiny abstention refiners while explicitly controlling correction strength. The resulting v4-tiny model contains 942,867 parameters and uses the same 16-channel input representation as the previous refiners. In addition to a bad-pixel confidence head and a bounded residual head, v4-tiny introduces a learned damping head. The final correction is therefore modulated as

$$D^{refined} = D^{raw} + P^{bad} \cdot \alpha \cdot \widehat{\Delta D},$$

where P^{bad} is the learned bad-pixel probability, $\alpha \in [0, 1]$ is a per-pixel damping factor, and $\widehat{\Delta D}$ is the bounded residual. The motivation is that the failure mode observed in v3.2c is not only where to apply a correction, but how aggressively to apply it once the abstention gate is open.

v4-tiny strongly validates the damping mechanism. On the pathological selected clips, the new-Bad3 rate from raw-good pixels drops from 15.77% for v3.2c to 0.33%, and the global frame-mean new-Bad3 drops from 5.36% to 0.55%. However, this safety comes at the cost of reduced accuracy. Selected-clip MAE worsens from 11.0054 px for v3.2c to 11.0669 px, and oracle-gap recovery drops from 7.03% to 5.67%. Moreover, v4-tiny regresses on the full-GT test split, increasing MAE from the raw S2M2 value of 4.6690 px to 4.7763 px. Thus, v4-tiny is not a new best model, but it demonstrates that learned damping is an effective mechanism for suppressing pathological overcorrection.

Lightweight suppression controller: SOG. We then tested whether the safety behavior of v4-tiny could be recovered without retraining the full refiner. The SOG branch freezes the v3.2c refiner and learns only a lightweight suppression/gating controller on top of its outputs. This branch has only 71k trainable parameters and adds a small runtime overhead, resulting in a combined runtime of approximately 1.65 ms/frame. Importantly, the model is initialized to reproduce v3.2c exactly, so the experiment starts from the current best practical operating point and only learns where to suppress harmful corrections.

SOG identifies a new useful operating point in the accuracy–safety trade-off. Compared with v3.2c, it reduces pathological new-Bad3 from 15.77% to 5.77%, while still improving over raw S2M2 on the full-GT test split (4.6690 → 4.6221 px). This is the first lightweight controller in the v3–v4 line to combine pathological safety below 8% with a full-GT test improvement over raw S2M2. However, this safety improvement still reduces oracle-gap recovery from 7.03% to 5.14%, and selected MAE worsens from 11.0054 px to 11.0909 px. Therefore, SOG is best interpreted as a practical safety controller rather than a true oracle-gap closing model.

Comparison of lightweight safety variants. The recent lightweight variants reveal a consistent trend. v3.2c remains the best accuracy-oriented lightweight refiner, with selected MAE 11.0054 px and 7.03% oracle-gap recovery, but it suffers from severe pathological overcorrection, with 15.77% new-Bad3 on the two pathological selected clips. v4-tiny nearly eliminates this pathological failure mode, reducing pathological new-Bad3 to 0.33%, but becomes too conservative and regresses on full-GT test performance. SOG provides an intermediate operating point: it reduces pathological new-Bad3 to 5.77% and still beats raw S2M2 on full-GT test, but it does not preserve the oracle-gap recovery of v3.2c.

These results indicate that the available low-level refiner features do contain signal for distinguishing beneficial and harmful corrections. However, small models and lightweight controllers primarily improve safety; they do not recover substantially more of the oracle/SAV gap. Closing that gap likely requires either higher-capacity refinement, richer temporal modeling, richer visual input, or stronger teacher distillation.

Event-Gated Boundary Memory Refiner. To test whether additional model capacity and explicit temporal reasoning can break the lightweight-refiner plateau, we also introduced an experimental Event-Gated Boundary Memory Refiner (EGBM). Unlike the previous variants, EGBM is not a simple dense residual head. It combines an event-gated temporal memory, a boundary-aware correction field, dynamic abstention, a mixture of correction experts, and an iterative energy-style

Table 8: Comparison of recent lightweight refiner variants on selected clips and full-GT test. Selected metrics are frame-mean over the selected oracle clips.

Metric	v3.2c	v4-tiny	v3.3 threshold	SOG
Selected MAE ↓	11.0054	11.0669	11.1062	11.0909
Oracle gap recovered ↑	7.03%	5.67%	4.80%	5.14%
Pathological new-Bad3 ↓	15.77%	0.33%	6.69%	5.77%
Clean new-Bad3 ↓	0.89%	0.64%	0.89%	–
Full-GT test MAE ↓	4.6145	4.7763	–	4.6221
Trainable parameters	–	943k	–	71k
Runtime	1.07 ms	2.74 ms	–	1.65 ms

update. The model has approximately 4.04M parameters in its standard configuration and remains within the online refiner budget.

The key design idea is to explicitly represent the two dominant failure modes identified in v3.2c: temporal flicker and boundary overcorrection. Temporal disparity inconsistencies are converted into event-like features and processed through a low-resolution recurrent memory. Boundary confidence is used to damp or stop correction propagation across likely disparity edges. A dynamic abstention module predicts local threshold offsets, while a mixture-of-experts decoder allocates pixels among identity preservation, smooth correction, boundary correction, and larger residual correction modes. The final update is applied in a small number of damped refinement steps rather than as a single unconstrained residual addition.

Early training logs suggest that EGBM has substantially stronger capacity than the lightweight variants. During detector pretraining, validation AUC peaks around 0.87, substantially above the final detector quality of v4-tiny. During the initial residual/damping stage, validation MAE improves from 5.1996 px to 5.1315 px with only 0.074% new-Bad3 and approximately 28.36% modified pixels. This indicates that a higher-capacity architecture can modify a larger fraction of pixels while keeping new errors low, a behavior not observed in the smaller models. The final selected-clip and full-GT test results are still required to determine whether EGBM becomes a new main-branch candidate.

Motivation for higher-capacity refiners. The objective of this line of work is not merely to build a lightweight post-processing module, but to approach the behavior of stronger offline video-level teachers such as SAV/oracle refinements. The results so far suggest that very small models can improve safety and reduce pathological overcorrection, but they are unlikely to close the teacher gap. In particular, v3.2c recovers only 7.03% of the selected-clip oracle gap, while v4-tiny and SOG improve safety at the cost of even lower gap recovery.

This motivates a second tier of refiners: medium-capacity, online-compatible models with richer temporal and boundary reasoning. Since S2M2 runs at roughly 58–62 ms/frame and the practical system budget is approximately 100 ms/frame, a refiner in the range of 10–35 ms/frame remains compatible with online use. This budget allows models in the 5M–30M parameter range, substantially larger than the 195k-parameter v3 refiner or the 943k-parameter v4-tiny model. Such models may provide enough capacity to recover more of the SAV/oracle gap while still preserving the safety constraints learned from the lightweight experiments.

Current interpretation. Overall, the recent experiments separate the refinement problem into two regimes. Small refiners and frozen-controller variants are effective for safety: they can reduce pathological new-Bad3 and prevent harmful corrections on raw-good pixels. However, they do not substantially increase oracle-gap recovery. In contrast, the early EGBM results suggest that additional capacity and explicit temporal memory may be necessary to move beyond the v3.2c plateau. The current best practical baseline remains v3.2c, SOG provides the best lightweight safety-oriented operating point, and EGBM represents the first serious higher-capacity attempt to

Table 9: Final comparison of the main refiner variants. Selected metrics are frame-mean over the selected oracle clips. Full-GT metrics are reported on the held-out test split.

Metric	v3.2c	v4-tiny	SOG	EGBM
Selected MAE ↓	11.0054	11.0669	11.0909	**10.4032**
Oracle gap recovered ↑	7.03%	5.67%	5.14%	**20.37%**
Pathological new-Bad3 ↓	15.77%	**0.33%**	5.77%	1.30%
Clean new-Bad3 ↓	0.89%	0.64%	–	0.81%
Global frame-mean new-Bad3 ↓	5.36%	**0.55%**	–	0.96%
Pixel-weighted new-Bad3 ↓	0.71%	–	–	**0.21%**
Full-GT test MAE ↓	4.6145	4.7763	4.6221	**4.5226**
Full-GT test Bad-3 ↓	–	–	–	**32.941%**
Parameters	195k	943k	71k trainable	4.04M
Runtime	1.07 ms	2.74 ms	1.65 ms	6.25 ms

close the gap toward SAV-like behavior.

Event-Gated Boundary Memory Refiner. The final refiner is an Event-Gated Boundary Memory Refiner (EGBM), a failure-aware post-hoc controller designed to refine frozen S2M2 disparities in surgical stereo sequences. Unlike the previous tiny refiners, which predict a single gated residual, EGBM decomposes refinement into several coupled decisions: whether a pixel should be corrected, how strongly the correction should be applied, which correction mode should be used, and whether temporal or boundary evidence should suppress the update.

EGBM combines four components. First, an event-gated temporal memory processes temporal disparity inconsistencies and provides a low-resolution memory of unstable regions. Second, a boundary-aware branch predicts edge-sensitive features to reduce residual bleeding across depth discontinuities. Third, a dynamic abstention module locally modulates the correction policy. Finally, a mixture-of-experts decoder combines smooth, boundary-aware, strong-correction, and identity-preserving experts. The resulting update is modulated by a learned damping factor, allowing the model to attenuate harmful corrections while preserving strong corrections on oracle-beneficial pixels.

Final EGBM results. EGBM is the first refiner to substantially break the v3.2c plateau. On the selected oracle clips, raw S2M2 obtains a frame-mean MAE of 11.3231 px, while EGBM reduces this to 10.4032 px. This corresponds to 20.37% oracle-gap recovery, compared with 7.03% for v3.2c. At the same time, EGBM reduces pathological new-Bad3 from 15.77% for v3.2c to 1.30%, while keeping clean-clip new-Bad3 at 0.81%. The global frame-mean new-Bad3 is 0.96%, and the pixel-weighted new-Bad3 is only 0.21%.

Importantly, the improvement is not limited to the selected oracle clips. On the full-GT test split, EGBM improves raw S2M2 from 4.6690 px to 4.5226 px and reduces Bad-3 from 33.536% to 32.941%. It also outperforms the previous best v3.2c test MAE of 4.6145 px. With 4.04M parameters and a refiner runtime of approximately 6.25 ms/frame, the estimated S2M2+EGBM system runtime is 68.25 ms/frame, remaining within the 100 ms online budget.

Damping interpretability. The learned damping behavior provides evidence that EGBM does not simply apply a globally aggressive correction. On the high-temporal-flicker pathological clip, the mean damping is 0.387 on hard-negative pixels and 0.733 on hard-positive pixels. On the high-boundary-error clip, this separation is even stronger: damping is 0.186 on hard negatives and 0.792 on hard positives. Thus, EGBM learns to suppress correction strength on pixels where refinement would create new Bad-3 errors, while preserving high correction strength on pixels where the oracle indicates that correction is beneficial.

Position with respect to prior work. Temporal aggregation, edge-aware refinement, confidence estimation, and mixture-of-experts have all been explored in stereo, depth estimation, or 3D reconstruction. Therefore, the novelty of EGBM is not any single component in isolation. Instead, EGBM contributes a failure-aware post-hoc refinement formulation for surgical stereo. Unlike video stereo methods that directly estimate temporally consistent disparity, EGBM operates on top of a frozen stereo matcher and explicitly targets the domain-specific failure modes observed in surgical scenes: temporal flicker and boundary overcorrection. The model integrates temporal event memory, boundary-aware damping, dynamic abstention, and correction experts to close the teacher/oracle gap while suppressing safety-critical overcorrection.

Current conclusion. The progression from v3.2c to v4-tiny, SOG, and EGBM reveals two distinct regimes. Lightweight controllers are effective at reducing pathological overcorrection, but they do not substantially increase oracle-gap recovery. EGBM is the first model to improve both sides of the trade-off: it recovers a much larger fraction of the oracle gap while reducing pathological new-Bad3 and improving the full-GT test split. We therefore treat EGBM as the new main branch for online surgical stereo refinement.

D.22 EGBM-v2-CARE: Change-Aware Reliability Encoding

Building on the EGBM-v1 refiner, we introduce EGBM-v2-CARE, a change-aware extension designed to improve temporal flicker handling without sacrificing online applicability. The motivation is that temporal inconsistency in surgical stereo is inherently ambiguous. A large change between consecutive frames may indicate a stereo artifact, such as disparity flicker or boundary overcorrection, but it may also correspond to a real scene change, such as tool insertion, tissue deformation, camera motion, occlusion, or specular/liquid appearance changes. Therefore, treating every temporal prediction error as an artifact can suppress legitimate surgical dynamics.

EGBM-v2-CARE augments the EGBM-v1 correction controller with a predictive temporal memory and a change-aware reliability head. Instead of only accumulating recent disparity changes, the model predicts the expected reliability features of the current frame from the past and interprets the resulting prediction error as a temporal surprise signal. CARE then classifies this surprise into lightweight change-type representations that modulate damping, dynamic abstention, expert routing, and bad-pixel confidence.

Predictive temporal reliability memory. For each frame t, the current raw stereo state is compressed into a low-resolution reliability embedding,

$$z_t = f(D_t, V_t, \nabla D_t, \Delta D_t), \tag{23}$$

where D_t is the raw S2M2 disparity, V_t is the validity mask, ∇D_t denotes disparity-gradient features, and ΔD_t represents local temporal disparity change. A causal memory state M_{t-1} summarizes past stereo behavior. From this state, the model predicts the expected current-frame embedding:

$$\hat{z}_t = g(M_{t-1}). \tag{24}$$

The discrepancy between the predicted and observed embeddings defines a temporal surprise signal:

$$e_t = \rho(z_t - \hat{z}_t), \tag{25}$$

where $\rho(\cdot)$ is a robust absolute-error operator. This signal is more informative than a raw frame-to-frame disparity difference: predictable camera or tissue motion can produce large disparity changes but low prediction error, whereas unstable stereo flicker tends to produce high prediction error.

Change-aware reliability encoding. The CARE module interprets temporal surprise rather than treating it as an artifact by default. Given the observed embedding z_t, predicted embedding $\hat{z}_t$, surprise e_t, and memory state M_{t-1}, CARE predicts low-resolution change-type probabilities:

$$c_t = \mathrm{CARE}(z_t, \hat{z}_t, e_t, M_{t-1}), \tag{26}$$

where c_t contains five soft change categories:

$$c_t = \{c_t^{\mathrm{art}}, c_t^{\mathrm{real}}, c_t^{\mathrm{tool}}, c_t^{\mathrm{bdry}}, c_t^{\mathrm{stable}}\}. \tag{27}$$

These correspond respectively to artifact-like change, real scene change, tool/occlusion-like change, boundary change, and stable predictable motion. The goal is not to obtain semantic tool labels, but to provide the correction controller with a reliability representation that separates stereo failure modes from legitimate scene dynamics.

CARE-modulated correction. The original EGBM controller predicts a bad-pixel probability, a bounded residual field, a damping factor, a local abstention threshold, and a mixture-of-experts routing distribution. EGBM-v2-CARE injects the memory, surprise, and change-type features into these heads. The refined disparity is computed as

$$D_t^{\mathrm{ref}} = D_t^{\mathrm{raw}} + G(p_t^{\mathrm{bad}}, \tau_t) \cdot \alpha_t \cdot R_t, \tag{28}$$

where $G(\cdot)$ is the dynamic abstention gate, p_t^{bad} is the predicted bad-pixel probability, τ_t is the local threshold, α_t is the learned damping factor, and R_t is the mixture-of-experts residual. In EGBM-v2-CARE, these quantities are conditioned on the CARE context:

$$(\alpha_t, \tau_t, R_t) = h(\Phi_t, M_{t-1}, e_t, c_t), \tag{29}$$

where Φ_t denotes the standard EGBM spatial, temporal, and boundary features. Artifact-like surprise encourages lower damping, higher abstention, and stronger identity routing. Real scene or tool-like change encourages memory adaptation and avoids blindly suppressing the correction. Boundary-change features reinforce edge-aware damping to prevent residual bleeding across depth discontinuities.

Causal memory update. After refining the current frame, the memory is updated causally:

$$M_t = U(M_{t-1}, z_t, e_t, c_t). \tag{30}$$

In the current implementation, the memory is replayed causally over the existing short temporal context window, preserving compatibility with the EGBM-v1 training pipeline and avoiding future-frame access. The recurrence itself is state-compatible and can be extended to a persistent streaming state over full video sequences. This makes the model online-compatible: at frame t, it uses only the current frame and the compressed history up to $t-1$.

Proxy supervision for CARE. Because explicit labels for artifacts, tools, occlusions, and real scene changes are not available, CARE is trained using lightweight proxy supervision derived from existing signals. Hard-negative pixels, defined as raw-good pixels that would become Bad-3 after refinement, encourage high artifact-like scores. Oracle beneficial pixels encourage lower artifact-like scores and higher correction trust. Boundary-change proxies are obtained from regions with both strong disparity gradients and temporal disparity changes. Coherent large temporal changes are used as weak proxies for real scene or occlusion-like change, while low temporal-change regions encourage the stable-predictable category.

The auxiliary loss is intentionally kept small so that the main refinement and safety losses dominate:

$$\begin{aligned} \mathcal{L}_{\mathrm{CARE}} =& \lambda_{\mathrm{pred}}\mathcal{L}_{\mathrm{pred}} + \lambda_{\mathrm{art}}\mathcal{L}_{\mathrm{art}} + \lambda_{\mathrm{bdry}}\mathcal{L}_{\mathrm{bdry}} \\ &+ \lambda_{\mathrm{real}}\mathcal{L}_{\mathrm{real}} + \lambda_{\mathrm{stable}}\mathcal{L}_{\mathrm{stable}} - \lambda_{\mathrm{ent}}\mathcal{H}(c_t). \end{aligned} \tag{31}$$

Table 10: Runtime and parameter comparison between EGBM-v1 and EGBM-v2-CARE. Both models are benchmarked at 256×320 resolution on an NVIDIA H100 GPU in FP32.

Model	Parameters	Runtime	Peak VRAM
EGBM-v1	4.04M	6.26 ms/frame	8.9 GB
EGBM-v2-CARE	4.35M	6.65 ms/frame	9.2 GB
CARE overhead	+0.31M	+0.39 ms/frame	+0.3 GB

Here, $\mathcal{L}_{\text{pred}}$ trains the next-feature predictor, $\mathcal{L}_{\text{art}}$ separates hard negatives from hard positives, $\mathcal{L}_{\text{bdry}}$ encourages boundary-change responses near moving depth discontinuities, $\mathcal{L}_{\text{real}}$ encourages real-change responses on coherent large temporal changes, and $\mathcal{H}(c_t)$ is a small entropy regularizer used to prevent collapse to a single change type.

Warm-start and computational cost. EGBM-v2-CARE is initialized from the EGBM-v1 checkpoint. All shape-compatible parameters are loaded directly, while the widened CARE-conditioned layers are initialized such that their additional CARE input channels initially contribute zero. This preserves the EGBM-v1 behavior exactly at initialization, ensuring that any improvement is attributable to the learned CARE pathway rather than to a different random initialization.

The resulting model adds only a small overhead. EGBM-v1 contains 4.04M parameters and runs in approximately 6.26 ms/frame on an H100 GPU. EGBM-v2-CARE contains 4.35M parameters and runs in approximately 6.65 ms/frame, adding only 0.39 ms/frame. Assuming the frozen S2M2 matcher runs at approximately 58–62 ms/frame, the full S2M2+CARE pipeline remains below the 100 ms online budget.

CARE diagnostics. In addition to the standard selected-oracle and full-GT metrics, we evaluate whether CARE learns the intended reliability structure. We report temporal surprise statistics by failure mode, CARE change-type scores on hard negatives and hard positives, damping separation on pathological clips, and entropy of the change-type distribution. In particular, the high-temporal-flicker clip is used to test the core hypothesis: EGBM-v1 has weaker damping separation on this mode than on boundary errors. CARE should reduce damping on temporal hard negatives while preserving high damping on oracle-beneficial hard positives.

Toward stateful streaming CARE. The current EGBM-v2-CARE implementation replays the predictive memory over the same short temporal window used by EGBM-v1. This provides a controlled architecture test while preserving the existing training and evaluation protocol. However, the CARE recurrence naturally supports a stronger stateful online formulation. In a streaming version, the memory state would persist across the full video sequence:

$$M_0 = 0, \qquad (D_t^{\text{ref}}, M_t) = \text{CAREStep}(D_t, V_t, M_{t-1}). \tag{32}$$

This would allow the model to summarize arbitrary-length past stereo behavior into a fixed-size causal reliability state. The computational cost per frame would remain constant, while the effective temporal context would grow with the sequence length. Such a stateful formulation is particularly attractive for robotic surgery, where regions may remain unstable over long periods and where new instruments or occlusions must be incorporated into the scene memory rather than suppressed indefinitely.

Potential advantage of persistent memory. A persistent CARE state would allow the model to distinguish transient stereo flicker from stable scene changes over longer horizons. For example, an entering tool may initially produce high temporal surprise, but if the change remains spatially coherent and temporally persistent, the memory update gate can adapt and treat it as part of the new scene state. Conversely, temporally alternating or spatially incoherent disparity changes can be retained as artifact-like signals and suppressed through damping and abstention. This gives the refiner an online reliability memory whose size is independent of the sequence length.

Interpretation if CARE improves over EGBM-v1. CARE improves over EGBM-v1 by resolving a key ambiguity in temporal refinement: not all temporal inconsistency is an artifact. By predicting the expected current-frame reliability features and interpreting the resulting surprise as artifact-like, boundary-like, stable, or real-scene-like change, CARE enables the controller to suppress harmful flicker without blindly damping legitimate surgical dynamics. This makes EGBM-v2-CARE the new main branch for online surgical stereo refinement.

Interpretation if CARE does not improve aggregate performance. Even when CARE does not immediately improve the aggregate oracle-gap metric, its diagnostics remain informative. A reduction in hard-negative damping on high-temporal-flicker regions, together with non-collapsed change-type scores, would indicate that predictive change-aware reliability contains useful signal for a future stateful streaming formulation. In this case, CARE should be interpreted as a temporal reliability controller whose full benefit may require persistent sequence-level memory rather than short-window replay.

D.23 EGBM-v3-CARE-S: Stateful Change-Aware Reliability Refinement

EGBM-v3-CARE-S extends the EGBM family with a predictive, change-aware temporal memory designed to improve correction safety in temporally unstable surgical stereo sequences. The model is motivated by a limitation observed in EGBM-v1: while the original event-gated memory and boundary-aware controller substantially improved oracle-gap recovery, temporal changes were still treated mainly as generic instability. This is problematic because temporal disparity variation is intrinsically ambiguous. A large change may correspond to stereo flicker, boundary overcorrection, or invalid matching, but it may also represent a legitimate scene transition caused by camera motion, tissue deformation, occlusion, tool insertion, or specular appearance changes.

The goal of EGBM-v3-CARE-S is therefore not simply to retain more frames, but to predict what the current temporal state should look like and to interpret the difference between predicted and observed behavior. This predictive change-aware representation is then used to control damping, abstention, expert routing, and memory updates.

Predictive reliability encoding. At each frame t, the model compresses the current stereo state into a low-resolution reliability embedding:

$$z_t = f(D_t, V_t, \nabla D_t, \Delta D_t), \tag{33}$$

where D_t is the raw disparity map, V_t is the validity mask, ∇D_t denotes spatial disparity gradients, and ΔD_t represents short-term temporal disparity variation.

A causal temporal state M_{t-1} summarizes the preceding sequence context. From this state, the model predicts the expected current embedding:

$$\hat{z}_t = g(M_{t-1}). \tag{34}$$

The temporal surprise signal is then defined as

$$e_t = |z_t - \hat{z}_t|. \tag{35}$$

Unlike a simple frame-to-frame difference, temporal surprise can remain low during predictable motion and become high when the observed stereo behavior deviates from the learned temporal dynamics. This enables the model to distinguish predictable scene evolution from unstable stereo artifacts.

Change-aware reliability representation. The CARE head processes the observed embedding, predicted embedding, surprise, and temporal state:

$$c_t = \mathrm{CARE}\left(z_t, \hat{z}_t, e_t, M_{t-1}\right). \tag{36}$$

The output c_t is a soft change-type representation containing five channels:

$$c_t = \left\{c_t^{\mathrm{artifact}}, c_t^{\mathrm{real}}, c_t^{\mathrm{tool}}, c_t^{\mathrm{boundary}}, c_t^{\mathrm{stable}}\right\}. \tag{37}$$

These channels represent artifact-like change, real scene change, tool/occlusion-like change, boundary change, and stable predictable motion.

The objective is not semantic scene understanding. Instead, CARE provides a failure-aware reliability representation that helps the controller distinguish between changes that should suppress refinement and changes that should be accepted as legitimate scene dynamics.

Correction controller. As in EGBM-v1, the model predicts a bad-pixel probability, a mixture of bounded residual corrections, an identity expert, a dynamic abstention threshold, a boundary-aware correction field, and a learned damping factor. The final refinement is expressed as

$$D_t^{\mathrm{ref}} = D_t^{\mathrm{raw}} + G\left(p_t^{\mathrm{bad}}, \tau_t\right) \cdot \alpha_t \cdot R_t, \tag{38}$$

where $G(\cdot)$ is the dynamic correction gate, p_t^{bad} is the predicted bad-pixel confidence, τ_t is the local abstention threshold, α_t is the damping factor, and R_t is the mixture-of-experts residual.

In EGBM-v3-CARE-S, these quantities are conditioned on the CARE context:

$$(\alpha_t, \tau_t, R_t, r_t) = h\left(\Phi_t, M_{t-1}, e_t, c_t\right), \tag{39}$$

where Φ_t denotes the standard spatial, temporal, and boundary features, and r_t is the expert-routing distribution.

Artifact-like regions encourage reduced damping, stronger abstention, and identity-expert routing. Boundary changes reinforce edge-aware correction, while coherent real-scene changes are allowed to modify the temporal state rather than being automatically suppressed.

Stateful streaming memory. EGBM-v3-CARE-S introduces an explicit persistent state that is updated frame-by-frame:

$$S_t = \left\{M_t^{\mathrm{EGBM}}, M_t^{\mathrm{CARE}}, \hat{z}_{t+1}, D_t^{\mathrm{LR}}, V_t^{\mathrm{LR}}\right\}. \tag{40}$$

The state includes the original EGBM temporal memory, the predictive CARE memory, the next predicted embedding, and the previous low-resolution disparity and validity maps.

Streaming inference follows

$$\left(D_t^{\mathrm{ref}}, S_t\right) = \mathrm{CAREStep}\left(X_t, S_{t-1}\right), \tag{41}$$

where X_t contains the current feature stack.

The temporal state has fixed spatial and channel dimensions. Therefore, both the memory footprint and computational cost per frame remain constant with respect to the sequence length:

$$\mathcal{O}(\text{memory}) = \mathcal{O}(1), \qquad \mathcal{O}(\text{cost per frame}) = \mathcal{O}(1). \tag{42}$$

Change-aware memory preservation. The streaming formulation includes a learned local memory-preservation gate. Given the current embedding, surprise, CARE probabilities, and previous memory, the model predicts

$$k_t = \sigma\left(q\left(z_t, e_t, c_t, M_{t-1}\right)\right), \tag{43}$$

where high k_t values preserve the previous memory and low values allow stronger adaptation to the current observation. The state is updated as

$$M_t = k_t \odot M_{t-1} + (1 - k_t) \odot \widetilde{M_t}. \tag{44}$$

This mechanism is intended to prevent temporally unstable artifact-like observations from contaminating the persistent state, while still allowing rapid adaptation to coherent scene changes.

Window and streaming modes. The architecture supports two inference configurations.

In *window mode*, the temporal memories are reconstructed causally from a four-frame local context for each target frame:

$$\{X_{t-3}, X_{t-2}, X_{t-1}, X_t\} \to D_t^{\mathrm{ref}}. \tag{45}$$

In *streaming mode*, the state persists across the full sequence:

$$S_0 = 0, \qquad \left(D_t^{\mathrm{ref}}, S_t\right) = \mathrm{CAREStep}\left(X_t, S_{t-1}\right). \tag{46}$$

This dual formulation enables a controlled comparison between predictive change-aware encoding with short temporal replay and predictive encoding with long-horizon persistent memory.

Training with truncated backpropagation through time. The model is trained on contiguous temporal chunks rather than independent frames. For each sequence chunk of length L, the state is initialized at the chunk boundary and propagated causally:

$$S_0 = 0, \qquad S_t = U\left(S_{t-1}, X_t\right), \quad t = 1, \ldots, L. \tag{47}$$

Losses are accumulated across the chunk and the state is detached between chunks, implementing truncated backpropagation through time. The default chunk length is 16 frames, with an initial burn-in period whose loss contribution is downweighted.

The training protocol follows three stages.

1. **Stage 1: predictive detector adaptation.** The backbone is initialized from EGBM-v1, while the bad-pixel detector and CARE pathway are trained using detector supervision and predictive-memory loss.

2. **Stage 2: full-GT streaming refinement.** The model is trained on full-GT temporal chunks using disparity reconstruction, preservation, new-Bad3, damping, detector, and temporal prediction objectives.

3. **Stage 3: oracle and hard-negative refinement.** Training mixes full-GT chunks, oracle-beneficial chunks, and pathological hard-negative chunks. This stage explicitly penalizes unsafe corrections, encourages identity routing on vulnerable pixels, and suppresses memory contamination by hard negatives.

CARE supervision. Explicit labels for artifact, tool, occlusion, and scene-change classes are not available. CARE therefore uses weak proxy supervision derived from available signals. Hard negatives encourage artifact-like responses, while oracle-beneficial pixels discourage artifact classification. Boundary-change responses are encouraged in regions with both high spatial gradients and high temporal variation. Coherent large temporal changes act as weak proxies for real-scene or occlusion-like transitions, while low temporal variation supports the stable-predictable category.

The CARE auxiliary objective is

$$\begin{aligned} \mathcal{L}_{\mathrm{CARE}} = &\lambda_{\mathrm{pred}}\mathcal{L}_{\mathrm{pred}} + \lambda_{\mathrm{artifact}}\mathcal{L}_{\mathrm{artifact}} + \lambda_{\mathrm{boundary}}\mathcal{L}_{\mathrm{boundary}} \\ &+ \lambda_{\mathrm{real}}\mathcal{L}_{\mathrm{real}} + \lambda_{\mathrm{stable}}\mathcal{L}_{\mathrm{stable}} - \lambda_{\mathrm{entropy}}\mathcal{H}(c_t). \end{aligned} \tag{48}$$

The auxiliary losses are kept small so that the main refinement and safety objectives remain dominant.

Table 11: Comparison of the main temporal refinement variants. Selected metrics are frame-mean over 502 frames from six selected oracle clips. Full-GT metrics are reported on the held-out test split.

Model	Selected MAE ↓	Oracle gap ↑	Patho nB3 ↓	Clean nB3 ↓	Full-GT test ↓
v3.2c	11.0054	7.03%	15.77%	0.89%	4.6145
v4-tiny	11.0669	5.67%	0.33%	0.64%	4.7763
SOG	11.0909	5.14%	5.77%	–	4.6221
EGBM-v1	10.4032	20.37%	1.30%	0.81%	4.5226
EGBM-v2-CARE	**10.3866**	**20.73%**	2.89%	0.80%	**4.5090**
EGBM-v3-CARE-S window	10.4066	20.29%	0.67%	**0.50%**	4.5753
EGBM-v3-CARE-S streaming	10.4125	20.16%	**0.47%**	**0.50%**	4.5880

Safety-aware refinement objective. The complete training objective combines full-GT reconstruction, detector supervision, residual regression, raw-good preservation, new-Bad3 suppression, damping regularization, hard-negative preservation, identity routing, and CARE auxiliary losses:

$$\begin{aligned}\mathcal{L} =&\lambda_{\text{full}}\mathcal{L}_{\text{full}} + \lambda_{\text{det}}\mathcal{L}_{\text{det}} + \lambda_{\text{res}}\mathcal{L}_{\text{res}} \\ &+ \lambda_{\text{pres}}\mathcal{L}_{\text{pres}} + \lambda_{\text{nB3}}\mathcal{L}_{\text{nB3}} + \lambda_{\text{damp}}\mathcal{L}_{\text{damp}} \\ &+ \lambda_{\text{id}}\mathcal{L}_{\text{id}} + \lambda_{\text{memory}}\mathcal{L}_{\text{memory}} + \mathcal{L}_{\text{CARE}}.\end{aligned} \tag{49}$$

The hard-negative terms target pixels that are correct in the raw disparity but would become erroneous after refinement. This directly trains the model to suppress unsafe corrections.

Initialization and computational cost. EGBM-v3-CARE-S is warm-started from EGBM-v1. All compatible parameters are loaded directly, while newly introduced CARE and streaming modules are initialized safely. The widened decoder and damping layers copy the original EGBM-v1 weights into their existing input channels, while the additional CARE channels are initialized to zero. Consequently, the initial window-mode behavior matches the original EGBM policy before CARE influence is learned.

The final model contains approximately 4.35M parameters. On an NVIDIA H100 GPU at 256×320 resolution, the window and streaming configurations run at approximately 7.41 ms/frame and 7.13 ms/frame, respectively. Combined with the frozen S2M2 matcher, the estimated full system latency is approximately 69 ms/frame, remaining below the 100 ms online budget.

Accuracy–safety trade-off. EGBM-v3-CARE-S operates in essentially the same aggregate accuracy regime as EGBM-v1. In window mode, selected MAE changes from 10.4032 px to 10.4066 px and oracle-gap recovery changes from 20.37% to 20.29%. In streaming mode, the model obtains 10.4125 px and 20.16%, respectively.

Although these differences are small, the safety metrics improve substantially. Pathological new-Bad3 decreases from 1.30% in EGBM-v1 to 0.67% in window mode and 0.47% in streaming mode. Clean-clip new-Bad3 decreases from 0.81% to 0.50% in both EGBM-v3 configurations.

Thus, the main benefit of EGBM-v3-CARE-S is not increased aggregate disparity accuracy, but reduced unsafe correction behavior.

Damping behavior on temporal flicker. The strongest evidence for the intended CARE mechanism is the damping response on the high-temporal-flicker clip. EGBM-v1 applies a mean damping of 0.387 on hard-negative pixels and 0.733 on hard-positive pixels. EGBM-v3-CARE-S reduces hard-negative damping to approximately 0.127 while preserving high damping on oracle-beneficial hard positives.

The hard-negative versus hard-positive damping separation therefore increases from approximately

$$0.733 - 0.387 = 0.346 \tag{50}$$

for EGBM-v1 to approximately

$$0.690 - 0.127 = 0.563 \tag{51}$$

for EGBM-v3-CARE-S.

This indicates that CARE learns a more selective correction policy: harmful updates are strongly attenuated while beneficial corrections remain active.

Source of the safety improvement. The safety improvement should not be attributed to persistent long-horizon memory. The windowed model, which reconstructs temporal context from four local frames, achieves nearly the same safety gains as the streaming model. Moreover, the window configuration does not use the persistent sequence-level memory gate during inference.

The improvement therefore originates primarily from two factors:

$$\text{predictive change-aware local encoding} \tag{52}$$

and

$$\text{safety-aware sequential training.} \tag{53}$$

CARE enables the controller to interpret short-term temporal variation rather than treating all changes as generic instability. Chunk-based training then converts this representation into a conservative correction policy through hard-negative preservation, new-Bad3 penalties, identity routing, damping supervision, and causal temporal optimization.

The resulting interpretation is

$$\text{Safety gain} \approx \text{change-aware short context} + \text{sequential safety training}, \tag{54}$$

rather than

$$\text{Safety gain} \approx \text{arbitrarily long temporal memory.} \tag{55}$$

Long-horizon memory ablation. Persistent streaming inference does not outperform short-window replay. The windowed model obtains a selected MAE of 10.4066 px and recovers 20.29% of the oracle gap, compared with 10.4125 px and 20.16% for streaming inference. The same trend appears on the full-GT test split, where window mode achieves 4.5753 px and streaming mode achieves 4.5880 px.

Streaming also fails to improve the clean or pathological safety metrics enough to justify the additional state-management complexity. Although pathological new-Bad3 is slightly lower in streaming mode, the gain is small relative to the added implementation burden and the mild degradation in aggregate accuracy.

These results do not indicate that temporal reasoning is unnecessary. Rather, they show that the relevant temporal evidence in this dataset is mostly captured within a short local horizon. The primary gain comes from learning how to interpret recent change, not from preserving an arbitrarily long history.

Why persistent memory did not help. Several explanations are consistent with the observed result. First, the selected validation clips may not contain temporal dependencies that require context beyond a few frames. Second, the predictive state may already summarize most useful information within the four-frame local window. Third, the weak proxy supervision used for CARE may be sufficient for local reliability classification but not strong enough to train meaningful long-horizon scene adaptation. Finally, persistent memory may introduce mild temporal smoothing or state inertia that reduces responsiveness to rapidly changing surgical content.

Importantly, empirical analysis did not support a simple long-sequence drift failure. Both window and streaming modes showed similar degradation on difficult sequence segments, suggesting that some regions are intrinsically harder rather than being degraded by persistent-state accumulation.

Practical deployment recommendation. For deployment, the window-mode EGBM-v3-CARE-S configuration offers the best practical trade-off. It retains almost the same disparity accuracy and oracle-gap recovery as EGBM-v1, while approximately halving pathological new-Bad3 and reducing clean-clip new-Bad3.

The streaming configuration provides slightly stronger pathological safety, but does not improve aggregate accuracy, full-GT generalization, or runtime enough to justify persistent state management. The recommended operational configuration is therefore:

$$\boxed{\text{EGBM-v3-CARE-S in window mode}} \tag{56}$$

for safety-oriented robotic deployment.

The EGBM-v2-CARE model remains the accuracy-oriented variant, as it achieves the best selected MAE, oracle-gap recovery, and full-GT test MAE, but its higher pathological new-Bad3 makes it less suitable when unsafe overcorrection is the primary concern.

Scientific interpretation. The architectural progression supports three conclusions.

First, increased capacity alone is insufficient. Larger models and candidate fusion did not consistently outperform EGBM-v1.

Second, failure-aware temporal interpretation is beneficial. CARE substantially improves the separation between harmful and useful corrections, especially on temporal flicker.

Third, long-horizon persistent memory is not required for the observed gain. Short-window predictive encoding combined with safety-aware sequential training captures most of the useful temporal structure.

The central finding is therefore:

> The dominant benefit does not arise from retaining a longer temporal history, but from learning how to interpret short-term temporal change and suppress unsafe corrections.

Final conclusion. EGBM-v3-CARE-S is not a clean accuracy winner over EGBM-v1, but it establishes a substantially safer refinement policy at almost unchanged disparity accuracy. The model reduces pathological new-Bad3 from 1.30% to 0.67% in window mode and 0.47% in streaming mode, while maintaining approximately 20% oracle-gap recovery.

The long-context hypothesis is not confirmed: persistent sequence-level memory does not outperform short-window replay. However, the predictive CARE representation and sequence-aware safety training clearly improve correction selectivity. EGBM-v3-CARE-S should therefore be interpreted as a safety-oriented temporal refinement architecture whose main contribution is more reliable correction control rather than longer temporal memory.

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ARGOS Temporal Stereo Refinement

From Universal Refinement to Lightweight Domain Adaptation August 17, 2026

E Original Objective

The original objective was to develop a lightweight temporal refinement module on top of a frozen stereo model, initially S2M2, to:

- reduce frame-to-frame disparity flickering;
- improve temporal coherence;
- correct large geometric errors;
- preserve already-correct raw predictions;

- operate causally and in real time;
- generalize to unseen surgical domains without retraining.

The intended system was:

$$\{D_{t-k}^{\text{raw}}, \ldots, D_t^{\text{raw}}\} \xrightarrow{\text{temporal refiner}} D_t^{\text{refined}}, \tag{57}$$

where the stereo backbone remained frozen and only the temporal refinement module was trained.

F Initial Experimental Findings

F.1 Refinement on SCARED

The developed refiners successfully improved S2M2 on the SCARED domain. The strongest conservative variants achieved a favorable compromise between accuracy and safety, while remaining lightweight and fast.

Table 12: Summary of representative refiner variants on the primary SCARED setup.

Method	Oracle gap	Pathological new-Bad3	Full-GT MAE	Runtime
Raw S2M2	0%	0%	4.6690	–
EGBM-v1	20.37%	1.30%	4.5226	6.25 ms
EGBM-v2 CARE	20.73%	2.89%	**4.5090**	6.65 ms
EGBM-v3 CARE window	20.29%	0.67%	4.5753	7.41 ms
EGBM-v3 CARE streaming	20.16%	**0.47%**	4.5880	7.13 ms

The short causal window was at least as effective as persistent long-term state. This suggests that the useful temporal signal is mostly local and that long memory is not automatically beneficial for surgical stereo refinement.

F.2 Oracle-Gap Analysis

A detailed oracle analysis showed that the main limitation was not detecting where the raw prediction was wrong, nor predicting the direction of the correction.

Approximately:

- 95% of the remaining oracle gap required corrections larger than 3 px;
- more than 92% required corrections larger than 6 px;
- approximately 86% required corrections larger than 12 px;
- correction support was approximately 76%;
- correction sign accuracy was approximately 94–95%;
- applied correction magnitude was only approximately 15–17% of the oracle magnitude when the sign was correct.

The main bottleneck was therefore correction magnitude.

Table 13: Large-correction experiments.

Method	Selected MAE	Oracle gap	Pathological new-Bad3	Full-GT MAE
MPC	9.1788	47.47%	7.33%	4.6386
CPV	9.2977	44.84%	5.05%	4.5939
ASSR	9.2431	46.05%	6.02%	4.5997

F.3 Large-Magnitude Proposal Branch

The MPC, CPV, and ASSR branches were introduced to allow large corrections.

These results confirmed that large-magnitude corrections could close a substantial part of the oracle gap. However, they also exposed a strong safety trade-off:

> Increasing correction magnitude improved difficult frames, but caused over-correction of predictions that were already geometrically accurate.

A conservative post-hoc configuration reduced pathological errors to approximately 0.43%, but also collapsed oracle-gap recovery to approximately 12.48%.

G What Did Not Work

G.1 Universal Zero-Shot Refinement

The original assumption was that a temporal refiner trained on SCARED could be placed on top of S2M2 and improve its outputs in other surgical domains.

This was disproven by the SERV-CT zero-shot evaluation.

Table 14: Zero-shot behavior on SERV-CT. Positive MAE change indicates degradation.

Method	Δ MAE	new-Bad3	Harmful correction rate
EGBM-v1	+0.53	8.5%	0.70
EGBM-v2	+0.60	8.5%	0.68
v3.2c	+0.54	13.3%	0.47
MPC	+5.35	44.2%	0.83
CPV	+5.06	39.3%	0.82

On SERV-CT, raw S2M2 already achieved a MAE of approximately 1.28 px. The refiner had been trained in a regime where significant corrections were often required. When transferred to a domain where the raw matcher was already accurate, the learned correction policy continued to activate and degraded the result.

G.2 Interpretation

The failure is not simply caused by insufficient model capacity.

A disparity-only refiner observes patterns such as:

- disparity gradients;
- temporal differences;
- boundaries;
- validity masks;
- local instability.

However, the same pattern may correspond to:

- a real geometric discontinuity;
- a stereo-matching failure;
- true object or tissue motion;
- temporal flicker;
- a prediction that is already correct.

Without access to the original matching evidence, cost volume, photometric consistency, or calibrated matcher confidence, the problem may be partially non-identifiable.

G.3 Why More SCARED Data Alone Is Not Sufficient

Even a large number of correlated frames does not guarantee domain coverage.

SCARED contains approximately 20,000 frames but only a limited number of independent sequences and acquisition conditions. Consecutive frames have high correlation, so the effective number of independent examples is much lower than the raw frame count.

More frames from the same error distribution do not necessarily teach the refiner how to behave when:

- the camera changes;
- calibration changes;
- the stereo backbone changes;
- raw predictions become much more accurate;
- the distribution of occlusions, specularities, and boundaries changes.

H Revised Scope

H.1 New Central Idea

The refiner is no longer presented as a universal zero-shot correction module.

The revised concept is:

> A lightweight causal temporal adapter that specializes a frozen stereo backbone to the error dynamics of a target domain using a limited amount of target data.

The full system becomes:

$$\underbrace{(I_t^L, I_t^R) \xrightarrow{\text{frozen stereo backbone}} D_t^{\text{raw}}}_{\text{general stereo estimation}} \xrightarrow[\text{domain adapter}]{\text{lightweight temporal}} D_t^{\text{refined}}. \tag{58}$$

Only the temporal adapter is trained or fine-tuned for the target domain.

H.2 Main Claim

> A small causal temporal adapter, with approximately four million trainable parameters, can efficiently specialize frozen stereo backbones to domain-specific error dynamics, improving geometric accuracy and temporal stability without retraining a large stereo network.

H.3 Claims Explicitly Avoided

The project will not claim that:

- one checkpoint improves every stereo backbone;
- one checkpoint improves every unseen domain zero-shot;
- the complete system contains only four million parameters;
- the method universally outperforms all large stereo models;
- sparse D4D keyframes provide dense temporal ground truth.

The approximately four million parameters refer only to the additional trainable adapter.

I Research Questions

RQ1 Can a lightweight causal temporal adapter improve S2M2 on SCARED in both geometric accuracy and temporal consistency?

RQ2 Can the same adapter architecture be trained successfully on top of different frozen stereo backbones?

RQ3 How much target-domain data is required before the adapter becomes effective?

RQ4 Does pretraining the adapter on a source domain reduce the target data required compared with training from scratch?

RQ5 Can an efficient stereo backbone plus a small adapter approach or outperform substantially larger online stereo systems on a target domain?

RQ6 Does the adapter improve true temporal stability, or does it merely learn a static dataset-specific geometric correction?

J Dataset Strategy

J.1 SCARED

SCARED remains the primary dataset.

Its role is:

- dense supervised training;
- dense temporal evaluation;
- oracle analysis;
- architectural ablations;
- flicker and jitter evaluation;
- boundary stability;
- temporal safety analysis.

SCARED is the dataset that directly supports the temporal-refinement claim.

J.2 SERV-CT

SERV-CT contains only a small collection of non-consecutive stereo images and is therefore not suitable for dense temporal-consistency evaluation.

Its role is restricted to:

- static cross-domain geometric evaluation;
- raw-good preservation;
- analysis of zero-shot false activation;
- optional few-shot geometric recalibration.

SERV-CT should not be used to support claims about flicker reduction.

J.3 D4D

D4D contains long stereo sequences and sparse Zivid structured-light geometry.

The raw organization is:

```
dataset/D4D/raw/extracted/
    specimen_N/
        session_timestamp/
            left_images/
            right_images/
            camera_info/
            depth_images/
            pointcloud/
            color_images/
            snr_images/
            masks/
            tf/
            clips.json
```

The stereo images are not originally rectified. Each session contains approximately 500 stereo frames, while Zivid depth and point clouds are available only at sparse start/end geometry anchors.

D4D cannot provide exact per-frame dense temporal ground truth because the scene is non-rigid.

Its scientifically valid role is:

- independent sparse-keyframe geometric evaluation;
- structured-light target-domain supervision;
- specimen-disjoint adaptation experiments;
- session-level few-shot adaptation;
- cross-backbone evaluation at real geometric anchors.

J.4 D4D Transform Chain

The Zivid geometry and the stereo endoscope are connected through optical tracking. For specimen 1, the validated marker-bridge transformation is:

$$T_{\mathrm{cam}\leftarrow\mathrm{zivid}} = T^{-1}_{\mathrm{ps}\leftarrow\mathrm{cam}} T_{\mathrm{ps}\leftarrow\mathrm{MiRe45}} T^{-1}_{\mathrm{polaris}\leftarrow\mathrm{MiRe45}} T_{\mathrm{polaris}\leftarrow\mathrm{zivid}}. \tag{59}$$

The full-dataset analysis revealed that transform conventions differ between specimens:

- `mire45_bridge`: used for specimen 1;
- `direct_ps`: camera and Zivid both represented in `polaris_spectra`;
- `direct_polaris`: camera and Zivid directly represented in `polaris`.

The convention is detected per session rather than forced globally.

J.5 D4D Benchmark Status

At the current frozen state:

Table 15: Current D4D sparse-keyframe benchmark.

Specimen	Candidate anchors	Valid anchors	TF convention
Specimen 1	82	60	MiRe45 bridge and direct Polaris
Specimen 2	39	31	Direct Polaris Spectra
Specimen 3	76	51	Direct Polaris Spectra and direct Polaris
Total	197	142 valid + 24 warning	166 usable

Additional properties:

- structured-light GT coverage: approximately 19–34%, median 27%;
- stereo–Zivid synchronization: approximately 1–55 ms, median 20 ms;
- exact depth–disparity numerical round trip;
- per-session calibration loading;
- focal length around 798 px for specimen 1 and 834 px for specimen 2;
- stereo baseline around 4.24 mm for specimen 1 and 3.97 mm for specimen 2;
- deterministic, resumable, and idempotent processing;
- session-disjoint, specimen-disjoint, leave-one-specimen-out, and few-shot splits;
- 47 automated validation checks passing.

Specimen 4 is pending complete extraction. Specimen 5 is currently blocked by a corrupted archive.

K Proposed Model Scope

K.1 Main Architecture

The main candidate is a consolidated version of EGBM-CARE with a short causal window.

The architecture contains:

- causal temporal context;
- residual disparity prediction;
- boundary-aware processing;
- identity/no-op expert;

- correction damping;
- change-aware reliability;
- safety-oriented gating;
- approximately four million trainable parameters;
- approximately seven milliseconds of additional runtime.

K.2 Role of MPC and Large-Correction Models

MPC, CPV, and ASSR are not the main deployable systems.

They are retained as analytical variants showing that:

- correction magnitude is the main oracle-gap bottleneck;
- large corrections can recover substantially more oracle error;
- unrestricted correction magnitude causes severe over-correction;
- safety and accuracy must be balanced explicitly.

L Backbone Strategy

The adapter should be evaluated on multiple frozen stereo backbones.

A practical initial set is:

1. S2M2 as the primary efficient backbone;
2. RAFT-Stereo or CREStereo as an architecturally different matcher;
3. Fast-FoundationStereo or FoundationStereo as a large modern reference.

For each backbone, three settings are evaluated.

L.1 Zero-Shot Cross-Backbone Transfer

$$\text{adapter trained on S2M2} \longrightarrow \text{different frozen backbone.} \tag{60}$$

This tests whether the learned correction policy transfers directly.

A negative result is still informative because it quantifies backbone-specific error distributions.

L.2 Backbone-Specific Adapter Training

The same adapter architecture is trained from scratch on predictions generated by the new frozen backbone.

This tests whether the architecture itself is reusable across matchers.

L.3 Lightweight Fine-Tuning

A source-pretrained adapter is fine-tuned using a small amount of target data.

This is the main proposed operating mode.

M Data-Efficiency Study

The target-domain sweep should be performed at the level of complete sessions or sequences, not individual randomly sampled frames.

Suggested fractions are:

$$0,\ 1,\ 2.5,\ 5,\ 10,\ 25,\ 50,\ 100\%. \tag{61}$$

For D4D, more interpretable units are:

$$1,\ 2,\ 4,\ 8 \text{ sessions}, \tag{62}$$

in addition to percentage-based subsets.

For every subset size, compare:

- raw frozen backbone;
- simple residual refiner;
- adapter trained from scratch;
- source-pretrained adapter;
- source-pretrained adapter fine-tuned on target data;
- full-data target adapter;
- large stereo baseline.

At small subset sizes, multiple deterministic seeds should be used.

N Experimental Protocol

N.1 Experiment A: SCARED Main Evaluation

Evaluate:

- raw S2M2;
- classical temporal filters;
- simple convolutional refiner;
- EGBM-v1;
- EGBM-v2 CARE;
- EGBM-v3 CARE window;
- streaming variant;
- published temporal-refinement baseline, where compatible.

N.2 Experiment B: Cross-Backbone Adaptation

For every frozen backbone:

1. generate raw disparities on the same sequences;
2. build backbone-specific oracle targets;
3. evaluate source-trained adapter zero-shot;
4. train the adapter from scratch;
5. fine-tune a pretrained adapter;
6. compare sample-efficiency curves.

N.3 Experiment C: D4D Few-Shot Adaptation

The temporal adapter receives causal frame context, but supervision is applied only at the sparse Zivid keyframe:

$$\left(D_{t-3}^{\text{raw}}, D_{t-2}^{\text{raw}}, D_{t-1}^{\text{raw}}, D_{t}^{\text{raw}}\right) \xrightarrow{\text{adapter}} D_t^{\text{refined}}, \tag{63}$$

with loss:

$$\mathcal{L}_t = \mathcal{L}_{\text{geometry}}\left(D_t^{\text{refined}}, D_t^{\text{Zivid}}\right) \tag{64}$$

only at valid Zivid anchors.
This provides sparse but highly accurate target-domain supervision.

N.4 Experiment D: SERV-CT Static OOD Evaluation

SERV-CT is used to measure:

- raw-preservation;
- geometric MAE;
- Bad-1 and Bad-3;
- new-Bad3;
- harmful correction rate;
- effect of limited target calibration.

No temporal consistency claim is made from SERV-CT.

O Evaluation Metrics

O.1 Geometric Metrics

- disparity MAE;
- depth MAE;
- Bad-1;
- Bad-3;
- Bad-5;

- new-Bad3;
- raw-good preservation;
- oracle-gap recovery;
- boundary and interior error;
- SNR-stratified error on D4D.

O.2 Temporal Metrics

- temporal disparity jitter;
- temporal error jitter;
- high-frequency temporal energy;
- motion-compensated consistency;
- boundary jitter;
- residual flicker;
- 3D point or surface jitter;
- motion lag;
- oversmoothing;
- drift over a sequence.

O.3 Safety Metrics

- harmful correction rate;
- pathological new-Bad3;
- clean-region degradation;
- fraction of already-correct pixels modified;
- modified-pixel ratio;
- correction magnitude distribution.

O.4 Efficiency Metrics

- total parameters;
- trainable parameters;
- frozen parameters;
- runtime;
- peak VRAM;
- adaptation GPU-hours;
- amount of target data;
- number of target sessions;
- performance per trainable parameter.

P Required Baselines

A strong evaluation requires:

- identity/no-refinement baseline;
- exponential temporal averaging;
- temporal median filtering;
- edge-aware temporal filtering;
- small generic U-Net or residual CNN;
- EGBM variants;
- at least one existing temporal depth/stereo refiner;
- raw frozen backbone;
- large stereo or video-stereo reference model.

The generic small refiner is particularly important. Without it, it is impossible to determine whether the proposed architecture is genuinely more sample-efficient than a standard lightweight network.

Q Main Paper Narrative

The proposed paper narrative is:

1. Modern stereo models provide strong per-frame geometry but may produce temporally unstable predictions in surgical video.
2. A lightweight temporal refiner can improve the target surgical domain without modifying the main stereo backbone.
3. Oracle analysis shows that conservative refiners are limited mainly by correction magnitude, while aggressive refiners introduce unsafe over-correction.
4. Zero-shot tests demonstrate that learned correction policies do not automatically transfer across domains or stereo backbones.
5. Instead of claiming universal refinement, the method is reframed as a lightweight temporal domain adapter.
6. The adapter is fine-tuned using only a small fraction of target-domain sequences while the large stereo backbone remains frozen.
7. SCARED validates dense temporal improvement, D4D validates sparse structured-light cross-domain adaptation, and SERV-CT measures geometric raw-preservation under domain shift.
8. The resulting system offers a favorable trade-off between adaptation cost, runtime, geometric accuracy, and temporal stability compared with retraining or replacing a large stereo network.

R Expected Contributions

1. A lightweight causal temporal adapter for frozen stereo backbones.
2. An oracle-driven analysis of correction support, sign, magnitude, and safety in learned stereo refinement.
3. A data-efficiency study across target-domain subset sizes and stereo backbones.
4. A processed D4D sparse-keyframe benchmark with structured-light ground truth, session/specimen-safe splits, and per-session transform conventions.
5. A systematic evaluation separating geometric accuracy, temporal consistency, correction safety, and adaptation cost.
6. Evidence that domain-specific lightweight adaptation is a practical alternative to retraining large stereo models.

S Minimum Viable ICRA Paper

The minimum viable paper requires:

- clear improvement of S2M2 on SCARED;
- direct improvement in temporal metrics;
- controlled new-error and harmful-correction rates;
- successful training of the same adapter architecture on at least one additional stereo backbone;
- a target-data learning curve;
- evidence that pretraining or the proposed architecture improves sample efficiency;
- D4D sparse-keyframe external evaluation;
- runtime and trainable-parameter comparison;
- at least one strong external baseline.

T Strong Paper Criteria

A stronger submission would additionally show:

- two or three stereo backbones;
- effective adaptation using only 5–10% of target data;
- specimen-disjoint D4D adaptation;
- improvement over a generic lightweight refiner;
- favorable comparison with a large stereo/video model;
- 3D surface-stability improvement;
- a multi-backbone adapter or leave-one-backbone-out experiment.

U Immediate Experimental Plan

1. Complete D4D specimen 4 extraction and conversion.
2. Freeze D4D benchmark version, quality thresholds, manifests, and splits.
3. Freeze the main SCARED architecture, provisionally EGBM-v3 CARE window.
4. Finalize dense temporal metrics on SCARED.
5. Select one second stereo backbone already supported by the ARGOS benchmark pipeline.
6. Generate its raw predictions on SCARED and D4D.
7. Train the same adapter architecture from scratch on the second backbone.
8. Test the S2M2-trained adapter zero-shot on the second backbone.
9. Run a reduced pilot adaptation sweep:
$$0,\ 5,\ 10,\ 25,\ 100\%.$$
10. If the pilot succeeds, expand to:
$$0,\ 1,\ 2.5,\ 5,\ 10,\ 25,\ 50,\ 100\%.$$
11. Compare from-scratch training against source-pretrained fine-tuning.
12. Run D4D specimen/session-disjoint few-shot adaptation.
13. Add a third backbone only after confirming the hypothesis on the second.
14. Stop developing unrelated new architectures after the cross-backbone pilot.

V Provisional Title Options

- **Lightweight Temporal Adaptation for Frozen Stereo Backbones**
- **Sample-Efficient Temporal Refinement of Frozen Stereo Models for Surgical Reconstruction**
- **Adapting Stereo Foundation Models with Lightweight Temporal Refiners**
- **Sparse-Supervised Temporal Adaptation for Surgical Stereo Depth**
- **A Lightweight Temporal Adapter for Domain-Specific Stereo Reconstruction**

W Final Scope Statement

> We propose a lightweight causal temporal adapter for frozen stereo backbones. Rather than assuming that one learned correction policy transfers universally across domains and matchers, the adapter is efficiently specialized using a small amount of target-domain sequential data. The method is evaluated through dense temporal supervision on SCARED, independent sparse structured-light validation and few-shot adaptation on D4D, and static geometric domain-shift analysis on SERV-CT. The goal is to obtain target-specific geometric and temporal improvements while training only approximately four million additional parameters and preserving the large stereo backbone unchanged.

X Failure Analysis and Scope Revision

The initial objective was to develop a lightweight temporal refiner capable of improving a frozen stereo backbone across surgical domains.

The experiments showed that the current model does not satisfy this objective. Although it improves S2M2 on SCARED, the learned behavior does not transfer to SERV-CT or D4D and does not provide measurable temporal stabilization on unseen D4D sequences.

The main failure modes are summarized below.

X.1 Domain-Specific Overfitting

On SCARED, the proposed EGBM/CARE refiners improve the raw S2M2 predictions and recover approximately 20% of the available oracle gap while maintaining a relatively low rate of newly introduced large errors.

However, the same SCARED-trained checkpoints degrade the raw predictions when applied zero-shot to other surgical datasets.

Table 16: Zero-shot behavior across surgical domains. Positive ΔMAE denotes improvement; negative values denote degradation.

Dataset	**Raw S2M2 MAE**	**Best refiner ΔMAE**	**Outcome**
SCARED	4.67 px	positive	improvement
SERV-CT	1.28 px	approximately -0.5 px	degradation
D4D	4.09 px	approximately -0.37 px	degradation

This indicates that the refiner does not learn a domain-independent correction rule. Instead, it learns the error distribution of the specific combination of:

- SCARED imagery;
- S2M2 prediction statistics;
- SCARED boundary and occlusion patterns;
- the frequency and magnitude of large disparity errors;
- the spatial priors present in the training sequences.

When the raw-error distribution changes, the correction policy continues to activate even when the raw prediction is already accurate.

X.2 False Activation on Raw-Good Pixels

The D4D raw-error stratification exposes the failure mechanism directly.

For EGBM-v3-CARE-S:

Table 17: Zero-shot D4D performance stratified by raw S2M2 error. Positive ΔMAE indicates improvement.

Raw error bin	**Raw MAE**	**Refined MAE**	**ΔMAE**
< 1 px	0.476	1.186	-0.710
1–3 px	1.721	1.966	-0.245
3–6 px	4.266	4.361	-0.095
6–12 px	8.197	7.785	$+0.412$
> 12 px	26.369	24.726	$+1.643$

The model retains some useful correction capability in the large-error regime. However, it applies corrections too frequently to raw-good pixels.

Since most D4D pixels fall in the low- and medium-error regimes, the damage introduced on already accurate predictions dominates the improvements obtained on large errors.

The problem is therefore not only correction quality, but correction selectivity:

> The model can improve large errors, but it cannot reliably determine when a correction should be applied outside the source domain.

X.3 Calibration Recovers Safety but Collapses to Identity

A few-shot adaptation experiment was performed on D4D using only a small calibration subset of the network.

The calibration-only configuration updated approximately 10.9k parameters, corresponding to roughly 0.25% of EGBM-v3-CARE-S.

Table 18: D4D few-shot adaptation results. The frozen raw S2M2 test MAE is 3.50 px.

Configuration	**MAE**	**new-Bad3**	**Harmful**	$\boldsymbol{\Delta}_{<1\text{px}}$	$\boldsymbol{\Delta}_{>6\text{px}}$
Zero-shot	4.72	23.5	0.83	-1.44	$+0.86$
Calibration-only, 2 sessions	3.66	1.9	0.34	-0.05	$+0.08$
Calibration-only, 8 sessions	3.50	0.0	0.04	≈ 0	≈ 0
Full fine-tuning, 4 sessions	4.09	3.4	0.61	-0.23	$+0.58$
From scratch, 4 sessions	3.47	1.2	0.13	-0.03	$+0.14$

Calibration-only adaptation successfully suppresses false activation:

- raw-good degradation is reduced by approximately 96%;
- new-Bad3 falls from 23.5% to almost zero;
- the harmful correction rate decreases substantially.

However, this safety improvement is obtained by suppressing almost all corrections.

At eight sessions, the calibrated model approximately reproduces the raw S2M2 output:

$$D_t^{\text{refined}} \approx D_t^{\text{raw}}. \tag{65}$$

The model becomes safe, but no longer provides useful refinement.

Therefore, lightweight calibration transfers abstention, not beneficial correction.

X.4 Limited Evidence of Transferable Correction Skill

Full fine-tuning preserves more of the large-error correction ability:

$$\Delta_{\text{MAE},\,>6\text{px}} = +0.58 \text{ px}. \tag{66}$$

However, with only a few target anchors it still over-corrects raw-good regions and does not outperform the raw stereo backbone in aggregate MAE.

The from-scratch model reaches approximately the raw performance, while the pretrained model does not demonstrate a clear sample-efficiency advantage.

This weakens the original hypothesis that SCARED pretraining learns a transferable temporal correction skill.

The current evidence supports only the following weaker statement:

> A small target-domain calibration set can rapidly suppress unsafe corrections, but more data or a different objective is required to obtain net-beneficial refinement.

X.5 No Temporal-Stability Improvement on D4D

The most important negative result is that the refiner does not improve temporal stability on full D4D clips.

The evaluation used real consecutive stereo frames and motion-compensated temporal diagnostics based on optical flow and forward–backward occlusion filtering.

Table 19: D4D temporal evaluation. The temporal metrics are prediction-space or motion-compensated diagnostics, since dense temporal GT is unavailable.

Configuration	Anchor MAE	MC inconsistency	HF energy	Depth MC [mm]
Raw S2M2	3.50	**0.390**	**0.507**	**0.691**
Zero-shot	4.72	0.397	0.525	0.717
Calibration-only, 8 sessions	3.50	0.400	0.525	0.723
Full fine-tuning, 4 sessions	4.09	0.394	0.521	0.706
From scratch, 4 sessions	3.47	**0.390**	**0.507**	**0.691**

The temporal axis is effectively flat:

- motion-compensated inconsistency changes by only a few percent;
- high-frequency temporal energy slightly increases;
- motion-compensated depth inconsistency slightly increases;
- pretrained full fine-tuning is not more stable than the scratch identity solution;
- calibration-only adaptation reduces correction magnitude but does not reduce temporal instability.

The refiner therefore does not act as a temporal stabilizer on D4D.

Instead, it behaves primarily as a frame-wise spatial residual corrector.

X.6 The Architecture Is Not Forced to Use Time

The current architecture receives temporal inputs, but it can minimize the training objective without learning a genuinely temporal function.

Its inputs include:

- current disparity;
- previous disparities;
- temporal differences;
- spatial gradients;
- edge features;
- local statistics;
- validity masks.

However, the network can still rely mainly on the current-frame disparity and dataset-specific spatial patterns.

The training target is the full residual:

$$R_t^\star = D_t^{\mathrm{GT}} - D_t^{\mathrm{raw}}, \tag{67}$$

which contains both:

- static spatial error;
- temporal flicker;
- bias;
- boundary mismatch;
- large geometric failure.

Nothing in the objective explicitly forces the model to separate temporal inconsistency from generic per-frame error.

As a result, the easiest solution is a domain-specific spatial correction policy.

X.7 Longer Temporal Memory Did Not Solve the Problem

A persistent streaming variant was compared with a short causal replay window.

The longer-memory model did not improve over the four-frame window.

This suggests that the main limitation is not insufficient temporal context.

The failure is more fundamental:

> The model is not learning the correct temporal task, rather than merely lacking enough history.

X.8 Large-Correction Models Expose a Safety Trade-Off

The MPC, CPV, and ASSR variants confirmed that the correction-magnitude bottleneck is real.

Allowing larger corrections increased oracle-gap recovery from approximately 20% to approximately 45–47%.

However, this also caused:

- higher new-Bad3;
- more harmful corrections;
- worse full-GT performance;
- severe degradation under domain shift.

The resulting Pareto frontier is:

$$\text{large correction capacity} \quad \Longleftrightarrow \quad \text{higher risk of unsafe over-correction.} \tag{68}$$

Post-hoc damping restored safety only by reducing the model toward a near-identity solution.

X.9 Resolution Sensitivity of Large-Proposal Branches

MPC and CPV also exposed an implementation-level limitation.

Their large-context path assumed feature-map dimensions compatible with the SCARED training resolution.

On D4D, the odd target-grid height caused incompatible tensor sizes in the multi-scale fusion path.

Although this issue is technically fixable through padding or explicit alignment, it reveals that the large-proposal branch was not yet robust to arbitrary deployment resolutions.

These branches are therefore retained only as analytical evidence for the magnitude bottleneck, not as deployable models.

X.10 Summary of Negative Findings

The current experiments support the following conclusions:

1. The refiner improves S2M2 on SCARED but overfits strongly to the SCARED error distribution.
2. Zero-shot transfer fails on both SERV-CT and D4D.
3. The model helps large errors but damages already accurate predictions.
4. Minimal calibration restores safety mainly by suppressing the correction toward identity.
5. Full fine-tuning retains some large-error capability but does not yet produce a net target-domain benefit.
6. SCARED pretraining does not provide a measurable temporal-stability advantage on D4D.
7. The current architecture behaves primarily as a spatial residual corrector, despite receiving temporal inputs.
8. Longer temporal memory does not solve the problem.
9. Increasing correction magnitude improves oracle recovery but worsens safety and OOD behavior.
10. The current evidence does not support the claim of a transferable temporal refiner.

X.11 Implication for the Next Model

The failure analysis motivates a strict redesign.

The next model should not predict an unrestricted per-frame residual.

Instead, it should be explicitly constrained to operate on the motion-aligned temporal inconsistency:

$$R_t^{\text{temp}} = D_t^{\text{raw}} - \mathcal{W}\left(D_{t-1}^{\text{raw}}, F_{t-1\to t}\right), \tag{69}$$

where $\mathcal{W}$ denotes warping according to image motion and $F_{t-1\to t}$ is the estimated optical flow.

The refined disparity should take the form:

$$D_t^{\text{refined}} = D_t^{\text{raw}} - G_t \odot R_t^{\text{temp}}, \tag{70}$$

where G_t is a bounded confidence gate.

This formulation restricts the model to removing temporally unexplained components instead of learning arbitrary spatial corrections.

The next architecture must also be validated with mandatory temporal ablations:

- full temporal model;
- current-frame-only model;
- shuffled previous-frame model;
- warped-history model without learning;
- identity baseline.

If removing or shuffling the temporal context does not reduce performance, the model cannot be considered a temporal refiner.

Y Does Explicit Temporal Supervision Induce Genuine Temporal Reasoning?

After observing that the original EGBM-based refiners behaved primarily as domain-specific spatial residual correctors, we investigated whether their failure was caused by the absence of explicit temporal supervision.

To isolate this question, we implemented a lightweight causal baseline inspired by video depth post-processing methods. The model consisted of a shared geometric encoder followed by a causal ConvGRU and a bounded residual head. The stereo backbone remained frozen, and the refiner received only disparity-derived inputs.

The main hypothesis was:

> If the model is trained with dense temporal-gradient supervision, then it should learn to exploit causal history and improve temporal consistency beyond what can be achieved by a frame-wise regularizer.

The initial pilot appeared to support this hypothesis. However, its ablation design was later found to be confounded: only the full temporal model was trained with the temporal loss, while the current-frame and shuffled-history variants were trained without it. Therefore, the observed separation could have originated from the loss rather than from genuine use of temporal history.

A dedicated confirmation study was consequently conducted.

Y.1 Temporal Gradient Matching

The temporal supervision was implemented using a Temporal Gradient Matching loss.

For two consecutive refined disparity maps $\hat{D}_{t-1}$ and $\hat{D}_t$, and the corresponding ground-truth maps D^{GT}_{t-1} and D^{GT}_t, the loss is defined as

$$\mathcal{L}_{\mathrm{TGM}} = \frac{1}{|\Omega_t|} \sum_{\mathbf{x} \in \Omega_t} \left| \Big(\hat{D}_t(\mathbf{x}) - \hat{D}_{t-1}(\mathbf{x}) \Big) - \big(D^{GT}_t(\mathbf{x}) - D^{GT}_{t-1}(\mathbf{x}) \big) \right|, \tag{71}$$

where Ω_t denotes pixels with valid ground truth in both frames.

The complete objective was

$$\mathcal{L} = \mathcal{L}_{\mathrm{spatial}} + \lambda_{\mathrm{TGM}} \mathcal{L}_{\mathrm{TGM}} + \lambda_{\mathrm{safe}} \mathcal{L}_{\mathrm{safe}}, \tag{72}$$

where

- $\mathcal{L}_{\mathrm{spatial}}$ supervises per-frame disparity accuracy;
- $\mathcal{L}_{\mathrm{TGM}}$ supervises frame-to-frame disparity changes;
- $\mathcal{L}_{\mathrm{safe}}$ discourages unnecessary modification of raw-good pixels.

The refined disparity was predicted through a bounded residual:

$$\hat{D}_t = D^{\mathrm{raw}}_t + \alpha \tanh\left(R_t\right), \tag{73}$$

where α limits the maximum correction magnitude.

Y.2 Factorial Experimental Design

To separate temporal input structure from loss supervision, we decoupled two factors:

1. temporal input mode;
2. loss mode.

The temporal input modes were:

- **full history**: the ConvGRU receives the correct causal sequence;
- **current frame only**: the recurrent state is reset at every frame;
- **shuffled history**: the current frame remains the target, while the historical frames are randomly reordered.

The loss modes were:

- spatial supervision only;
- spatial supervision plus TGM.

The core factorial comparison was therefore

Table 20: Confirmation-study matrix. The decisive comparisons use the same TGM loss, same architecture, and same training budget.

Configuration	**Temporal input**	**Loss**
Full history + spatial	Correct causal history	Spatial only
Full history + TGM	Correct causal history	Spatial + TGM
Current frame only + TGM	No temporal memory	Spatial + TGM
Shuffled history + TGM	Corrupted history order	Spatial + TGM

All configurations used:

- the same causal ConvGRU architecture;
- clip length of eight frames;
- frozen S2M2 stereo predictions;
- the same SCARED sequence-disjoint train/validation/test splits;
- identical optimizer and training duration;
- three independent random seeds.

This design tests whether the improvement is due to

$$\text{temporal supervision,} \tag{74}$$

or specifically due to

$$\text{temporal supervision} + \text{correct causal history.} \tag{75}$$

Y.3 SCARED Results

Table 21 reports the temporal-gradient error on the frozen SCARED test split.
The first comparison confirms that TGM is beneficial:

$$1.7023 \rightarrow 1.6456. \tag{76}$$

Thus, adding dense temporal-gradient supervision reduces the average temporal error.
However, the comparisons against the temporal ablations are not robust.
The full-history model only marginally outperforms shuffled history:

Table 21: SCARED temporal-gradient error across three seeds. Lower is better.

Configuration	Mean	Std.	Seed values
Full history + spatial	1.7023	0.0117	1.6934, 1.6946, 1.7189
Full history + TGM	**1.6456**	0.0534	1.6121, 1.7210, 1.6038
Current frame only + TGM	1.6816	0.0033	1.6781, 1.6861, 1.6806
Shuffled history + TGM	1.6501	0.0107	1.6380, 1.6641, 1.6482

$$1.6456 \quad \text{vs.} \quad 1.6501, \tag{77}$$

corresponding to an average difference of approximately 0.3%.

Moreover, the seed distributions overlap substantially:

$$\max\left(E_{\text{full}}\right) = 1.7210 > \min\left(E_{\text{shuffled}}\right) = 1.6380. \tag{78}$$

The worst full-history run is also worse than every current-frame-only run.

Therefore, the group mean follows the expected direction, but the result is not stable enough to support the claim that the ConvGRU reliably exploits causal history.

Y.4 Geometric Accuracy and Safety

The temporal objective did not produce a clear geometric separation between the three TGM configurations.

Table 22: Mean geometric behavior of the three TGM configurations on SCARED.

Configuration	MAE [px]	Interpretation
Full history + TGM	5.130	no clear advantage
Current frame only + TGM	5.131	effectively equivalent
Shuffled history + TGM	5.150	within experimental noise

The harmful-correction rate also showed high variance across seeds and no stable ordering between the configurations.

This indicates that the full-history architecture did not learn a reliably superior correction policy.

Y.5 TGM Weight Sweep

To investigate whether the high variance was caused by an overly aggressive temporal weight, we tested

$$\lambda_{\text{TGM}} \in \{0.2,\ 0.5,\ 1.0\}. \tag{79}$$

Table 23: Effect of the TGM weight for the full-history model.

λ_{TGM}	TGM error	MAE [px]	Harmful rate
0.2	1.6887 ± 0.0147	5.118 ± 0.009	0.303 ± 0.068
0.5	1.6619 ± 0.0081	5.137 ± 0.050	0.327 ± 0.042
1.0	1.6456 ± 0.0534	5.130 ± 0.079	0.234 ± 0.167

The highest weight produced the best mean temporal score but also the highest variance.

Lower weights were more stable, suggesting a trade-off between nominal temporal improvement and optimization robustness.

Importantly, none of the tested weights produced a clear and stable separation from the current-frame or shuffled-history variants.

Y.6 D4D Zero-Shot Evaluation

The temporal-usage hypothesis was further tested on D4D.

Because D4D does not provide dense temporal ground truth, the evaluation used prediction-space motion-compensated diagnostics and sparse Zivid ground-truth anchors.

The primary temporal diagnostic was motion-compensated disparity inconsistency.

For each configuration and seed, we measured

$$\Delta E_{\mathrm{MC}} = E_{\mathrm{MC}}^{\mathrm{refined}} - E_{\mathrm{MC}}^{\mathrm{raw}}, \tag{80}$$

where negative values indicate improvement.

Table 24: D4D motion-compensated inconsistency relative to raw S2M2. Values are mean paired differences with bootstrap confidence intervals.

Configuration	**Seed 0**	**Seed 1**	**Seed 2**
Full history	+0.0001	0.0000	+0.0029
Current frame only	−0.0020	−0.0020	−0.0038
Shuffled history	−0.0006	+0.0018	+0.0001

The full-history model does not show a stable improvement.

Its effect is approximately zero or slightly negative in all three seeds.

Unexpectedly, the current-frame-only model shows the most consistent reduction in motion-compensated inconsistency.

This strongly suggests that the small temporal improvement observed on SCARED is not caused by robust exploitation of causal history.

Instead, TGM appears to regularize the output even when no temporal memory is available.

Y.7 Sparse Geometric Transfer on D4D

All three TGM variants remained geometrically safe on the sparse Zivid anchors.

The raw S2M2 MAE was approximately

$$4.090 \text{ px}. \tag{81}$$

The different configurations produced small improvements of similar magnitude.

Table 25: D4D sparse-anchor zero-shot behavior. Positive ΔMAE indicates improvement over raw S2M2.

Configuration	**ΔMAE [px]**	**Harmful rate**	**Modified ratio**
Full history	0.00–0.04	0.00–0.02	0.00–0.02
Current frame only	0.04–0.05	0.00–0.01	0.01–0.02
Shuffled history	0.03–0.05	0.01–0.08	0.01–0.02

The variants are all conservative and do not exhibit the catastrophic false-activation observed with EGBM.

However, their similar behavior again suggests that the transferable component is a conservative correction prior rather than a learned causal temporal mechanism.

Y.8 Five-Gate Decision

The confirmation study used five predefined gates.

The resulting verdict is:

Marginal — genuine causal-history usage is not confirmed.

Table 26: Temporal-usage decision gates.

Gate	Criterion	Outcome
1	Full history + TGM beats full history + spatial-only	Pass
2	Full history + TGM robustly beats current-frame-only + TGM	Marginal
3	Full history + TGM robustly beats shuffled-history + TGM	Marginal
4	Full-history improvement transfers to D4D with stable sign	Fail
5	No identity collapse or unsafe false activation	Pass

Y.9 Interpretation

The study supports two distinct conclusions.

First, explicit temporal supervision is useful:

$$\mathcal{L}_{\mathrm{TGM}} \quad \text{reduces temporal-gradient error.} \tag{82}$$

Second, the causal ConvGRU does not demonstrate a reliable ability to exploit the correct temporal history:

$$E_{\mathrm{full+TGM}} \approx E_{\mathrm{shuffled+TGM}} \approx E_{\mathrm{current+TGM}}. \tag{83}$$

The most plausible interpretation is that TGM acts as an output regularizer.

It encourages conservative frame-to-frame changes, but the network is not forced to build accurate temporal correspondences.

The current architecture can therefore satisfy the objective through a mostly frame-wise correction policy.

Y.10 Implication for the Model Design

The result clarifies the next methodological step.

TGM should be retained as a training objective, but the disparity-only ConvGRU should not be used as the final temporal architecture.

The next refiner should explicitly construct correspondences between the target frame and causal reference frames.

A suitable formulation is:

$$\mathbf{q}_t = \phi\left(I_t, D_t^{\mathrm{raw}}\right), \tag{84}$$

$$\mathbf{k}_{t-\tau}, \mathbf{v}_{t-\tau} = \phi\left(I_{t-\tau}, D_{t-\tau}^{\mathrm{raw}}\right), \qquad \tau \in \{1, 2, 3\}, \tag{85}$$

followed by local target-to-history matching:

$$\mathbf{z}_t = \mathrm{CrossAttention}\left(\mathbf{q}_t, \{\mathbf{k}_{t-\tau}\}, \{\mathbf{v}_{t-\tau}\}\right). \tag{86}$$

The final correction should remain bounded:

$$\hat{D}_t = D_t^{\mathrm{raw}} + G_t \odot \tanh\left(R_t\right), \tag{87}$$

where G_t is a learned confidence gate.

The training objective should combine:

$$\mathcal{L} = \mathcal{L}_{\mathrm{spatial}} + \lambda_{\mathrm{TGM}}\mathcal{L}_{\mathrm{TGM}} + \lambda_{\mathrm{warp}}\mathcal{L}_{\mathrm{warp}} + \lambda_{\mathrm{safe}}\mathcal{L}_{\mathrm{safe}}. \tag{88}$$

Here, $\mathcal{L}_{\text{warp}}$ should supervise motion-aligned temporal consistency:

$$\mathcal{L}_{\text{warp}} = \frac{1}{|\Omega_t^{\text{flow}}|} \sum_{\mathbf{x}\in\Omega_t^{\text{flow}}} \left| \hat{D}_t(\mathbf{x}) - \mathcal{W}\left(\hat{D}_{t-1}, F_{t-1\to t}\right)(\mathbf{x}) \right|, \tag{89}$$

where

- $\mathcal{W}$ denotes warping;
- $F_{t-1\to t}$ is optical flow;
- Ω_t^{flow} excludes occlusions and unreliable flow.

This design makes temporal matching explicit rather than expecting it to emerge implicitly from recurrent state.

Y.11 Final Conclusion

The confirmation study revises the interpretation of the original pilot.

The initial result did not demonstrate robust temporal reasoning.

Instead, the experiments show that:

1. TGM is a useful temporal regularizer;
2. the current ConvGRU does not reliably exploit causal history;
3. shuffled and current-frame variants perform similarly to the full model;
4. D4D does not show stable temporal transfer for the full-history model;
5. the conservative geometric behavior transfers better than the temporal mechanism;
6. the next model requires explicit target-to-history correspondence.

Therefore, the disparity-only ConvGRU branch is retained as a diagnostic baseline, while the next model will combine explicit temporal matching, motion-aware supervision, bounded correction, and surgical-domain visual features.

Z ARGOS v2

ARGOS v2 extends the original frame-wise stereo evaluation into a unified framework for temporally consistent surgical depth estimation. The objective is not only to reduce frame-to-frame flickering, but also to preserve or improve geometric accuracy under camera motion, tissue deformation, occlusions, specularities, and instrument interaction.

A central limitation of existing surgical stereo datasets is that no single benchmark simultaneously provides long stereo sequences, dense metric depth independently measured at every frame, realistic tissue deformation, and sufficient scale for supervised temporal training. ARGOS v2 therefore adopts a complementary multi-dataset protocol in which each dataset is assigned a clearly defined role.

Z.1 Dataset Roles

SCARED. The original SCARED dataset provides 45 structured-light keyframes with directly measured geometric ground truth. These keyframes are used for static geometric evaluation. Since datasets 4 and 5 contain known calibration issues, the main clean evaluation protocol uses the remaining 35 keyframes, while the complete 45-keyframe benchmark is reported separately. The intermediate stereo video frames are retained only for no-reference temporal analysis, as their original depth annotations were propagated using inaccurate robot kinematics.

Table 27: Role of each dataset in the ARGOS v2 evaluation protocol.

Dataset	Geometric reference	Temporal sequences	Dense temporal GT	ARGOS v2 role
SCARED	Structured light at 45 keyframes	Yes	No reliable per-frame GT	Static geometry and no-reference temporal test
SCARED-C	Structured-light-derived metric pseudo-GT	Yes	16,921 curated frames	Supervised temporal train/validation/test
D4D	Zivid structured-light anchors	Yes	Sparse anchors only	OOD geometry and deformation test
SERV-CT	CT-derived geometry on 16 stereo pairs	No	No	Static OOD geometry test
StereoMIS	No dense depth GT	Yes, in vivo	No	Unsupervised adaptation and held-out in-vivo test

SCARED-C. SCARED-C replaces the original kinematics-based camera poses with poses estimated using Structure-from-Motion and subsequently aligned to metric scale using the structured-light keyframe depth. The resulting temporal depth is therefore not independently measured at every frame, but is derived by reprojecting the structured-light keyframe geometry through the corrected camera poses.

The official release contains 17,135 RGB-D pairs. After sequence-level quality control, the ARGOS v2 curated subset contains 16,921 temporally aligned stereo frames with metric pseudo-ground truth. SCARED-C is consequently used as the primary dataset for supervised training, validation, and in-domain evaluation of the temporal refinement module.

D4D. D4D provides sparse but high-quality Zivid structured-light geometry anchors within long sequences containing substantial non-rigid deformation and camera motion. It is used as an out-of-distribution benchmark for evaluating geometric fidelity at measured anchor frames and temporal robustness between anchors.

SERV-CT. SERV-CT contains 16 stereo pairs with geometry derived from registered CT data. Although too small for training, it provides an independent static out-of-distribution test of geometric accuracy and consistency.

StereoMIS. StereoMIS provides long in-vivo stereo sequences containing respiration, tissue deformation, specularities, occlusions, and surgical instruments, but no dense per-frame depth ground truth. It is therefore used for label-free domain adaptation and no-reference temporal evaluation. To preserve a meaningful in-vivo generalization test, adaptation and evaluation are separated at the subject level.

Z.2 Multi-Backbone Temporal Refinement

The temporal module is trained as a post-processing refiner over predictions produced by multiple frozen stereo backbones. For each SCARED-C training clip, the input disparity sequence is sampled from a heterogeneous pool of frame-wise stereo models. The refiner therefore learns from different error distributions rather than specializing to a single backbone.

Given the current stereo pair (I_t^L, I_t^R), the raw backbone disparity $\hat{d}_t^{\text{raw}}$, and the previous refined state, the temporal module predicts a residual correction Δd_t and an update gate g_t:

$$\hat{d}_t^{\text{ref}} = \hat{d}_t^{\text{raw}} + g_t \odot \Delta d_t, \tag{90}$$

where $g_t \in [0, 1]$ controls how strongly temporal information is allowed to modify the current frame. This formulation enables the model to preserve an already accurate frame-wise estimate while correcting unstable or temporally inconsistent regions.

The main training objective combines geometric accuracy, temporal consistency, edge preservation, and regression safety:

$$\mathcal{L} = \lambda_{\text{geo}}\mathcal{L}_{\text{geo}} + \lambda_{\text{temp}}\mathcal{L}_{\text{temp}} + \lambda_{\text{edge}}\mathcal{L}_{\text{edge}} + \lambda_{\text{safe}}\mathcal{L}_{\text{safe}} + \lambda_{\text{upd}}\mathcal{L}_{\text{upd}}. \tag{91}$$

The safety term explicitly penalizes cases in which temporal refinement makes the prediction less accurate than the original backbone output:

$$\mathcal{L}_{\text{safe}} = \max\left(0, \left|\hat{d}_t^{\text{ref}} - d_t^*\right| - \left|\hat{d}_t^{\text{raw}} - d_t^*\right| - m\right), \tag{92}$$

where d_t^* denotes the SCARED-C metric pseudo-ground truth and m is a small tolerance margin.

Z.3 Training and Evaluation Protocol

The complete ARGOS v2 protocol is divided into four stages:

1. **Supervised temporal training.** The refiner is trained on SCARED-C using cached disparity predictions from multiple frozen stereo backbones.
2. **In-domain model selection.** Architecture selection, loss weighting, checkpoint selection, and early stopping are performed exclusively on a sequence-disjoint SCARED-C validation split.
3. **Label-free in-vivo adaptation.** The trained refiner may be adapted on selected StereoMIS subjects using stereo photometric consistency, left–right consistency, occlusion-aware temporal consistency, and teacher–student regularization. The frame-wise stereo backbone remains frozen.
4. **Frozen external evaluation.** The final model is evaluated without additional tuning on held-out SCARED-C sequences, held-out StereoMIS subjects, SCARED structured-light keyframes, SERV-CT, and D4D.

Z.4 Evaluation Questions

ARGOS v2 is designed to answer the following questions:

1. Does temporal refinement reduce flickering while preserving or improving metric geometric accuracy?
2. Does a refiner trained with predictions from multiple stereo backbones generalize to an unseen backbone?
3. Does the learned temporal prior transfer zero-shot from SCARED-C to independent surgical datasets?
4. Can label-free adaptation on in-vivo StereoMIS sequences improve temporal behavior without causing geometric degradation on SCARED, SERV-CT, or D4D?
5. Does the method remain stable around depth discontinuities, occlusions, surgical instruments, and non-rigid tissue deformation?

Z.5 Evaluation Metrics

On datasets providing geometric reference, the evaluation reports disparity EPE, Bad-3, depth MAE, AbsRel, RMSE, and δ_1. Temporal performance is evaluated using motion-compensated disparity consistency, temporal error variance, the fraction of frames improved over the raw backbone, and the frequency and magnitude of geometric regressions.

For datasets without dense depth ground truth, temporal evaluation relies on stereo photometric consistency, left–right consistency, motion-compensated temporal residuals, edge stability, and qualitative analysis of ghosting, occlusions, and newly visible regions.

The primary claim of ARGOS v2 is therefore not merely that the refined depth is smoother, but that temporal refinement produces more stable predictions without sacrificing geometric fidelity, and that this behavior generalizes across stereo backbones and surgical domains.

Positioning ARGOS v2 Against the Temporal Stereo State of the Art

The recent literature on temporally consistent depth estimation contains several strong and partially overlapping lines of research. However, the existing approaches address different subsets of the problem targeted by ARGOS v2. In particular, three recent methods are especially relevant: EndoStreamDepth, BiDAStabilizer, and PPMStereo.

.1 EndoStreamDepth: Causal Hierarchical Temporal Modeling for Endoscopy

EndoStreamDepth is a streaming monocular depth estimation framework designed specifically for endoscopic videos. Its architecture combines a Depth Anything V2-based encoder, a DPT-style decoder, and hierarchical temporal Mamba modules inserted at multiple decoder resolutions. Each decoder level maintains a recurrent hidden state that is propagated sequentially across frames, allowing the model to process arbitrarily long video streams in a causal manner.

The main architectural contribution is therefore a multi-scale temporal memory:

$$\mathbf{H}_t^{(l)} = \mathcal{M}^{(l)}\left(\mathbf{F}_t^{(l)}, \mathbf{H}_{t-1}^{(l)}\right), \tag{93}$$

where $\mathbf{F}_t^{(l)}$ denotes the current feature representation at scale l, and $\mathbf{H}_{t-1}^{(l)}$ is the temporal hidden state propagated from the previous frame.

This hierarchical structure allows the network to model both fine anatomical boundaries and coarse global geometry. EndoStreamDepth further introduces endoscopy-specific transformations, including rotations, blur, defocus, brightness and gamma perturbations, fog, and motion artifacts, in order to improve robustness to surgical imaging conditions.

The method is highly relevant to ARGOS v2 because it demonstrates that:

- causal streaming inference is feasible for endoscopic depth estimation;
- temporal memory should operate at multiple spatial resolutions;
- endoscopy-specific data augmentation can substantially improve robustness;
- temporal consistency and anatomical boundary preservation must be optimized jointly.

However, EndoStreamDepth is not a plug-in temporal refiner. It modifies the full monocular depth network and relies on backbone-specific encoder and decoder features. Consequently, it cannot be directly applied to arbitrary stereo predictors such as S2M2, RAFT-Stereo, StereoAnywhere, or FoundationStereo. In addition, it does not explicitly include an identity-preserving mechanism that prevents modifications when the input prediction is already correct.

Its temporal regularization is based on direct frame-to-frame variation of normalized depth:

$$\mathcal{L}_{\text{temp}} = \frac{1}{(T-1)N} \sum_{t=1}^{T-1} \sum_{i=1}^{N} \left| \bar{D}_{t+1}(i) - \bar{D}_t(i) \right|, \tag{94}$$

which may encourage smoothness, but does not explicitly compensate for camera motion, tissue deformation, or newly appearing structures.

.2 BiDAStabilizer: Plug-and-Play Stereo Stabilization

BiDAStabilizer is the closest existing method to the ARGOS v2 plug-in formulation. It receives disparity predictions from a frozen image-based stereo network and transforms them into temporally consistent disparity sequences without retraining the original stereo matcher.

The pipeline can be summarized as follows:

$$\{d_{t-1}, d_t, d_{t+1}\} \rightarrow \text{flow-based alignment} \rightarrow \text{temporal propagation} \rightarrow \Delta d_t, \tag{95}$$

with final prediction

$$d_t^{\text{out}} = d_t^{\text{raw}} + \Delta d_t. \tag{96}$$

The method first estimates bidirectional optical flow, then aligns neighboring disparity maps toward the central frame. A forward hidden state and a backward hidden state are propagated across the sequence and fused to predict the final disparity residual.

BiDAStabilizer establishes several important principles:

- temporal aggregation should operate on motion-aligned disparity information;
- a residual plug-in can improve existing frozen stereo models;
- local alignment and global propagation provide complementary benefits;
- temporal processing can improve both accuracy and consistency across datasets.

However, BiDAStabilizer is fundamentally non-causal. It uses future disparity d_{t+1} and a backward propagation pass from the end of the sequence toward the beginning. Therefore, it is unsuitable for strict online deployment in a robotic control loop.

Furthermore, BiDAStabilizer predicts an unconstrained residual and does not include:

- an explicit error-detection gate;
- a bounded correction magnitude;
- an identity-preserving initialization;
- a dedicated penalty on predictions that were already correct;
- an unseen-backbone evaluation protocol;
- surgical cross-domain validation.

These limitations are particularly important in light of the ARGOS v1 results, where temporal refiners trained on SCARED-specific error distributions generated false positive corrections on SERV-CT and strongly degraded already accurate predictions.

.3 PPMStereo: Reliability-Aware Long-Term Memory

PPMStereo introduces a Pick-and-Play Memory mechanism for dynamic stereo matching. Instead of using only adjacent frames or storing the complete sequence indiscriminately, the model maintains a memory buffer and dynamically selects a compact subset of high-quality references.

The memory selection process combines three signals:

$$S_t = S_t^{\text{conf}} + S_t^{\text{rel}}, \tag{97}$$

where S_t^{conf} represents confidence and S_t^{rel} combines semantic similarity and redundancy awareness.

The top-K memory entries are retained:

$$\mathcal{I}_t = \{i \mid \text{rank}(S_t[i]) \leq K\}, \tag{98}$$

and then adaptively weighted during the play stage:

Table 28: Positioning of ARGOS v2 relative to closely related temporal depth and stereo methods.

Method	Plug-in	Causal	Long-term memory	Backbone-agnostic	Safety on clean inputs	Surgical validation
EndoStreamDepth	No	Yes	Hierarchical Mamba	No	No explicit mechanism	Yes
BiDAStabilizer	Yes	No	Bidirectional propagation	Partially	No explicit mechanism	No
PPMStereo	No	No	Selective memory buffer	No	No explicit mechanism	No
ARGOS v2	**Yes**	**Yes**	**Selective causal memory**	**Yes**	**Explicit identity preservation**	**Yes**

$$\tilde{S}_t[i] = \frac{S_t[i]}{\sum_{j \in \mathcal{I}_t} S_t[j]}. \tag{99}$$

An attention-based read-out aggregates the selected memory values:

$$\mathbf{F}_t^{\text{agg}} = \mathbf{F}_t^{\text{cost}} + \alpha \, \text{Softmax}\left(\frac{\mathbf{q}_t \mathbf{k}_m'^{\top}}{\sqrt{D_k}}\right) \mathbf{v}_m', \tag{100}$$

where α is initialized to zero.

The main lesson from PPMStereo is that memory quantity alone is not sufficient. Naively retaining all frames, the latest frames, or random frames performs worse than selecting references based on reliability, relevance, and redundancy. The reported ablations indicate that performance saturates around $K = 5$, suggesting that a compact, high-quality memory is preferable to unrestricted temporal accumulation.

However, PPMStereo is not a plug-in refiner. It relies on:

- backbone-specific feature maps;
- internal cost volumes;
- context encoders;
- iterative GRU-based disparity updates;
- a DynamicStereo-style full stereo architecture.

It therefore cannot be directly applied to cached disparity predictions from heterogeneous frozen stereo networks. Moreover, its formulation is not a pure online causal system, because the vanilla memory is constructed over the full sequence. Finally, it does not explicitly include a mechanism that guarantees identity preservation on already correct predictions.

.4 Combined Comparison

Table 28 summarizes the main differences between these methods and the target formulation of ARGOS v2.

.5 Resulting Design Space for ARGOS v2

The analysis above suggests that ARGOS v2 should not be designed as a simple convolutional smoothing module or as a larger ConvGRU. Instead, it should combine the strongest principles from the existing literature while addressing the limitations that remain unsolved.

The proposed formulation is a:

> **Safe, causal, backbone-agnostic temporal stereo refiner with reliability-aware selective memory.**

The architecture should contain five main components.

.5.1 Universal Evidence Encoder

The refiner must rely only on signals that are available for any stereo backbone:

$$\mathbf{x}_t = \left[I_t^L, I_t^R, d_t^{\text{raw}}, r_t^{\text{stereo}}, \nabla d_t^{\text{raw}}, \tilde{d}_{t-1}^{\text{ref}}, d_t^{\text{raw}} - \tilde{d}_{t-1}^{\text{ref}} \right], \tag{101}$$

where:

- I_t^L and I_t^R are the left and right images;
- d_t^{raw} is the disparity predicted by the frozen backbone;
- r_t^{stereo} is a stereo photometric residual;
- $\tilde{d}_{t-1}^{\text{ref}}$ is the previous refined disparity aligned to the current frame.

No internal backbone-specific feature map, correlation volume, or confidence representation should be required.

.5.2 Causal Motion Alignment

Following the strongest result from BiDAStabilizer, memory should be aligned before temporal fusion. For a previous disparity d_{t-k}^{ref}, the aligned memory is:

$$\tilde{d}_{t-k\to t}^{\text{ref}} = \mathcal{W}\left(d_{t-k}^{\text{ref}}, f_{t-k\to t} \right), \tag{102}$$

where $\mathcal{W}$ denotes differentiable warping and $f_{t-k\to t}$ is a motion field.

Only past frames are used:

$$\mathcal{M}_t = \{ m_{t-1}, m_{t-2}, \dots, m_{t-K} \}, \tag{103}$$

ensuring strict causal inference.

.5.3 Reliability-Aware Pick-and-Play Memory

Inspired by PPMStereo, memory entries should be ranked according to reliability rather than temporal proximity alone.

A generic memory score may be defined as:

$$s_{t,k} = \lambda_q q_{t,k} + \lambda_s s_{t,k}^{\text{sim}} - \lambda_r r_{t,k}^{\text{red}} - \lambda_o o_{t,k} - \lambda_a a_{t,k}, \tag{104}$$

where:

- $q_{t,k}$ is estimated memory quality;
- $s_{t,k}^{\text{sim}}$ is similarity to the current frame;
- $r_{t,k}^{\text{red}}$ measures redundancy;
- $o_{t,k}$ measures occlusion or warp invalidity;
- $a_{t,k}$ encodes memory age.

The top-K memories are selected and adaptively weighted before aggregation.

.5.4 Hierarchical Temporal State

Following EndoStreamDepth, temporal information should be represented at multiple spatial scales:

$$\mathbf{H}_t = \left\{ \mathbf{H}_t^{1/4}, \mathbf{H}_t^{1/8}, \mathbf{H}_t^{1/16} \right\}. \tag{105}$$

The finest scale captures local corrections and disparity boundaries, while coarser scales preserve global shape, motion context, and long-term structural information.

The temporal operator may be implemented using Mamba blocks, recurrent gated units, or a hybrid design. The scientific contribution should not be framed merely as the use of Mamba, since hierarchical temporal Mamba has already been introduced for endoscopic depth estimation.

.5.5 Safe Identity-Preserving Output

The final correction should be explicitly bounded:

$$d_t^{\text{ref}} = d_t^{\text{raw}} + g_t c_t \tau_t \tanh\left(\Delta d_t\right), \tag{106}$$

where:

- $g_t \in [0, 1]$ is the probability that the raw disparity is incorrect;
- $c_t \in [0, 1]$ is the confidence in the temporal correction;
- τ_t is a local maximum update magnitude;
- Δd_t is the proposed residual correction.

The network should be initialized such that:

$$g_t \approx 0, \qquad \Delta d_t \approx 0, \tag{107}$$

and therefore:

$$d_t^{\text{ref}} \approx d_t^{\text{raw}} \tag{108}$$

at the beginning of training.

This identity-preserving property must be structural rather than being left entirely to the optimization objective.

.6 Novelty Boundary

The novelty of ARGOS v2 cannot be stated as temporal memory alone, since selective long-term memory has already been introduced by PPMStereo. It cannot be stated as hierarchical Mamba modeling for endoscopy, since this is already present in EndoStreamDepth. It also cannot be stated merely as a stereo stabilization plug-in, since BiDAStabilizer already establishes that paradigm.

The contribution instead lies in the combination of properties that has not been demonstrated jointly:

- plug-in refinement of frozen stereo backbones;
- strict causal and streaming operation;
- backbone-independent input representation;
- supervised training on multiple heterogeneous stereo predictors;
- evaluation on completely unseen stereo backbones;

- reliability-aware selective temporal memory;
- explicit identity preservation on already accurate predictions;
- zero-shot evaluation across surgical datasets;
- validation on both geometric ground truth and in-vivo unlabeled sequences.

The resulting central claim of ARGOS v2 is therefore:

> *ARGOS v2 introduces a causal, safe, and backbone-agnostic temporal stereo refiner trained across heterogeneous frozen stereo predictors. The model uses reliability-aware selective memory to improve temporal consistency while preserving accurate geometry, and is evaluated under unseen-backbone and surgical cross-domain shifts.*

.7 Core Scientific Gap

The central unresolved problem is not simply how to smooth temporally inconsistent stereo predictions. The key challenge is determining when temporal correction is justified.

A successful temporal refiner must distinguish between:

$$\text{prediction variation caused by stereo flicker} \tag{109}$$

and

$$\text{prediction variation caused by real scene motion, deformation, or occlusion.} \tag{110}$$

The failure mode observed in ARGOS v1 demonstrates that a refiner trained on one backbone and one dataset can learn a specific error signature rather than a transferable temporal prior. When evaluated on an out-of-domain dataset in which the raw stereo prediction is already accurate, the refiner may trigger false positive corrections and significantly degrade the result.

ARGOS v2 is therefore motivated by a stricter principle:

The temporal refiner must earn the right to modify the raw prediction.

This principle motivates the combination of multi-backbone training, selective memory, uncertainty-aware gating, bounded updates, and explicit safety evaluation on clean predictions.

.8 Experimental Consequences

The proposed formulation requires evaluation along three independent axes.

Seen-backbone performance. The refiner must improve or preserve all stereo backbones used during training.

Unseen-backbone generalization. A backbone that is completely excluded from training and model selection must be evaluated on the same test domain.

Cross-domain generalization. The frozen refiner must then be evaluated on unseen surgical datasets, including SERV-CT and D4D, and on unlabeled in-vivo sequences such as StereoMIS.

Particular attention should be given to already accurate predictions. Let:

$$\Omega_{\text{clean}} = \left\{ i \mid \left| d_i^{\text{raw}} - d_i^{\text{gt}} \right| < \epsilon \right\}. \tag{111}$$

The safety error on clean pixels can be defined as:

$$\mathcal{L}_{\text{clean}} = \frac{1}{|\Omega_{\text{clean}}|} \sum_{i \in \Omega_{\text{clean}}} \left| d_i^{\text{ref}} - d_i^{\text{raw}} \right|. \tag{112}$$

Additional safety metrics should include:

- new-Bad3 introduced by the refiner;
- percentage of clean pixels degraded;
- false update rate;
- mean update magnitude on clean predictions;
- fraction of frames with worse EPE after refinement.

These metrics are necessary to verify that the model improves temporal consistency without reproducing the out-of-domain degradation observed in ARGOS v1.

.9 BiDA-Style Motion Alignment as the First Validated Building Block

The first validated component of ARGOS v2 is a causal BiDA-style motion-alignment module. For each current frame t, a frozen optical-flow network estimates the temporal motion between the current and previous left images. The previous disparity prediction is then warped into the current-frame coordinate system:

$$\hat{d}_{t-1\to t} = \mathcal{W}\left(d_{t-1}, F_{t\to t-1}\right),$$

where d_{t-1} is the previous frame-wise stereo prediction, $F_{t\to t-1}$ is the current-to-previous optical flow, and $\mathcal{W}$ denotes target-to-source bilinear warping using the BiDA convention.

The validation shows that motion alignment alone is not sufficient to improve the current disparity through direct replacement, fixed blending, or a hand-crafted forward–backward and photometric gate. However, the aligned temporal memory contains complementary geometric evidence. In particular, a per-pixel oracle that selects the lower-error value between the current raw disparity and the aligned previous disparity improves the result across all evaluated sequences and stereo backbones.

This leads to the following interpretation:

$$\text{Optical flow} \neq \text{automatic disparity improvement},$$

but rather

$$\text{Optical flow} \to \text{temporally aligned evidence}.$$

The remaining scientific problem is therefore not how to transport information from the past, but how to determine whether the current prediction or the aligned temporal memory is more reliable at each spatial location.

Accordingly, the BiDA-style alignment block is retained as the first architectural building block of ARGOS v2:

$$\left(I_{t-1}^L, I_t^L, d_{t-1}, d_t\right) \to \left(\hat{d}_{t-1\to t}, m_{\text{warp}}, e_{\text{FB}}, e_{\text{photo}}, \Delta d_t\right),$$

where:

- $\hat{d}_{t-1\to t}$ is the aligned previous disparity;
- m_{warp} is the valid warp-support mask;

- e_{FB} is the forward–backward flow-consistency error;
- e_{photo} is the photometric residual;
- $\Delta d_t = d_t - \hat{d}_{t-1\to t}$ is the temporal disparity disagreement.

These signals will be passed to a learned, identity-preserving error detector and refinement module. The intended output remains:

$$d_t^{\text{ref}} = d_t + g_{\text{error}} \cdot c_{\text{memory}} \cdot \tau \cdot \tanh(\delta_t),$$

where the learned gate must determine when the aligned memory provides trustworthy corrective evidence.

The current experimental conclusion is therefore:

> BiDA-style causal alignment is a **GO** as a universal temporal evidence module, while direct memory replacement, fixed blending, and heuristic gating remain a **NO-GO**.

.10 Milestone: Validation of Causal BiDA-Style Temporal Alignment

Status: Major Scientific Milestone Achieved

The first major milestone of ARGOS v2 was successfully completed.

Rather than directly developing a new temporal refiner, we first isolated and validated the core temporal alignment mechanism inspired by BiDAStabilizer in order to determine whether motion-compensated temporal evidence actually contains useful corrective information for surgical stereo.

The study produced three important findings:

- A causal adaptation of the BiDA alignment module successfully transfers temporal information from previous frames using only past observations, making it compatible with real-time surgical deployment.
- Oracle experiments demonstrate that the aligned previous disparity consistently contains complementary geometric information capable of improving the current prediction across multiple frozen stereo backbones.
- Simple heuristic selectors based on forward-backward consistency and photometric consistency are insufficient to exploit this information safely, motivating the need for a learned refinement strategy.

This represents an important scientific result because it demonstrates that temporal alignment itself is valuable, while identifying the true research problem:

> *The challenge is not obtaining temporal information, but learning when that information should be trusted.*

Building on this observation, the first learned identity-preserving refiner was developed.

The learned bounded residual model:

- improves all three training backbones;
- generalizes without tuning to the unseen Fast-FoundationStereo backbone;
- recovers a substantial fraction of the oracle improvement;
- remains lightweight (approximately 39k parameters).

Although the current selector is not yet sufficiently reliable to satisfy the desired safety constraints (clean-input preservation and low false-update rate), these experiments validate the central hypothesis of ARGOS v2:

> *Causally aligned temporal evidence can improve frozen stereo predictors in a backbone-agnostic manner; the remaining challenge is reliable intervention rather than temporal information extraction.*

This milestone establishes the first building block of the final ARGOS v2 architecture and justifies subsequent research on learned confidence estimation, selective memory, and safety-aware temporal refinement.

.11 Milestone: Long-Range Temporal Evidence and the Limits of Explicit Pick-and-Play Memory

Status: Long-range temporal evidence validated; current PPMStereo-style selector not promoted.

After validating causal BiDA-style alignment and a learned bounded residual refiner based on the immediately preceding frame, we investigated whether longer temporal memory could provide additional corrective information.

A causal memory bank was constructed using BiDA-aligned disparity predictions from:

$$t-1, \quad t-2, \quad t-4, \quad t-8.$$

The study compared the existing learned $t-1$ refiner against deterministic and learned PPMStereo-inspired memory-selection strategies.

At cache-grid coverage 0.50, the following aggregate results were obtained over the three seen stereo backbones:

Method	EPE
Raw disparity	0.64384
Oracle using $t-1$	0.57495
Oracle using any memory	0.47672
Learned BiDA $t-1$ refiner	0.61971
Learned long-memory refiner	0.62951

The multi-memory oracle therefore provides an additional improvement of:

$$0.57495 - 0.47672 = 0.09823 \text{ px}$$

beyond the $t-1$ oracle.

This demonstrates that older aligned memories contain substantial complementary geometric evidence. In particular, frames at $t-2$, $t-4$, and $t-8$ repeatedly contain correct disparity information that is unavailable in the current prediction and in the immediately preceding frame.

However, the tested PPMStereo-style selectors were unable to exploit this additional evidence safely.

The learned long-memory model remained worse than the validated $t-1$ refiner on all evaluated backbones, including the unseen Fast-FoundationStereo model:

Backbone	Raw EPE	Learned $t-1$	Long Memory
S2M2-S	0.69611	0.67702	0.67913
RAFT-Stereo	0.61744	0.59070	0.60420
StereoAnywhere	0.61795	0.59143	0.60520
Fast-FoundationStereo	0.64515	0.62332	0.63578

The learned selector assigned approximately 72% of its total mass to temporal memory and distributed this mass almost uniformly across memory ages. It therefore failed to perform reliable selective memory access or abstention.

The resulting safety behaviour was insufficient:

- clean-pixel degradation of approximately 36%;
- false-update rate above 41%;
- approximately $39\% - 41\%$ of frames worsened;
- degraded performance relative to the learned $t-1$ refiner on both seen and unseen backbones.

The correct interpretation is therefore not that long temporal memory is uninformative. On the contrary, the oracle results demonstrate a substantially higher temporal correction ceiling.

The current limitation is the inability of universal disparity- and flow-based features to identify which memory is trustworthy at each spatial location.

The resulting decision is:

GO: long-range aligned memory as a source of complementary temporal evidence.

NO-GO: the current deterministic and learned PPMStereo-style top-K selectors for inclusion in the first ARGOS v2 refiner.

This milestone reveals that the bottleneck is no longer the availability of temporal information, but its representation and safe selection.

The PPMStereo memory-bank infrastructure is retained for future experiments, while the first ARGOS v2 refiner remains based on the validated causal BiDA $t-1$ pathway.

.12 Milestone 4 – Evaluation of Foundation Visual Features (DINOv3)

A natural hypothesis was that richer semantic visual representations could substantially improve temporal memory selection.

To test this, we integrated the official frozen DINOv3 ViT-L/16 foundation model as an alternative feature extractor for the temporal selector while keeping the remainder of the pipeline unchanged. Multiple feature configurations were evaluated, including intermediate transformer layers and multi-layer feature fusion.

The objective was to determine whether high-level semantic descriptors provide better evidence for deciding when historical temporal information should be trusted compared with purely geometric cues.

The controlled experiments led to a clear negative result.

Although DINOv3 slightly increased the ranking AUROC, all practically relevant decision metrics deteriorated:

$$\text{Geometry-only selector} > \text{DINOv3-enhanced selector}$$

The DINO-based selector showed:

- lower Average Precision,
- lower Top-1 retrieval accuracy,
- higher decision regret,
- no measurable improvement in temporal refinement quality.

Consequently, DINOv3 was not promoted to the main ARGOS v2 architecture.

This experiment provides an important scientific insight: for plug-and-play temporal stereo refinement, deciding whether historical disparity should be trusted appears to depend primarily on geometric consistency rather than on high-level semantic image understanding.

As a result, ARGOS v2 continues to rely on universal geometric evidence (disparity disagreement, optical-flow alignment, validity masks and temporal consistency) instead of expensive frozen foundation features.

This milestone substantially narrows the architectural search space and suggests that future improvements should focus on better temporal reasoning mechanisms rather than richer image representations.

.13 Milestone: Cross-Memory Consensus Reveals Correlated Temporal Errors

After observing that long-range aligned memories contain substantial oracle improvement, we investigated whether reliable temporal corrections could be identified without learning a per-memory selector.

We introduced a zero-parameter causal baseline, termed *Cross-Memory Consensus Correction* (CMC). Given the BiDA-aligned disparities from previous frames

$$\mathcal{M}_t = \left\{ \tilde{d}_{t-1\to t}, \tilde{d}_{t-2\to t}, \tilde{d}_{t-4\to t}, \tilde{d}_{t-8\to t} \right\},$$

CMC computes a validity-aware median:

$$d_{\text{med}}(p) = \text{median} \left\{ \tilde{d}_{t-k\to t}(p) \mid v_{t-k}(p) = 1 \right\},$$

and a robust dispersion estimate using the median absolute deviation:

$$s_{\text{MAD}}(p) = \text{median} \left\{ \left| \tilde{d}_{t-k\to t}(p) - d_{\text{med}}(p) \right| \mid v_{t-k}(p) = 1 \right\}.$$

The current raw disparity is corrected only when:

$$n_{\text{valid}}(p) \geq n_{\min},$$

$$s_{\text{MAD}}(p) \leq \tau_s,$$

and

$$|d_{\text{raw}}(p) - d_{\text{med}}(p)| \geq \tau_d + \kappa s_{\text{MAD}}(p).$$

The final correction is bounded:

$$d_{\text{CMC}}(p) = d_{\text{raw}}(p) + g_{\text{CMC}}(p)\, \text{clip}\left(d_{\text{med}}(p) - d_{\text{raw}}(p), -B, B\right),$$

with $B = 3$ cache-grid pixels.

The hypothesis was that several temporally separated memories agreeing tightly with each other while disagreeing with the current prediction would indicate that the current prediction is an outlier.

A predeclared sweep over 36 configurations was performed exclusively on the training sequences and three seen stereo backbones, covering approximately 30 million evaluated pixels.

The predefined promotion criteria required:

$$\Delta\text{EPE} \geq 0.005 \text{ px},$$

$$\text{FalseUpdateRate} \leq 20\%,$$

and

$$\text{CleanDegradation} \leq 15\%.$$

CMC did not satisfy these criteria. The best configuration achieved only:

$$\Delta\text{EPE} = 0.0013 \text{ px},$$

with:

$$\text{FalseUpdateRate} = 22.8\%.$$

Aggressive configurations reached false-update rates as high as 86%.

However, the mechanism remained highly conservative on already accurate predictions, with a clean-pixel degradation rate of approximately:

$$0.04\%.$$

The most important result is therefore not the failure of the median-based correction itself, but the structure of the failure.

When all temporal memories agree against the current prediction, they are frequently wrong together. Persistent stereo artifacts, specularities, occlusion errors, and backbone-specific failure modes remain correlated across multiple memory ages, even after optical-flow alignment.

The CMC mechanism oracle obtained a maximum gain of only:

$$0.0297 \text{ px},$$

whereas the unrestricted multi-memory oracle previously achieved approximately:

$$0.098 \text{ px}$$

beyond the $t-1$ oracle.

Therefore, consensus captures only about:

$$\frac{0.0297}{0.098} \approx 30\%$$

of the available long-memory improvement.

The remaining 70% is concentrated in cases where only one temporal age contains the correct evidence, while the remaining memories and the current prediction may be jointly incorrect.

This leads to the following conclusion:

> Temporal agreement is not equivalent to temporal correctness. Long-range stereo errors are strongly correlated across memory ages, and the useful evidence is frequently contained in a minority temporal candidate rather than in the consensus.

CMC is therefore not promoted as the main ARGOS v2 correction mechanism. It is retained as:

- a zero-parameter temporal baseline;
- an ultra-conservative safety prior;
- a source of high-precision pseudo-labels;
- a hard-positive mining mechanism for future quality estimation.

Together with the PPMStereo, DINOv3, and latent-state experiments, this result motivates explicit prediction of candidate error and uncertainty rather than similarity, consensus, or latent-state aggregation alone.

.14 Milestone: Absolute Quality Estimation Does Not Enable Reliable Memory Selection

We investigated whether safe temporal refinement could be obtained by explicitly predicting the expected disparity error of the current prediction and four causally aligned temporal candidates:

$$\mathcal{D}_t = \left\{ d_{\text{raw}}, \tilde{d}_{t-1\to t}, \tilde{d}_{t-2\to t}, \tilde{d}_{t-4\to t}, \tilde{d}_{t-8\to t} \right\}.$$

For each candidate i, the target quality was defined as its absolute disparity error:

$$e_i(p) = |d_i(p) - d_{\text{GT}}(p)|.$$

A lightweight backbone-agnostic Quality Prediction Module estimated the expected error and uncertainty of every candidate:

$$\hat{e}_i(p), \qquad \hat{\sigma}_i(p).$$

The model used only universal geometric and temporal signals. No RGB features, backbone identity, stereo cost volumes, internal hidden states, or future frames were provided.

The quality predictor learned a meaningful measure of absolute scene difficulty. Pearson correlations between predicted and true candidate errors ranged approximately from 0.60 to 0.70, while uncertainty remained positively correlated with actual prediction error.

However, this signal did not translate into reliable relative candidate ranking. The main results were:

Method	Top-1 Accuracy	Diagnostic Regret	Raw-vs-Memory AUROC
Constant baseline	27.9%	0.150	0.506
PPMStereo selector	28.3%	0.119	0.422
CMC spread heuristic	11.2%	0.154	0.679
Q0 quality predictor	19.7%	0.153	0.514

The proposed quality predictor therefore performed worse than both the constant baseline and the previous PPMStereo-style selector for candidate ranking.

The failure was particularly evident where a minority temporal candidate contained the correct evidence:

$$R^{Q0}_{\text{minority}} = 0.280, \qquad R^{PPM}_{\text{minority}} = 0.194.$$

The model predominantly learned a shared difficulty component:

$$\hat{e}_{\text{raw}} \approx \hat{e}_{t-1} \approx \hat{e}_{t-2} \approx \hat{e}_{t-4} \approx \hat{e}_{t-8},$$

rather than identifying which specific candidate retained correct geometry.

Risk–coverage analysis did not provide a useful intervention regime. At 10% coverage, regret decreased to approximately 0.049, but 98.1% of the selected pixels were already raw-clean and intervention precision was only 1.33%.

The central finding is therefore:

> Absolute prediction difficulty is learnable, whereas relative temporal candidate correctness is not reliably recoverable from the tested universal evidence.

Together with the PPMStereo, DINOv3, latent-state, and cross-memory consensus experiments, this result motivates abandoning multi-age candidate selection for the first ARGOS v2 model.

The next architecture therefore focuses on the more tractable safety question:

Is the current raw disparity sufficiently unreliable to justify intervention?

The resulting design direction is:

$$\boxed{\text{causal BiDA } t-1 + \text{raw-error detection} + \text{abstention} + \text{bounded residual correction}}$$

.15 Milestone: Safe Intervention via Raw Error Detection and Abstention

The central objective of ARGOS v2 is not to maximize temporal smoothing, but to determine when temporal correction is actually justified.

Following the failure of multiple temporal memory selection strategies (PPMStereo-inspired memory selection, DINOv3 features, latent temporal state, cross-memory consensus, and explicit quality prediction), we reformulated the problem as a binary safety decision:

Is the current raw disparity sufficiently unreliable to justify any temporal correction?

Instead of learning which temporal candidate should replace the current prediction, ARGOS v2 separates temporal refinement into two independent stages:

$$\boxed{\text{Correction Proposal} \rightarrow \text{Intervention Authorization}}$$

The correction proposal is generated by the previously validated bounded BiDA $t-1$ residual refiner,

$$d_{\text{proposal}} = d_{\text{raw}} + g \cdot c \cdot \tau \tanh(\Delta),$$

while a lightweight Raw Error Detector predicts

$$P(\text{raw is wrong}), \qquad \mu_{\text{raw}}, \qquad \sigma_{\text{raw}},$$

where
μ_{raw} is the expected raw disparity error and σ_{raw} represents predictive uncertainty.

The detector authorizes refinement only when the estimated error is sufficiently large and the uncertainty remains below a calibrated threshold. Otherwise the model abstains:

$$d_{\text{out}} = d_{\text{raw}}.$$

Importantly, the detector contains only 1,107 trainable parameters and uses exclusively backbone-independent geometric evidence.

On held-out SCARED-C sequences, the resulting system achieved:

Method	*EPE*	*False Update*	*Clean Degradation*	*Coverage*
Raw	0.9616	0%	0%	0%
A2	0.9190	35.1%	13.2%	40.0%
Authorized A2	0.9347	1.9%	0.97%	5.6%

Although intervention coverage is intentionally conservative, the detector preserves approximately

$$63.2\%$$

of the original A2 refinement gain while reducing false updates by nearly an order of magnitude.

Most importantly, after freezing all model parameters and operating thresholds, the detector generalized without retraining to the unseen Fast-FoundationStereo backbone:

$$0.9757 \rightarrow 0.9523$$

while maintaining virtually identical safety metrics.

This experiment validates the central hypothesis of ARGOS v2:

> Safe temporal stereo refinement is better formulated as an intervention authorization problem than as a direct temporal replacement problem.

The resulting architecture becomes

$$\boxed{\text{BiDA causal alignment} + \text{bounded residual proposal} + \text{raw-error detector} + \text{calibrated abstention}}$$

which constitutes the first promoted ARGOS v2 architecture.

.16 Milestone: Backbone Transfer Succeeds, but Cross-Domain Authorization Remains Unsolved

After promoting the balanced ARGOS v2 authorization policy, we froze the complete inference system before observing any out-of-distribution result.

The frozen system comprised:

$$\text{SEA-RAFT alignment} + \text{bounded BiDA } t-1 \text{ proposal} + \text{Raw Error Detector} + \text{calibrated abstention}.$$

All model parameters, detector temperature, intervention thresholds, and source files were frozen and recorded through SHA-256 hashes. No optimization or post-hoc tuning was performed on any OOD dataset.

The evaluation therefore tested two distinct forms of generalization:

1. transfer to an unseen stereo backbone within the SCARED-C domain;
2. transfer across unseen surgical datasets and acquisition conditions.

The main results were:

Dataset / protocol	Raw EPE	Refined EPE	ΔEPE	False Update	Verdict
CREStereo, unseen backbone	1.1003	1.0625	-0.0378	2.08%	GO
SERV-CT	1.0454	1.1015	$+0.0561$	38.41%	NO-GO
SCARED structured-light	2.9026	2.9132	$+0.0106$	0.71%	Preservation
D4D anchors	3.7498	3.7004	-0.0494	29.15%	NO-GO

Here, negative ΔEPE denotes improvement.

On the second unseen stereo backbone, CREStereo, ARGOS v2 reduced EPE by:

$$3.43\%,$$

while maintaining:

$$\text{FalseUpdate} = 2.08\%, \qquad \text{CleanDegradation} = 1.00\%,$$

and an intervention precision of:

$$81.67\%.$$

This confirms that the promoted refiner does not merely memorize the error distribution of the three training backbones. The combination of universal evidence, bounded correction, and intervention authorization transfers to a previously unseen stereo architecture within the same surgical domain.

However, cross-dataset evaluation exposed a different limitation.

On SERV-CT, the frozen detector authorized incorrect updates too frequently:

$$\text{FalseUpdate} = 38.41\%, \qquad \text{CleanDegradation} = 22.69\%,$$

causing EPE to increase from:

$$1.0454 \to 1.1015.$$

On D4D structured-light anchors, mean EPE improved slightly:

$$3.7498 \to 3.7004,$$

but safety remained unacceptable:

$$\text{FalseUpdate} = 29.15\%, \qquad \text{CleanDegradation} = 17.95\%,$$

and Bad3 increased despite the lower average EPE.

This demonstrates that average geometric improvement is insufficient evidence of safe refinement.

The temporal evaluations further confirmed that improved temporal smoothness does not necessarily imply improved geometry. D4D temporal disagreement decreased, while anchor-based safety metrics worsened. Similarly, StereoMIS showed sparse bounded interventions, but aggregate no-reference temporal consistency did not improve:

$$\Delta E_{\text{temporal}} = +0.00149.$$

The principal finding is therefore:

> ARGOS v2 generalizes across unseen stereo backbones within the training domain, but the intervention authorization policy remains sensitive to cross-dataset calibration shift.

The failure is localized primarily in the authorization stage rather than in the causal alignment or bounded proposal mechanism:

$$\text{backbone transfer} \quad \checkmark$$

$$\text{bounded causal proposal} \quad \checkmark$$

$$\text{within-domain authorization} \quad \checkmark$$

$$\text{cross-domain authorization calibration} \quad \times$$

This distinction is central to the ARGOS v2 contribution. The model does not fail because temporal evidence is absent; it fails because the detector assigns excessive confidence to intervention decisions under unseen acquisition distributions.

The defensible conclusion is therefore:

> Safe temporal refinement requires both error detection and awareness of whether the detector itself remains within its calibrated operating domain.

Consequently, the next ARGOS v2 milestone is not a more complex temporal refiner, but a domain-shift-aware authorization mechanism capable of abstaining when its own evidence distribution is unreliable.

.17 Milestone: Cross-Domain Failure Is Explained by Feature-Support Shift

After validating safe authorization on held-out SCARED-C sequences and two unseen stereo backbones, we investigated why the same frozen authorization policy failed on SERV-CT and D4D.

The complete ARGOS v2 pipeline was kept frozen:

$$\text{SEA-RAFT alignment} + \text{bounded A2 proposal} + \text{Raw Error Detector} + \text{balanced abstention policy}.$$

No retraining, threshold tuning, or architectural modification was performed. The detector, A2 checkpoint, BiDA implementation, temperature, and operating thresholds were verified through frozen SHA-256 hashes before inference.

The analysis compared the detector evidence distribution of each test domain against held-out SCARED-C using:

- nearest-neighbour support overlap;
- Mahalanobis distance;
- detector calibration;
- raw-error discrimination;
- authorization frequency;
- clean-pixel safety.

The results revealed a clear separation between unseen-backbone transfer and unseen-domain transfer:

Dataset	NN Support Overlap	Mean Mahalanobis	AUROC	ECE
SCARED-C reference	–	22.4	0.843	0.059
CREStereo	92.8%	21.3	0.859	0.056
SERV-CT	0.0%	581.8	0.487	0.409
D4D	7.8%	168.8	0.553	0.290
StereoMIS	42.8%	66.3	n/a	n/a

CREStereo, despite being an unseen stereo architecture, remained strongly inside the calibrated SCARED-C feature support. Its detector discrimination and calibration remained comparable to the reference distribution, explaining the successful transfer observed previously.

In contrast, SERV-CT was entirely outside the training support:

$$\text{SupportOverlap}_{\text{SERV-CT}} = 0\%,$$

$$D_{\text{Mahalanobis}} = 581.8.$$

Under this shift, the Raw Error Detector became effectively non-discriminative:

$$\text{AUROC} = 0.487,$$

while remaining strongly overconfident:

$$\text{ECE} = 0.409.$$

It authorized intervention on approximately:

$$48.3\%$$

of evaluated pixels, and 64.2% of GT-paired authorizations were diagnostically incorrect.

D4D exhibited the same failure mode at lower severity:

$$\text{SupportOverlap}_{\text{D4D}} = 7.8\%, \qquad D_{\text{Mahalanobis}} = 168.8,$$

with:

$$\text{AUROC} = 0.553, \qquad \text{ECE} = 0.290.$$

A stricter global authorization threshold was insufficient. The previously frozen ultra-safe operating point continued to authorize approximately:

$$43.3\%$$

of SERV-CT pixels and:

$$28.5\%$$

of D4D pixels.

Therefore, the failure cannot be explained solely by scalar miscalibration. The detector is confidently applying an in-domain decision rule to evidence lying outside its learned support.

The central finding is:

> ARGOS v2 transfers across unseen stereo backbones when the universal evidence remains inside the calibrated training support, but authorization becomes unsafe when the evidence distribution undergoes substantial cross-domain shift.

This establishes the distinction:

$$\boxed{\text{unseen backbone} \neq \text{unseen evidence distribution}}$$

and localizes the remaining limitation to support awareness rather than correction capacity.

The promoted architecture currently performs:

$$\text{raw-error authorization}$$

but does not yet answer:

Is the authorization evidence itself within a region where the detector is reliable?

The next ARGOS v2 mechanism is therefore a second abstention layer:

$$a_{\text{final}} = a_{\text{support}} \cdot a_{\text{error}},$$

where:

$$a_{\text{support}} = \mathbb{1}\left[s_{\text{OOD}} < \tau_{\text{support}}\right].$$

When evidence lies outside the calibrated feature support, the complete system must preserve the raw disparity:

$$a_{\text{support}} = 0 \implies d_{\text{out}} = d_{\text{raw}}.$$

This motivates a feature-support-aware authorization policy rather than a larger refiner or further temporal-memory complexity.

.18 Milestone: Supervised Multi-Domain Exposure Does Not Yield Third-Domain Authorization Robustness

After identifying cross-domain calibration shift as the main limitation of the ARGOS v2 Raw Error Detector, we tested whether supervised exposure to an additional surgical domain was sufficient to produce domain-general authorization.

The correction pipeline was kept unchanged:

$$\text{frozen SEA-RAFT} + \text{causal BiDA alignment} + \text{frozen A2 bounded proposal} + \text{trainable Raw Error Detector} + \text{bit-ex}$$

Only the Raw Error Detector and its calibration policy were retrained. No new architecture, temporal module, support detector, or feature extractor was introduced.

Two leave-one-domain-out experiments were conducted:

$$\text{M1}: \quad \text{SCARED-C} + \text{D4D} \to \text{SERV-CT},$$

$$\text{M2}: \quad \text{SCARED-C} + \text{SERV-CT} \to \text{D4D}.$$

All final-domain evaluations were performed only after freezing the detector, calibration parameters, operating thresholds, and artifact hashes.

At GT coverage 0.50, the main results were:

Experiment	Dataset	Raw EPE	Output EPE	Gain	False Update	Coverage
M1	SCARED-C	0.9616	0.9491	+0.0125	0.88%	2.33%
M1	Fast-FoundationStereo	0.9757	0.9656	+0.0101	0.79%	2.19%
M1	CREStereo	1.1003	1.0862	+0.0141	0.89%	2.52%
M1	SERV-CT unseen	1.0463	1.4140	−0.3678	49.08%	51.02%
M2	SCARED-C	0.9616	0.9421	+0.0195	0.25%	1.97%
M2	Fast-FoundationStereo	0.9757	0.9581	+0.0177	0.19%	1.97%
M2	CREStereo	1.1003	1.0752	+0.0251	0.20%	2.08%
M2	D4D unseen	2.9055	2.9001	+0.0054	1.80%	3.31%

The primary hypothesis was rejected.

In M1, adding D4D supervision did not improve transfer to SERV-CT. Instead, the detector failed more severely than the original SCARED-C-only detector:

$$\text{FalseUpdate}_{\text{M1,SERV}} = 49.08\%,$$

compared with approximately 38.3% for the previous SCARED-C-only model.

The SERV-CT EPE increased by:

$$1.4140 - 1.0463 = 0.3678 \text{ px},$$

and 78.6% of evaluated frames were degraded.

The detector remained effectively non-discriminative:

$$\text{AUROC}_{\text{M1,SERV}} = 0.468.$$

Therefore, exposure to SCARED-C and D4D did not produce a transferable definition of raw stereo error for SERV-CT.

M2 behaved more conservatively. Training with SCARED-C and SERV-CT reduced the risk on unseen D4D:

$$\text{FalseUpdate}_{\text{M2,D4D}} = 1.80\%,$$

with a small aggregate gain:

$$\Delta\text{EPE} = +0.0054 \text{ px}.$$

However, this behaviour was obtained at only 3.31% intervention coverage and was not consistent across specimens:

D4D specimen	Gain
1	+0.00765 px
2	−0.00036 px
3	−0.01269 px

Thus, M2 primarily learned conservative abstention rather than a useful and repeatable cross-domain authorization rule.

The main scientific conclusion is:

> Supervised exposure to two heterogeneous surgical domains is not sufficient for the current Raw Error Detector to learn an authorization rule that transfers safely to a third domain.

This result rejects the simple hypothesis:

$$\text{more supervised domains} \quad \Longrightarrow \quad \text{domain-invariant raw-error detection.}$$

Instead, the meaning of "raw prediction is wrong" remains strongly dependent on the acquisition and error distribution of each domain.

The experiments indicate that the current detector learns:

$$P\left(e_{\text{raw}} > \epsilon \mid \text{training-domain evidence}\right),$$

rather than a universal intervention rule.

This explains why:

- it generalizes across unseen stereo backbones inside SCARED-C;
- it preserves reasonable performance on domains represented during training;
- it can remain confidently wrong on a third unseen domain;
- adding domains may reduce coverage without improving decision quality.

The limitation is therefore not solved by simply concatenating supervised datasets.

The next design step should stop estimating raw error as the final authorization target. Instead, authorization should be conditioned on the specific correction proposal:

Will applying this bounded proposal improve the current prediction?

Define the proposal utility:

$$u(p) = e_{\text{raw}}(p) - e_{\text{proposal}}(p).$$

A positive value indicates that the proposed correction is beneficial:

$$u(p) > 0 \quad \Longrightarrow \quad e_{\text{proposal}}(p) < e_{\text{raw}}(p).$$

The next ARGOS v2 decision module should estimate either:

$$\hat{u}(p),$$

or the probability:

$$P\left(u(p) > \epsilon\right),$$

with an explicit neutral region and conservative abstention.
This changes the decision problem from:

Is the raw disparity wrong?

to:

$$\boxed{\textit{Is this particular bounded temporal correction expected to help?}}$$

This proposal-conditioned authorization is more directly aligned with the actual safety objective of ARGOS v2.

.19 Milestone: Proposal Utility Becomes Effective When Used as a Conditional Safety Veto

The dual-stage authorization experiment evaluated whether the frozen P4 proposal-applicability model could improve the safety of the existing Raw Error authorization policy.

The cascade was defined as:

$$a_{\text{final}}(p) = a_{\text{raw}}(p)\left(1 - v_{\text{P4}}(p)\right),$$

where a_{raw} is the frozen Raw Error authorization and v_{P4} can only reject an already authorized A2 proposal.

The final output remained structurally safe:

$$d_{\text{out}}(p) = \begin{cases} d_{\text{A2}}(p), & a_{\text{final}}(p) = 1, \\ d_{\text{raw}}(p), & a_{\text{final}}(p) = 0. \end{cases}$$

Therefore, abstention returned the raw disparity bit-exactly, while accepted interventions returned the frozen A2 proposal bit-exactly.

The selected cascade used the rule:

$$v_{\text{P4}}(p) = \mathbb{1}\left[\hat{u}_{\text{P4}}(p) \leq 0\right],$$

where $\hat{u}_{\text{P4}}$ is the predicted proposal utility.

The final held-out SCARED-C results were:

Metric	Raw Error authorization	P4 cascade	Target
EPE	0.934660	0.934845	$\approx$ baseline
Gain	0.026946	0.026760	$\geq 80\%$
Gain retained	–	99.31%	$\geq 80\%$
False update	1.90%	1.53%	$< 1.25\%$
Clean degradation	0.98%	0.73%	$< 0.60\%$
Intervention precision	75.63%	79.46%	$> 80\%$
Coverage	5.62%	4.72%	> 0

The cascade retained:

$$99.31\%$$

of the original authorized-A2 gain, while reducing false updates by:

$$\frac{1.90 - 1.53}{1.90} = 19.3\%,$$

and clean-pixel degradation by:

$$\frac{0.98 - 0.73}{0.98} = 25.5\%.$$

The EPE penalty relative to the original Raw Error authorization was only:

$$0.934845 - 0.934660 = 0.000185 \text{ px}.$$

A rejection-rate-matched random veto produced an EPE of:

$$0.940068,$$

demonstrating that the P4 veto contains genuine information and that the safety improvement is not explained solely by reduced intervention coverage.

All three seen stereo backbones remained improved relative to their raw predictions:

Backbone	Raw	Raw Error authorization	P4 cascade
S2M2-S	1.028950	1.009592	1.009830
RAFT-Stereo	0.925869	0.897324	0.897360
StereoAnywhere	0.929996	0.897062	0.897347

Although the predeclared promotion criteria were narrowly missed, the experiment demonstrates that raw-error authorization and proposal-utility estimation provide complementary decision signals.

The conditional audit revealed that, among proposals already authorized by the Raw Error Detector:

$$56.91\%$$

were helpful,

$$32.00\%$$

were harmful, and:

$$11.09\%$$

were indifferent.

This conditional distribution is substantially different from the global proposal-utility distribution, where harmful proposals represented only approximately 1.93% of all pixels.

P4 was originally trained on the global distribution, but was deployed as a veto on the restricted distribution:

$$P\left(\text{proposal harmful} \mid a_{\text{raw}} = 1\right).$$

Despite this training–deployment mismatch, P4 retained 94.68% of useful proposals on validation and 91.52% on the final test, while rejecting 27.42% of harmful proposals.

The central finding is therefore:

> Proposal-utility prediction provides a useful safety signal when conditioned on proposals already authorized by the Raw Error Detector, but a globally trained applicability model is not optimally matched to this conditional veto task.

This motivates a dedicated Conditional Harmfulness Veto trained exclusively on the decision subset produced by the frozen Raw Error authorization policy.

For each pixel satisfying:

$$a_{\text{raw}}(p) = 1,$$

the conditional target is:

$$y_{\text{harm}}(p) = \mathbb{1}\left[e_{\text{A2}}(p) > e_{\text{raw}}(p) + \epsilon\right].$$

The resulting architecture remains:

$$\boxed{\text{Raw Error authorization} \rightarrow \text{Conditional Harmfulness Veto} \rightarrow \text{frozen bounded A2 proposal}}$$

with:

$$a_{\text{final}}(p) = a_{\text{raw}}(p)\left(1 - \hat{y}_{\text{harm}}(p)\right).$$

The new veto is not required to discover useful interventions or estimate raw disparity error. Its only task is to identify dangerous corrections among proposals that the first-stage policy has already decided to apply.

This conditional decomposition targets the remaining safety gap directly while preserving the gain and bit-exact abstention properties of the existing ARGOS v2 pipeline.

.20 Current Status of ARGOS v2

At this stage, the core architecture of ARGOS v2 has stabilized around a clear design principle:

Temporal evidence is useful; the fundamental challenge is deciding when it is safe to use it.

Extensive controlled investigations have been conducted to evaluate multiple temporal refinement strategies.

Validated components (GO).

- **Causal BiDA alignment** successfully provides temporally aligned disparity evidence without introducing future information.
- **Bounded residual proposal (A2)** consistently improves disparity estimation using only geometric cues while guaranteeing bounded corrections.
- **Raw Error Detector** is the first promoted authorization module, reducing false updates from approximately 35% to below 2% while preserving more than 60% of the refinement gain.
- The complete pipeline generalizes across multiple unseen stereo backbones (Fast-FoundationStereo and CREStereo) without retraining.

Rejected research directions (NO-GO). The following hypotheses were experimentally validated and rejected:

- Long-term memory (PPMStereo-inspired)
- Latent recurrent state (EndoStreamDepth-inspired)
- Frozen semantic features (DINOv3)
- Cross-memory consensus correction

- Quality prediction for memory ranking
- Proposal-only authorization
- Lightweight support guard
- Multi-domain supervised training

Although several approaches demonstrated useful signals, none produced a safer or more general solution than the current bounded proposal plus authorization strategy.

Current scientific understanding. The accumulated evidence consistently indicates that temporal refinement itself is no longer the limiting factor.

Instead, the dominant challenge is reliable *decision making*:

$$\boxed{\text{Should this particular correction be applied?}}$$

rather than

$$\boxed{\text{How should the correction be computed?}}$$

This shifts the contribution of ARGOS v2 from a temporal refinement network toward a *safe decision framework* for plug-and-play stereo refinement.

Current architecture.

Frozen Stereo → SEA-RAFT → Causal BiDA Alignment → Bounded Proposal (A2) → Raw Error Authorizatio

The proposal-conditioned veto remains an optional safety-enhanced variant. Although it preserves over 99% of the authorized gain while reducing false updates and clean degradation, it narrowly misses the pre-registered promotion criteria and is therefore retained as a positive ablation rather than the primary model.

Current research direction. The next phase no longer focuses on inventing new temporal modules.

Instead, the objective is to understand and improve the authorization mechanism itself, investigating whether harmful proposals can be identified more accurately without sacrificing the gain already achieved by the bounded residual refiner.

.21 From Temporal Oracle Gain to Deployable Selective Refinement

Our experiments show that causally propagated disparity memory contains substantial complementary geometric information. Given the current stereo estimate d_t^{raw} and the temporally aligned memory estimate d_t^{mem}, a ground-truth oracle that selects the better candidate independently at each pixel consistently outperforms the raw stereo prediction. Across the stereo backbones used during development, the oracle reduces the mean end-point error (EPE) from 0.2666 to 0.2210, corresponding to a relative improvement of 17.1%. The gain remains positive on unseen backbones, where the oracle provides a relative EPE reduction of 10.5%. These results indicate that the aligned temporal memory is not merely temporally smoother, but contains genuine corrections to errors made by the current-frame stereo estimator.

However, the existence of a strong oracle ceiling does not imply that this information can be exploited reliably without ground truth. A broad set of deterministic and learned selection strategies was investigated, including direct replacement, fixed blending, forward–backward flow consistency, photometric gating, PPMStereo-inspired cues, DINO features, recurrent state representations, consensus signals, quality prediction, support guards, and multi-domain selector training. None of these approaches recovered the oracle gain with sufficient reliability across sequence-disjoint evaluation splits.

The final selector was evaluated using three independent training seeds and a strict sequence-disjoint protocol. The raw stereo baseline achieved an EPE of

$$\mathrm{EPE}_{\mathrm{raw}} = 0.5337,$$

whereas the selective raw-versus-memory policy obtained

$$\mathrm{EPE}_{\mathrm{sel}} = 0.5218 \pm 0.0031.$$

This corresponds to an absolute improvement of

$$\Delta\mathrm{EPE} = 0.0119 \pm 0.0031.$$

Although the average change is positive, the selector recovers only

$$22.7\% \pm 5.8\%$$

of the oracle improvement. Moreover, the effective intervention coverage is only

$$0.97\% \pm 1.07\%,$$

while

$$9.0\% \pm 5.5\%$$

of the evaluated frames are degraded relative to the raw baseline.

These results lead to a clear interpretation. The selector is conservative enough to remain beneficial on average, but it intervenes too rarely and captures too little of the available oracle gain. It therefore does not satisfy our predefined requirement of recovering at least 50% of the oracle improvement. Although all three development backbones improve slightly, the magnitude of the gain is insufficient to justify scaling the method to additional backbones as a final stereo-improvement result.

Earlier positive results obtained with smaller development protocols should consequently be interpreted as exploratory evidence rather than confirmatory results. Those experiments were useful for identifying promising signals and failure modes, but they are superseded for final claims by the stricter sequence-disjoint, multi-seed evaluation.

.21.1 Interpretation of the Bottleneck

The main bottleneck is not the existence of useful temporal information, nor does the evidence suggest that optical-flow alignment is the dominant limitation. Instead, the unresolved problem is decision-making: determining, from observable inference-time cues, whether d_t^{mem} is more accurate than d_t^{raw} at a given pixel or region.

Many investigated cues identify ambiguity, disagreement, poor flow support, or possible inconsistency. However, such signals do not necessarily reveal the sign of the relative error

$$\Delta e_t(x) = e(d_t^{\mathrm{raw}}(x)) - e(d_t^{\mathrm{mem}}(x)),$$

where $e(\cdot)$ denotes the unknown geometric error with respect to ground truth. In particular, disagreement between the two candidates may indicate that one estimate is wrong, but does not identify which estimate is more accurate. This distinction explains why a selector may achieve non-trivial ranking performance while still producing limited or even negative geometric utility at an operational threshold.

The out-of-distribution experiments further indicate that this limitation cannot be reduced to backbone shift alone. CREStereo remained relatively close to the training support, whereas SERV-CT and D4D exhibited considerably stronger distribution shifts. Neither global confidence thresholds nor feature-space guards based on Mahalanobis distance or k-nearest neighbours produced a satisfactory trade-off: conservative guards reduced the intervention coverage almost to zero, while less restrictive settings retained harmful failures.

.21.2 Relation to Prior Work and Novelty

The underlying candidate-selection problem is closely related to CODD, which already considers causal fusion between a current stereo estimate and an aligned temporal estimate, and supervises its fusion weights using the relative errors of the two candidates. Consequently, the generic idea of learning whether current or temporally propagated disparity is preferable cannot by itself constitute the main novelty of ARGOS v2.

A defensible contribution would instead require an explicit selective-decision formulation with a keep-current default, together with calibrated control of harmful interventions. Relevant distinctions from conventional temporal fusion include:

- an explicit decision between using temporal memory and retaining the current estimate;
- abstention when the predicted benefit is insufficiently reliable;
- optimization for net geometric utility rather than temporal smoothness alone;
- evaluation through intervention coverage, harm rate, oracle recovery, and risk–coverage behaviour;
- strict sequence-disjoint and cross-backbone validation.

The present results do not yet establish such a method. They instead identify a robust scientific gap between the amount of complementary information available in temporal memory and the fraction of that information that can be safely recovered by an inference-time selector.

.21.3 Current Conclusion

The findings can be summarized as follows:

1. BiDA-aligned temporal memory contains genuine and substantial complementary geometric information.
2. Blind replacement or blending of the raw stereo estimate is not reliable.
3. The tested selectors do not yet recover a sufficient fraction of the oracle gain.
4. The current ARGOS v2 pipeline is therefore not ready to be presented as a general stereo-improvement method.
5. The central open problem is safe prediction of the relative utility of current and propagated disparity estimates.

Accordingly, the most accurate current framing is not that temporal refinement has been solved, but that temporal memory exhibits a strong and reproducible oracle advantage while safe per-pixel exploitation of that advantage remains unresolved. Future work should therefore focus on determining whether the limitation arises from target construction, sampling and calibration, or from an intrinsic lack of sufficiently informative observable cues. Until this gap is closed under the strict evaluation protocol, further architectural scaling or additional out-of-distribution experiments would not provide a reliable basis for a final method claim.

.22 Relation to Selective Prediction and Temporal Memory Arbitration

The problem studied in ARGOS v2 is related to several partially overlapping research areas: temporal stereo fusion, confidence estimation, learning to defer, selective prediction, and quality-aware temporal memory. However, these areas differ substantially in the quantity being predicted and in the operational objective.

Classical confidence and uncertainty estimation methods associate a reliability score with a single prediction. Such a score can identify potentially inaccurate disparity estimates, but it does not necessarily determine which of two competing candidates is preferable. In our setting, the relevant quantity is instead the signed difference between the errors of the current and propagated estimates,

$$\Delta e_t(x) = \left| d_t^{\mathrm{raw}}(x) - d_t^{\mathrm{gt}}(x) \right| - \left| d_t^{\mathrm{mem}}(x) - d_t^{\mathrm{gt}}(x) \right|. \tag{113}$$

A positive value indicates that replacing the current estimate with temporal memory would be beneficial, whereas a negative value indicates a harmful intervention. Consequently, generic confidence ranking is insufficient unless it is also informative about this comparative error.

This distinction is consistent with results from learning-to-defer and cascade-deferral theory. Optimal arbitration between two predictors generally depends on their relative expected losses rather than on the confidence of either predictor in isolation okati2021triage,jitkrittum2023confidence. This is particularly relevant when the alternative predictor behaves as a specialist: temporally propagated disparity can be highly accurate in static and well-aligned regions, yet unreliable around occlusions, newly visible surfaces, motion boundaries, or previously corrupted estimates.

Selective prediction further introduces a reject option, allowing the system to avoid an intervention when its expected benefit is uncertain geifman2017selective,geifman2019selectivenet,zaoui2020regression. In ARGOS v2, rejection is naturally interpreted as retaining the current stereo estimate. The objective is therefore not merely to identify uncertain pixels, but to maximize geometric benefit while controlling the risk that accepted temporal updates worsen the baseline.

.22.1 Closest Prior Art: CODD

The closest prior work is CODD, which performs causal online depth estimation by combining a current stereo disparity with an aligned temporal estimate li2023codd. CODD already contains several elements that closely match the problem considered here:

- a current-frame stereo candidate and an aligned temporal candidate;
- causal per-pixel arbitration;
- supervision based on the relative errors of the two candidates;
- margin regions in which the candidates are treated as similarly accurate;
- a structural fallback to the current estimate;
- an empirical oracle demonstrating the potential gain from selecting the better candidate.

Its fused prediction can be expressed as

$$d_t^{\mathrm{fused}} = \left(1 - w_t^{\mathrm{reset}} w_t^{\mathrm{fusion}}\right) d_t^{\mathrm{raw}} + w_t^{\mathrm{reset}} w_t^{\mathrm{fusion}} d_t^{\mathrm{mem}}. \tag{114}$$

Thus, the general concept of learning whether temporal memory is preferable to a current stereo estimate cannot be considered novel by itself.

There are nevertheless important differences between CODD and the intended ARGOS v2 formulation. CODD performs continuous soft fusion and is primarily motivated by temporally consistent online depth. It uses a comparatively rich and tightly integrated evidence set, including learned scene motion, per-pixel transformations, internal stereo features, appearance correlations, visibility, semantic information, and previously fused memory. By contrast, ARGOS v2 investigates whether the relative utility of memory can be estimated from lightweight, frozen, and largely backbone-independent signals.

Moreover, CODD does not formulate deployment as an explicit risk-controlled switch with a reject option, nor does it report intervention coverage, harmful-update rate, oracle recovery, or risk–coverage curves against a never-update baseline. The potential contribution space is therefore not generic temporal candidate selection, but the narrower problem of safe and calibrated temporal intervention under limited observable evidence.

.22.2 Related Gating Mechanisms in Other Video Tasks

Several adjacent fields contain mechanisms that reject or attenuate temporal information. Dynamic video segmentation methods choose between warped previous predictions and current-frame computation xu2018dvsnet,nilsson2018grfp,jain2019accel. Video object segmentation methods regulate which observations are written to memory or restrict the memory bank to reduce degradation from stale information liu2022qdmn,cheng2022xmem,zhou2024rmem. Tracking systems similarly learn whether an appearance model should be updated, since incorrect updates can cause irreversible drift zhang2019updatenet,dai2020ltmu.

These methods establish that temporal memory can be conditionally harmful and that learned gating is useful. However, most predict absolute quality, optimize the downstream task implicitly, or operate at frame or region level. They generally do not learn an explicit pixel-wise comparative error between two finished dense-regression candidates.

Video restoration provides additional evidence that propagation can amplify previous errors. Recurrent super-resolution systems exploit temporal recurrence effectively, while later work explicitly observes that real-world degradations and artifacts may accumulate through propagation chan2021basicvsr,chan2022basicvsrpp,chan2022realbasicvsr. These works motivate memory rejection, but their mitigation strategies typically involve learned reconstruction, pre-cleaning, or recurrent feature refinement rather than selective replacement between two disparity maps.

.22.3 Risk and Net-Utility Evaluation

The final operational value of a selector cannot be inferred from AUROC or AUPRC alone. A ranking model may separate useful and harmful updates reasonably well, yet still yield negative or negligible benefit at the threshold used for deployment. This discrepancy is closely related to decision-curve analysis, in which prediction models are evaluated by their net benefit relative to default policies rather than only by discrimination metrics vickers2006decision,vickers2016netbenefit.

For temporal disparity selection, the natural default policy is

$$\pi_0(x) = d_t^{\mathrm{raw}}(x), \tag{115}$$

corresponding to never applying temporal memory. A selective policy with threshold τ is

$$\pi_\tau(x) = \begin{cases} d_t^{\mathrm{mem}}(x), & s_t(x) \geq \tau, \\ d_t^{\mathrm{raw}}(x), & s_t(x) < \tau, \end{cases} \tag{116}$$

where $s_t(x)$ estimates the expected relative utility of the temporal candidate.

Evaluation should therefore report, as a function of τ,

- intervention coverage;
- mean geometric gain;
- harmful-update rate;
- fraction of degraded frames;
- recovery of oracle improvement;
- risk–coverage or utility–coverage behaviour.

Distribution-free calibration approaches such as Learn-then-Test, conformal risk control, and conformal decision theory provide possible tools for selecting operating points under explicit risk constraints angelopoulos2021learntest,angelopoulos2022crc,lekeufack2024cdt. Their direct application to dense, temporally correlated regression is not immediate, however, because the calibration unit, dependence structure, and definition of acceptable geometric risk must first be specified.

.22.4 Novelty Positioning

Based on the reviewed literature, the following claims would not be defensible:

- that ARGOS v2 introduces the first learned selection between current and temporally propagated disparity;
- that pairwise relative-error supervision is itself novel;
- that per-pixel disparity candidate selection is new;
- that temporal memory rejection has not been previously studied.

A more defensible contribution would require the combination of:

1. a hard keep-current-versus-memory decision;
2. explicit abstention to the current estimate;
3. optimization and calibration for net geometric utility;
4. direct measurement of harmful temporal interventions;
5. risk–coverage evaluation under sequence-disjoint testing;
6. lightweight and backbone-independent inference signals.

The experiments presented in this work do not yet establish such a successful method. Instead, they reveal a reproducible gap between the oracle utility contained in temporal memory and the amount of utility recoverable from lightweight causal signals.

.23 Capacity and Observability as Competing Explanations

The failure of the lightweight selector admits two fundamentally different explanations. First, the available signals may contain sufficient information, but the selected architecture, loss, sampling strategy, or calibration procedure may fail to extract it. Second, the winner between the current and temporal candidates may be only weakly identifiable from the available causal inputs.

To distinguish these cases, a capacity probe should preserve the candidates, input channels, target construction, data split, and calibration protocol, while increasing only model capacity and spatial receptive field. The interpretation is then:

- If a larger model substantially increases oracle recovery on sequence-disjoint test data, the limitation lies primarily in representation capacity, optimization, or calibration.
- If the model fits the training set but fails to generalize, the observable signals are sequence- or domain-specific, or the comparative target is excessively noisy.
- If even training performance remains weak, the current input representation is insufficient to identify relative utility.
- If ranking metrics improve while deployment utility remains low, the main limitation lies in thresholding and cost-sensitive decision-making rather than representation.

This test cannot establish that temporal candidate selection is impossible in general. It can only establish whether the currently available causal, backbone-independent signals contain sufficient generalizable information. A negative result would imply that further progress requires additional evidence, such as internal stereo matching features, cost-volume information, learned 3D motion, additional viewpoints, or stronger geometric supervision, rather than merely a larger selector.

Table 29: Capacity-only comparison of the raw-versus-memory utility selector. The larger model uses the same observable evidence and differs only in capacity and receptive field. Values are aggregated over three seeds.

Metric	64×4 selector	128×8 selector
Parameters	0.304 M	2.379 M
Receptive field	19×19	35×35
EPE gain	0.01189 ± 0.00306	0.01411 ± 0.00242
Oracle recovery	$22.7\% \pm 5.8\%$	$27.0\% \pm 4.6\%$
Intervention coverage	0.97%	0.99%
False-update rate	0.60%	0.55%
Clean-pixel degradation	0.289%	0.286%
Intervention precision	39.4%	47.3%
Degraded frames	9.0%	10.7%
Worst-frame degradation	0.445 px	1.065 px

.24 Capacity Scaling Does Not Close the Temporal Oracle Gap

The limited oracle recovery of the utility selector may arise from two fundamentally different causes. First, the available causal signals may contain sufficient information, while the selector lacks the capacity or spatial context required to extract it. Second, the relative correctness of the current and propagated disparities may be only weakly identifiable from the available backbone-independent observations.

To distinguish these explanations, we conducted a controlled capacity probe. The data, candidate disparities, thirteen input signals, target construction, sampling strategy, training objective, sequence-disjoint split, checkpoint selection, and calibration protocol were kept fixed. Only the capacity and receptive field of the selector were increased.

The original selector used 64 channels and four residual blocks, corresponding to 303,747 parameters and an approximate receptive field of 19×19 pixels. The capacity probe increased the model to 128 channels and eight residual blocks, corresponding to 2,379,011 parameters and a receptive field of 35×35 pixels. Both configurations were evaluated using three independent seeds on the same held-out sequence-disjoint test set.

Increasing the parameter count by approximately a factor of eight produces a measurable but limited improvement. Mean oracle recovery increases by only 4.3 percentage points, from 22.7% to 27.0%, while the EPE gain increases by approximately 0.00222 pixels. The larger selector improves all three evaluated stereo backbones across all three seeds, indicating that additional capacity can extract some further signal from the existing inputs.

However, the operational behaviour remains essentially unchanged. Intervention coverage stays below 1%, oracle recovery remains far below the predefined 50% promotion requirement, and every seed continues to fail the promotion gate. Moreover, although average false-update and clean-pixel degradation rates remain low, the worst-frame degradation increases from 0.445 to 1.065 pixels. Thus, greater capacity does not eliminate the harmful tail of the intervention distribution.

These results indicate that model capacity is not the dominant bottleneck. The available raw-disparity, propagated-memory, optical-flow, support, and consistency signals contain some predictive information, as demonstrated by the improvement of the larger model. Nevertheless, they do not contain enough generalizable evidence to reliably determine which candidate is geometrically more accurate over a sufficiently large fraction of pixels.

The result should not be interpreted as proving that temporal candidate selection is impossible. Rather, it establishes a narrower observability result:

> Given the currently available causal and backbone-independent signals, scaling the selector alone does not recover a substantial fraction of the temporal oracle advantage.

Table 30: Strict three-seed result of the CODD-style fusion module on the held-out dataset-7 split. The raw-versus-memory oracle is computed using the same candidate pair and evaluation support.

Metric	CODD-style BiDA fusion
Raw stereo EPE	0.53317
Fused EPE	0.50357 ± 0.00176
Absolute EPE gain	0.02960 ± 0.00176
Raw-versus-memory oracle gain	0.05233
Oracle recovery	$56.56\% \pm 3.36\%$
Harmful-update rate	$0.658\% \pm 0.074\%$
Clean-pixel degradation	$3.41\% \pm 0.28\%$
Degraded frames	27.4%
Worst-frame degradation	0.681 ± 0.327 px

Further progress therefore requires an information probe rather than another capacity increase. In particular, the selector should be provided with additional evidence that is directly related to stereo matching ambiguity, such as candidate-conditioned matching costs, local cost-volume shape, confidence margins, internal stereo features, or an independent geometric signal. A controlled information ablation can then determine whether the remaining oracle gap is caused by missing observable evidence or by a deeper generalization limitation in relative utility prediction.

.25 Milestone: CODD-Style Fusion Recovers More Than Half of the Temporal Oracle

The preceding experiments established that the aligned temporal memory contains genuine complementary geometric information, but that this information cannot be recovered effectively through lightweight post-hoc selection alone. Increasing the selector capacity from 0.304 to 2.379 million parameters raised oracle recovery only from 22.7% to approximately 27.0%. Similarly, augmenting the selector with compact candidate-conditioned census evidence produced no material improvement.

These negative probes motivated a more substantial change in how temporal evidence is represented and consumed. We therefore implemented a CODD-style causal fusion module inspired by li2023codd, while retaining the existing BiDA-aligned temporal memory. The fused disparity is defined as

$$d_t^F = \left(1 - w_t^{\text{reset}} w_t^{\text{fusion}}\right) d_t^S + w_t^{\text{reset}} w_t^{\text{fusion}} d_t^M, \tag{117}$$

where d_t^S is the current stereo prediction, d_t^M is the aligned temporal memory, and both weights are bounded to $[0, 1]$.

The reset weight is supervised to reject the temporal candidate when its error is substantially larger than that of the current estimate. The fusion weight is supervised to favour the lower-error candidate outside a dead-band and to approach equal averaging when the two candidates have similar errors. In addition, the final fused disparity is directly supervised against ground truth using a robust regression loss.

Unlike the previous hard selector, the CODD-style module receives learned stereo-matching and spatio-temporal evidence, including candidate-conditioned left–right feature distances, local disparity and appearance self-correlations, cross-frame correlations, motion magnitude, flow validity, visibility, and support information. Training is performed through a four-frame causal unrolling procedure in which subsequent frames consume the previous fused estimate, thereby reducing the mismatch between training and online inference.

The CODD-style configuration reduces EPE from 0.53317 to 0.50357 ± 0.00176, corresponding to an average gain of 0.02960 ± 0.00176 pixels. Most importantly, it recovers $56.56\% \pm 3.36\%$ of the raw-versus-memory oracle advantage, exceeding the predefined promotion requirement of 50%.

Table 31: Progressive recovery of the temporal oracle under the strict sequence-disjoint protocol.

Configuration	Oracle recovery
Lightweight 64×4 hard selector	$22.72\% \pm 5.84\%$
Capacity-scaled 128×8 hard selector	$26.96\% \pm 4.63\%$
128×8 selector with local census evidence	$27.24\% \pm 1.85\%$
CODD-style recurrent soft fusion	$\mathbf{56.56\% \pm 3.36\%}$

The improvement is observed across all three evaluated stereo backbones, indicating that it is not produced by a single backbone-specific failure mode.

This progression changes the interpretation of the previous negative results. The raw-versus-memory decision is not fundamentally unobservable. Rather, it is poorly identified by simple post-hoc cues and hard selective replacement. The substantial increase obtained by the CODD-style configuration indicates that the missing ingredients were a combination of richer learned matching evidence, explicit spatial and temporal correlations, recurrent training with fused memory, direct supervision of the fused disparity, and a continuous fusion mechanism.

The result represents the first ARGOS v2 configuration to convert the strong BiDA oracle ceiling into a substantial deployable improvement. It therefore promotes temporal fusion from an unresolved diagnostic problem to a promising method direction.

Difference from CODD. The experiment should be described as CODD-style rather than as a faithful reproduction of CODD. The temporal candidate is aligned using the existing BiDA and optical-flow pipeline rather than a learned per-pixel SE(3) motion network. Moreover, because the cached stereo backbones do not expose a common internal feature representation, the fusion module uses a shared frozen feature encoder instead of the native stereo features used by CODD. The experiment therefore isolates whether CODD's fusion principles transfer to the ARGOS setting without simultaneously introducing its learned 3D motion module.

Remaining safety limitation. Passing the oracle-recovery gate does not imply that the method is fully ready for deployment. Although the average geometric gain is substantial, approximately 27.4% of frames are degraded, and the worst-frame degradation remains non-negligible. The next stage must therefore focus on reproducibility, common-support verification, component ablation, cross-backbone transfer, and risk-controlled operation rather than immediately adding a more complex motion model.

Milestone conclusion. The principal result of this phase is that BiDA memory can be exploited effectively when it is embedded in a CODD-style recurrent fusion architecture. ARGOS v2 therefore has a credible temporal stereo improvement mechanism for the first time. The central open question shifts from whether the temporal oracle is recoverable to whether the recovered gain can be made robust across frames, backbones, and deployment conditions.

.26 CODD-Style Recurrent Fusion with BiDA-Aligned Memory

The preceding experiments demonstrated that causally aligned temporal disparity contains substantial complementary geometric information, but that lightweight post-hoc selectors recover only a limited fraction of this potential. In particular, increasing the capacity of the hard selector and adding compact census-based candidate evidence both resulted in approximately 27% recovery of the historical raw-versus-memory oracle.

To determine whether this limitation originated from the fusion formulation and observable evidence rather than from the temporal candidate itself, we implemented a causal CODD-style fusion module inspired by li2023codd. The configuration does not retrain the stereo predictors and does not modify the validated BiDA/SEA-RAFT alignment pipeline. Instead, it replaces the previous hard raw-versus-memory decision with a continuous, evidence-conditioned fusion of the current stereo estimate and a causally propagated temporal state.

.26.1 Current and Temporal Disparity Candidates

Let

$$d_t^S \tag{118}$$

denote the current raw disparity produced by a frozen stereo backbone at time t. The temporal state used at the previous time step is defined as

$$d_{t-1}^{\text{state}} = \begin{cases} d_{t-1}^S, & \text{at the first step of a training clip,} \\ d_{t-1}^F, & \text{at subsequent steps,} \end{cases} \tag{119}$$

where d_{t-1}^F is the preceding fused prediction.

SEA-RAFT estimates target-to-source optical flow between the current and previous left images. The validated causal BiDA operator then aligns the preceding state with the current frame:

$$d_t^M = \text{BiDAWarp}\left(d_{t-1}^{\text{state}}, F_{t\to t-1}\right). \tag{120}$$

The aligned memory is accompanied by warp support, disparity validity, forward–backward flow consistency, and visibility information. No future frame is used.

.26.2 Dual-Weight Soft Fusion

The fusion module predicts two spatially varying weights,

$$w_t^{\text{reset}},\ w_t^{\text{fusion}} \in [0,1]^{H\times W}. \tag{121}$$

The effective temporal contribution is

$$w_t^T = w_t^{\text{reset}} w_t^{\text{fusion}}. \tag{122}$$

The output disparity follows the CODD fusion equation:

$$d_t^F = \left(1 - w_t^T\right) d_t^S + w_t^T d_t^M. \tag{123}$$

Therefore,

$$w_t^T = 0 \quad \Longrightarrow \quad d_t^F = d_t^S, \tag{124}$$

whereas

$$w_t^T = 1 \quad \Longrightarrow \quad d_t^F = d_t^M. \tag{125}$$

Intermediate values produce a convex interpolation between the current stereo prediction and the aligned temporal memory. Unlike the previous ARGOS hard selector, the primary configuration does not threshold w_t^T during inference.

.26.3 CODD-Style Comparative Supervision

Let the current and temporal disparity errors be

$$e_t^S = \left|d_t^S - d_t^{GT}\right|, \qquad e_t^M = \left|d_t^M - d_t^{GT}\right|, \tag{126}$$

and define their relative difference as

$$\Delta e_t = e_t^M - e_t^S. \tag{127}$$

The reset head is trained to suppress temporal memory when it is substantially worse than the current prediction:

$$\mathcal{L}_{\text{reset}} = \begin{cases} w_t^{\text{reset}}, & \Delta e_t > \tau_{\text{reset}}, \\ 1 - w_t^{\text{reset}}, & \Delta e_t < -\tau_{\text{reset}}, \\ 0, & \text{otherwise}. \end{cases} \tag{128}$$

The fusion head uses a narrower dead-band:

$$\mathcal{L}_{\text{fusion}} = \begin{cases} w_t^{\text{fusion}}, & \Delta e_t > \tau_{\text{fusion}}, \\ 1 - w_t^{\text{fusion}}, & \Delta e_t < -\tau_{\text{fusion}}, \\ \alpha_{\text{reg}} \left| w_t^{\text{fusion}} - 0.5 \right|, & |\Delta e_t| \leq \tau_{\text{fusion}}. \end{cases} \tag{129}$$

CODD defines $\tau_{\text{reset}} = 5$ and $\tau_{\text{fusion}} = 1$ in native-resolution disparity pixels. The SCARED-C images have width 1280, whereas the cached disparity maps have width 180. The thresholds were consequently converted while preserving their image-space meaning:

$$\tau_{\text{reset}} = 5\frac{180}{1280} = 0.703125, \tag{130}$$

$$\tau_{\text{fusion}} = 1\frac{180}{1280} = 0.140625. \tag{131}$$

The fused output is additionally supervised using a Huber loss:

$$\mathcal{L}_{\text{disp}} = \text{Huber}\left(d_t^F, d_t^{GT}\right). \tag{132}$$

The complete objective is

$$\mathcal{L} = \alpha_{\text{disp}}\mathcal{L}_{\text{disp}} + \alpha_{\text{reset}}\mathcal{L}_{\text{reset}} + \alpha_{\text{fusion}}\mathcal{L}_{\text{fusion}}, \tag{133}$$

with

$$\alpha_{\text{disp}} = \alpha_{\text{reset}} = \alpha_{\text{fusion}} = 1, \qquad \alpha_{\text{reg}} = 0.2. \tag{134}$$

The two heads therefore perform related but distinct functions. The reset head learns coarse outlier rejection, whereas the fusion head estimates how strongly temporal information should be incorporated when memory is not clearly invalid.

.26.4 Candidate-Conditioned Stereo Evidence

The fusion network receives 142 input channels. These include the thirteen universal geometric channels used by the previous ARGOS selectors, together with explicit stereo matching, spatial correlation, temporal correlation, and contextual evidence.

Stereo matching at the candidate disparities. A frozen ImageNet-pretrained ResNet-18 layer-1 encoder extracts feature maps from the current rectified left and right images. For both d_t^S and d_t^M, the right feature map is sampled at the corresponding disparity location. Matching evidence is also evaluated at offsets

$$\{-1, 0, +1\} \tag{135}$$

in the feature grid.

The candidate-conditioned matching cost is based on the channel-wise feature distance:

$$C(d) = \frac{1}{C} \left\| \phi_L(x, y) - \phi_R(x - d, y) \right\|_1. \tag{136}$$

Because the ResNet feature map operates at one-quarter resolution, an offset of one corresponds to one feature-grid pixel rather than one cached-disparity pixel.

This evidence allows the network to observe not only that the raw and temporal disparities disagree, but also which candidate is better supported by the current left–right stereo correspondence.

Local disparity correlations. Pixel-to-patch correlations are computed in 3×3 neighborhoods with dilation 2 for both current and temporal disparity. The similarity between a central disparity and a neighboring value is represented as

$$S_d(d_c, d_n) = 1 - \frac{\min(|d_n - d_c|, 4)}{4}. \tag{137}$$

These features describe local continuity, boundaries, isolated disparity errors, and structural disagreement between the two candidates.

Appearance and cross-frame correlations. The input also includes:

- current-frame feature self-correlation;
- aligned previous-frame feature self-correlation;
- cross-frame correlation between current features and BiDA-warped past features;
- local cross-correlation between current and memory disparity structures.

The cross-frame appearance features use the same validated target-to-source BiDA warp as the temporal disparity. No alternative flow convention is introduced.

Geometric and contextual cues. The remaining channels include:

- current raw disparity d_t^S;
- aligned memory disparity d_t^M;
- signed and absolute raw–memory disagreement;
- horizontal and vertical optical flow;
- flow magnitude;
- forward–backward flow confidence;
- warp support and aligned-disparity validity;
- current RGB;
- 64 frozen ResNet context channels.

The principal normalization ranges are:

$$C(d) \in [0, 1], \tag{138}$$
$$S_{\text{cosine}} \in [-1, 1], \tag{139}$$
$$F_x, F_y \in [-32, 32], \tag{140}$$
$$d \in [0, 64], \tag{141}$$
$$I_{\text{RGB}} \in [-1, 1]. \tag{142}$$

The ResNet feature extractor contains 157,504 frozen parameters. Only the fusion head is optimized.

.26.5 Fusion Architecture

The trainable fusion module contains 177,338 parameters, approximately matching the small scale of the original CODD fusion component.

Its fine-resolution branch consists of:

$$\mathrm{Conv}_{3\times 3}(142, 24) \to \mathrm{GroupNorm} \to \mathrm{SiLU} \to \text{residual block}. \tag{143}$$

A coarse branch performs spatial downsampling and increases the representation to 48 channels:

$$\mathrm{AvgPool} \to \mathrm{Conv}_{s=2} \to 3 \times \text{residual block}. \tag{144}$$

The fusion head predicts w_t^{fusion} at reduced resolution and upsamples it bilinearly. The reset head combines fine features with upsampled coarse context and predicts w_t^{reset} directly on the cached disparity grid.

The reset-head bias is initialized to -2, giving

$$w_t^{\text{reset}} \approx \sigma(-2) \approx 0.119. \tag{145}$$

The fusion-head bias is initialized to zero:

$$w_t^{\text{fusion}} = \sigma(0) = 0.5. \tag{146}$$

The initial temporal contribution is therefore approximately

$$w_t^T \approx 0.119 \times 0.5 \approx 0.0595. \tag{147}$$

The network consequently begins close to the raw prediction rather than initially replacing it with temporal memory.

.26.6 Four-Frame Causal Training

Training samples consist of four-frame causal clips. At the first step, temporal memory is constructed from the previous raw disparity:

$$d_{t-1}^{\text{state}} = d_{t-1}^S. \tag{148}$$

At subsequent steps, the fused output becomes the propagated state:

$$d_t^{\text{state}} = d_t^F. \tag{149}$$

The model therefore consumes its own preceding fused prediction during the second, third, and fourth steps of the clip. Gradients propagate through the complete four-step unroll, and the clip loss is the mean of the individual frame losses:

$$\mathcal{L}_{\text{clip}} = \frac{1}{4}\sum_{k=1}^{4} \mathcal{L}_{t+k}. \tag{150}$$

The state is reset at clip boundaries. Thus, the current experiment establishes bounded four-frame recurrence rather than arbitrary long-sequence streaming.

.26.7 Training and Evaluation Protocol

The experiment uses a strictly dataset-ID-disjoint protocol:

The seen stereo backbones are S2M2-S, RAFT-Stereo, and StereoAnywhere. The model is trained for 12 epochs with AdamW, learning rate 2×10^{-4}, weight decay 10^{-4}, batch size 4 clips, automatic mixed precision, and gradient clipping at 1.0. Checkpoints are selected exclusively by validation fused EPE.

All comparisons use the common support

$$\mathcal{M} = \mathcal{M}_{GT} \cap \mathcal{M}_{\text{raw}} \cap \mathcal{M}_{\text{aligned}} \cap \mathcal{M}_{\text{warp}}. \tag{151}$$

Raw disparity, temporal memory, fused disparity, and oracle metrics are therefore evaluated on the same valid pixels.

Table 32: Dataset roles for the CODD-style Phase-1 experiment.

Role	Dataset IDs
Training	$1, 3, 6$
Validation and checkpoint selection	2
Final test	7

Table 33: Three-seed CODD-style Phase-1 result on the held-out dataset-7 split.

Metric	Result
Raw EPE	0.53317
Fused EPE	0.50357 ± 0.00176
Absolute EPE gain	0.02960 ± 0.00176
Historical raw-versus-memory oracle gain	0.05233
Historical normalized gain	$56.56\% \pm 3.36\%$
Harmful-update rate	$0.658\% \pm 0.074\%$
Clean-pixel degradation	$3.41\% \pm 0.28\%$
Frames with increased EPE	27.4%
Worst-frame degradation	0.681 ± 0.327 px

.26.8 Main Result

The CODD-style module almost doubles the normalized gain obtained by the previous hard selectors:

$$\sim 27\% \longrightarrow 56.56\%. \tag{152}$$

The improvement is also observed in temporal metrics. Raw warped disparity variation is approximately 0.10696, whereas the three fused models obtain 0.06459, 0.06236, and 0.06040. Similarly, raw TEPE is 0.10883, compared with 0.07483, 0.07126, and 0.07037 for the three seeds. The reduction in temporal variation is therefore accompanied by an improvement in geometric EPE rather than being produced solely by blind smoothing.

.26.9 Why the Configuration Works

The improvement cannot yet be attributed to one isolated component because the configuration changes several aspects of the previous selector simultaneously. Nevertheless, the results support several complementary mechanisms.

Candidate quality becomes directly observable. The former selector mainly observed disparity disagreement, flow, and support. These cues can reveal that the candidates differ, but not necessarily which candidate is geometrically supported by the current stereo pair. Candidate-conditioned left–right feature costs give the model direct evidence about whether d_t^S or d_t^M corresponds to a plausible stereo match.

Spatial correlation distinguishes noise from structure. Disparity disagreement alone cannot distinguish a stereo error from a real depth boundary. Local disparity and appearance correlations provide the context required to recognize smooth surfaces, discontinuities, isolated errors, and occlusion boundaries.

Cross-frame evidence tests the alignment itself. A propagated disparity can be plausible geometrically but originate from an incorrect temporal correspondence. Cross-frame appearance and disparity correlations indicate whether the aligned past observation remains compatible with the current image.

Reset and fusion solve different subproblems. A single hard score must simultaneously detect catastrophic memory failures and estimate the preferred candidate. The dual-head design separates these tasks. The reset head removes large temporal outliers, while the fusion head controls aggregation in less decisive regions.

Soft fusion can correct both candidates. A hard selector is restricted to

$$d_t^S \quad \text{or} \quad d_t^M. \tag{153}$$

The CODD-style module can instead predict any point on the segment joining them. When d_t^{GT} lies between the two candidates, a convex combination can outperform both. Part of the observed gain may therefore originate from interpolation rather than from correct endpoint selection.

Direct disparity supervision optimizes the actual output. The previous selector was primarily optimized to classify relative candidate utility. The CODD-style objective also supervises the final d_t^F. This permits the network to learn a weight that minimizes geometric error directly rather than only reproducing a binary candidate label.

Recurrent training reduces train–inference mismatch. During deployment, temporal memory may contain an earlier fused output rather than an independent raw prediction. Four-frame causal unrolling exposes the model to this feedback during training and allows earlier corrections to propagate into subsequent states.

The most defensible current conclusion is therefore:

> The failure of the previous hard selectors was not caused by an absolute absence of causal temporal information. A compact recurrent fusion model can exploit BiDA-aligned memory when it receives explicit stereo-matching and spatio-temporal correlation evidence, is trained with comparative reset/fusion supervision, and is allowed to perform continuous convex correction.

.26.10 Interpretive Caveats

The current configuration should be described as *CODD-style*, not as a complete reproduction of CODD. CODD uses a learned per-pixel SE(3) motion module based on RAFT3D and native stereo-backbone features. ARGOS v2 instead uses the existing SEA-RAFT/BiDA alignment and a common frozen ResNet-18 feature space.

The reported 56.56% value also uses a historical oracle based on

$$\operatorname{warp}\big(d_{t-1}^S\big)\,, \tag{154}$$

whereas the model uses

$$\operatorname{warp}\big(d_{t-1}^F\big) \tag{155}$$

after the first step. Furthermore, the historical oracle selects the better endpoint, while the model can interpolate between the endpoints. The value should therefore be reported as a *historical-selection normalized gain*, not as the exact fraction of the full convex-fusion ceiling recovered.

Finally, the continuous sigmoid weights do not provide bit-exact abstention. Approximately 27.4% of frames are worsened despite the strong mean improvement. The experiment is therefore a strong validation of the fusion mechanism, but not yet a demonstration of a clinically safe operating policy.

Current milestone. This configuration constitutes the first robust ARGOS v2 method that converts BiDA-aligned temporal information into substantial improvements in both geometric and temporal accuracy. The next analysis must determine how much of the gain arises from learned stereo evidence, convex interpolation, and recurrent candidate improvement, and whether the four-frame state remains stable during continuous sequence-level streaming.

Table 34: Bounded-memory ablations on the held-out dataset-7 split. Gain is computed relative to the corresponding raw stereo prediction. All disparity values are expressed in cache-grid pixels.

Configuration	EPE	Gain	Harmful updates	Clean degradation	Worst frame
Full, fixed $H = 4$	0.50613	+0.02755	2.85%	3.72%	0.41
No recurrent fused state	0.51211	+0.02157	1.80%	2.28%	2.17
No learned stereo evidence	**0.50348**	**+0.03020**	2.59%	3.10%	0.80
Hard endpoint, $H = 4$	0.50887	+0.02482	5.55%	7.91%	0.41
Adaptive hybrid, $H_{\max} = 8$	0.51160	+0.02208	3.85%	4.80%	0.44

.27 Bounded Recurrence Reveals a Temporal-Memory Provenance Problem

The preceding CODD-style experiment established that BiDA-aligned temporal information can improve frame-wise stereo disparity. However, the initial experiment did not determine whether the improvement originated from recurrent state propagation, learned stereo evidence, soft interpolation, or the periodic four-frame reset used during training and evaluation.

We therefore conducted a controlled bounded-memory study in which the principal components were independently removed or modified while preserving the frozen stereo predictions, SEA-RAFT/BiDA alignment, common-support masks, optimization protocol, and causal four-frame construction.

.27.1 Component Ablations

Table 34 reports the results on the held-out dataset-7 split.

The full bounded configuration improves the raw prediction by

$$G_{\text{full}} = 0.02755. \tag{156}$$

When the preceding fused output is never propagated and temporal memory is always constructed from the previous raw prediction,

$$d_t^M = \text{BiDAWarp}\left(d_{t-1}^S\right), \tag{157}$$

the gain remains positive:

$$G_{\text{no-rec}} = 0.02157. \tag{158}$$

The additional benefit attributable to bounded fused-state recurrence is therefore approximately

$$G_{\text{full}} - G_{\text{no-rec}} = 0.00598. \tag{159}$$

Thus, recurrence is useful but is not the sole source of temporal improvement.

.27.2 Short Recurrence Helps, Indefinite Recurrence Fails

The bounded model propagates the preceding fused prediction according to

$$d_t^M = \text{BiDAWarp}\left(d_{t-1}^F\right), \tag{160}$$

and produces

$$d_t^F = (1 - w_t)\, d_t^S + w_t d_t^M. \tag{161}$$

With a reset every four frames, this recurrent correction remains beneficial:

$$\text{EPE}_{H=4} = 0.50613. \tag{162}$$

When the same state is propagated continuously without periodic re-anchoring, performance collapses:

$$\mathrm{EPE}_{\text{continuous}} = 1.6259, \tag{163}$$

$$G_{\text{continuous}} = -1.0922. \tag{164}$$

In continuous streaming, 63.5% of the evaluated frames are degraded and the worst-frame degradation reaches 22.81 EPE. The mean geometric correction increases from approximately 0.06 pixels under bounded recurrence to approximately 1.50 pixels under unrestricted recurrence.

This behavior indicates a recurrent feedback process:

$$\text{state error} \to \text{temporal warp} \to \text{incorrect fusion} \to \text{larger state error}. \tag{165}$$

The preceding fused prediction is therefore useful as a short-lived temporal candidate, but it is unsafe as an indefinitely persistent representation of the complete history.

.27.3 Learned Stereo Evidence Is Not the Dominant Factor

The no-evidence ablation removes:

- candidate-conditioned learned left–right matching costs;
- learned appearance self-correlation;
- learned cross-frame appearance correlation;
- frozen ResNet context features.

It retains disparity disagreement, flow, support, validity, motion information, dual reset/fusion supervision, bounded recurrence, and soft disparity fusion.

This reduced configuration obtains

$$\mathrm{EPE}_{\text{no-features}} = 0.50348, \tag{166}$$

corresponding to a gain of

$$G_{\text{no-features}} = 0.03020. \tag{167}$$

In the canonical run, removing the learned stereo evidence therefore does not reduce performance. This result does not prove that learned stereo features are universally harmful, because the ablation was not replicated across multiple random initializations. It does establish that they are not necessary for obtaining the current temporal gain.

The evidence consequently suggests that the principal bottleneck is not solely feature availability. The lifecycle and provenance of the temporal state are at least as important as the representation used to score it.

.27.4 Soft Correction Is Preferable to Hard Replacement

The hard endpoint variant restricts the output to

$$d_t^H \in \left\{d_t^S, d_t^M\right\}. \tag{168}$$

Although it preserves a positive gain of 0.02482, its harmful-update rate increases to 5.55% and its clean-pixel degradation increases to 7.91%.

The soft model instead performs a bounded correction:

$$d_t^F = d_t^S + w_t\left(d_t^M - d_t^S\right). \tag{169}$$

The benefit of soft fusion is not primarily due to predicting values that outperform both endpoints. The measured aggregate gain from such cases is only approximately 0.00072 EPE. Rather, the continuous weight allows the model to move conservatively toward a useful temporal candidate without fully replacing the current stereo estimate.

.27.5 Fixed Re-Anchoring Generalizes Better than the Adaptive Policy

A validation-only policy search evaluated fixed horizons

$$H \in \{1, 2, 4, 6, 8, 12, 16\}, \tag{170}$$

as well as cumulative-update, evidence-based, and hybrid reset policies.

The hybrid policy with maximum age $H_{\max} = 8$ was preferred on the validation split. After being frozen and evaluated on dataset 7, however, it produced

$$\mathrm{EPE}_{\mathrm{hybrid}} = 0.51160, \tag{171}$$

which is worse than the fixed $H = 4$ result.

Thus, the available reset cues do not reliably detect state corruption before it affects the output. Fixed four-frame re-anchoring remains the most defensible operating policy.

.27.6 Temporal-Memory Interpretation

The combined ablations reveal a temporal-memory provenance problem.

A raw disparity prediction

$$d^S_{t-k} \tag{172}$$

is an independent observation produced directly by the stereo backbone. A corrected prediction

$$d^F_{t-k} \tag{173}$$

may be more accurate locally, but it also contains the accumulated effect of preceding warps and fusions.

The two types of memory therefore have different properties:

$$d^F : \text{high short-term utility, increasing recurrent risk,} \tag{174}$$

$$d^S : \text{lower correction level, independent long-term anchor.} \tag{175}$$

This motivates a provenance-typed temporal memory:

$$\mathcal{M}_t = \mathcal{M}_t^{\mathrm{corr}} \cup \mathcal{M}_t^{\mathrm{raw}} \cup \mathcal{B}_t^{\mathrm{emb}}, \tag{176}$$

where

$$\mathcal{M}_t^{\mathrm{corr}} = \left\{d^F_{t-1}, d^F_{t-2}\right\} \tag{177}$$

contains short-lived corrected states,

$$\mathcal{M}_t^{\mathrm{raw}} = \left\{d^S_{t-1}, d^S_{t-2}, d^S_{t-4}, d^S_{t-8}\right\} \tag{178}$$

contains immutable raw anchors, and $\mathcal{B}_t^{\mathrm{emb}}$ stores compact temporal evidence for consensus, retrieval, flicker detection, and memory-confidence estimation.

Under this formulation, corrected predictions are never promoted to permanent long-term memory. They expire after a short temporal lifetime, whereas raw observations preserve access to a longer causal history without accumulating fusion drift.

.27.7 Validated Scope

The reported results establish held-out generalization to dataset ID 7 within the SCARED-C experimental protocol. They do not yet establish external out-of-distribution generalization.

The refiner was trained and evaluated using predictions from S2M2-S, RAFT-Stereo, and StereoAnywhere. Consequently, the experiment demonstrates compatibility across multiple heterogeneous but seen stereo backbones. It does not establish transfer to a stereo backbone excluded from refiner training.

Similarly, dataset ID 7 belongs to the same surgical dataset family used for training and validation. No external surgical dataset, different acquisition system, or unseen backbone has yet been evaluated.

The current claim is therefore:

Table 35: Post-hoc temporal-memory oracle audit on dataset 7. The oracle selects the candidate with the smallest disparity error independently at each valid pixel. These results measure information availability and do not represent the performance of a deployable model.

Candidate bank	Oracle EPE	Gain versus raw
Current raw	0.54773	–
Current + CS1	0.49537	0.05236
Current + CF1/CF2	0.45401	0.09372
Current + CS1/CS2/CS4/CS8	0.40821	0.13952
Full hybrid	0.40331	0.14442
Full hybrid, convex oracle	**0.38879**	**0.15894**

> ARGOS v2 provides a bounded-horizon causal temporal refinement mechanism that generalizes to held-out SCARED-C sequences and multiple seen stereo backbones. Backbone-agnostic transfer and external-dataset robustness remain unverified.

Main finding. The principal requirement for stable temporal refinement is not an indefinitely expressive recurrent state. It is a controlled memory lifecycle: corrected states provide additional short-term gain, raw predictions provide stable re-anchoring, and soft fusion prevents aggressive endpoint replacement. This finding motivates a hybrid causal memory in which short-lived corrected estimates and immutable raw anchors are stored separately.

.28 Long-Range Causal History Is Preserved by Immutable Raw Anchors

The preceding bounded-recurrence experiments showed that short-term propagation of corrected disparity improves temporal refinement, while unrestricted recurrent propagation causes severe long-horizon drift. This raised a separate question: whether distant causal history contains useful geometric information that is lost by the current single-state representation.

We therefore performed a post-hoc multi-anchor oracle audit using direct current-to-anchor BiDA alignment. No model was trained during this experiment. At frame t, the candidate bank contained the current stereo estimate

$$d_t^S, \tag{179}$$

short-term corrected candidates

$$\mathcal{M}_t^F = \left\{\hat{d}_{t-1}^F, \hat{d}_{t-2}^F\right\}, \tag{180}$$

and immutable raw anchors

$$\mathcal{M}_t^S = \left\{\hat{d}_{t-1}^S, \hat{d}_{t-2}^S, \hat{d}_{t-4}^S, \hat{d}_{t-8}^S\right\}, \tag{181}$$

where every temporal candidate was independently warped into the current frame using direct current-to-anchor optical flow.

.28.1 Multi-Anchor Oracle Ceiling

Table 35 reports the availability-aware oracle results on dataset 7. The evaluation contains 28,031,346 valid pixels.

The immutable raw-anchor bank reduces the oracle EPE from

$$0.54773 \tag{182}$$

Table 36: Incremental oracle gain when raw anchors are added in increasing temporal order.

Candidate added	Incremental gain
CS1	0.05236
CS2	0.02841
CS4	0.02787
CS8	0.03088
CF1	0.00274
CF2	0.00215

to

$$0.40821, \tag{183}$$

corresponding to a ceiling gain of

$$G_{\text{raw bank}} = 0.13952. \tag{184}$$

Adding the corrected short-term candidates improves the full hybrid ceiling to

$$G_{\text{hybrid}} = 0.14442. \tag{185}$$

Consequently, the raw anchors explain

$$\frac{G_{\text{raw bank}}}{G_{\text{hybrid}}} = 96.6\% \tag{186}$$

of the full hybrid selection ceiling. The two corrected candidates contribute only

$$G_{\text{CF1+CF2}|\text{raw bank}} = 0.00489, \tag{187}$$

or approximately 3.4% of the complete gain.

.28.2 Incremental Value of Distant Raw Anchors

The raw-first incremental audit produced the following gains:

The gain does not saturate after the immediately preceding frame. In particular, the $t-8$ raw anchor contributes an additional

$$0.03088 \tag{188}$$

EPE of oracle gain after the inclusion of CS1, CS2, and CS4.

Although CS8 has a direct candidate EPE of

$$\text{EPE}_{\text{CS8}} = 0.61433, \tag{189}$$

which is worse than the current raw prediction, it provides a selective oracle gain of

$$G_{\text{CS8}} = 0.09137. \tag{190}$$

CS8 outperforms the raw estimate on 41.91% of its valid pixels and contributes to 82.20% of evaluated frames. It is therefore not a reliable global prediction, but a highly informative local specialist.

.28.3 Raw and Corrected Memory Have Asymmetric Complementarity

The provenance-complementarity audit found that:

- a raw anchor is useful while both corrected candidates fail on 14.34% of evaluated pixels;
- CS4 or CS8 recovers a failure of short corrected memory on 9.08%;
- a corrected candidate is useful while all raw anchors fail on only 0.85%;
- both provenance families contain some useful candidate on 60.24%;
- the current raw prediction remains optimal on 24.58%.

The complementarity is therefore strongly asymmetric. Corrected states provide a small residual benefit, whereas immutable raw observations carry nearly all of the long-range temporal signal.

.28.4 The Gain Is Not Caused by Support Differences

Temporal support decreases only moderately with anchor age:

$$\text{coverage(CS1)} = 99.06\%, \tag{191}$$
$$\text{coverage(CS2)} = 98.73\%, \tag{192}$$
$$\text{coverage(CS4)} = 97.99\%, \tag{193}$$
$$\text{coverage(CS8)} = 96.54\%. \tag{194}$$

Under strict-intersection support, where every compared candidate must be valid, coverage remains 96.40%. The full hybrid selection oracle still achieves

$$\text{EPE}_{\text{strict}} = 0.39393 \tag{195}$$

and a gain of

$$G_{\text{strict}} = 0.14396. \tag{196}$$

The observed long-range signal therefore cannot be explained by incompatible masks or candidate-specific support advantages.

.28.5 Consistency Across Backbones and Sequences

The additional full-hybrid gain over the best short-term family is positive for every evaluated backbone:

$$G_{\text{RAFT-Stereo}} = 0.05847, \tag{197}$$
$$G_{\text{S2M2-S}} = 0.04469, \tag{198}$$
$$G_{\text{StereoAnywhere}} = 0.04893. \tag{199}$$

It is also positive for each dataset-7 sequence, ranging from

$$0.02666 \tag{200}$$

to

$$0.07624. \tag{201}$$

Thus, the signal is not confined to a single seen stereo backbone or individual sequence.

.28.6 Memory-Failure Detection

Compact temporal statistics were evaluated as diagnostic predictors of raw error, temporal-memory failure, and harmful corrected updates. An Isolation Forest trained only on training datasets 1, 3, and 6 achieved:

$$\mathrm{AUROC}_{\text{memory failure}} = 0.812, \tag{202}$$

$$\mathrm{AUROC}_{\text{harmful corrected}} = 0.828. \tag{203}$$

These results indicate that candidate failure is partially observable from compact temporal statistics. However, the anomaly detector remains a validation-only diagnostic and is not used as a fusion or authorization policy.

.28.7 Architectural Implication

The audit supports a non-recurrent causal memory bank:

$$\mathcal{M}_t^S = \left\{ \hat{d}_{t-1}^S, \hat{d}_{t-2}^S, \hat{d}_{t-4}^S, \hat{d}_{t-8}^S \right\}. \tag{204}$$

A deployable model should retrieve one candidate

$$k^\star = \underset{k \in \{1,2,4,8\}}{\arg\max}\ s_t^{(k)} \tag{205}$$

and perform a bounded soft correction:

$$d_t^{\text{out}} = (1 - w_t)\, d_t^S + w_t \hat{d}_{t-k^\star}^S. \tag{206}$$

When no temporal candidate is sufficiently reliable, the model must return

$$d_t^{\text{out}} = d_t^S \tag{207}$$

with a bit-exact raw fallback.

Corrected candidates may be added as an explicitly typed memory with maximum lifetime two frames:

$$\mathcal{M}_t^F = \left\{ \hat{d}_{t-1}^F, \hat{d}_{t-2}^F \right\}, \tag{208}$$

but they should be treated as a controlled ablation rather than as the primary memory representation.

Main finding. Long-range causal history is highly informative for surgical stereo, but its utility is preserved by independently anchored raw observations rather than by recurrent propagation of corrected disparity. This motivates a causal multi-anchor retrieval and soft-fusion adapter, not an unrestricted recurrent or generic embedded memory bank.

Scope. The result is an oracle ceiling analysis rather than deployable model performance. It is validated on held-out SCARED-C sequences and three seen stereo backbones. Transfer to unseen stereo estimators and external surgical datasets remains unverified.

.29 Multi-Anchor Raw Memory Outperforms Bounded Recurrent Fusion

The oracle analysis established that long-range causal history contains substantial geometric information, primarily in the form of independently generated raw disparity anchors. We therefore evaluated whether this signal could be recovered by a deployable model without recurrently propagating refined disparity.

Table 37: Frozen dataset-7 results for the realizable raw multi-anchor experiment. Gain versus $H = 4$ is measured against the bounded CODD-style recurrent reference on common support.

Configuration	EPE	Gain vs. raw	Gain vs. $H = 4$
CS1 soft	0.53240	0.01552	-0.01515
Hard multi-anchor retrieval	0.51420	0.03372	$+0.00306$
Soft multi-anchor fusion	**0.51047**	**0.03745**	$+$**0.00679**
Temporal consensus	0.54100	0.00693	-0.02374

At frame t, the proposed adapter constructs the immutable raw-anchor bank

$$\mathcal{M}_t^S = \left\{\hat{d}_{t-1}^S, \hat{d}_{t-2}^S, \hat{d}_{t-4}^S, \hat{d}_{t-8}^S\right\}, \tag{209}$$

where every past raw disparity is independently aligned to the current frame using direct current-to-anchor SEA-RAFT flow and a causal BiDA-style pull warp.

A shared candidate-wise module predicts a utility score and a fusion weight for every valid anchor:

$$\left(s_t^{(k)}, w_t^{(k)}\right) = f_\theta\left(x_t^{(k)}\right), \qquad k \in \{1, 2, 4, 8\}. \tag{210}$$

The highest-scoring valid candidate is retrieved:

$$k^\star = \underset{k\in\{1,2,4,8\}}{\arg\max}\ s_t^{(k)}, \tag{211}$$

and softly fused with the current raw stereo prediction:

$$d_t^{\text{out}} = \left(1 - w_t^{(k^\star)}\right) d_t^S + w_t^{(k^\star)} \hat{d}_{t-k^\star}^S. \tag{212}$$

The refined output is never written back into the raw-anchor bank. Consequently, the method accesses long-range causal history without introducing a recurrent disparity state.

.29.1 Frozen Dataset-7 Results

Table 37 compares the learned multi-anchor adapter with the one-step reference, hard candidate retrieval, and a deterministic temporal-consensus baseline. All comparisons use the same evaluation support.

The soft multi-anchor model improves upon the bounded recurrent $H = 4$ reference by

$$G_{\text{multi-anchor } vs.\ H4} = 0.00679. \tag{213}$$

The improvement is positive for all three evaluated stereo backbones and all four dataset-7 sequences. Moreover, the learned model recovers

$$26.8\% \tag{214}$$

of the raw-bank selection-oracle gain and

$$24.2\% \tag{215}$$

of the corresponding convex-oracle gain.

The one-step CS1 model does not match the $H = 4$ reference, while both hard and soft multi-anchor models outperform it. This confirms that the improvement is not explained by the immediately preceding frame alone. The additional ages

$$\{t - 2, t - 4, t - 8\} \tag{216}$$

provide realizable complementary information.

Soft fusion also outperforms hard retrieval:

$$0.51047 < 0.51420. \tag{217}$$

Thus, selecting an informative temporal anchor is useful, but retaining a bounded convex correction toward the current raw estimate remains beneficial.

.29.2 Selective-Intervention Limitation

Despite its geometric improvement, the current policy does not yet provide sufficiently safe intervention behavior. On dataset 7, it produces:

$$\begin{aligned}
\text{HarmfulUpdateRate} &= 35.1\%, && (218)\\
\text{CleanDegradation} &= 4.86\%, && (219)\\
\text{InterventionPrecision} &= 49.1\%, && (220)\\
\text{DegradedFrameFraction} &= 26.2\%. && (221)
\end{aligned}$$

The model therefore identifies enough useful temporal corrections to reduce aggregate EPE, but approximately half of its accepted interventions are not beneficial. The remaining limitation is not the availability of temporal information, but the reliability of the intervention-authorization policy.

Finding. Immutable multi-age raw disparity anchors provide a realizable improvement over bounded recurrent CODD-style fusion. A lightweight retrieval and pairwise soft-fusion adapter recovers a meaningful fraction of the multi-anchor oracle ceiling and improves every evaluated backbone and sequence. However, the current candidate acceptance policy remains insufficiently selective, motivating an explicit risk-aware reject option before deployment.

Scope. The result demonstrates multi-backbone compatibility across three stereo estimators seen during training. It does not yet establish unseen-backbone transfer, external-dataset generalization, or safety for clinical deployment.

.30 Milestone: Transfer to Unseen Stereo Backbones

A central objective of ARGOS v2 is to develop a temporal refinement module that operates independently of the internal architecture of the underlying stereo matcher. To test this property, we evaluated the frozen bounded CODD-style $H = 4$ refiner on stereo backbones that were completely excluded from refiner training.

Training and evaluation protocol. The temporal refiner was trained using disparity predictions produced by three heterogeneous frozen stereo estimators:

$$\mathcal{B}_{\text{train}} = \{\text{S2M2-S}, \text{RAFT-Stereo}, \text{StereoAnywhere}\}. \tag{222}$$

After model selection, the canonical temporal checkpoint was frozen. No backbone-specific fine-tuning, threshold calibration, feature normalization adaptation, or parameter update was performed during the transfer experiment.

The frozen refiner was then applied to:

$$\mathcal{B}_{\text{unseen}} = \{\text{CREStereo}, \text{Fast-FoundationStereo}\}, \tag{223}$$

neither of which contributed predictions during temporal-refiner training.

Table 38: Frozen transfer of the bounded CODD-style $H = 4$ refiner to stereo backbones excluded from temporal-refiner training. Lower EPE is better. Gain is computed relative to the corresponding raw backbone.

Backbone	Raw EPE	Refined EPE	Gain
CREStereo	0.57046	0.54823	+0.02224
Fast-FoundationStereo	0.53227	0.51754	+0.01473

For every backbone, the temporal module retains the same bounded correction rule:

$$\alpha_t = w_t^{\text{reset}} w_t^{\text{fusion}}, \tag{224}$$

$$d_t^F = (1 - \alpha_t)\, d_t^S + \alpha_t d_t^M, \tag{225}$$

where d_t^S is the current raw stereo disparity and d_t^M is the causally aligned temporal memory. The recurrent state is re-anchored after a fixed horizon of four frames to prevent indefinite error accumulation.

Canonical reproduction. Before opening the unseen-backbone results, the original evaluation path was reproduced using the canonical checkpoint. The like-for-like reproduction differed from the validated reference by at most:

$$\epsilon_{\text{reproduction}} = 2.72 \times 10^{-6}, \tag{226}$$

confirming that the transfer experiment reused the unchanged model, feature-construction path, fusion rule, and reset protocol.

Unseen-backbone results. Table 38 reports the frozen SCARED-C results under strict common support.

The frozen refiner improves both unseen stereo estimators:

$$G_{\text{CREStereo}} = 0.02224, \tag{227}$$

$$G_{\text{FastFoundationStereo}} = 0.01473. \tag{228}$$

For context, the same checkpoint also improves all three stereo estimators represented during training:

$$G_{\text{S2M2}-\text{S}} = 0.01941, \tag{229}$$

$$G_{\text{RAFT}-\text{Stereo}} = 0.02928, \tag{230}$$

$$G_{\text{StereoAnywhere}} = 0.03397. \tag{231}$$

The resulting evaluation therefore covers five substantially different stereo architectures:

$$\left\{ \begin{array}{c} \text{S2M2-S, RAFT-Stereo, StereoAnywhere,} \\ \text{CREStereo, Fast-FoundationStereo} \end{array} \right\}. \tag{232}$$

Backbone-independent interface. The temporal refiner does not require internal backbone-specific representations such as stereo cost volumes, recurrent hidden states, matching-network features, or proprietary confidence outputs. Instead, it operates through an external interface constructed from signals available for heterogeneous stereo matchers:

$$x_t = \left[d_t^S, d_t^M, d_t^S - d_t^M, F_{t \to t-1}, C_t^{\text{FB}}, V_t, S_t, I_t^L, I_t^R\right], \tag{233}$$

where F denotes temporal motion, C^{FB} forward–backward confidence, V validity, and S warp support.

This interface allows the same temporal checkpoint to operate on predictions from stereo estimators with substantially different architectures.

Finding. The experiment provides direct evidence that the bounded CODD-style ARGOS v2 refiner is backbone-agnostic across the evaluated stereo estimators within the SCARED-C domain. A single frozen checkpoint improves three training-seen backbones and two stereo backbones excluded from training, without backbone-specific adaptation.

More precisely, the result demonstrates:

$$\boxed{\text{same-domain transfer to unseen stereo backbones}}. \tag{234}$$

It does not yet demonstrate:

$$\boxed{\text{joint unseen-backbone and external-domain transfer}}. \tag{235}$$

The term *backbone-agnostic* is therefore used to describe the architecture and its verified transfer across the five evaluated stereo estimators, rather than as a universal claim covering every possible matcher and acquisition domain.

Research significance. This result establishes the bounded CODD-style model as a robust architectural foundation for ARGOS v2. Subsequent work can focus on improving correction selection, temporal-memory utilization, and intervention safety without requiring a redesign tied to a specific stereo backbone.

The immediate research path is:

$$\boxed{\begin{array}{c}\text{validated backbone-independent temporal fusion}\\ \Downarrow \\ \text{safety-aware intervention selection}\\ \Downarrow \\ \text{stronger training and multi-anchor refinement}\end{array}} \tag{236}$$

If the current spatial safety selector successfully identifies harmful temporal updates, it can be integrated as an abstention layer over the multi-anchor architecture. If it does not, the bounded CODD-style $H = 4$ system remains a validated and transferable base architecture that can be further improved through larger-scale training, stronger safety supervision, and controlled architectural refinement.

Scope. The current evidence supports backbone-independent transfer within SCARED-C. External-domain geometric generalization remains unresolved because the available D4D structured-light reference does not currently satisfy the required stereo-disparity compatibility contract. SERV-CT does not contain temporally consecutive frames and therefore cannot support an $H = 4$ temporal evaluation. No claim of universal cross-domain or clinical robustness is made.

.31 Conditional Milestone: Spatially Selective Multi-Anchor Refinement

The immutable multi-anchor temporal refiner demonstrated a realizable geometric advantage over bounded recurrent fusion, but its ungated intervention policy introduced an excessive number of harmful updates. We therefore investigated whether a lightweight spatial comparative critic could identify when the multi-anchor proposal should replace the current raw stereo estimate.

Frozen proposal and selective intervention. Let d_t^S denote the current raw stereo disparity and d_t^P the frozen multi-anchor proposal. The critic does not modify the proposal, the selected anchor, or the fusion weight. It only predicts whether the proposal should be accepted:

$$d_t^{\text{out}} = \begin{cases} d_t^P, & a_t = 1, \\ d_t^S, & a_t = 0. \end{cases} \tag{237}$$

The fallback is implemented through exact tensor selection, ensuring that rejected interventions reproduce the raw disparity bit-exactly.

The critic predicts separate heteroscedastic error estimates for the raw and proposed disparities:

$$\left(\mu_t^S, \sigma_t^S\right), \qquad \left(\mu_t^P, \sigma_t^P\right), \tag{238}$$

together with a direct harmful-update probability p_t^{harm}.

The predicted proposal gain is:

$$\widehat{\Delta}_t = \mu_t^S - \mu_t^P, \tag{239}$$

with comparative uncertainty:

$$\sigma_t^\Delta = \sqrt{\left(\sigma_t^S\right)^2 + \left(\sigma_t^P\right)^2}. \tag{240}$$

A conservative lower confidence bound is defined as:

$$\text{LCB}_t^\Delta = \widehat{\Delta}_t - \lambda_{\text{unc}} \sigma_t^\Delta. \tag{241}$$

The frozen intervention rule is:

$$a_t = 1\left[\text{LCB}_t^\Delta > \tau_\Delta \wedge p_t^{\text{harm}} < \tau_{\text{harm}} \wedge V_t = 1\right], \tag{242}$$

where V_t denotes proposal validity.

Spatial comparative critic. The selected critic is a lightweight fully convolutional network using Group Normalization, SiLU activations, and dilated convolutions with dilation pattern:

$$[1, 2, 4, 2, 1]. \tag{243}$$

The resulting receptive field is approximately 43 pixels and the selected stereo-evidence configuration contains approximately 0.40 million trainable parameters.

The critic receives spatially structured evidence describing:

- current raw, selected-anchor, and proposed disparities;
- raw–anchor and raw–proposal disagreement;
- proposal fusion weight and validity;
- temporal motion and forward–backward consistency;
- local disparity gradients and spatial consensus;
- temporal photometric residuals;
- stereo photometric residuals for both the raw and proposed hypotheses.

The multi-anchor proposal generator remains completely frozen throughout critic training.

Protocol. Training, validation, and testing use dataset-disjoint SCARED-C splits:

$$\mathcal{D}_{\text{train}} = \{1, 3, 6\}, \qquad \mathcal{D}_{\text{val}} = \{2\}, \qquad \mathcal{D}_{\text{test}} = \{7\}. \tag{244}$$

The critic checkpoint and intervention thresholds were selected exclusively on dataset 2. The complete checkpoint, feature schema, thresholds, and policy were then frozen and hashed before the single evaluation on dataset 7.

No test-time calibration, threshold adjustment, or model selection was performed.

Table 39: Frozen dataset-7 evaluation of the spatial safety critic. Lower EPE is better.

Method	EPE	Gain over raw
Raw stereo	0.5479	0.0000
Bounded CODD-style $H = 4$	0.5173	0.0306
Ungated immutable multi-anchor	0.5105	0.0374
Multi-anchor + spatial critic	**0.5072**	**0.0407**

Table 40: Safety and intervention behavior on frozen dataset 7.

Metric	Ungated	Spatial critic
Harmful interventions	35.1%	24.7%
Intervention precision	49%	62%
Clean-pixel degradation	4.9%	2.1%
Degraded frames	26.0%	16.7%
Intervention coverage	100%	7.9%

Frozen test results. Table 39 compares the principal methods on the held-out dataset-7 test split.

The frozen critic improves over the bounded CODD-style baseline by:

$$0.5173 - 0.5072 = 0.0101 \text{ px.} \tag{245}$$

It also improves over the ungated multi-anchor proposal:

$$0.5105 - 0.5072 = 0.0033 \text{ px.} \tag{246}$$

Consequently, the selective policy retains more than the complete geometric gain of the ungated proposal:

$$\text{GainRetained} = 108.8\%. \tag{247}$$

This indicates that rejecting harmful proposals increases the net geometric benefit rather than merely trading accuracy for safety.

Safety behavior. Table 40 compares the ungated and selectively gated multi-anchor variants.

The critic satisfies the predefined clean-pixel and degraded-frame constraints:

$$\text{CleanDegradation} = 2.1\% < 3\%, \tag{248}$$

$$\text{DegradedFrames} = 16.7\% < 25\%. \tag{249}$$

However, the accepted harmful-update rate remains:

$$\text{HarmRate} = 24.7\%, \tag{250}$$

which exceeds the predefined 10% target.

The policy therefore improves safety substantially but does not yet provide fully risk-controlled intervention.

Validation-to-test transfer. A central limitation of the previous scalar gate was the collapse of its validation-selected operating point on the test split. The spatial critic does not exhibit the same failure.

The harmful-update rate changes only slightly from validation to test:

$$26\% \rightarrow 24.7\%, \tag{251}$$

while intervention coverage remains non-trivial:

$$\text{Coverage}_{\text{test}} = 7.9\%. \tag{252}$$

The critic improves all three evaluated stereo backbones and all four held-out dataset-7 sequences. Approximately 51% of accepted interventions originate from the distant $CS4$ and $CS8$ raw anchors, demonstrating that the policy does not reduce the multi-anchor model to an adjacent-frame refiner.

The harm-discrimination AUROC improves from approximately 0.57 for the previous scalar gate to approximately:

$$\text{AUROC}_{\text{harm}} \approx 0.69 \tag{253}$$

on the frozen test split.

Finding. The experiment establishes that spatial comparative evidence provides a substantially stronger intervention signal than pointwise scalar statistics. The frozen critic:

- transfers from dataset 2 to dataset 7 without coverage collapse;
- improves the raw prediction and the bounded CODD-style baseline;
- improves over the ungated multi-anchor proposal;
- reduces harmful updates and clean-region degradation;
- preserves useful long-range $CS4/CS8$ interventions;
- improves every evaluated backbone and sequence.

Nevertheless, the critic does not meet the primary accepted-harm target. The resulting verdict is therefore:

$$\boxed{\text{Geometry GO} \quad + \quad \text{Safety CONDITIONAL} \quad = \quad \text{Overall CONDITIONAL GO}}. \tag{254}$$

Scientific interpretation. The remaining limitation is no longer the availability of useful temporal geometry, validation-to-test transfer, or coverage collapse. Instead, the dominant bottleneck is the fine-grained discrimination and calibration of harmful interventions.

The current result supports the following architectural progression:

$$\boxed{\begin{array}{c} \text{immutable raw multi-anchor retrieval} \\ \Downarrow \\ \text{pairwise soft temporal fusion} \\ \Downarrow \\ \text{spatial comparative error prediction} \\ \Downarrow \\ \text{exact raw abstention} \end{array}} \tag{255}$$

The post-hoc veto formulation improves both accuracy and safety, but a fully risk-controlled system will likely require the predicted intervention utility to influence anchor retrieval and fusion magnitude during training rather than only accepting or rejecting an already constructed proposal.

Scope. These results demonstrate held-out same-domain generalization across the three stereo backbones represented during critic training. They do not establish external-dataset robustness, unseen-backbone transfer of the safety critic, clinical safety, or formal statistical coverage guarantees.

Immutable Raw Multi-Anchor Temporal Refinement

.1 Overview

We consider causal temporal refinement of a frozen stereo estimator. At time t, a stereo backbone S predicts a raw disparity map

$$d_t^{\text{raw}} = S\left(I_t^L, I_t^R\right), \tag{256}$$

where I_t^L and I_t^R are the current rectified stereo images. The temporal module is required to operate without future frames and without accessing backbone-specific cost volumes, hidden states, or internal feature representations.

Our final geometric architecture is composed of five stages:

$$\boxed{\begin{array}{l}\text{frozen stereo prediction}\\ \to \text{direct causal motion alignment}\\ \to \text{immutable multi-age raw-anchor bank}\\ \to \text{learned candidate retrieval}\\ \to \text{pairwise soft fusion with exact raw fallback.}\end{array}} \tag{257}$$

The central design choice is that long-term memory contains only independently generated raw stereo predictions. Temporally fused predictions are never written back into the long-term bank.

.2 Motivation: Bounded Corrected Memory

We first investigated a CODD-style recurrent formulation [?]. A previous disparity estimate is aligned to the current frame and softly fused with the current stereo prediction:

$$d_t^F = \left(1 - w_t^{\text{reset}} w_t^{\text{fusion}}\right) d_t^{\text{raw}} + w_t^{\text{reset}} w_t^{\text{fusion}} d_t^{\text{mem}}. \tag{258}$$

This formulation is effective when the corrected state is maintained only over a short horizon. In our experiments, a four-frame memory horizon, denoted $H = 4$, provides stable improvements. In contrast, indefinite recurrence causes prediction errors to be repeatedly aligned, fused, and propagated, leading to severe drift.

This result motivates a separation between:

- **short-lived corrected memory**, which can be useful locally but must be bounded; and
- **long-lived raw memory**, which should preserve independent observations without recursive contamination.

The proposed architecture therefore removes long-term corrected-state recurrence entirely.

.3 Immutable Raw Anchor Bank

At time t, we maintain a causal bank of raw disparity predictions generated at different temporal ages:

$$\mathcal{B}_t = \left\{d_{t-1}^{\text{raw}}, d_{t-2}^{\text{raw}}, d_{t-4}^{\text{raw}}, d_{t-8}^{\text{raw}}\right\}. \tag{259}$$

We denote these candidates by

$$\text{CS1}, \quad \text{CS2}, \quad \text{CS4}, \quad \text{CS8}. \tag{260}$$

Each anchor satisfies the following properties:

1. it is generated independently by the frozen stereo backbone;
2. it is stored without modification;
3. it is never replaced by a fused prediction;
4. it is aligned directly to the current frame;
5. it has explicit temporal age and validity provenance.

The bank update is therefore

$$\mathcal{B}_{t+1} = \mathrm{Update}\,(\mathcal{B}_t, d_t^{\mathrm{raw}})\,, \tag{261}$$

and not

$$\mathcal{B}_{t+1} \neq \mathrm{Update}\left(\mathcal{B}_t, d_t^{\mathrm{out}}\right). \tag{262}$$

This prevents recursively accumulated temporal errors from contaminating future long-range memory.

.4 Direct Causal Alignment

For every available anchor of age $k \in \{1, 2, 4, 8\}$, a frozen SEA-RAFT model [?] estimates target-to-source optical flow:

$$F_{t\to t-k} = M\left(I_t^L, I_{t-k}^L\right). \tag{263}$$

The historical disparity is brought into the current coordinate system using BiDA-style pull warping [?]:

$$\widetilde{d}_t^{(k)} = \mathcal{W}\left(d_{t-k}^{\mathrm{raw}}, F_{t\to t-k}\right). \tag{264}$$

Every anchor is aligned independently from the current frame to its original source frame. Flow fields are not recursively composed across time.

For each aligned anchor, we construct a validity mask

$$V_t^{(k)} = V_{t,k}^{\mathrm{bounds}} \wedge V_{t,k}^{\mathrm{flow}} \wedge V_{t,k}^{\mathrm{FB}} \wedge V_{t-k}^{\mathrm{disp}}, \tag{265}$$

where:

- V^{bounds} indicates that the sampling location lies inside the source image;
- V^{flow} indicates finite and valid optical flow;
- V^{FB} represents forward–backward flow consistency;
- V^{disp} represents valid source disparity.

.5 Shared Candidate Representation

Each aligned anchor is described by a shared candidate representation

$$x_t^{(k)} = \Phi\left(d_t^{\mathrm{raw}}, \widetilde{d}_t^{(k)}, I_t^L, I_{t-k}^L, F_{t\to t-k}, V_t^{(k)}, k\right). \tag{266}$$

The validated implementation uses 17 universal channels per candidate. These channels contain only externally available evidence, including:

- current raw disparity;
- aligned anchor disparity;
- signed raw–anchor disparity difference;
- absolute raw–anchor disparity difference;
- local disparity gradients;
- warp support and validity;
- forward–backward flow consistency;
- optical-flow magnitude;
- temporal photometric agreement;
- local robust statistics;
- anchor age and provenance.

The same encoder is applied to every candidate:

$$z_t^{(k)} = E_\theta\left(x_t^{(k)}\right). \tag{267}$$

No backbone identifier, backbone-specific feature map, cost volume, or hidden state is provided to the model.

.6 Candidate Retrieval

The shared encoder predicts a candidate score

$$s_t^{(k)} = G_\theta\left(z_t^{(k)}\right). \tag{268}$$

Unavailable or invalid anchors are assigned

$$s_t^{(k)} = -\infty. \tag{269}$$

The selected anchor age is

$$k^\star = \arg\max_{k\in\{1,2,4,8\}} s_t^{(k)}. \tag{270}$$

Only one anchor is retrieved for each output location. The model does not perform an unconditional average over the complete memory bank.

This distinction is important because distant anchors may be globally less accurate than the current prediction while remaining locally useful in regions where the current stereo backbone fails.

.7 Pairwise Soft Fusion

After retrieval, a lightweight fusion head compares the current raw prediction and the selected aligned anchor:

$$w_t = \sigma\left[H_\theta\left(d_t^{\text{raw}}, \widetilde{d}_t^{(k^\star)}, z_t^{(k^\star)}\right)\right], \tag{271}$$

where $w_t \in [0,1]$ is a spatially varying fusion weight.

The proposed disparity is

$$d_t^{\text{prop}} = (1-w_t)\, d_t^{\text{raw}} + w_t \widetilde{d}_t^{(k^\star)}. \tag{272}$$

The retrieval and fusion network contains approximately

$$60\,739 \tag{273}$$

trainable parameters.

The stereo backbone, SEA-RAFT model, and all external feature extractors remain frozen.

.8 Exact Raw Fallback

If no valid temporal candidate is available, or if the frozen retrieval policy rejects the candidate, the method returns the current raw prediction:

$$d_t^{\text{out}} = \begin{cases} d_t^{\text{prop}}, & a_t = 1, \\ d_t^{\text{raw}}, & a_t = 0, \end{cases} \tag{274}$$

where a_t denotes the frozen candidate-acceptance decision.

The fallback is implemented using exact tensor selection, rather than an arithmetic blend with a zero weight. Consequently, rejected output pixels are bit-identical to the original raw stereo prediction.

TODO: Insert the exact frozen probability and utility thresholds from the validated experiment manifest.

.9 Causal Inference Procedure

[t] [1] Current stereo pair (I_t^L, I_t^R), raw anchor bank $\mathcal{B}_t$ $d_t^{\text{raw}} \leftarrow S(I_t^L, I_t^R)$ $k \in \{1,2,4,8\}$ d_{t-k}^{raw} is available $F_{t\to t-k} \leftarrow M(I_t^L, I_{t-k}^L)$ $\widetilde{d}_t^{(k)} \leftarrow \mathcal{W}(d_{t-k}^{\text{raw}}, F_{t\to t-k})$ Construct $V_t^{(k)}$ Construct candidate representation $x_t^{(k)}$ $s_t^{(k)} \leftarrow G_\theta(E_\theta(x_t^{(k)}))$ $s_t^{(k)} \leftarrow -\infty$ no valid candidate exists $d_t^{\text{out}} \leftarrow d_t^{\text{raw}}$ $k^\star \leftarrow \arg\max_k s_t^{(k)}$ Predict pairwise fusion weight w_t Construct d_t^{prop} using Eq. (272) Apply frozen acceptance policy Select d_t^{out} using Eq. (274) Store d_t^{raw} in $\mathcal{B}_{t+1}$ Never store d_t^{out} in long-term memory d_t^{out}

.10 Backbone-Independent Interface

The complete temporal adapter consumes only:

- rectified RGB images;
- raw disparity predictions;
- optical flow;
- warp support;
- validity masks;

Table 41: Same-domain transfer to stereo estimators excluded from temporal-refiner training. Results use strict common support across the raw prediction, bounded $H = 4$ refiner, and immutable multi-anchor refiner. Lower EPE is better.

Backbone	Raw	$H = 4$	Multi-anchor	Gain vs. raw	Gain vs. $H = 4$
CREStereo	0.5784	0.5555	**0.5437**	**+0.0347**	**+0.0119**
Fast-FoundationStereo	0.5388	0.5237	**0.5209**	**+0.0179**	**+0.0028**

- forward–backward consistency;
- externally computed local evidence.

It does not consume:

- stereo cost volumes;
- backbone confidence heads;
- recurrent hidden states;
- backbone-specific features;
- backbone identifiers.

The same frozen retrieval and fusion checkpoint can therefore be applied to heterogeneous stereo estimators without architectural modification.

Transfer to Stereo Backbones Excluded from Training

We evaluate the frozen immutable raw multi-anchor refiner on two stereo estimators excluded from temporal-refiner training:

- CREStereo;
- Fast-FoundationStereo.

The evaluation is performed within SCARED-C using dataset 7, which is held out from the frozen training and calibration protocol. No model parameter, threshold, or preprocessing configuration is tuned on these two backbones.

The canonical frozen multi-anchor checkpoint is

```
results/raw_multi_anchor_temporal_refiner/
soft_fusion/checkpoints/best_validation.pt
```

with SHA-256

```
40526a32ef6e9a62a3ea2b59e6751a60
c441b8190f9b96522e3b12b35895d5cd
```

The frozen SEA-RAFT checkpoint has SHA-256

```
1a21575ed6ca2c6945fb8e25c4169d24
1cf59ee5d12b8802c01c965206268cac
```

The frozen multi-anchor model improves both unseen stereo estimators relative to their raw predictions and relative to bounded recurrent fusion.

The improvement is consistent across sequences: all four evaluated sequences improve for CREStereo and all four improve for Fast-FoundationStereo.

Long-range memory remains active after transfer. The combined fraction of selected CS4 and CS8 anchors is

$$53.4\% \tag{275}$$

for CREStereo and

$$55.8\% \tag{276}$$

for Fast-FoundationStereo. The model therefore does not collapse to an adjacent-frame-only solution when applied to stereo estimators excluded from training.

.1 Evaluation Support

The comparison uses strict pixelwise common support across:

- the raw stereo prediction;
- the bounded $H = 4$ prediction;
- all required multi-anchor candidates;
- ground-truth disparity.

The resulting common pixel support is approximately

$$97.31\%. \tag{277}$$

Frames with no pixels in the complete strict-support intersection are recorded explicitly and excluded from metric aggregation. A total of 1 226 such frame entries are reported in the evaluation artifacts.

This exclusion is caused by an empty intersection among the required temporal and $H = 4$ support masks, rather than by post-hoc error filtering.

.2 Supported Claim

The experiment supports the following claim:

> The frozen immutable raw multi-anchor refiner transfers to stereo estimators excluded from its training within SCARED-C. The same retrieval and soft-fusion module improves both raw stereo predictions and bounded recurrent refinement on every evaluated sequence, while retaining substantial use of long-range temporal anchors.

The result supports a backbone-independent input interface with demonstrated same-domain transfer across the evaluated stereo estimators.

It does not establish:

- universal backbone agnosticism;
- external-domain generalization;
- clinical safety;
- risk-controlled intervention;
- real-time operation on deployment hardware.

Selective Intervention as an Open Problem

The immutable multi-anchor model provides the strongest geometric performance among the evaluated temporal architectures. However, an ungated temporal correction can still degrade locally accurate raw predictions.

We investigated a post-hoc spatial comparative critic that estimates whether a frozen multi-anchor proposal should be accepted. The critic improves average disparity accuracy and reduces several degradation metrics, but the strict validation target

$$\text{harmful accepted updates} \leq 10\% \tag{278}$$

at

$$\text{coverage} \geq 5\% \tag{279}$$

was not achieved.

Hard-negative training, calibration, and an ensemble reduced the accepted harmful-update rate, but the resulting validation operating point remained approximately

$$\text{harm} = 20.1\%, \qquad \text{coverage} = 2.06\%. \tag{280}$$

We therefore treat strict risk-controlled authorization as an open problem rather than as a solved component of the proposed geometric method.

A natural extension is to jointly optimize

$$\boxed{\text{anchor retrieval} + \text{fusion magnitude} + \text{harm prediction} + \text{abstention}} \tag{281}$$

instead of applying a binary veto to an already constructed proposal.

Architecture Summary

The two principal ARGOS v2 temporal designs can be summarized as:

Bounded corrected memory.

$$\boxed{\text{frozen stereo} \rightarrow \text{SEA-RAFT} \rightarrow \text{BiDA alignment} \rightarrow \text{CODD-style fusion} \rightarrow \text{reset every four frames}} \tag{282}$$

This design is robust and provides the controlled baseline demonstrating that corrected state is useful only over bounded horizons.

Immutable raw multi-anchor memory.

$$\boxed{\text{frozen stereo} \rightarrow \text{directly aligned CS1/CS2/CS4/CS8} \rightarrow \text{shared retrieval} \rightarrow \text{pairwise soft fusion} \rightarrow \text{raw-only mem}} \tag{283}$$

This design provides the strongest geometric performance, exploits long-range temporal evidence, and transfers to stereo estimators excluded from training within the tested domain.

Together, the two architectures support the following design principle:

> Corrected disparity memory is effective when short-lived and bounded, whereas long causal history is better preserved as independently generated immutable raw observations.

Immutable Raw Multi-Anchor Temporal Refinement

.1 Problem Formulation

Let

$$\left(I_t^L, I_t^R\right) \tag{284}$$

denote the current rectified stereo pair at time t. An external frozen stereo estimator S produces the positive-left disparity map and its validity mask:

$$d_t^S,\, M_t = S\left(I_t^L, I_t^R\right), \tag{285}$$

where

$$d_t^S \in \mathbb{R}^{B\times1\times H\times W}, \qquad M_t \in \{0,1\}^{B\times1\times H\times W}. \tag{286}$$

ARGOS v2 operates exclusively on the externally available RGB images, raw disparity predictions, optical flow, and geometric validity information. It does not consume backbone identifiers, stereo cost volumes, confidence heads, hidden states, or internal backbone features.

The temporal adapter implements the following causal pipeline:

$$\boxed{\begin{array}{l}\text{frozen stereo prediction}\\ \to \text{immutable multi-age raw memory}\\ \to \text{direct current-to-anchor motion estimation}\\ \to \text{causal pull alignment}\\ \to \text{shared utility-aware candidate encoding}\\ \to \text{pixelwise single-anchor retrieval}\\ \to \text{pairwise soft correction}\\ \to \text{geometric acceptance and exact raw fallback.}\end{array}} \tag{287}$$

The right image I_t^R is used by the external stereo estimator. Once d_t^S has been produced, the temporal adapter operates on the current left image I_t^L, the raw disparity, and the causal memory bank.

.2 Immutable Raw Temporal Memory

Before processing frame t, the temporal memory contains only independently generated raw stereo records from previous frames:

$$\mathcal{M}_t = \{R_j : \max(0, t-8) \le j \le t-1\}, \tag{288}$$

where each record is

$$R_j = \left(d_j^S, M_j, I_j^L, \mathrm{id}_j, j, \tau_j\right). \tag{289}$$

The stored disparity, validity mask, and RGB image are detached and cloned before insertion. Frame indices must be strictly increasing, and valid disparity values must be finite and positive. Each record carries the provenance label

$$\texttt{independent\_frozen\_stereo}. \tag{290}$$

The canonical anchor-age set is

$$\mathcal{A} = \{1, 2, 4, 8\}. \tag{291}$$

For an age $k \in \mathcal{A}$, the corresponding raw anchor is

$$d_{t-k}^S, \tag{292}$$

which we denote as CSk. Thus, CS1, CS2, CS4, and CS8 are raw observations produced independently by the frozen stereo estimator. They are not recursively fused states.

If the source frame $t-k$ is unavailable, the corresponding candidate is represented by a zero-supported dummy tensor and is excluded from retrieval through explicit masking.

After the current output has been constructed, the bank is updated as

$$\mathcal{M}_{t+1} = \text{EvictAndAppendRaw}\left(\mathcal{M}_t, R_t\right), \tag{293}$$

with

$$R_t = \left(d_t^S, M_t, I_t^L, \text{id}_t, t, \tau_t\right). \tag{294}$$

Importantly,

$$d_t^{\text{out}} \notin \mathcal{M}_{t+1}. \tag{295}$$

Consequently, an error introduced by a temporal correction cannot become the source of a later temporal correction. Long-range information is therefore preserved without recursively propagating corrected-state errors.

.3 Direct Causal Motion Estimation

For every available age $k \in \mathcal{A}$, a frozen SEA-RAFT model estimates optical flow independently in both directions:

$$F_{t\to t-k} = M_{\text{flow}}\left(I_t^L, I_{t-k}^L\right), \tag{296}$$

$$F_{t-k\to t} = M_{\text{flow}}\left(I_{t-k}^L, I_t^L\right). \tag{297}$$

The frozen flow model uses four recurrent iterations and returns flow in pixel units:

$$F_{a\to b} \in \mathbb{R}^{B\times 2\times H\times W}. \tag{298}$$

All flow estimates are computed directly between the current frame and the original source frame. No flow fields are composed through intermediate time steps:

$$F_{t\to t-k} \neq F_{t\to t-1} \circ \cdots \circ F_{t-k+1\to t-k}. \tag{299}$$

For $K=4$ available anchors, the architecture requires $2K$ directional flow estimates per frame. These evaluations may be batched in implementation, but they remain geometrically independent.

SEA-RAFT is a frozen external motion estimator and is not an ARGOS contribution.

.4 BiDA-Style Causal Pull Alignment

The historical disparity is aligned to the current image grid using target-to-source pull warping.

For current-grid coordinate (x, y), define the source sampling coordinate:

$$q_x^{(k)}(x,y) = x + F_{t\to t-k}^x(x,y), \tag{300}$$

$$q_y^{(k)}(x,y) = y + F_{t\to t-k}^y(x,y). \tag{301}$$

The normalized sampling coordinates used by `grid_sample` are

$$g_x^{(k)} = \frac{2q_x^{(k)}}{W-1} - 1, \tag{302}$$

$$g_y^{(k)} = \frac{2q_y^{(k)}}{H-1} - 1. \tag{303}$$

The aligned candidate disparity is

$$\widetilde{d}_t^{(k)} = \mathcal{W}\left(d_{t-k}^S, F_{t\to t-k}\right), \tag{304}$$

or, pixelwise,

$$\widetilde{d}_t^{(k)}(x,y) = \text{grid\_sample}\left(d_{t-k}^S, g_x^{(k)}(x,y), g_y^{(k)}(x,y)\right). \tag{305}$$

The frozen implementation uses:

- bilinear interpolation;
- zero padding;
- `align_corners=True`.

Disparity is treated as a scalar field and sampled directly. No horizontal-flow correction is added after warping.

The geometric support mask is

$$S_t^{(k)}(x,y) = \mathbb{I}\left[0 \le q_x^{(k)}(x,y) \le W-1\right] \mathbb{I}\left[0 \le q_y^{(k)}(x,y) \le H-1\right]. \tag{306}$$

The historical source-validity mask is warped with the same grid and thresholded at 0.999:

$$\widetilde{M}_{t-k}^{(k)} = \mathbb{I}\left[\mathcal{W}\left(M_{t-k}, F_{t\to t-k}\right) \ge 0.999\right]. \tag{307}$$

The candidate availability mask used by retrieval is

$$V_t^{(k)} = M_t \wedge \widetilde{M}_{t-k}^{(k)} \wedge S_t^{(k)}. \tag{308}$$

.5 Forward–Backward Motion Confidence

The reverse flow is pulled onto the current grid:

$$\overline{F}_{t-k\to t} = \mathcal{W}\left(F_{t-k\to t}, F_{t\to t-k}\right). \tag{309}$$

The forward–backward residual is

$$e_t^{(k)} = \left\|F_{t\to t-k} + \overline{F}_{t-k\to t}\right\|_2. \tag{310}$$

The adaptive consistency scale is

$$\theta_t^{(k)} = 0.5 + 0.01\left(\|F_{t\to t-k}\|_2 + \left\|\overline{F}_{t-k\to t}\right\|_2\right). \tag{311}$$

A continuous forward–backward confidence map is then computed as

$$C_t^{(k)} = S_t^{(k)} \exp\left[-\frac{e_t^{(k)}}{\max\left(\theta_t^{(k)}, 10^{-6}\right)}\right]. \tag{312}$$

The implementation also computes the binary consistency diagnostic

$$B_t^{(k)} = S_t^{(k)} \wedge \left[e_t^{(k)} \le \theta_t^{(k)}\right]. \tag{313}$$

However, $B_t^{(k)}$ does not directly determine candidate availability in Eq. (308). Motion consistency is instead provided to the learned model through the continuous confidence channel $C_t^{(k)}$.

.6 Robust Cross-Anchor Statistics

The four aligned candidates provide both local candidate evidence and robust bank-level context.

Let

$$n_t(x) = \sum_{j\in\mathcal{A}} V_t^{(j)}(x) \tag{314}$$

denote the number of available anchors at pixel x.

The candidate median over the available anchors is denoted by

$$m_t(x) = \mathrm{Median}\left\{\widetilde{d}_t^{(j)}(x) : V_t^{(j)}(x) = 1\right\}. \tag{315}$$

Table 42: Exact candidate-evidence channels used by the frozen ARGOS v2 geometry-v1 checkpoint. The ordering is part of the checkpoint contract.

Index	Channel	Definition
1	Current raw disparity	$\mathrm{clip}(r_t, 0, 64)/64$
2	Aligned candidate disparity	$\mathrm{clip}(c_t^{(k)}, 0, 64)/64$
3	Signed raw–candidate residual	$\mathrm{clip}(\epsilon_t^{(k)}, -16, 16)/16$
4	Absolute raw–candidate residual	$\left\|\epsilon_t^{(k)}\right\|/16$
5	Local absolute-residual mean	$\mathrm{clip}\left(\mu_5(\|\epsilon_t^{(k)}\|), 0, 8\right)/8$
6	Local residual standard deviation	$\mathrm{clip}\left(\sigma_5(\epsilon_t^{(k)}), 0, 8\right)/8$
7	Normalized anchor age	$k/8$
8	Candidate availability	$V_t^{(k)}$
9	Geometric warp support	$S_t^{(k)}$
10	Forward–backward confidence	$C_t^{(k)}$
11	Valid-witness fraction	$n_t/\|\mathcal{A}\|$
12	Candidate median	$\mathrm{clip}(m_t, 0, 64)/64$
13	Candidate MAD	$\mathrm{clip}(q_t, 0, 8)/8$
14	Candidate-to-median deviation	$\mathrm{clip}(\|c_t^{(k)} - m_t\|, 0, 8)/8$
15	Raw-to-median deviation	$\mathrm{clip}(\|r_t - m_t\|, 0, 8)/8$
16	Agreement fraction	$A_t/\|\mathcal{A}\|$
17	Raw-anchor provenance	$R_t^{(k)} = 0$ in the canonical pipeline

The corresponding median absolute deviation is

$$q_t(x) = \mathrm{Median}\left\{\left|\widetilde{d}_t^{(j)}(x) - m_t(x)\right| : V_t^{(j)}(x) = 1\right\}. \tag{316}$$

The number of agreeing anchors is

$$A_t(x) = \sum_{j \in \mathcal{A}} \mathbb{I}\left[V_t^{(j)}(x) = 1\right] \mathbb{I}\left[\left|\widetilde{d}_t^{(j)}(x) - m_t(x)\right| \le q_t(x) + 0.10\right]. \tag{317}$$

These statistics provide cross-anchor context without using temporal attention, 3D convolutions, or recurrent hidden states.

.7 Exact Candidate Evidence

For clarity, let

$$r_t = d_t^S, \qquad c_t^{(k)} = \widetilde{d}_t^{(k)}, \qquad \epsilon_t^{(k)} = c_t^{(k)} - r_t. \tag{318}$$

For each candidate age k, the model receives an exactly ordered 17-channel evidence tensor:

$$x_t^{(k)} = \left[x_{t,1}^{(k)}, \ldots, x_{t,17}^{(k)}\right]. \tag{319}$$

The channels are:

The local operators μ_5 and σ_5 use a 5×5 window, unit stride, and padding two. The local variance is clamped to zero before applying the square root.

Table 43: Layer-by-layer architecture of the shared candidate encoder.

Component	Operation	Parameters
Stem convolution	3×3, $17 \to 32$, padding 1, no bias	4,896
Stem normalization	$\mathrm{GroupNorm}(8, 32)$	64
Stem activation	SiLU	0
Residual block 1	Two 3×3, $32 \to 32$ convolutions with GN	18,560
Residual block 2	Two 3×3, $32 \to 32$ convolutions with GN	18,560
Residual block 3	Two 3×3, $32 \to 32$ convolutions with GN	18,560
Output head	1×1, $32 \to 3$	99
Total		**60,739**

For $K = 4$ anchors, the complete evidence tensor has shape

$$X_t \in \mathbb{R}^{B \times K \times 17 \times H \times W}. \tag{320}$$

The provenance channel is retained for checkpoint compatibility and is identically zero for all canonical immutable raw anchors.

The alignment utility additionally computes photometric residual, optical-flow magnitude, disparity-disagreement, and gradient helper quantities. These quantities are not included in the 17-channel tensor passed to the CNN.

.8 Shared Candidate Encoder

The candidate dimension is folded into the batch dimension:

$$X_t : [B, K, 17, H, W] \longrightarrow [BK, 17, H, W]. \tag{321}$$

A single shared CNN E_θ is therefore applied independently to every candidate age:

$$Z_t^{(k)} = E_\theta\left(x_t^{(k)}\right). \tag{322}$$

The candidates share all learned weights. They interact only through the robust bank-level evidence described in Section .6 and through the subsequent pixelwise retrieval operation.

The exact network is reported in Table 43.

Each residual block applies

$$h' = \mathrm{SiLU}\left[h + \mathrm{GN}_2\left(\mathrm{Conv}_2\left(\mathrm{SiLU}\left[\mathrm{GN}_1\left(\mathrm{Conv}_1(h)\right)\right]\right)\right)\right]. \tag{323}$$

The output head is initialized with zero weights and bias

$$[-2,\ 0,\ -3]. \tag{324}$$

The theoretical local receptive field of the CNN is 15×15 pixels.

.9 Utility-Aware Pixelwise Retrieval

For each candidate k, the three output channels are transformed into:

$$\pi_t^{(k)} = \sigma\left(z_{t,0}^{(k)}\right), \tag{325}$$

$$\widehat{\Delta}_t^{(k)} = 8\tanh\left(z_{t,1}^{(k)}\right), \tag{326}$$

$$w_t^{(k)} = \sigma\left(z_{t,2}^{(k)}\right). \tag{327}$$

Here:

- $\pi_t^{(k)}$ is a learned helpfulness output;
- $\widehat{\Delta}_t^{(k)}$ is the predicted signed geometric improvement in pixels;
- $w_t^{(k)} \in (0,1)$ is the candidate-specific fusion weight.

The retrieval utility is

$$u_t^{(k)} = \pi_t^{(k)} \widehat{\Delta}_t^{(k)}. \tag{328}$$

Unavailable candidates are assigned

$$u_t^{(k)} = -\infty \qquad \text{where } V_t^{(k)} = 0. \tag{329}$$

The selected age is determined independently for every output pixel:

$$k_t^\star(x) = \arg\max_{k \in \mathcal{A}} u_t^{(k)}(x). \tag{330}$$

The retrieval operation selects exactly one candidate per pixel. There is no softmax over ages and no unconditional averaging of the candidate bank.

The output $\pi_t^{(k)}$ is not interpreted as a calibrated probability of geometric correctness. It is a learned helpfulness quantity used jointly with the predicted improvement.

.10 Pairwise Soft Correction

Let

$$\widetilde{d}_t^\star = \widetilde{d}_t^{(k_t^\star)} \tag{331}$$

and

$$w_t^\star = w_t^{(k_t^\star)} \tag{332}$$

denote the selected aligned disparity and fusion weight.

The proposed corrected disparity is

$$d_t^{\text{prop}} = d_t^S + w_t^\star \left(\widetilde{d}_t^\star - d_t^S \right) \tag{333}$$

$$= (1 - w_t^\star)\, d_t^S + w_t^\star \widetilde{d}_t^\star. \tag{334}$$

The two expressions are numerically equivalent. The implementation uses the residual form in Eq. (333).

This operation constitutes a CODD-style pairwise soft correction, but the complete architecture is not a full CODD model.

.11 Geometric Acceptance and Exact Raw Fallback

Let

$$V_t^\star = V_t^{(k_t^\star)}, \qquad \pi_t^\star = \pi_t^{(k_t^\star)}, \qquad u_t^\star = u_t^{(k_t^\star)}. \tag{335}$$

The frozen acceptance mask is

$$a_t = V_t^\star \wedge [\pi_t^\star \geq 0.9] \wedge [u_t^\star \geq 0.1]\,. \tag{336}$$

The final output is

$$d_t^{\text{out}} = \text{where}\left(a_t, d_t^{\text{prop}}, d_t^S\right). \tag{337}$$

The effective fusion weight is therefore

$$w_t^{\text{eff}} = a_t w_t^\star. \tag{338}$$

Equation (337) is implemented using exact tensor selection. Rejected pixels are consequently bit-identical to the input raw disparity.

A pixel falls back to raw when:

- no temporal anchor is available;
- the selected candidate lacks valid geometric support;
- the predicted helpfulness is below 0.9;
- the predicted utility is below 0.1 pixels.

An accepted correction can still remain close to raw when $w_t^\star$ is small.

The acceptance mechanism provides geometric abstention, but it does not guarantee that accepted corrections are safe or non-degrading.

.12 Canonical Training Objective

Let g_t denote the ground-truth disparity and G_t its coverage map. For candidate k, define the raw and candidate errors:

$$e_t^S = \left|d_t^S - g_t\right|, \tag{339}$$

$$e_t^{(k)} = \left|\widetilde{d}_t^{(k)} - g_t\right|. \tag{340}$$

The true candidate improvement is

$$\Delta_t^{(k)} = e_t^S - e_t^{(k)}. \tag{341}$$

The supervised-valid mask is

$$Q_t^{(k)} = [G_t > 0.5] \wedge M_t \wedge V_t^{(k)} \wedge \text{finite}\left(\Delta_t^{(k)}\right). \tag{342}$$

A candidate is considered helpful when it improves raw disparity by more than 0.10 pixels:

$$h_t^{(k)} = Q_t^{(k)} \wedge \left[\Delta_t^{(k)} > 0.10\right]. \tag{343}$$

Let N_+ and N_- denote the positive and negative supervised counts. The positive-class weight is

$$\rho = \text{clip}\left(\frac{N_-}{\max(N_+, 1)}, 1, 50\right). \tag{344}$$

The helpfulness-classification loss is

$$\mathcal{L}_{\text{cls}} = \text{MaskedMean}_Q\left[\text{BCEWithLogits}\left(z_0, h; \text{pos\_weight} = \rho\right)\right]. \tag{345}$$

The target improvement is clipped to

$$\overline{\Delta}_t^{(k)} = \text{clip}\left(\Delta_t^{(k)}, -8, 8\right). \tag{346}$$

The improvement-regression loss is

$$\mathcal{L}_{\text{reg}} = \text{MaskedMean}_Q\left[\text{SmoothL1}\left(\widehat{\Delta}, \overline{\Delta}\right) \text{where}\,(h, \rho, 1)\right]. \tag{347}$$

For a valid ordered pair of candidates (i, j), define

$$o_{ij} = \text{sign}\left(\Delta^{(i)} - \Delta^{(j)}\right). \tag{348}$$

Pairs are used when

$$\left|\Delta^{(i)} - \Delta^{(j)}\right| > 0.10. \tag{349}$$

Table 44: Canonical optimization configuration.

Setting	Value
Training sequence IDs	$1, 3, 6$
Validation sequence ID	2
Training backbones	S2M2-S, RAFT-Stereo, StereoAnywhere
Crop size	64×80
Batch size	12
Epochs	10
Optimizer	AdamW
Initial learning rate	2×10^{-3}
Weight decay	10^{-4}
Scheduler	Cosine annealing
Minimum learning rate	$0.05\eta_0$
Gradient clipping	5
Mixed precision	Enabled on CUDA

The ranking loss is

$$\mathcal{L}_{\text{rank}} = \text{MaskedMean}\left[\max\left(0, 0.10 - o_{ij}\left(\widehat{\Delta}^{(i)} - \widehat{\Delta}^{(j)}\right)\right)\right]. \tag{350}$$

During training, the differentiable proposal for each candidate is

$$q_t^{(k)} = d_t^S + w_t^{(k)}\left(\tilde{d}_t^{(k)} - d_t^S\right). \tag{351}$$

The fusion loss is

$$\mathcal{L}_{\text{fuse}} = \text{MaskedMean}_Q\left[\text{SmoothL1}\left(q_t^{(k)}, g_t\right) \text{where} \left(h_t^{(k)}, \rho, 1\right)\right]. \tag{352}$$

The harmful-update regularizer penalizes fusion weight on candidates that do not improve raw by more than 0.10 pixels:

$$\mathcal{L}_{\text{harm}} = \text{MaskedMean}_{Q \wedge [\Delta \leq 0.10]}\left[w\right]. \tag{353}$$

The complete canonical soft-training objective is

$$\boxed{\mathcal{L} = 1.0\mathcal{L}_{\text{cls}} + 0.5\mathcal{L}_{\text{reg}} + 0.25\mathcal{L}_{\text{rank}} + 1.0\mathcal{L}_{\text{fuse}} + 0.2\mathcal{L}_{\text{harm}}} \tag{354}$$

SEA-RAFT, pull alignment, frozen stereo predictions, and raw disparity anchors are outside the learned gradient path. Only the 60,739-parameter candidate retrieval and fusion network is optimized.

.13 Canonical Training Configuration

The validated training configuration uses:

Training samples are grouped by backbone and sequence. Each group is shuffled deterministically using the run seed and epoch index. Shorter groups are repeated to match the longest group and the groups are then interleaved.

The model does not receive the identity of the stereo backbone. Multi-backbone learning is achieved through the shared external evidence representation.

.14 Computational Structure

For $K = 4$ available anchors, one current frame requires:

- K current-to-anchor flow estimates;
- K anchor-to-current flow estimates;
- K disparity pull warps;
- K reverse-flow pull warps;
- K 17-channel candidate evidence tensors;
- K shared candidate-encoder evaluations;
- one pixelwise candidate selection;
- one pairwise fusion and fallback operation.

Ignoring the external optical-flow network, evidence construction and the learned adapter scale as

$$\mathcal{O}(KHW), \tag{355}$$

with $K = 4$.

The learned component contains only

$$60\,739 \tag{356}$$

parameters. Temporal memory scales with the retained raw disparity, validity, and left-image records over an eight-frame horizon.

A dedicated runtime benchmark is required before making a real-time deployment claim.

.15 Architectural Invariants

The frozen geometry-v1 implementation enforces the following invariants:

1. only past frames are addressable;
2. anchor ages are exactly $\{1, 2, 4, 8\}$;
3. every anchor is aligned directly from the current frame;
4. optical-flow chains are never composed;
5. long-term memory contains only independent raw stereo outputs;
6. fused outputs are never written back into memory;
7. no recurrent corrected disparity state is maintained;
8. invalid candidate scores are set to $-\infty$;
9. rejected corrections return the raw tensor exactly;
10. disparity follows the positive-left convention;
11. the temporal model receives no backbone identifier;
12. the temporal model receives no internal stereo features;
13. no spatial critic or safety gate belongs to the frozen geometry core;
14. frozen inference is deterministic and parity-tested against the validated implementation.

.16 Relation to Bounded Corrected Memory

The bounded CODD-style $H = 4$ model maintains a corrected disparity state within a short recurrent horizon. Periodic reset limits the accumulation of corrected-state errors.

In contrast, geometry-v1 does not propagate a corrected state. It retrieves from multiple independently generated raw anchors and applies a single pairwise correction to the current raw prediction.

The two designs therefore represent different forms of causal memory:

$$\boxed{\begin{aligned} \text{bounded } H = 4 &: \text{short-lived corrected memory,} \\ \text{geometry-v1} &: \text{long-lived immutable raw memory.} \end{aligned}} \tag{357}$$

The bounded $H = 4$ model is retained as a strong architectural baseline, whereas immutable raw multi-anchor retrieval constitutes the principal ARGOS v2 geometric method.

.17 Scope and Limitations

The architecture supports the following claims:

- causal and online framewise operation;
- direct multi-age temporal alignment;
- immutable raw long-term memory;
- no recursive propagation of fused-state errors;
- a backbone-independent external input interface;
- pixelwise use of long-range temporal anchors;
- exact raw fallback on rejected corrections.

The current evidence does not establish:

- universal backbone agnosticism;
- external-domain or OOD generalization;
- clinical safety;
- strict risk-controlled intervention;
- guaranteed non-degradation on accepted pixels;
- real-time deployment performance.

Exact raw fallback protects only rejected pixels. It does not imply that accepted temporal corrections are always beneficial.

.18 Geometric refinement across stereo backbones

The bounded H4 temporal refiner consistently improves geometric accuracy over the frozen raw stereo predictions on SCARED-C D2. Across five stereo backbones, the aggregate EPE decreases from 0.302805 to 0.289806, corresponding to a relative reduction of approximately 4.29%.

Importantly, the improvement is not confined to the stereo estimators used during temporal-refiner training. H4 also improves CREStereo and Fast-FoundationStereo, which were excluded from temporal-refiner training, with relative EPE reductions of 3.77% and 3.52%, respectively. Overall,

Table 45: Geometric refinement on SCARED-C D2. Relative improvement is computed as $(\mathrm{EPE}_{\mathrm{raw}} - \mathrm{EPE}_{\mathrm{H4}})/\mathrm{EPE}_{\mathrm{raw}}$.

Backbone	**Raw EPE**	**H4 EPE**	**Relative reduction**
S2M2-S	0.294882	0.285021	3.34%
RAFT-Stereo	0.302213	0.291390	3.58%
StereoAnywhere	0.318889	0.296360	7.07%
CREStereo	0.294341	0.283241	3.77%
Fast-FoundationStereo	0.303701	0.293015	3.52%
Aggregate	**0.302805**	**0.289806**	**4.29%**

Table 46: Summary of the current transfer evidence for the frozen H4 temporal module. "OOD geometry" is intentionally left unclaimed where a reliable metric reference is unavailable.

Evaluation	Shift	Result	Evidence
SCARED-C D2	Backbone	−4.29% EPE	5/5 backbones improve
SCARED-C D2	Unseen backbone	Positive	CREStereo + Fast-FoundationStereo
SCARED-C D7	Held-out sequence	−4.42% EPE	5/5 backbones improve
D4D	Cross-dataset	−54.3% temporal inconsistency	No-reference
SERV-CT	Cross-dataset static	Identity	Temporal test N/A

H4 improves 14/15 backbone–sequence groups, with an aggregate frame-level win rate of 57.62% and a sequence-level win rate of 93.3%.

These results indicate that the temporal module provides a consistent geometric refinement effect across heterogeneous frozen stereo estimators. We therefore describe this result as *backbone transfer within SCARED-C*, rather than universal backbone agnosticism.

Temporal consistency remains unresolved. The geometric improvement does not yet translate into improved GT-referenced temporal dynamics. Although H4 reduces the no-reference motion-compensated temporal inconsistency (NR-TCE) across all evaluated backbones and temporal spans, GT-referenced temporal consistency (GT-TCE) worsens for four of the five stereo backbones. StereoAnywhere is the only backbone for which GT-TCE improves consistently across $k \in \{1, 2, 4, 8\}$.

This discrepancy suggests that improved self-consistency may partly originate from temporal smoothing or lag rather than more faithful tracking of the true geometric dynamics. Consequently, we consider the geometric refinement result established, while temporal-consistency improvement remains an open objective.

The current result can therefore be summarized as:

$$\boxed{\text{H4: } \approx 4.3\% \text{ mean EPE reduction} \quad + \quad 5/5 \text{ backbone improvement} \quad + \quad \text{transfer to unseen stereo estimators}}$$

while a successful temporal refiner should ultimately satisfy both

$$\Delta\mathrm{EPE} = \mathrm{EPE}_{\mathrm{raw}} - \mathrm{EPE}_{\mathrm{refined}} > 0$$

and

$$\Delta\mathrm{TCE} = \mathrm{TCE}_{\mathrm{raw}} - \mathrm{TCE}_{\mathrm{refined}} > 0.$$

.19 Current Experimental Status

We evaluate a single frozen causal temporal refinement model, denoted **Canonical H4**, using the same checkpoint and bounded recurrent policy across stereo backbones and datasets. The model was

Table 47: Cross-backbone transfer of Canonical H4 on SCARED-C D2. Values correspond to the primary macro-sequence EPE aggregation. “Unseen” denotes stereo estimators excluded from temporal-refiner training.

Backbone	Status	Raw EPE ↓	H4 EPE ↓	Rel. gain
S2M2-S	Seen	0.270422	0.264544	2.17%
RAFT-Stereo	Seen	0.269113	0.261388	2.87%
StereoAnywhere	Seen	0.302181	0.276674	8.44%
CREStereo	Unseen	0.266808	0.257823	3.37%
Fast-FoundationStereo	Unseen	0.272485	0.264712	2.85%

Table 48: Held-out SCARED-C D7 evaluation using the unchanged Canonical H4 checkpoint.

Backbone	Status	Raw EPE ↓	H4 EPE ↓	Rel. gain
S2M2-S	Seen	0.517635	0.500710	3.27%
RAFT-Stereo	Seen	0.467080	0.438604	6.10%
StereoAnywhere	Seen	0.477363	0.444676	6.85%
CREStereo	Unseen	0.516462	0.495545	4.05%
Fast-FoundationStereo	Unseen	0.481883	0.466750	3.14%

trained using disparity streams from S2M2-S, RAFT-Stereo, and StereoAnywhere, while CREStereo and Fast-FoundationStereo were excluded from temporal-refiner training.

The current evidence addresses three complementary questions: (i) whether the same temporal refiner transfers across stereo architectures; (ii) whether it remains beneficial on held-out and external surgical data; and (iii) how it compares with existing temporal stabilization methods.

.19.1 Cross-Backbone Transfer on SCARED-C

On SCARED-C D2, Canonical H4 improves disparity accuracy for all five evaluated stereo estimators. Importantly, the improvement extends to CREStereo and Fast-FoundationStereo, whose predictions were never observed during temporal-refiner training.

Under the canonical pixel-weighted aggregation, the corresponding overall D2 EPE decreases from

$$0.302805 \to 0.289806,$$

which corresponds to a relative reduction of approximately **4.29%**.

.19.2 Held-Out In-Domain Evaluation

The same frozen checkpoint also improves all five backbones on SCARED-C D7, without backbone-specific retraining.

The previously established pixel-weighted aggregate decreases from approximately

$$0.540759 \to 0.516854,$$

corresponding to an overall improvement of approximately **4.42%**.

.19.3 External-Domain Evaluation

DRENDS. We next evaluate Canonical H4 without retraining on a chronological DRENDS ex-vivo recording containing 1,499 synchronized frames. The disparity reference is derived from the metric ToF reference and is therefore not an independent stereo disparity ground truth. Accordingly, DRENDS is treated as an external metric evaluation with this explicit caveat.

Table 49: Zero-shot evaluation on the 1,499-frame DRENDS `Vid14_Pancreas_High` recording. Depth is evaluated using the corrected DRENDS protocol.

Metric	Raw	Canonical H4	Change
Disparity EPE ↓	1.137046 px	1.084080 px	−4.66%
Depth MAE ↓	14.3581 mm	13.8997 mm	−3.19%
Depth median ↓	13.4207 mm	13.0666 mm	−2.64%
Depth P95 ↓	28.4528 mm	26.8138 mm	−5.76%
Depth RMSE ↓	16.6314 mm	17.9957 mm	+8.20%

Table 50: No-reference motion-compensated temporal inconsistency on D4D. These values measure prediction-space temporal stability and must not be interpreted as geometric ground-truth error.

Backbone	Raw ↓	H4 ↓	Rel. change
RAFT-Stereo	0.509406	0.245049	−51.9%
StereoAnywhere	6.927071	3.151616	−54.5%

After correcting the DRENDS depth-evaluation protocol, both disparity EPE and depth MAE improve.

The improvement in median, MAE, and P95 depth error indicates that the benefit is not restricted to disparity space. However, depth RMSE increases, revealing a remaining tail-risk failure mode that is not visible from mean disparity EPE alone. DRENDS therefore provides encouraging preliminary evidence of cross-dataset geometric transfer, while also motivating explicit tail-risk and safety-aware evaluation.

D4D. D4D cannot currently provide a trustworthy metric geometric comparison because of an unresolved inconsistency between the available structured reference and the cached stereo geometry. We therefore use it only as a no-reference temporal-transfer experiment.

.19.4 Comparison with Existing Temporal Stabilizers

We additionally reproduced two external temporal stabilization methods on the SCARED-C D2 development split.

BiDAStabilizer. The official BiDAStabilizer pipeline was evaluated on 4,249 D2 frames using RAFT-Stereo predictions. The upstream disparities were verified to be numerically identical to the corresponding raw baseline inputs. Its pixel-weighted EPE changes from

$$0.302862 \rightarrow 0.302109,$$

corresponding to only a small geometric improvement. The method improves two of the three evaluated D2 sequences and slightly degrades the third.

In contrast, Canonical H4 obtains a substantially larger geometric reduction under its canonical D2 evaluation:

$$0.302805 \rightarrow 0.289806.$$

The exact numerical difference between the two raw EPE values reflects their independently validated evaluation paths and should therefore not be interpreted as a direct subtraction between methods.

Table 51: Current positioning with respect to external refinement and stabilization approaches. Metrics are intentionally kept in their appropriate prediction spaces.

Method	Setting	Primary result	Interpretation
Canonical H4	Causal stereo	−4.29% EPE	Metric refinement
BiDAStabilizer	Bidirectional stereo	Small EPE gain	Direct temporal competitor
NVDS	Bidirectional monocular depth	−53.4% OPW	Temporal-stability reference

NVDS. We also reproduced the official bidirectional NVDS pipeline using its intended monocular relative-depth setting. Because NVDS does not operate in metric stereo-disparity space, it is not compared to H4 using disparity EPE.

Across all 4,249 D2 frames, NVDS reduces its official-style motion-based OPW diagnostic from

$$0.127475 \to 0.059383,$$

corresponding to approximately a

$$\boxed{53.4\%}$$

reduction in relative-depth temporal inconsistency. NVDS therefore serves as a strong temporal-stability reference, rather than as a direct metric-stereo competitor.

.19.5 Failed Safety Variants and Current Model Selection

Several post-hoc hard/gated variants of Canonical H4 were explored on the D2 development split. Although some variants improved mean geometric metrics, none satisfied the complete promotion criteria across geometry, safety, unseen backbones, and external-domain behavior.

In particular, a hard threshold of 0.35 reduced D2 mean disparity and depth errors and also improved CREStereo and Fast-FoundationStereo. However, it slightly increased Bad1 and produced severe near-zero disparity outliers under DRENDS. More conservative relative and support-aware guards also failed the predefined D2 promotion criteria, primarily because of recurrent degradation on StereoAnywhere.

Consequently, none of these variants replaces Canonical H4.

$$\boxed{\text{Canonical H4 remains the frozen reference model.}}$$

This negative result is informative: apparently safe frame-wise gating rules can alter the recurrent state and create substantial downstream tail errors. Safety mechanisms must therefore be designed and evaluated as part of the recurrent dynamics rather than as purely post-hoc per-frame filters.

.19.6 Current Evidence Summary

Current takeaway. The results obtained so far indicate that temporal refinement learned on a limited set of stereo estimators is transferable across architectures and remains beneficial under held-out and external surgical data. Most importantly, the same frozen checkpoint improves two stereo backbones excluded from temporal-refiner training and shows preliminary zero-shot geometric improvement on DRENDS.

At the same time, the DRENDS RMSE and the failed post-hoc gating experiments show that improvements in average accuracy do not guarantee control of rare harmful corrections. This motivates the next stage of ARGOS: preserving the demonstrated cross-backbone and cross-dataset gains while explicitly controlling temporal-refinement risk and tail degradation.

STEER – D2 Experimental Closure

Date: 14 August 2026

Table 52: Current experimental evidence for ARGOS. A checkmark denotes directly supported evidence; "partial" indicates that the current experiment supports only part of the corresponding claim.

Question	Current evidence	Status
Improves seen stereo backbones	SCARED-C D2/D7	✓
Transfers to unseen backbones	CREStereo, Fast-FoundationStereo	✓
Held-out in-domain transfer	SCARED-C D7	✓
Cross-dataset temporal transfer	D4D	✓
Cross-dataset geometric transfer	DRENDS Vid14	Preliminary
Improves metric depth OOD	DRENDS MAE/Median/P95	Preliminary
Controls OOD tail risk	DRENDS RMSE	Not yet
Beats direct temporal competitors	BiDA D2	Promising
General temporal stability	NVDS comparison	Different prediction space

Protocol. The experimental closure was completed on SCARED-C D2 using: 15 temporal variants, 5 stereo backbones, and 3 temporal sequences. The final evaluation contains 225 standard evaluations plus 15 oracle diagnostics, for a total of 240 rows. All 241 recorded hashes were verified. Raw and refined predictions were evaluated on identical support.

The common raw macro EPE is:

$$\boxed{{}_{\text{raw}} = 0.276202}$$

Main ranking

Method	Refined EPE	EPE reduction	Improved groups
STEER $H = 6$	**0.264656**	4.18%	12/15
STEER $H = 8$	0.264912	4.09%	11/15
STEER $H = 4$	0.265028	4.05%	14/15
STEER $H = 2$	0.267265	3.24%	14/15
STEER $H = 1$	0.269001	2.61%	15/15
Fixed $w = 0.5$	0.274562	0.59%	10/15
EMA-2	0.274562	0.59%	10/15
Fixed $w = 0.3$	0.274852	0.49%	10/15
Fixed $w = 0.2$	0.275229	0.35%	10/15
EMA-3	0.275386	0.30%	6/15
Fixed $w = 0.1$	0.275684	0.19%	10/15
Warped raw previous	0.276298	-0.03%	5/15
FB confidence	0.282182	-2.17%	5/15
Warped recurrent $w = 1$	0.286035	-3.56%	5/15
Continuous recurrence	0.449349	-62.69%	0/15
Oracle raw/memory	0.242223	12.30%	15/15

Table 53: SCARED-C D2 experimental closure. Positive EPE reduction denotes improvement relative to the common raw prediction.

Learned head versus trivial blending. For the canonical $H = 4$ policy:

$$\Delta_{\text{STEER}} = 0.276202 - 0.265028 = 0.011174 \text{ px.}$$

For the best fixed blend:

$$\Delta_{\text{fixed}} = 0.276202 - 0.274562 = 0.001640 \text{ px.}$$

Therefore,

$$\boxed{\frac{\Delta_{\text{STEER}}}{\Delta_{\text{fixed}}} \approx 6.8}$$

i.e. the learned temporal policy recovers almost seven times the gain of the best fixed temporal blend.

The result is therefore not explained by simple averaging, EMA, warped-memory copying, or direct use of forward–backward confidence.

Horizon risk–gain trade-off

Method	EPE	Degraded frames	HUR	BUR	B^+/H^+
$H = 4$	0.265028	41.46%	0.4048	0.5074	**2.53**
$H = 6$	**0.264656**	42.18%	0.4053	0.5065	2.30
$H = 8$	0.264912	42.18%	0.4068	0.5047	2.04

Table 54: Risk–gain comparison for the three strongest bounded horizons.

The pure-EPE difference between $H = 4$ and $H = 6$ is:

$$0.265028 - 0.264656 = \boxed{0.000372 \text{ px}},$$

corresponding to approximately 0.14% of the $H = 4$ EPE.

However, $H = 4$ provides a better intervention ratio:

$$\boxed{B^+/H^+ = 2.53}$$

compared with 2.30 for $H = 6$ and 2.04 for $H = 8$.

Tail behaviour also favours $H = 4$:

	$H = 4$	$H = 6$	$H = 8$
$P95$ frame degradation	0.01715	0.02037	0.02441
NewBad1	0.0467%	0.0676%	0.1282%

Hence:

$$\boxed{H = 6: \text{ best development EPE} \qquad H = 4: \text{ selected risk–gain operating point}}$$

No significance claim is made for the $H = 4$ versus $H = 6$ EPE difference.

Bounded recurrence

Continuous corrected-state recurrence fails strongly:

$$0.276202 \to 0.449349,$$

corresponding to a 62.69% increase in disparity error.

This supports the conclusion that periodic raw re-anchoring is a structural component of STEER rather than a minor implementation detail.

Cross-backbone behaviour

Average gain:

$$\boxed{\text{seen backbones}: 4.65\%}$$

$$\boxed{\text{unseen backbones}: 3.11\%}$$

All six unseen-backbone sequence groups improve.

Backbone	Training status	EPE reduction	Improved groups
Stereo Anywhere	seen	8.44%	3/3
CREStereo	unseen	3.37%	3/3
RAFT-Stereo	seen	2.87%	3/3
Fast-FoundationStereo	unseen	2.85%	3/3
S^2M^2-S	seen	2.17%	2/3

Table 55: Canonical STEER $H = 4$ performance by stereo backbone.

Sequence behaviour

Sequence	EPE reduction	Improved backbones
Keyframe 4	4.84%	5/5
Keyframe 2	4.63%	5/5
Keyframe 3	2.48%	4/5

Table 56: Canonical STEER $H = 4$ performance by SCARED-C D2 sequence.

The only negative $H = 4$ backbone–sequence group is S^2M^2-S on keyframe 3.

Oracle headroom

The pixelwise raw-versus-aligned-memory oracle obtains:

$$0.276202 \to 0.242223,$$

corresponding to:

$$\boxed{12.30\% \text{ potential EPE reduction}}$$

The available oracle gain is:

$$\Delta_{\text{oracle}} = 0.033979 \text{ px}.$$

Canonical STEER $H = 4$ captures:

$$\frac{0.011174}{0.033979} = \boxed{32.9\%}$$

of this oracle improvement.

This indicates that motion-aligned memory contains substantially more useful information than the current learned policy is able to exploit.

Current interpretation

The D2 closure supports the following mechanism:

1. motion alignment creates useful but imperfect temporal candidates;
2. indiscriminate temporal fusion does not improve stereo substantially and can degrade it;
3. the learned STEER head provides the majority of the observed temporal gain by selectively weighting aligned memory;
4. bounded recurrence prevents error accumulation;
5. $H = 4$ sacrifices only 0.000372 px relative to the lowest development EPE at $H = 6$ while providing better tail and intervention behaviour.

The current concise interpretation is:

> **STEER does not behave as a simple temporal smoother. Its gain arises from the combination of geometric alignment, learned selective fusion, and bounded recurrent memory.**

Status. This experiment is development-only (SCARED-C D2). The frozen $H = 4$ policy must be confirmed on held-out D7 without changing the selection criterion.

Winning Configuration (current)

This section is maintained as the single source of truth for which configuration produces the paper numbers. It is updated whenever the winning configuration changes, and the superseded one is recorded immediately below it.

Item	Value
Temporal module	`canonical_h4_masked`
Module code	`model_design/comparison/canonical_h4_masked.py`
Parent module	`canonical_h4`, sha256 `4148ee8b4b6d...` (unmodified)
Checkpoint	`codd_style_h4_best_validation.pt`
Checkpoint sha256	`99c5745c164fd...0d4ed4305cd10609324d725`
Head parameters	177,338 (unchanged)
Evidence channels	142 (unchanged)
State lifetime	bounded $H = 4$ with raw re-anchor (unchanged)
Policy file	`codd_style_h4_policy.json` (unchanged)
Delta vs. parent	one line: `fused = where(support, fused, raw)`

Nothing was retrained. The winning configuration differs from the previously reported one only by confining the module output to the support mask that the module already computes, which is the same masking every matched baseline in `experimental_policies.py` has always applied.

Superseded. `canonical_h4` with unmasked output. Retained for provenance and for the published canonical-versus-masked comparison, but it must not be used for new paper numbers.

Root Cause: Unmasked Fusion Output

Metric depth regressed on DRENDS and D4D while disparity improved. This was root-caused to a missing mask rather than to geometry.

1. `causal_warp` uses `F.grid_sample(..., padding_mode="zeros")`, so an out-of-bounds warp returns an aligned past disparity of *exactly* 0.0.
2. `CODDStyleFusionHead.forward` computes $d^{\text{fused}} = (1 - w)\, d^{\text{raw}} + w\, \widetilde{m}$ over *every* pixel, with no support mask.
3. `canonical_h4.step` returns that value unmasked, while computing and reporting a `support` it never applies. Every baseline in `experimental_policies.py` does apply `torch.where(support, fused, raw)`.
4. Where the head places $w \approx 1$ on a zero-memory pixel, the fused disparity collapses to 0 and the prediction becomes invalid.

5. `drive()` feeds `result["disparity"]` back as the next state, so the zero propagates until the next raw re-anchor. The offender state-age histogram is dominated by ages 2–4 (Vid10: 10/103/135/52 at ages 1/2/3/4), and offending frames arrive in bursts of at most four consecutive frames. Bounded $H = 4$ contains the damage exactly as designed, but does not prevent the initial injection.
6. `MetricConfig.invalid_penalty_mm` = 10000.0 charges each such pixel.

Measured: Vid10 has 300 invalid pixels out of 38,828,160, Vid11 has 265 out of 38,439,360. The aligned memory of *every* offender has min = max = 0.0 with zero non-finite values. Diagnostic script: `scripts/diagnose_invalid_refined.py`.

Corrections to two earlier readings. The 1000 px and 10000 mm values appearing in the tables are *error penalties* for invalid predictions, not clamped predicted values. The regression is also *not* a $Z = fB/d$ conversion artefact; that earlier hypothesis was wrong.

Fix Verified and Adopted

Recording	invalid px, canonical	invalid px, masked
Vid10_Liver_Med	300 / 38,828,160	**0**
Vid11_Liver_High	265 / 38,439,360	**0**

DRENDS recompute, five recordings by three backbones:

Metric	canonical	masked
depth RMSE	regresses **14/14**, mean +110.4%	improves **14/14**, mean −3.2%
depth MAE	regresses 10/14, mean +4.2%	improves **14/14**, mean −3.7%
InvalidRate	up to 1.3×10^{-5}	0 everywhere
disparity EPE	−4.30%	**−5.02%**
disparity Bad3	−14.04%	**−18.09%**

Worst case: Vid10 with RAFT-Stereo, depth RMSE 69.12 (+256.5%) becomes 18.74 (−3.4%). Across 14 combinations and 5 metrics, **70 of 70 cells are equal or better and none regress**. Disparity improves as well, because masking the output also stops the zero from entering memory.

Consequence. `transfer_closure_v7` justified `external_ood_transfer: NOT_CONFIRMED` with "a severe refined tail" and "maxima hit 1,000 px / 10,000 mm". That tail was the missing line. The same defect drives the failing gates in `joint_d4d_v5`, where CREStereo on specimen_3 shows a depth MAE of 23,060 mm while depth P99 stays at 53 mm.

Protocol note. This is a protocol change, not a silent bug fix. The canonical-versus-masked comparison is itself a result and should be published: it shows that the support contract must be applied to the output, not merely declared.

Joint Unseen-Backbone and External-Domain Transfer

`transfer_closure_v7` recorded `joint_unseen_backbone_and_external_ood_transfer: UNAVAILABLE` on the grounds that no completed evaluation crossed both axes. That cell is now measured.

Canonical $H = 4$ was run on five DRENDS recordings (Vid10_Liver_Med, Vid11_Liver_High, Vid12_Pancreas_Ext, Vid13_Pancreas_Med, Vid14_Pancreas_High) with three stereo estimators, two of which were excluded from temporal-head training, for a total of 20,929 frames.

Disparity improved in **42 of 42 cells**, that is 14 recording-by-backbone combinations across EPE, Bad1 and Bad3, with mean EPE −4.31%, Bad1 −5.30% and Bad3 −14.06% (best −27.06%).

The per-recording pattern is determined by the recording rather than by the backbone. On Vid10 the EPE change is −3.95%, −3.92% and −4.25% for RAFT-Stereo (seen), CREStereo (unseen) and Fast-FoundationStereo (unseen) respectively. This is stronger evidence for backbone independence than any aggregate mean, because it shows the residual variance comes from the data and not from the estimator.

The new module reuses the pinned `drends_evaluation.py` (sha256 `38214918bd328ab2...`) unchanged and replaces only the stereo predictor: `model_design/comparison/drends_backbone_transfer.py`.

Selective Risk: w_t Does Not Rank Harm

The AUROC of the effective temporal weight w_t as a predictor of $\delta e > 0$ is 0.366–0.382 on D2 and 0.496–0.506 on D7. The raw-versus-memory disagreement and the update magnitude behave the same way. Risk–coverage analysis with w_t captures only 3.5% (D2) and 6.1% (D7) of the achievable oracle gain.

The reliability diagram explains why. For RAFT-Stereo the harm rate *falls* with w on D2 ($0.550 \to 0.324$) but *rises* with w on D7 ($0.260 \to 0.450$). The ordering inverts between development and held-out data, so w_t is not merely a weak risk score, it is a **sign-unstable** one and therefore not transferable.

Wording correction. The module intervenes on approximately 90% of pixels with a mean weight of 0.34. That is *graded convex intervention*, not "selective intervention". The conclusion's wording must be changed accordingly.

The Published Oracle Is the Wrong Ceiling

The reported oracle selects per pixel between raw and aligned memory, so it is the ceiling of $w \in \{0, 1\}$. STEER acts on $w \in [0, 1]$, whose ceiling is $w^\star = \mathrm{clip}\big((g - d^{\mathrm{raw}})/(\widetilde{m} - d^{\mathrm{raw}}),\ 0,\ 1\big)$.

Only 6.1% of D2 pixels and 11–14% of D7 pixels have the ground truth bracketed by the two candidates, so the continuous ceiling sits just below the binary one and the recovered fraction falls from 32.9% to approximately 30%. The correction is real but small: it is a matter of rigour, not a change of story.

Runtime

NVIDIA H100 PCIe, torch 2.5.1+cu121, 144×180 grid, 200 timed iterations after 30 warmup iterations.

Stage	median (ms)	p95 (ms)
SEA-RAFT forward	13.222	13.289
SEA-RAFT reverse	13.191	13.268
BiDA alignment	1.038	1.057
142-channel evidence (incl. frozen ResNet-18)	5.122	5.156
Fusion head	**1.027**	1.043
Total temporal overhead	**33.601**	
Peak VRAM		**124.2 MB**

End-to-end throughput ranges from 11.0 FPS with Fast-FoundationStereo down to 1.4 FPS with Stereo Anywhere. **No real-time claim is supported.**

The 177k-parameter head costs 1.03 ms, only 3% of the overhead, while 79% is the two frozen SEA-RAFT passes. The reverse pass costs 13.19 ms and feeds only the forward–backward confidence cue, and using that cue directly as the fusion weight *worsens* EPE by 2.17%. Removing it would cut overhead by 39%, which makes it a worthwhile ablation.

Matched-Protocol Audit

All 15 closure methods come from a single run under `paper_d2_strict_all_anchors`, sharing the bounded state lifetime, the re-anchor schedule, the SEA-RAFT alignment, the support definition and the fusion form; only the blending weight differs. The 225 reports correspond to 15 methods by 5 backbones by 3 sequences.

Qualification. The output *masking* was not matched. The baselines applied `where(support, fused, raw)` while the canonical module did not. On SCARED-C this does not bite, since the refined InvalidRate is zero there, so the 6.8× ratio stands. However the claim that "only w differs" was too strong before the masked variant. With `canonical_h4_masked` the protocol is genuinely matched.

An identity worth stating. `ema2_h4` and `fixed_w0.5_h4` are both `PolicySpec(horizon=4, weight=0.5)`, that is the same policy under two names, and `ema3_h4` is $w = 2/3$. The baseline family is therefore five fixed convex weights $w \in \{0.1, 0.2, 0.3, 0.5, 2/3\}$ over the shared bounded recurrent state, not two independent families. Reporting it that way is both more honest and stronger, because two "different" baselines producing an identical number otherwise looks like a bug.

Metrics Already Computed but Never Reported

The following are present in `definitive_table.csv` and absent from the paper.

Bad-pixel rates improve in 30 of 30 SCARED-C cells (2 splits by 5 backbones by {Bad1, Bad3, Bad5}). Stereo Anywhere on D2 gives Bad1 −19.7%, Bad3 −32.0% and Bad5 −45.3%. On D7, Bad5 improves more than Bad1 on all five backbones, which means the module removes large errors rather than smoothing small ones. This is the anti-smoothing argument in direct form.

Disparity RMSE improves on all five backbones on both splits. P99 improves on 10 of 10 and Max on 9 of 10. Stereo Anywhere on D2 has its Max fall from 33.27 to 14.88, a 55% reduction.

Harm and benefit per backbone (macro-sequence): B^+/H^+ lies between 1.76 and 4.59; RecoveredBad1 exceeds NewBad1 by factors of 2.9 to 10.4; IdentityPreservation is at least 0.9987 everywhere.

Temporal fidelity (grid DTCE, disparity, $k \in \{1, 2, 4, 8\}$) improves on **20 of 20** D7 cells, between −2.1% and −26.2%, while on D2 it worsens slightly on four of five backbones. That asymmetry is an asset rather than a weakness: on D2 geometry improves while grid temporal change does not, which directly supports the claim that temporal smoothness is not temporal correctness. Grid DTCE is not motion-compensated and must be labelled as such.

Open check. `TEPE_corr` and `OPW`, both described in the evaluation framework, were not found in the produced artefacts. They must either be located or removed from the framework description.

Suspected field bug. `FrameDegradation/CatastrophicFraction` equals `PositiveFraction` to five decimals on all ten SCARED-C rows. It should not be used until this is resolved.

Spread of the D2 Mean, and an Arithmetic Correction

On D2 the pooled gain is 4.05%, the mean of per-backbone percentages is 3.94%, the median is 2.87% and the range is 2.17–8.44%; excluding Stereo Anywhere the mean falls to 2.82%. On D7 the pooled gain is 4.64%, the median is 4.05% and the range is 3.14–6.85%; excluding Stereo Anywhere it is 4.14%, with RAFT-Stereo alone at 6.10%.

The held-out result does not depend on the outlier. The paper should report median and range, and state explicitly that 4.05% is pooled rather than a mean of per-backbone percentages.

Correction. Section V-E states that seen backbones improve by 4.65% on D2. The mean of the per-backbone seen gains is $(2.1737 + 2.8705 + 8.4412)/3 = \mathbf{4.4951}\%$. The unseen figure of 3.11% is correct, so the method is right and only the number is wrong; 4.65% appears to have been transcribed from the D7 pooled value of 4.639%.

Name Collision

"STEER" is already used in robotics: *STEER: Flexible Robotic Manipulation via Dense Language Grounding*, arXiv:2411.03409, November 2024, which targets the same venue population. No collision was found in the stereo or depth literature. The acronym is also never expanded in our own text. The name should be changed or at least expanded.

Hardening Pass: Closing the Submission TODOs

Started 2026-08-15 21:30. Goal: make every remaining submission-critical claim defensible rather than promised.

Seed variance, launched. The held-out table reported seed 0 only, which is the single largest rejection risk given per-backbone gains of 2–3%. Seeds 1 and 2 are now training, one per H100, in a single `p1i` job.

`train.py` hard-fails unless the locked provenance matches the canonical configuration byte for byte, including `seed: 0`. That guard is correct and was left in place. Instead `model_design/train_seed.py` imports `train.py` and reuses its `metadata()`, `verify_sources()`, `validate_output()` and `config()` unchanged, overriding exactly one field. It refuses any seed not listed in the pre-registration, and writes a `seed_provenance.json` recording the deviation.

Pre-registered in `model_design/multiseed_preregister.json` *before* launch, declaring the primary endpoint, the resampling unit (sequences, never pixels), and four prohibitions: no seed selection, no post-hoc threshold retuning, no change to the checkpoint-selection rule, and no use of D7 for any decision. The canonical seed-0 checkpoint remains the reported model; this is a variance estimate, not a selection experiment.

CatastrophicFraction: definitional, not a bug. Resolved. `PositiveFraction` is $\Pr(D_t > 0)$ and `CatastrophicFraction` is $\Pr(D_t > \tau_{\text{cat}})$, with `catastrophic_px` and `catastrophic_mm` both defaulting to 0.0 in `MetricConfig`. The evaluators construct `MetricConfig()` with defaults, so the two coincide by construction. The code is correct; the threshold was simply never set. The paper now reports the degraded-frame fraction with its definition stated, and omits CatastrophicFraction.

TEPE_corr and OPW: implemented, never executed. Both exist in `unified_metrics.py` (lines 834–881) but are gated on a `flow` argument that no caller of `evaluate_argos_prediction` supplies. They were therefore never computed. This is an opportunity rather than a defect: the driver already computes the SEA-RAFT flows used for warping, so passing them would upgrade the multi-horizon temporal section from a disclaimed grid diagnostic to genuine motion-compensated fidelity.

Degenerate D4D controls: the metric is won by doing nothing. Two controls were added in `model_design/comparison/degenerate_policies.py`, alongside the pinned `experimental_policies.py` rather than inside it. Over 504 frames the raw inconsistency of 3.7182 becomes 1.7128 under STEER (53.9%), 0.0986 under copy-previous (97.3%), and *exactly* 0.0000 under warped-previous (100%).

Warped-previous is zero by construction: its output *is* the motion-compensated previous frame, so it is trivially consistent with the motion-compensated previous frame. The ordering on this metric is therefore inverted with respect to geometric quality — the two policies that discard current stereo evidence beat the one that uses it. The paper now reports the controls next to the metric, so no reduction figure can be misread as accuracy.

TEPE_corr and OPW now computed. `scripts/evaluate_temporal_corrected.py` assembles the `alignment_by_horizon` contract from the SEA-RAFT flows the driver already produces. The convention was read off the consumer rather than guessed: `flow_t_to_tk[:, i]` is applied as `warp_with_flow(x[:, i+k], flow)`, so it lives on frame i and pulls $i + k$, which is the *backward* member of the pair. On a 40-frame smoke with S^2M^2-S, TEPE_corr improves 0.03030 → 0.02454 (−19.0%) and OPW 0.03288 → 0.02676 (−18.6%). Full D2 and D7 runs are in flight.

3D surface consistency, first numbers. `scripts/evaluate_3d_consistency.py` measures frame-to-frame surface stability without camera pose, which SCARED-C does not provide: each frame's disparity is back-projected to a metric cloud and compared against the previous frame's cloud pulled through the same SEA-RAFT correspondence. On a 12-frame smoke, point-to-point drift is 0.4896 mm raw, 0.4471 mm refined and 0.4040 mm for GT, so the *excess over the GT floor* halves (0.0855 → 0.0431 mm).

One caveat found immediately: GT surface normals are *noisier* than both predictions (11.83° versus 5.04° raw and 4.24° refined), because the pseudo-GT is sparse and central differences amplify it. The GT-floor interpretation is therefore valid for the point-to-point metric and *not* for the normal metric, which must be reported without it.

Stale TODO closed. The request to extend DRENDS to three further sequences was superseded by the five-recording, three-backbone run already reported.

Open Items

- **Requires training**, not started, and must be pre-registered: canonical seeds 1 and 2 with paired bootstrap; evidence-channel ablation; head-structure ablation (reset-only, fusion-only, single versus dual resolution); convex-constraint ablation.
- **Inference only**: degenerate D4D baselines (copy-previous, warped-previous); a common-support Raw / BiDAStabilizer / STEER table; a 3D consistency experiment; and a re-run of `joint_d4d` under the masked module.
- **Editorial**: Eq. (53) is truncated; the Fig. 1 caption overlaps the diagram; the qualitative figure is still a placeholder; BUR should be added to Table III (0.5074, 0.5065, 0.5047 for $H = 4, 6, 8$, falling as HUR rises).

Night of 15–16 August: The Comparison That Does Not Exist

The standing objection was the sharpest one available to a reviewer: TETHER is never compared to another method. We went looking for one. The honest answer is that in our exact category — causal, black-box, attached to a frozen surgical stereo network — no runnable published method exists, and we can now say so with receipts rather than with a shrug.

StereoDiffusion (MICCAI 2024). The only paper in our category: causal, black-box, frozen stereo, surgical, evaluated on SCARED. Its repository has a single commit, 7.43 KiB of git objects, and contains a `.gitignore`, a `LICENSE` and a twelve-byte `README.md`. No code, no weights, no other branch. There is nothing to run.

Neural Disparity Refinement (3DV 2021 / TPAMI 2024). Same input contract as ours, spatial only. The code clones fine; the weights do not exist any more. Both released Google Drive IDs return HTTP 404 from the download endpoint *and* from the plain file-view page — a restricted file answers 403 or an interstitial, so 404 on the view page means deleted. `docs.google.com/uc` agrees. Hugging Face has no mirror; the GitHub releases API returns an empty list. Our network is not the obstacle: the same `curl` reached huggingface.co and archive.org in the same session.

Lai et al. (ECCV 2018). The natural task-agnostic baseline. Its download script points at `vllab.ucmerced.edu`, which answers 200 for the directory and 404 for every model file. The Wayback CDX index has captures of the project page, its stylesheets and the 34 MB supplementary PDF, and zero captures of `ECCV18_blind_consistency.pth` — crawlers archive HTML and skip large binaries. No Hugging Face mirror, no GitHub release. The hosting rotted and nothing caught it.

TC-Stereo (ECCV 2024), and what we did about it. This one loads. We found a configuration that matches all three released checkpoints with zero missing and zero unexpected keys, and it imports cleanly in our environment. The blocker is scientific, not technical. Its temporal propagation warps the previous disparity by a rigid 6-DoF reprojection built from per-frame camera pose, intrinsics and baseline. SCARED-C supplies no per-frame pose, and deformable tissue violates the rigid-scene assumption regardless — so its temporal path is not merely unavailable on our data, it is the wrong model of the scene. Running it with `params=None` is possible but degenerates it to a single-frame stereo network.

So we ran exactly that, and labelled it honestly. `scripts/run_tcstereo_reference.py` drives TC-Stereo over the *same* frozen NPZ boundary the BiDAStabilizer comparison consumed, so the frames, the crops and the ground truth are bit-identical across methods. The row it produces is an integrated-architecture reference — 16.7 M parameters, trained end to end, evaluated zero-shot — not a temporal rival, and the manifest records the disabled temporal path and the reason. It answers the one question our own ablations cannot: whether one should simply use a newer integrated network instead of refining a frozen one.

What went into the paper. A new Related Work paragraph, "The comparison that does not exist", with five citations. It cost a page, and the page came back out of the external-interface, support-contract, closure and evaluation subsections without dropping a single result or table. The submission is at eight pages including the bibliography, which is the hard constraint.

Seed variance, pre-registered. `scripts/aggregate_seeds.py` implements the reporting rule written down before any seed result existed: mean and standard deviation across seeds $0, 1, 2$, plus a paired bootstrap in which *sequences*, never pixels, are the resampling unit. Pixels inside one endoscopic sequence are not independent samples and resampling them would manufacture significance. Seeds are averaged inside each resample, so seed choice is part of the method rather than a free parameter — which is what the pre-registered prohibition on selecting the best seed actually requires. Run on seed 0 alone as a plumbing check, the pooled D2+D7 Bad1 reduction is 5.79% with a 95% interval of $[4.15, 8.35]$ and no resample in which refinement failed to help. The real three-seed number replaces it when seeds 1 and 2 finish.

Seed 1 Lands, and Says Something About Checkpoint Selection

Seed 1 finished its D2 and D7 evaluation through the frozen support-masked path. Pairing it with seed 0 under the pre-registered rule — sequences as the resampling unit, seeds averaged inside each resample — gives the following relative Bad1 reductions.

Split	Backbone	mean%	std%	seed 0	seed 1
D2	CREStereo	7.44	1.17	8.26	6.62
D2	Fast-FoundationStereo	6.83	1.53	7.91	5.74
D2	RAFT-Stereo	6.17	1.10	6.95	5.40
D2	S2M2-S	7.05	0.50	7.40	6.70
D2	StereoAnywhere	17.80	2.62	19.65	15.95
D7	CREStereo	4.18	0.46	3.86	4.51
D7	Fast-FoundationStereo	4.02	0.08	3.97	4.08
D7	RAFT-Stereo	4.65	0.99	3.95	5.35
D7	S2M2-S	5.96	0.42	5.66	6.25
D7	StereoAnywhere	6.73	0.47	6.40	7.07
Pooled, 35 sequences		5.90	0.15	5.79	6.00

Every cell is a reduction on both seeds. The pooled standard deviation is 0.15 percentage points on a 5.90 point effect, and the paired sequence-level bootstrap gives $[4.26, 8.38]$ with no resample in which refinement failed to help.

The interesting part is the sign pattern, which is not noise. Seed 1 is worse than seed 0 on *all five* D2 backbones and better than seed 0 on *all five* D7 backbones. Ten out of ten is not a coin flip. The explanation is in our own protocol: the checkpoint-selection rule picks the best development (D2) fused EPE, so seed 0 is by construction the run that looked best on D2. Its D2 margin is partly a selection artefact, and the held-out split — the one that carries the claim — is where the other seed does slightly better.

This is a point in favour of the result rather than against it. The number we report on held-out data is not the one selection inflated, and it is stable across an independent training run. Seed 2 is in flight; the reported figure will be the three-seed mean with this same bootstrap, and this two-seed table stands as the record of what was known first.

Staleness Audit: Do the Tables Still Match the Files?

The support-mask fix changed enough numbers that a table could plausibly have been left behind. While the GPU jobs ran we re-derived the two largest tables straight from the report JSONs instead of trusting the manuscript.

Table **??** (cross-backbone SCARED-C) reproduces exactly from `results/scared_masked`: S^2M^2-S Bad1 .01394 → .01291 on D2 and .06440 → .06076 on D7, RAFT-Stereo .01603 → .01492 and .06085 → .05845. No drift.

Table **??** (zero-shot DRENDS) reproduces exactly from `results/drends_masked`, all fifteen recording–backbone cells, and so do the four pooled means quoted underneath it: −5.00% EPE, −17.35% Bad3, −3.68% depth MAE, −3.21% depth RMSE, with 60/60 cells improving.

One trap worth writing down, because it nearly caught us. `results/drends_multisequence` still exists and still contains four RAFT-Stereo reports whose numbers (−3.9% EPE on Vid10 against the paper's −5.7%) look like a contradiction. It is not: that directory is the pre-mask, single-backbone, four-recording run, superseded by `drends_masked`. Two result directories that differ by a bug fix are indistinguishable by filename, and the older one sorts first. The manifests disambiguate them; directory names do not.

Three Seeds, and a Selection Effect We Can Measure

Seed 2 landed. The full pre-registered analysis now runs on seeds $0, 1, 2$: pooled Bad1 reduction $5.69 \pm 0.37\%$ over 35 sequences, paired sequence-level bootstrap $[4.11, 8.02]$, $p = .000$; pooled EPE $4.62 \pm 0.18\%$, $[3.42, 5.80]$.

Split apart, the two halves behave completely differently:

	seed 0	seed 1	seed 2	spread
D2 Bad1	10.78	8.67	5.75	5.03
D7 Bad1	4.78	5.47	5.19	0.69
D2 EPE	4.05	3.43	3.17	0.88
D7 EPE	4.57	5.25	5.38	0.81

On D2 seed 0 is best and the sequence is monotone. On D7 seed 0 is *worst* on both metrics. Ten of ten per-backbone comparisons agree. This is the signature of our own checkpoint-selection rule, which picks best D2 fused EPE: seed 0 is by construction the run that looked best on the split it was selected on. The development margin carries selection optimism; the held-out margin, which is what the paper claims, does not.

We could have buried this. Reporting it is better: it converts a reviewer's likely objection — "your development gains look suspiciously large" — into a measured quantity with the right sign.

What it cost the manuscript. Adding the subsection pushed the paper to nine pages. Getting back to eight took four passes: merging "What Makes TETHER Work" with the 177k-parameter subsection, compressing the D4D degenerate section into prose with its table moved to the supplementary, tightening Cross-Backbone Development Behaviour, and finally dropping the seed table itself in favour of four inline mean±std figures. The last one was the right call and should have been the first: the table held twelve numbers the prose already stated.

One stale sentence died in the process, which is the point of doing this before submission. Discussion still read "held-out numbers are a single canonical seed; sequence-aware confidence intervals over additional seeds are required before the per-backbone margins of two to three percent can be called separated." Half of that is now false. It has been replaced by what is still true: the three-seed study bounds the pooled effect, so individual backbone rows remain unseparated from one another.

The full per-backbone seed table, including the three D2 rows whose intervals cross zero, is in the supplementary. Those wide intervals are a property of a three-sequence split, not of the method, and we prefer them to the narrow and meaningless intervals that resampling pixels would have produced.

Audit completed: all four main tables. Continuing the staleness pass while the GPU jobs ran, Table **??** (intervention quality) also reproduces from `results/scared_masked`: recomputing B^+/H^+ and RecoveredBad1/NewBad1 per split and backbone lands within 0.06 of every printed cell, the residual being the difference between a ratio of sequence means and the framework's macro-sequence aggregation rather than an error. The prose claim underneath it — identity preservation above 99.87% in every case — is correct and tight: the minimum over all ten split–backbone cells is 99.877%, on RAFT-Stereo D7.

The closure table needed a note rather than a correction. Its source is `definitive_evaluation/canonical_h4_b` the *unconfined* module, while every other table in the paper is the confined one. The tempting fix — update the $H = 4$ row to the confined number — would have been wrong: the other ten rows ($H = 6$, $H = 1$, the fixed blends, the oracle) are all unconfined ablations, so changing one row would have compared two implementations inside a table whose entire purpose is that only the policy differs. Confinement moves D2 EPE by 1.2×10^{-5} px, so the honest fix was to say in the text that the whole table runs unconfined for exactly that reason. Raw D2 pooled EPE is 0.276202 in both runs and the reduction is 4.05% in both, so no other figure in the paper depends on the choice.

The Comparison Exists Now

BiDAStabilizer finished on all three D2 sequences at common support. This is the number the whole night was for.

Method	Causal	EPE	Bad1	Bad3	RMSE
Raw RAFT-Stereo	—	.30286	2.087	.186	.44842
BiDAStabilizer	no	.30211	1.987	.171	.44026
TETHER	yes	**.28899**	**1.826**	**.157**	**.42562**

Same 4249 frames, same frozen RAFT-Stereo robust predictions, same prediction-independent support, 18.0M pixels. TETHER wins every metric and is the only causal entry: −4.58% EPE, −12.53% Bad1, −15.52% Bad3 against raw, where the bidirectional baseline manages −0.25%, −4.83%, −8.32%.

Worth remembering how close this came to being meaningless. The first comparison we had was not a comparison: the published reproduction ran RAFT-Stereo *robust* and we ran *middlebury*, and the two were scored under different support contracts — raw EPE 0.3029 against 0.2762. Subtracting those would have produced a confident number with no content. The fix was to stop generating our own inputs and read the stored NPZ boundary the reproduction had already written, so the RGB and the disparity BiDAStabilizer consumed are the same arrays ours consumes.

The cell we do not like. One sequence of three favours BiDAStabilizer on all three metrics, and on it our EPE is 0.27% *worse* than raw. It goes in the paper. With three sequences a single sequence-level disagreement is well inside what the split can produce, and a method allowed to look ahead ought to win somewhere; the claim the table supports is the pooled one. Hiding it would have been both dishonest and, given that a reviewer can run the same three sequences, stupid.

Page budget, again. The subsection plus table cost a full page. Recovered by: deleting the CatastrophicFraction derivation from the main text, where it duplicated the supplementary verbatim; folding the "Selective risk" capability list into one sentence; compressing D4D a second time; and tightening the $H = 4$, recurrence and cross-backbone prose. Nothing was dropped that is not still stated somewhere. Eight pages, comparison table included.

TC-Stereo Runs, and We Argue Against Our Own Number

The last job finished. TC-Stereo, single-frame path, SceneFlow weights, over the same boundary as everything else: EPE 0.5242, Bad1 9.05%, against raw RAFT-Stereo robust at 0.3029 and 2.087%. Seventy-three percent worse on EPE, four times worse on Bad1.

The tempting move is to put that in the main paper as a win. It is not one, and the reason is worth recording.

The stored boundary is 144×180 with a median ground-truth disparity of 8.04 px, range 5.57–15.53. TC-Stereo was trained at full resolution on disparity ranges an order of magnitude larger. We are asking a from-scratch cost-volume matcher to work two octaves below its training regime. It still functions — 91% of pixels land within 1 px, so this is a working network producing worse geometry, not a broken configuration or a scale bug — but the comparison is not fair to the architecture. A refiner is far less exposed to the same problem: both BiDAStabilizer and TETHER consume the raw disparity and correct it, so boundary resolution constrains them much less than it constrains a correlation pyramid.

We kept the boundary rather than re-running TC-Stereo at full resolution, because doing that would buy fairness to one method by destroying the property the entire table rests on: that every method sees identical frames and identical ground truth. So the row goes in the supplementary, with the caveat attached, supporting the claim it can support — swapping in a different integrated architecture zero-shot is not a drop-in substitute for refining a strong in-domain frozen model — and explicitly not the claim it cannot: that TETHER outperforms TC-Stereo.

A design that turned out vacuous. The runner scored two supports, ground truth alone and ground truth intersected with raw validity, on the theory that a method not consuming the backbone should not be charged for the backbone's refusals. On this boundary `raw_valid` is true everywhere, so both conditions select the same 17,999,475 pixels and the two tables are identical. The distinction was worth building — it would matter on a boundary where the backbone declines pixels — but here it bought nothing, and the supplementary says so rather than presenting two columns that look like independent confirmation.

16 August, Afternoon: The Ablations Answer, and One Answer Is No

Four architecture ablations were pre-registered this morning and launched on the two H100s. Two have finished training and evaluation, one is training, one is being evaluated. Two results are already worth writing down, and the first one costs us a sentence in the paper.

A4: the convexity claim is withdrawn. The Analysis said that confining the output to the interval between the two candidate geometries was "a plausible reason why one small head transfers to architectures absent from its training". That was an assertion with no experiment behind it, so we built one: 49 parameters, a zero-initialised additive term that lets the output leave the interval. Zero-initialised so training starts exactly at the canonical model and any departure has to be learned.

The pre-registration named Bad1 on held-out D7 as the primary endpoint and committed, in advance, to *withdrawing* rather than softening the claim if the relaxed variant matched or beat canonical there.

It beat it. On D7: Bad1 6.02% against 4.78%, and also better on Bad3, RMSE and worst-case error. The claim is withdrawn.

What the experiment actually shows is not what we expected, and is more specific. Mean EPE gain collapses from 4.57% to 0.05%, turning *negative* on three of five backbones — the module makes disparity worse than doing nothing. Our first reading was the obvious one, occasional large failures, and it is wrong: RMSE and the worst case both *improve*, and a quadratic metric would be the first to register outliers. The only consistent reading is the opposite. The relaxed variant fixes the few large errors and degrades a great many pixels that were already close. A linear mean registers that; thresholds and squared errors do not.

So the convex constraint does not buy safety from catastrophic corrections. It buys mean accuracy, by stopping the module from disturbing the majority of pixels the frozen estimator already had right. Narrower than what we claimed, and now supported.

A1: forty-five percent of the input is redundant. The evidence space carries 64 frozen ResNet-18 channels out of 142. But appearance self-correlation and cross-appearance correlation are computed from those same features — so the real question was never "do appearance features help" but "does raw appearance add anything *beyond* the correlations already derived from it".

On held-out D7, dropping them changes nothing: Bad1 4.88% against 4.78%, a difference of 0.10 points against a three-seed spread of 0.37. EPE, Bad3 and RMSE are all marginally better. By the pre-registered rule this is *indistinguishable*, and the pre-registration also said what to do about it: describe the channels as redundant.

One asymmetry is worth noting because it is the same one the seed study found. On D2, A1 is clearly worse on Bad1 (8.81% against 10.78%), and D2 is the split the checkpoint is selected on. Development says the channels matter; held-out says they do not.

What this does not buy. A1 shrinks the first convolution from 30,672 to 16,848 parameters, about 8% of the head, and does *not* remove the frozen extractor from the runtime — the appearance correlations still need it. Any latency argument belongs to A2, which removes the learned evidence entirely and is in evaluation now. A3, the single-resolution variant, is training: it keeps the parameter count exactly identical and moves only the receptive field, because deleting the context branch outright would have confounded "context needs low resolution" with "the head needs capacity".

A2: The Evidence Space Is Over-Specified

A2 removes *all* learned stereo evidence — the confidence curves and their supports, both appearance correlations, and the 64 frozen ResNet-18 channels — leaving 38 channels of disparity, motion and RGB geometry. Verified on the checkpoint: `cue_channels` is 38 and the head is 154,874 parameters instead of 177,338. Crucially it is also the only variant that removes the frozen extractor from the runtime, since nothing left in the stack needs it.

We expected this to be the ablation that hurt. On held-out D7 it is *better than canonical on every metric*:

D7 held out	EPE	Bad1	Bad3	RMSE
Canonical, 142 ch	4.57	4.78	4.07	5.33
A1, 78 ch	4.93	4.88	4.12	5.46
A2, **38** ch	**5.49**	**5.69**	**4.85**	**5.92**

Its Bad1 margin is +0.91 points against a three-seed spread of 0.37, so by the pre-registered rule this is *better*, not indistinguishable.

The three ablations line up into one pattern rather than three independent results. A1 removes the raw appearance channels but keeps the correlations computed from them: neutral. A2 removes the correlations too: better. The more appearance-derived evidence leaves the interface, the better the head generalises to backbones and sequences it never saw. That is the signature of evidence that ties the decision to the training domain — useful where it was fitted, a liability elsewhere.

What this does and does not cost the paper. It does not touch the interface idea. A 38-channel external evidence space is still an external evidence space, still carries no ground truth, no backbone identity and no future frame, and still lets one head serve heterogeneous estimators. What fails is the specific richness of our instantiation: we built 142 channels and 104 of them are, on held-out data, worse than useless.

Two honest qualifications. On D2 — the split the checkpoint is selected on — A2's EPE is *worse* than canonical (3.69 against 4.05), which is the third time this development-versus-held-out inversion has appeared today. And each ablation is a single seed, while the 0.37 point yardstick comes from three seeds of the canonical model; a margin of 0.91 clears it but not by an order of magnitude.

The uncomfortable implication. If A2 holds up, the canonical architecture we are submitting is not the best one we have measured. The honest options are to report the ablation as it is and say so plainly, or to re-run the seed study on the 38-channel variant before claiming anything stronger. We are not going to quietly promote A2 to canonical after the fact — the pre-registration prohibits exactly that, and selecting the best variant post hoc is how the selection effect we spent today measuring gets back in through the side door.

Correction: We Compared Against the Wrong Baseline

Everything written above about A1, A2 and A4 used `results/scared_masked` as the reference. That is canonical *seed 0*. And the seed study, run this same morning, established that seed 0 is the *weakest* of the three canonical seeds on held-out D7. So every ablation was being measured against the luckiest possible comparison, and every one of them looked better than it is.

Against the three-seed mean, which is the reference the pre-registration actually named:

	D2 EPE	D2 Bad1	D7 EPE	D7 Bad1
Canonical, 3 seeds	3.55	8.40	5.06	5.15
A1	4.18	8.81	4.93	4.88
A2	3.69	12.84	5.49	5.69
A4	5.53	12.44	0.05	6.02

Two statements from earlier in this diary are wrong and are withdrawn. "A2's D2 EPE is worse than canonical" compared 3.69 against seed 0's 4.05; against the three-seed mean of 3.55 it is better. And the margins quoted for A2 and A4 on held-out Bad1 (+0.91 and +1.23) are really +0.55 and +0.87.

The script now enforces the right baseline, prints the individual seed values, and refuses to call a one-seed margin a result when it falls inside the canonical seed range.

And a decision that was rejected too quickly. Asked whether the better variant should simply replace the canonical model, the answer given was a flat no, on the grounds that choosing it would burn the held-out split. That reasoning is correct and it is not the whole question. A2 is outside the canonical seed range on *both* splits: +1.8 standard deviations on D2 Bad1 and +1.6 on D7. D2 is the development split — deciding on it is what it exists for, and the pre-registration prohibits using D7 for decisions, not D2.

So there is a clean path, and refusing it would have been caution mistaken for rigour. A2 is being trained at seeds 1 and 2, giving it the same three-seed treatment the canonical model has. The comparison that may promote it is the three-seed *D2* mean. D7 is not consulted for that choice and stays held out; if A2 is promoted, its held-out numbers are reported afterwards and are not part of the decision. Declared in advance: if A2's three-seed D2 mean does not clearly exceed 8.40 with a comparable spread, A2 stays an ablation and the canonical model is submitted unchanged.

All Four Ablations: Nothing We Designed Is Load-Bearing

A3 landed, completing the set. Before reading anything into the numbers we checked the one thing that would invalidate the comparison: the raw baseline. It is identical to six decimals across all six runs — the three canonical seeds and the four variants — at 0.062051 on D7 Bad1 over twenty reports each. The comparison is apples to apples.

	Change	D7 EPE	D7 Bad1
Canonical (3 seeds)	—	5.06	5.15 ± 0.34
A1	$142 \to 78$ ch	4.93	4.88
A2	$142 \to 38$ ch	5.49	5.69
A3	full-resolution context	5.61	5.85
A4	unconstrained output	0.05	6.02

A1 is indistinguishable. A2, A3 and A4 sit 1.6, 2.0 and 2.5 standard deviations above the canonical mean, and A2 and A3 beat it on EPE as well. Every design decision we made is neutral or worse than its alternative.

Why this is suspicious, and what survives the suspicion. Three variants, each changing a *different* thing, all landing above the mean on the same side is not what seed noise looks like — noise scatters. The reading that fits is that the canonical design is over-specified, and that simplifying it in almost any direction helps generalisation. What does *not* survive is any claim that a particular one of our choices is responsible: each variant is one training run against a three-seed baseline, and two standard deviations from a single sample is a hint, not a finding.

A4 is set aside on its own terms rather than on that caution. It has the best held-out Bad1 in the table and an EPE gain of 0.05%: it fixes threshold crossings by disturbing everything else, which is not a trade we want.

A3 corrects the Method, not just the Analysis. The paper justified predicting the fusion weight at quarter resolution on the reasoning that deciding *how much* to intervene needs more context than deciding *whether*. A3 keeps the parameter count at exactly 177,338 and moves that branch to full resolution — and does not do worse. So the dual-resolution design is a compute choice, roughly 16× cheaper, and the Method now says that instead of implying a necessity. This is the ablation the agentic review asked for, and it answered in the opposite direction to our expectation.

What went into the paper. A four-row table, generated by `scripts/ablation_table.py` rather than transcribed, and a section that states plainly that none of the four decisions is load-bearing. It cost the per-recording DRENDS table — whose 60/60 claim and Vid10 example survive in prose — and four citations that were redundant inside groups keeping a representative each. Eight pages, twenty-seven references.

A2 is now training at seeds 1 and 2. The question it answers is whether the 38-channel variant is genuinely better or one lucky run, and by pre-registration that question is settled on D2 alone.

Night Audit: One Number Was Computed on a Subset

With the ablations queued and nothing to collect, the night went into checking every figure the paper quotes against the files it came from. Four checks, one real defect.

The defect. The paper claimed TETHER captures 32.9% of the raw-versus-memory oracle gain and 30.1% of the continuous one. Neither reproduces from `results/oracle_risk_analysis`. Recomputing pixel-weighted, macro over sequences, and as a ratio of summed gains all give the same pair: 37.2% and 34.5%.

The published numbers turn out to match exactly one computation — dropping StereoAnywhere, which is the outlier at 50.9% against 31–33% for the other four. That may well have been a deliberate robustness check at the time. It was quoted without the exclusion, which makes it wrong as written.

Corrected in both documents, along with the bracketed fraction from 6.1% to 6.2%. Worth recording that the correct figure is the *better* one: we were under-reporting our own result by four points. Errors of this kind are usually assumed to run the other way, which is probably why nobody re-derived it.

The three that held. D4D reproduces exactly — TETHER 53.93%, copy-previous 97.35%, warped-previous 100.00% against the 53.9, 97.3 and 100.0 printed. Runtime reproduces exactly: 33.60 ms overhead, 124.2 MB peak VRAM, and end-to-end rates of 10.98 and 1.38 FPS with overhead fractions of 36.9% and 4.6% at the two extremes. And every cross-document promise now has a section behind it: the fifteen closure policies, the DRENDS per-recording figures, the per-backbone intervention table, the ablation D2 figures, and — added tonight — the horizon sweep's risk columns, which the main paper pointed at and the supplementary did not carry.

Method. The audit is worth doing in this direction: recompute from the stored reports and compare to the manuscript, rather than reading the manuscript and asking whether it looks right. Every number here looked right. The one that was wrong looked right too.

A2 Wins the Pre-Registered Comparison, and We Submit the Other Model

Three seeds of A2 finished overnight. On D2 — the split the pre-registration named as the decision split, precisely so that a promotion could be made without touching held-out data:

D2 Bad1 reduction	seeds	mean	std
Canonical, 142 ch	10.78, 8.67, 5.75	8.40	2.53
A2, 38 ch	12.84, 12.38, 12.98	**12.73**	**0.32**

The ranges do not overlap: A2's worst seed beats the canonical model's best. The declared criterion — clearly exceeding 8.40 with a comparable spread — is met, and the spread is not comparable but eight times tighter.

That variance is the more interesting half of the result. The canonical model swings five points across seeds on D2, from 5.75 to 10.78; the 38-channel variant swings half a point. A head whose evidence is mostly appearance-derived is apparently fitting the training domain in a way that depends heavily on initialisation, and stripping that evidence removes the dependence along with it.

Held out, reported after the decision and not part of it: A2 gives $5.84 \pm 0.22\%$ against the canonical $5.15 \pm 0.34\%$.

And we are submitting the canonical model anyway. Not out of caution. Two reasons, and neither is that the result is doubtful.

First, promotion is not an edit. Every table in the paper — DRENDS zero-shot, the BiDAStabilizer common-support comparison, the transfer matrix, the intervention ratios, the runtime budget — was measured on the canonical checkpoint. A2 has been evaluated on SCARED-C only. Promoting it means re-running all of it, and a paper whose headline model is supported by one split while its tables come from another is worse than either option taken cleanly.

Second, and this is the part worth remembering: the comparison that favours A2 was deliberately built on the development split so that the held-out numbers would stay untouched by any choice of ours. Promoting on the strength of it and then reporting D7 for the promoted model would spend exactly the thing the design was protecting.

So the paper reports the ablation as it stands: a pre-registered variant that beats the submitted model on the split where such comparisons are allowed, with the numbers, the variance and the reason we did not act on it. That is a finding, not an embarrassment — and the honest place for it is the paper rather than a private note about what we would have done with another week.

17 August, Retracted Within the Hour: The Threefold DRENDS Win Was One Sequence Against Five

Everything in the first four paragraphs below is wrong. It is left standing because the way it was wrong is the useful part, and because the commit that carried it has been pushed. The retraction follows.

The reason we did not promote A2 was that it had been measured on SCARED-C only. So we measured it on DRENDS, the out-of-domain split, three seeds against three backbones. The result reverses cleanly along an axis we had not separated.

On SCARED, A2 is worse. Held-out D7 Bad1 reduction 5.84% against the canonical 5.15% — and broken out by backbone, A2 gives up 0.75 percentage points on the three the head was trained beside and 0.61 on the two it had never seen. Unseen *backbone* does not rescue it.

On DRENDS it is not close, and it is the same direction for every metric and every backbone:

Reduction on DRENDS	Canonical	A2
EPE	4.5–5.4%	17.4–18.5%
Bad1	4.7–5.6%	13.1–15.3%
Bad3	12–16%	59–67%
RMSE	6.5–7.0%	23–28%

Three to four times the reduction, out of domain, from the variant that loses in domain. The axis is not *unseen sequence* and not *unseen backbone* — A2 loses on both. It is *unseen domain.* The 64 frozen ResNet channels and the confidence machinery built on them let the head fit the appearance of ex-vivo porcine tissue; on synthetic colonoscopy that fit is a liability, and the variant that never had access to appearance has nothing to mislead it. Geometry and motion transfer. Appearance, even frozen appearance, does not.

One thing we caught before believing it. A2's seed spread on DRENDS is 0.001 percentage points, against 0.34 for the canonical model on SCARED. A spread that small usually means the same checkpoint was loaded three times. It was not: the three files hash differently and their provenance records carry seeds 0, 1 and 2. The spread is real and is itself informative — out of domain, three independently trained A2 heads converge to almost identical behaviour, which suggests what happens there is governed by the support and confinement machinery rather than by the learned weights.

And the comparison is still not matched. The seed study was run on SCARED only, so the canonical model has exactly one DRENDS number and A2 has three. Setting a three-seed mean against a one-seed baseline is precisely the error recorded two entries above, where canonical seed 0 turned out to be the weakest of the three and flattered every variant. The table above is therefore the seed-0 against seed-0 pairing, which is matched, and canonical seeds 1 and 2 are running on DRENDS now. The margin is large enough that the conclusion is unlikely to move, which is not a reason to skip the measurement.

Nothing in the paper has been changed on the strength of this. Promotion still means re-running every table, and the decision waits for the matched baseline.

Retraction. None of that survives. The reader who saw the table above and said it was impossible was right, and the reason is embarrassingly simple: **A2 had been evaluated on one DRENDS recording and the canonical model on five**. The comparison averaged Vid14_Pancreas_High alone against the mean of Vid10 through Vid14, and Vid14 is the easier of them. The threefold improvement was a statement about which sequences were in each average, and none of it was about the model.

Restricted to the one recording both sides actually cover, the two models are the same to three decimal places:

Bad1 on Vid14	Raw	Canonical	A2
CREStereo	0.499	0.473	0.473
Fast-FoundationStereo	0.487	0.465	0.465
RAFT-Stereo	0.538	0.512	0.511

Deltas of 0.001 percentage points, in both directions. Indistinguishable.

Two defects, and the second is the worse one. The first is in `drends_backbone_transfer.py`, which read

```
selected = recordings or ["Vid14_Pancreas_High"]
```

so a runner that omits `--recordings` silently evaluates a single sequence — and the report it writes still labels itself `complete_recordings`, because that field only distinguishes a frame cap from no frame cap. The canonical DRENDS run passed the flag; the A2 runner we wrote did not. The default is now the full five, and the constant carries the story.

The second defect is in the comparison script, and it is the one that allowed the first to matter. It *did* check that both sides shared identical raw predictions before believing any delta — and that check passed, because it ran on the intersection of the two sequence sets, which was one sequence, and on that sequence the raw predictions agreed perfectly. The check then handed off to a mean computed over each side's *own* sequences. A guard that verifies the overlap and then reports the union is not a guard. Both sides are now reduced over the shared set only, and any coverage mismatch prints a warning before the table.

The earlier interpretation — "frozen appearance features fit ex-vivo tissue and mislead out of domain" — was a tidy story invented to explain an artefact. It explained the artefact well. That is exactly what made it dangerous, and it is worth noting that the story arrived within seconds of the number and never once prompted a check of what the number was averaged over.

The A2 DRENDS evaluation is running again on all five recordings. The canonical seed study on DRENDS is cancelled: with a delta of 0.001 percentage points there is nothing for a seed baseline to resolve.

17 August, Afternoon: What A2 Actually Does Out of Domain

Seed 0 finished on all five DRENDS recordings against all three backbones, 7,476 frames per backbone, matched sequence for sequence against the canonical run. This is the comparison the morning's retracted table was pretending to be.

	Canonical		A2	
Reduction on DRENDS	Min	Max	Min	Max
EPE	4.52	5.42	4.75	5.74
Bad1	4.78	5.69	5.14	6.17
Bad3	11.14	16.17	11.61	17.04
RMSE	6.43	6.96	6.37	7.18

Across the twelve comparisons — three backbones by four metrics — A2 is ahead in eleven and behind in one, and every margin lies between 0.000 and 0.005 in absolute terms. That is a consistent direction at a magnitude nobody should care about.

The finding, stated the way it survives. A2 removes 157,504 parameters of frozen ResNet-18 from the deployed system, three forward passes per frame with it, and 104 of the 142 evidence channels. Out of domain it costs nothing: eleven of twelve margins favour it, none exceeds 0.005.

The sentence that stood here was backwards and is corrected in the evening entry below. It read that A2 "costs 0.69 percentage points of Bad1 reduction on held-out D7". These figures are relative *reductions*, so a larger number is a better result: canonical is 5.15% and A2 is 5.84% across three seeds. A2 does not pay 0.69 points in domain, it *gains* them, and it gains 4.44 on D2. Every conclusion drawn from the wrong sign is withdrawn with it.

A third defect, caught before it mattered. The comparison script was fixed at midday to reduce both sides over their shared sequences. That fix intersected across *every* seed present — so the moment seed 1 started writing its first recording, seed 0's finished five-sequence comparison silently collapsed to the one sequence seed 1 had reached. A guard that over-corrects into a new way of comparing unequal things is not fixed. Each seed is now intersected with canonical on its own, incomplete seeds are named in a warning and excluded from the spread, and the header carries n per seed.

Seeds 1 and 2 are still running and add A2's own spread, not a second baseline. Nothing in the paper has changed.

Eleven of twelve is not evidence, so we bootstrapped it. Counting how often A2 wins settles nothing: the twelve backbone-metric comparisons share five sequences and are not independent draws. So the difference was resampled at the level that actually varies. Sequences are the resampled unit, the paired difference is formed per sequence before resampling — which cancels sequence difficulty exactly, the variance that manufactured this morning's phantom — and all three backbones of a resampled sequence move together, preserving their correlation rather than averaging it away.

Metric	Mean difference	95% interval	P(A2 better)
EPE	−0.0042	[−0.0072, −0.0016]	1.000
Bad1	−0.0026	[−0.0041, −0.0013]	1.000
Bad3	−0.0004	[−0.0007, −0.0001]	0.992
RMSE	−0.0017	[−0.0062, +0.0026]	0.767

Three of four intervals exclude zero. The advantage is therefore real and detectable — and it is 0.004 pixels of EPE. Both halves of that sentence are load-bearing: this is what a genuine effect of no consequence looks like, and it is worth having a name for, because an interval that excludes zero is otherwise very easy to report as though it mattered.

One limit stated plainly: the interval covers variation across *sequences* only. The canonical side is a single training run and the A2 side currently pools one seed, so seed variance is not in it — and A2's seed-to-seed spread on DRENDS is of the same order as the differences above. The reading stays *detectable and negligible*, not *better*. The script prints that caveat with its own output so the number cannot travel without it.

17 August, Evening: The Third Seed Removes the Effect

The caveat above was the right caveat and it was still too generous. Seeds 1 and 2 finished, all nine runs at five recordings, and pooling them moves every interval across zero:

Metric	Mean difference	95% interval	P(A2 better)
EPE	−0.0028	[−0.0056, +0.0002]	0.962
Bad1	−0.0015	[−0.0032, +0.0001]	0.971
Bad3	−0.0002	[−0.0005, +0.0000]	0.930
RMSE	−0.0031	[−0.0059, +0.0009]	0.942

Three of four intervals excluded zero on seed 0, three of four still excluded it with two seeds pooled, and *none* of four excludes it with three. The per-seed numbers say why, and they are monotone:

Bad1, RAFT-Stereo	Canonical	A2 seed 0	seed 1	seed 2
DRENDS	0.555	0.552	0.553	0.555

A2 seed 2 lands exactly on the canonical value, on EPE as well (1.241 against 1.241). The advantage was a property of seed 0, and the earlier sentence — "the advantage is therefore real and detectable" — does not survive. The finding is now the flat one: out of domain, A2 and the canonical model are **indistinguishable**.

What this does not need. The canonical DRENDS seed study was cancelled earlier today on the reasoning that a 0.001 point difference left a seed baseline nothing to resolve. That reasoning was based on the one-sequence numbers and was wrong at the time: A2's real seed spread here is 0.003, the same size as the gap it was being compared against. The conclusion survives the error only because it runs the safe way — adding canonical seed variance widens the intervals, and every interval already includes zero. Had the three seeds gone the other way, the missing baseline would have been load-bearing.

17 August, Late: A Sign Error, and the Conclusion It Was Hiding

Every in-domain statement written today had the sign backwards. The ablation table reports *relative reduction*, computed as 100 (raw − refined)/raw, so a larger number is a better result. Read correctly:

Bad1 reduction (%)	Canonical	A2
D2 development	8.40	**12.84**
D7 held out	5.15 ± 0.34	$\mathbf{5.84 \pm 0.22}$
DRENDS	indistinguishable	

A2 does not lose in domain. It *wins* in domain, by 0.69 points on held-out D7 and by 4.44 on D2, and it is neutral out of domain. Three commits and two diary paragraphs today asserted the opposite, including the phrases "A2 loses in domain" and "more than a hundred times the largest DRENDS margin". All of it is withdrawn.

What was actually being measured. With the sign fixed, the result stops being about domains and becomes a statement about the architecture. The 64 frozen ResNet-18 channels, the confidence curves built from them and both appearance correlations — 104 of 142 channels, 157,504 frozen parameters, three forward passes per frame — buy accuracy *nowhere*. In domain the model is better without them. Out of domain it is unchanged.

The pre-registration allowed for exactly this and named the consequence before any of it ran: if the appearance channels match within the seed spread they "are redundant given their own correlations and should be dropped from the architecture description".

The uncomfortable version. A2 is not alone. On held-out D7, A3 (single resolution) reaches 5.85 and A4 (relaxed convexity) 6.02, both above canonical's 5.15; only A1 is below, at 4.88. Every simplification we pre-registered does at least as well as the model we designed. The narrow reading is that the frozen extractor is unnecessary. The wider one is that the canonical head is over-specified and its parts are not earning their places — and that is a finding about our design process, not about temporal refinement.

Two limits keep this from being stronger than it is. Each ablation is a *single* seed against a three-seed canonical, and A2's +0.55 on one seed sits only modestly above the 0.34 canonical spread. And the D2 margin, being the selection split, is exactly where a variant is most flattered.

The whole day in one line. A2 removes the frozen ResNet-18 from the runtime, is better in domain and indistinguishable out of it. The out-of-domain claim went threefold better, then detectably better, then neither; the in-domain claim went the other way, from a loss to a gain, because of a sign. Each revision came from checking something that had been assumed — and the sign was assumed longest, because it was the half nobody was arguing about.